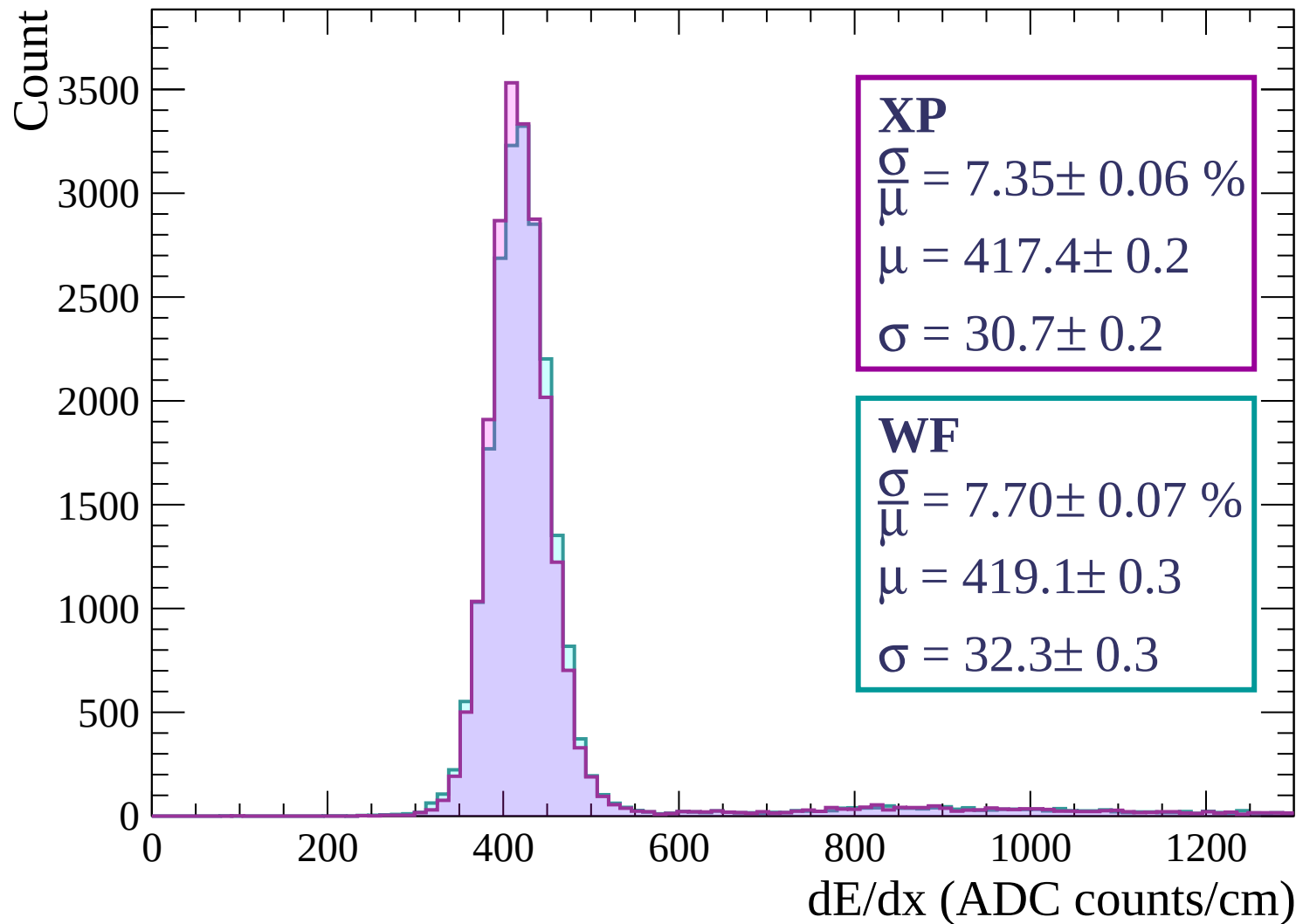
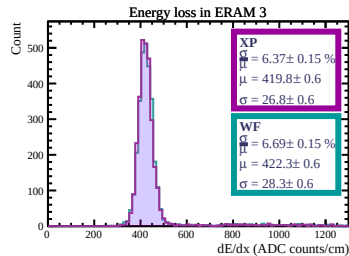
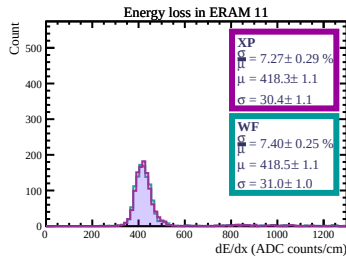
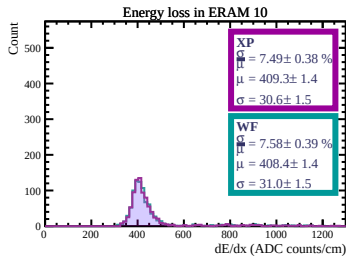
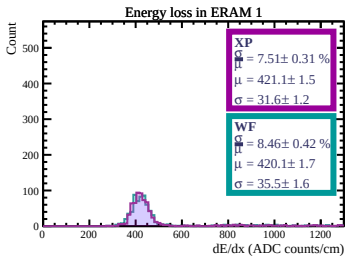
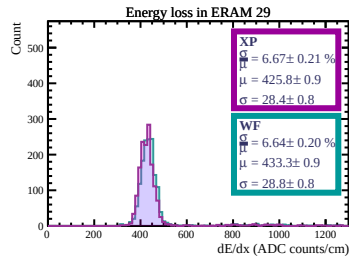
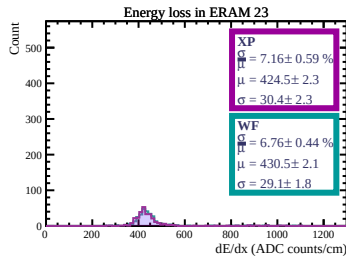
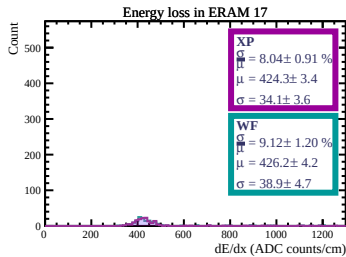
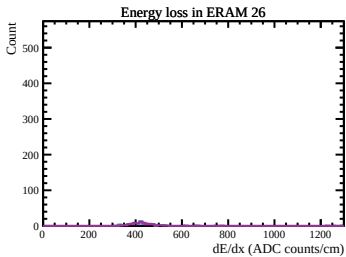
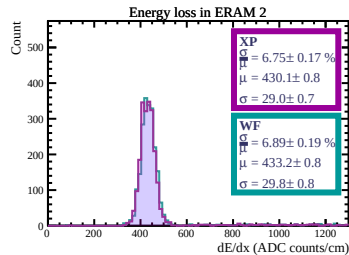
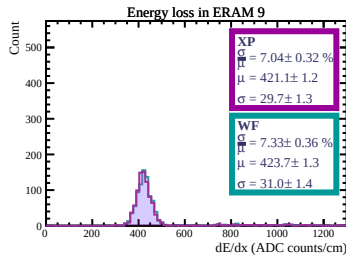
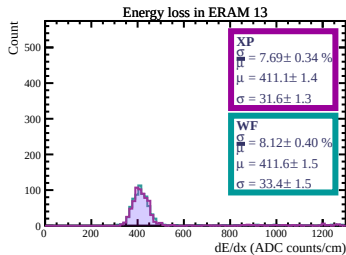
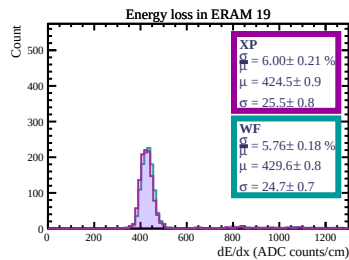
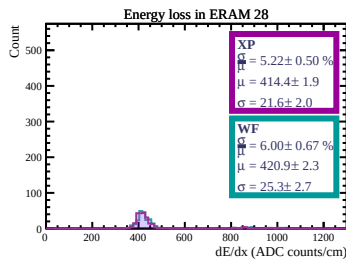
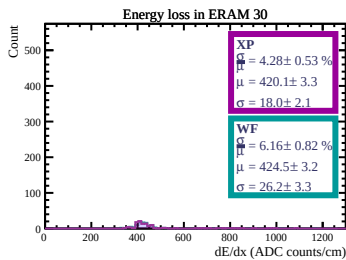
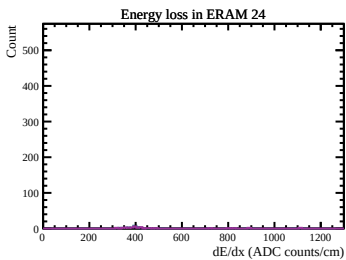
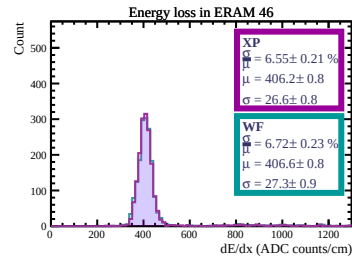
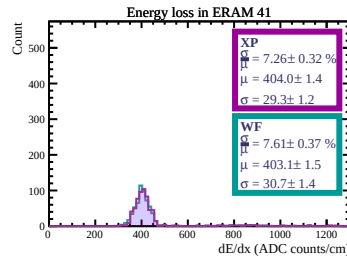
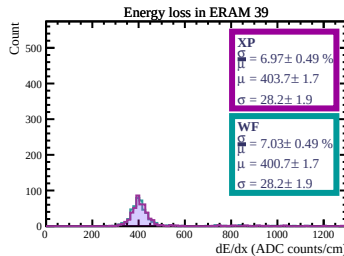
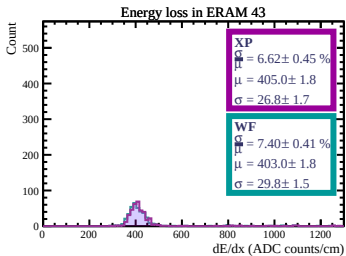
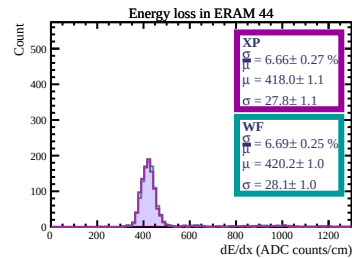
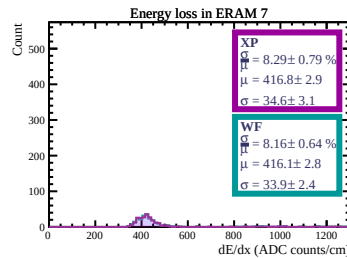
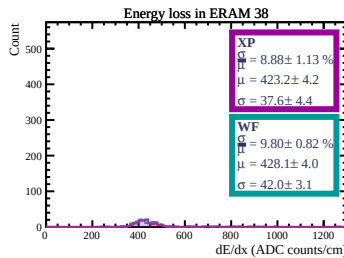
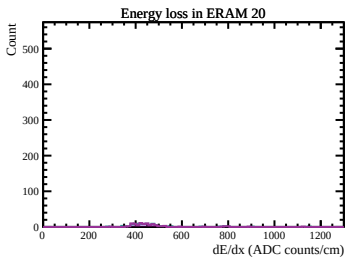
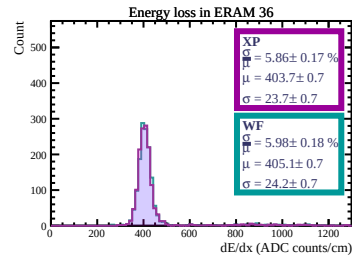
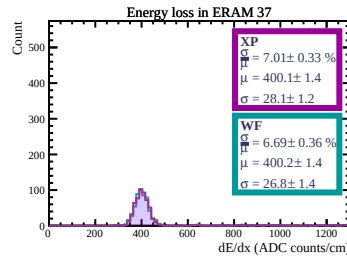
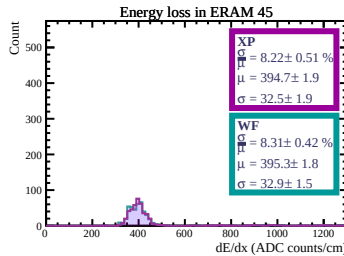
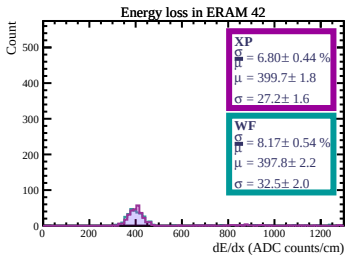
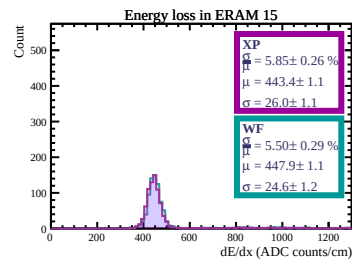
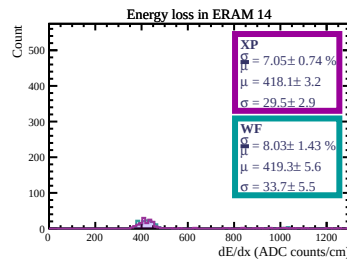
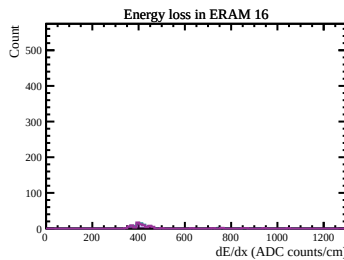
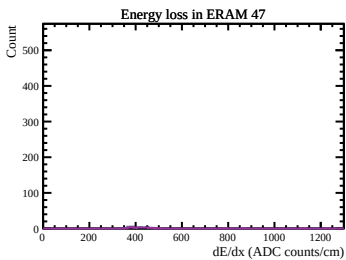
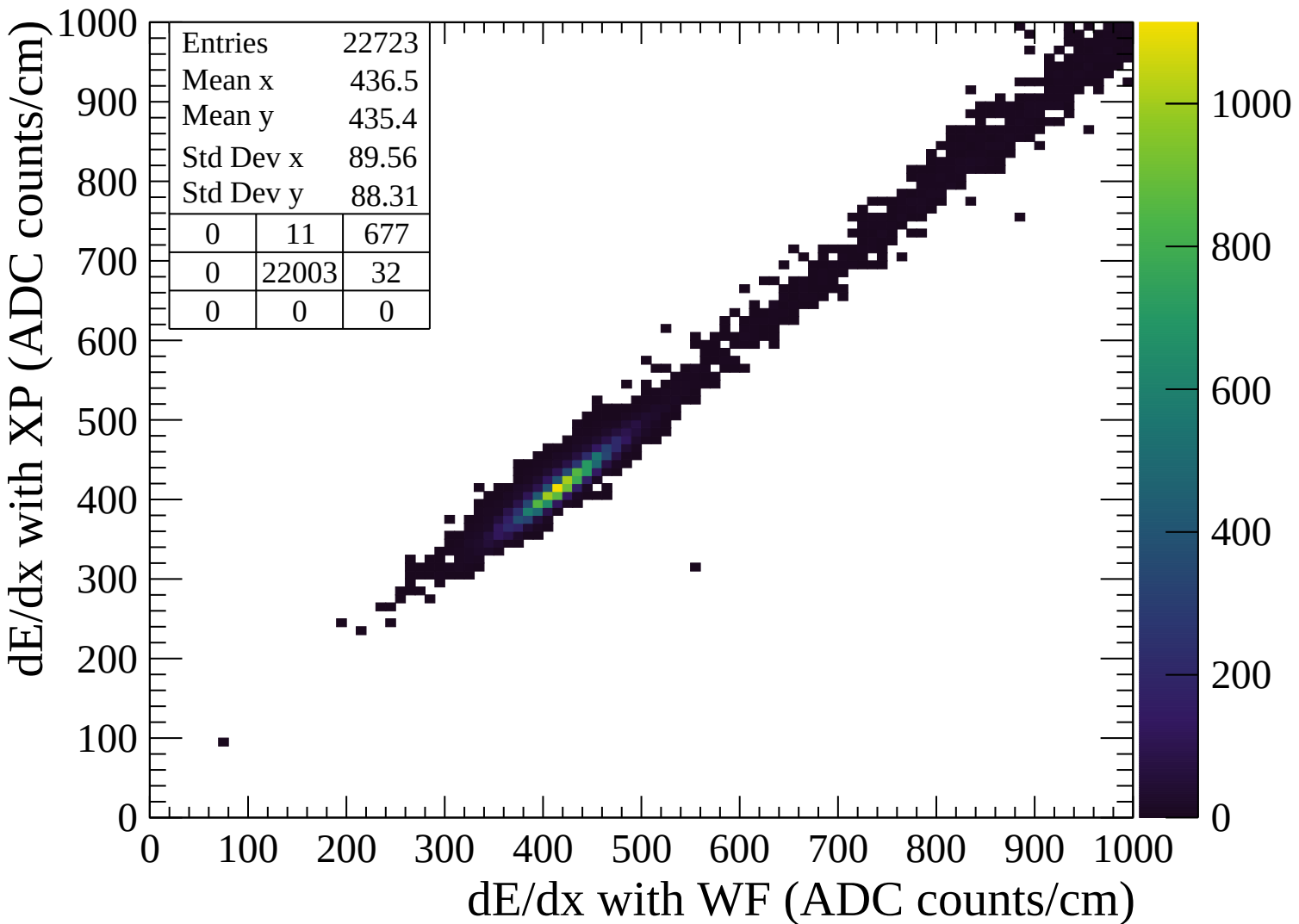
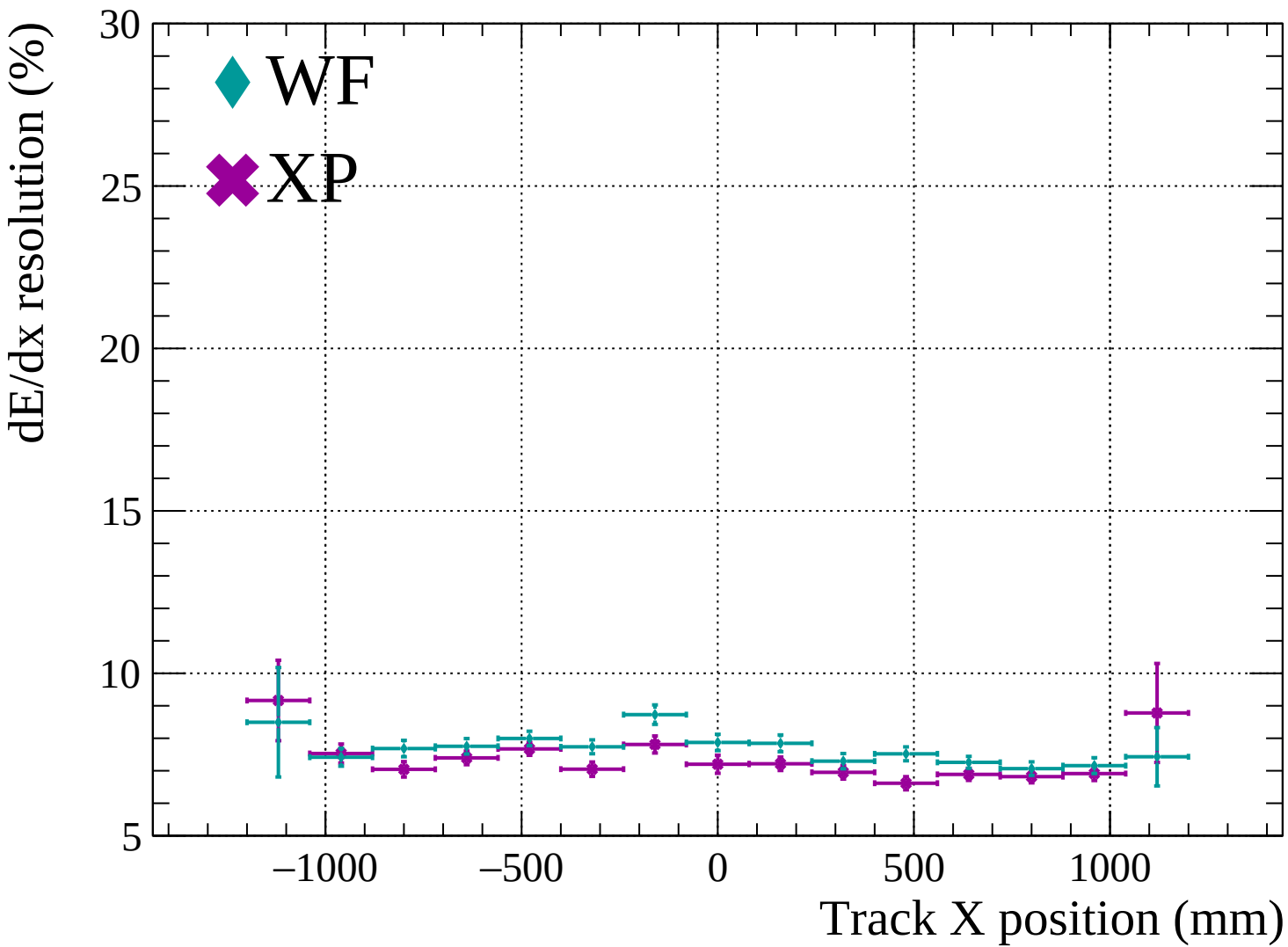


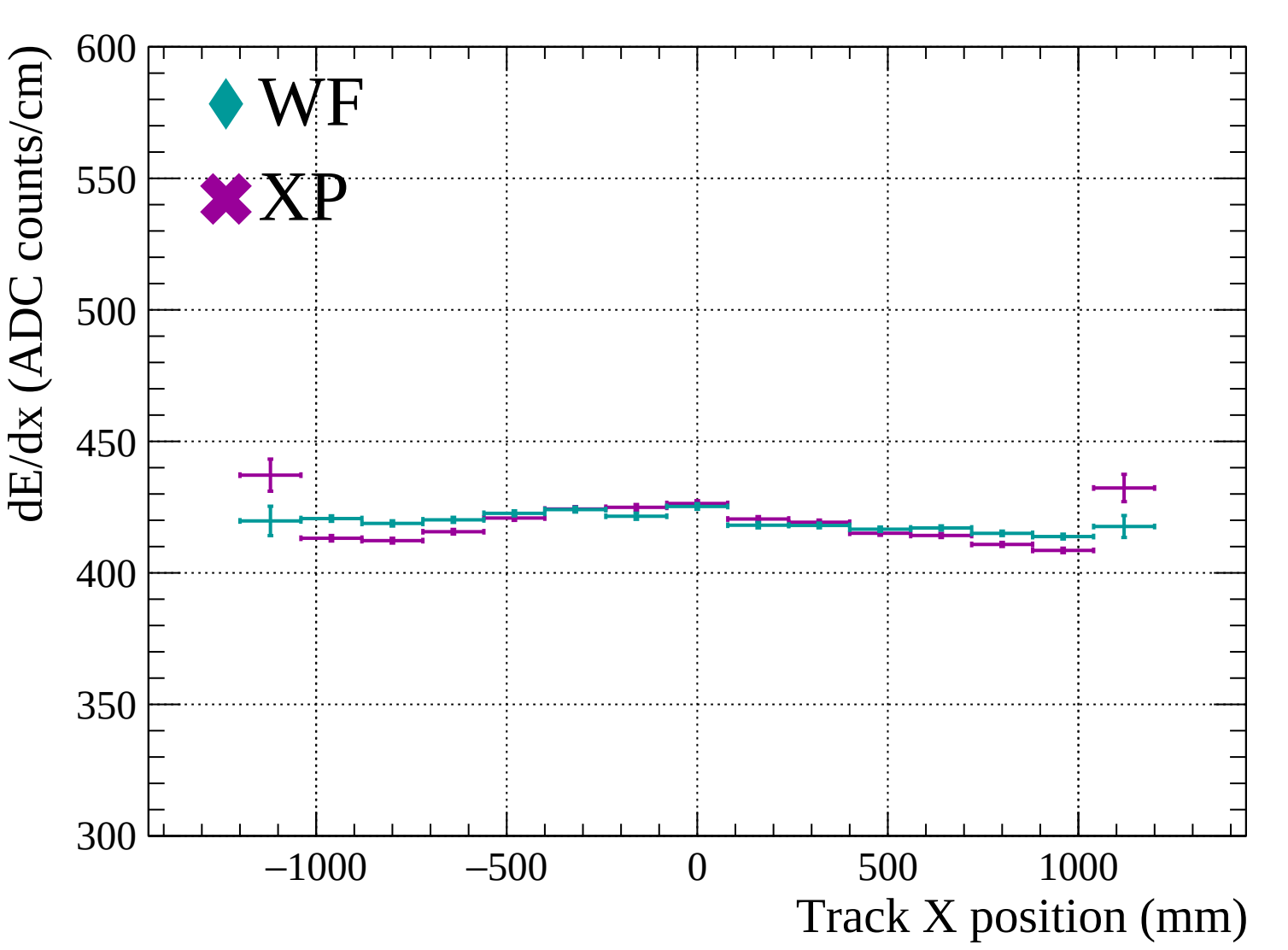


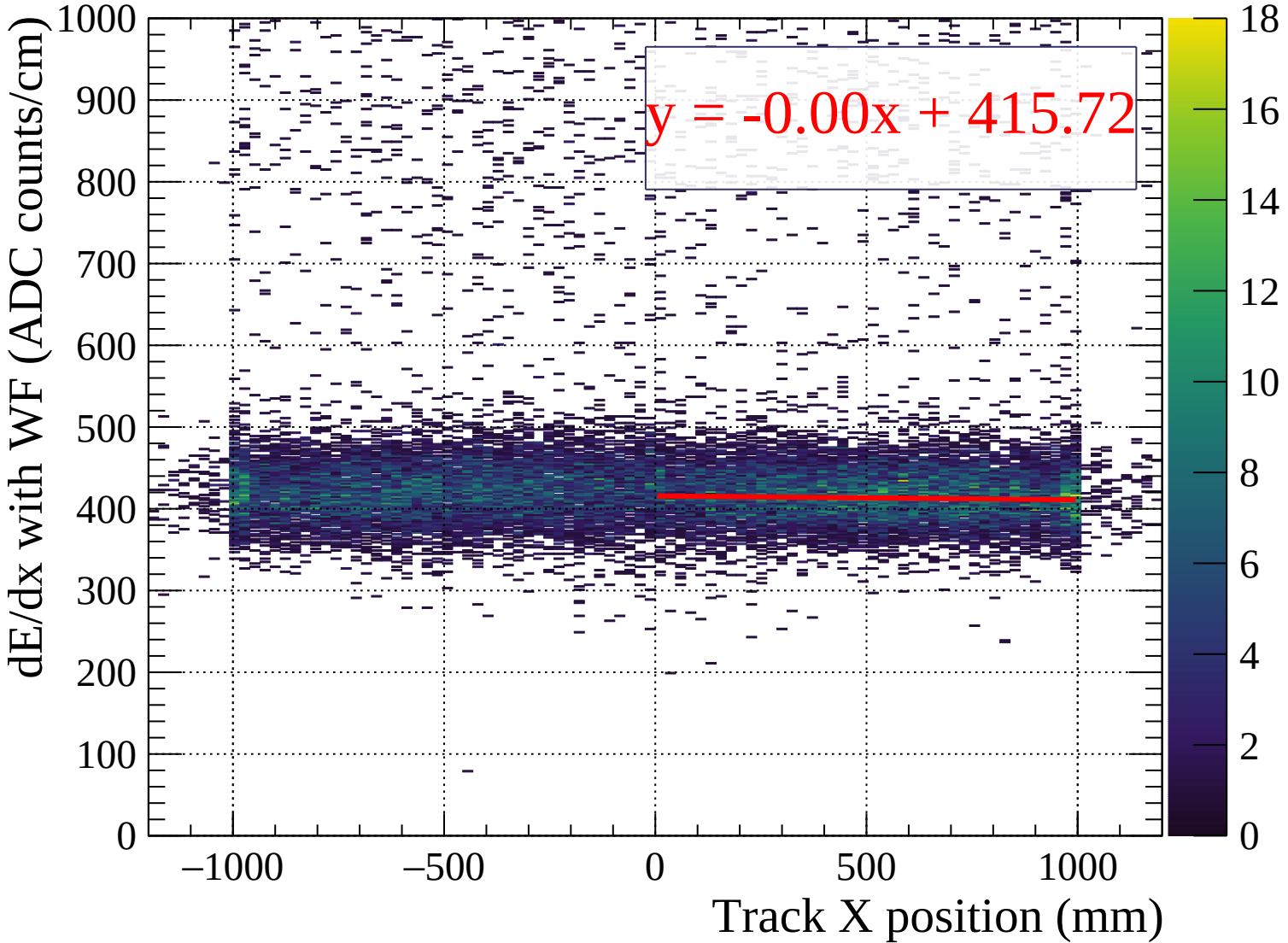


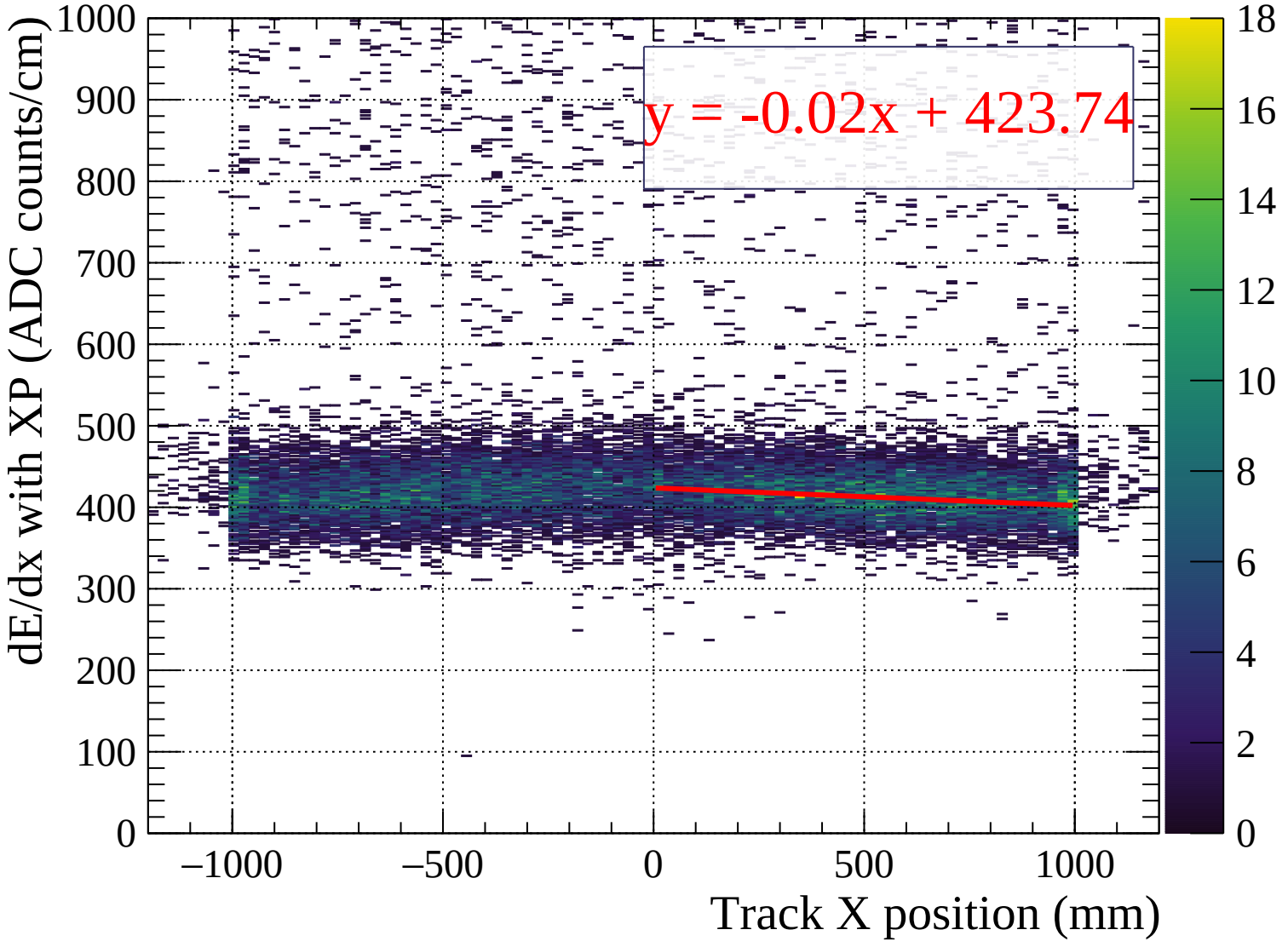


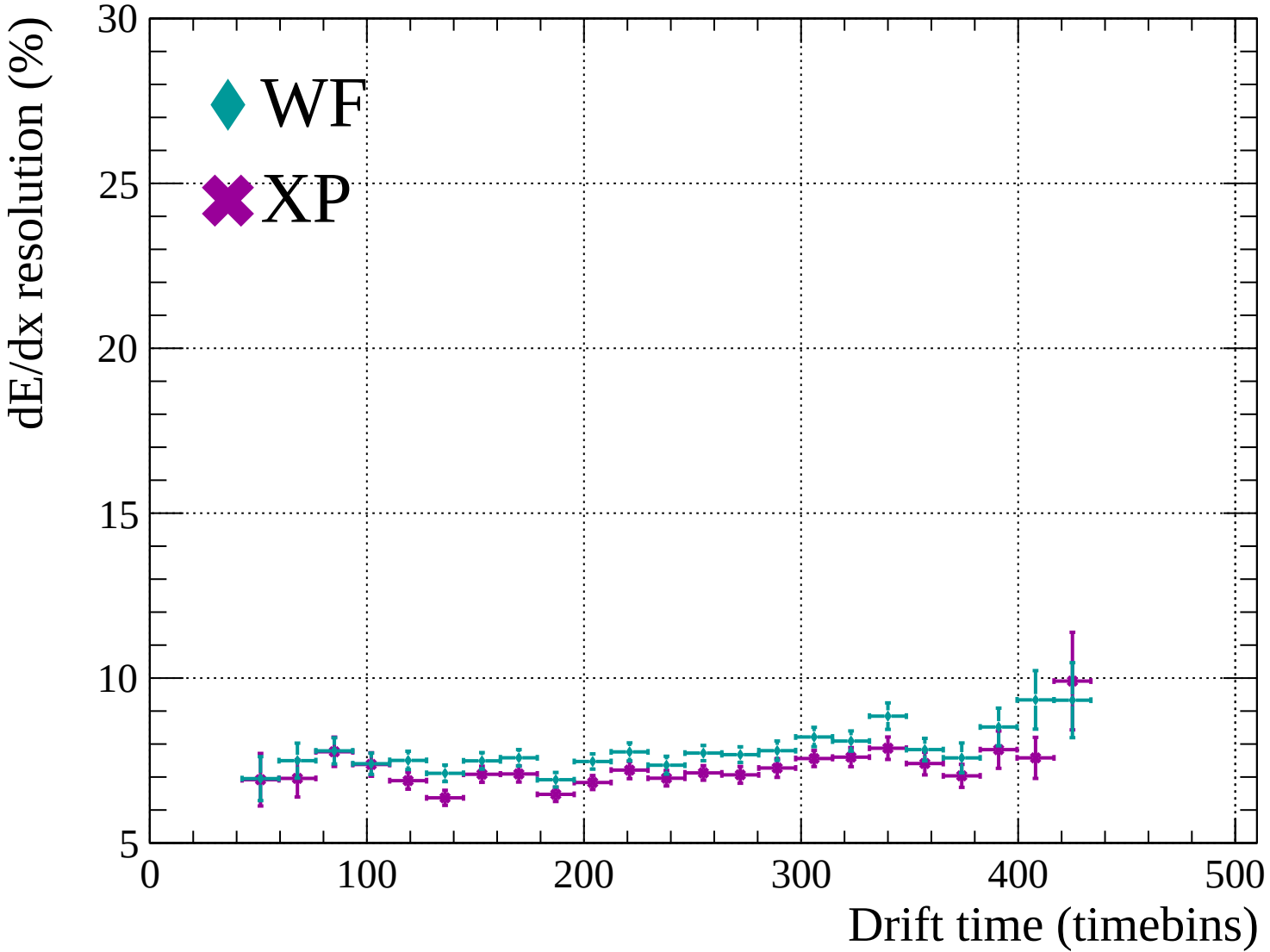


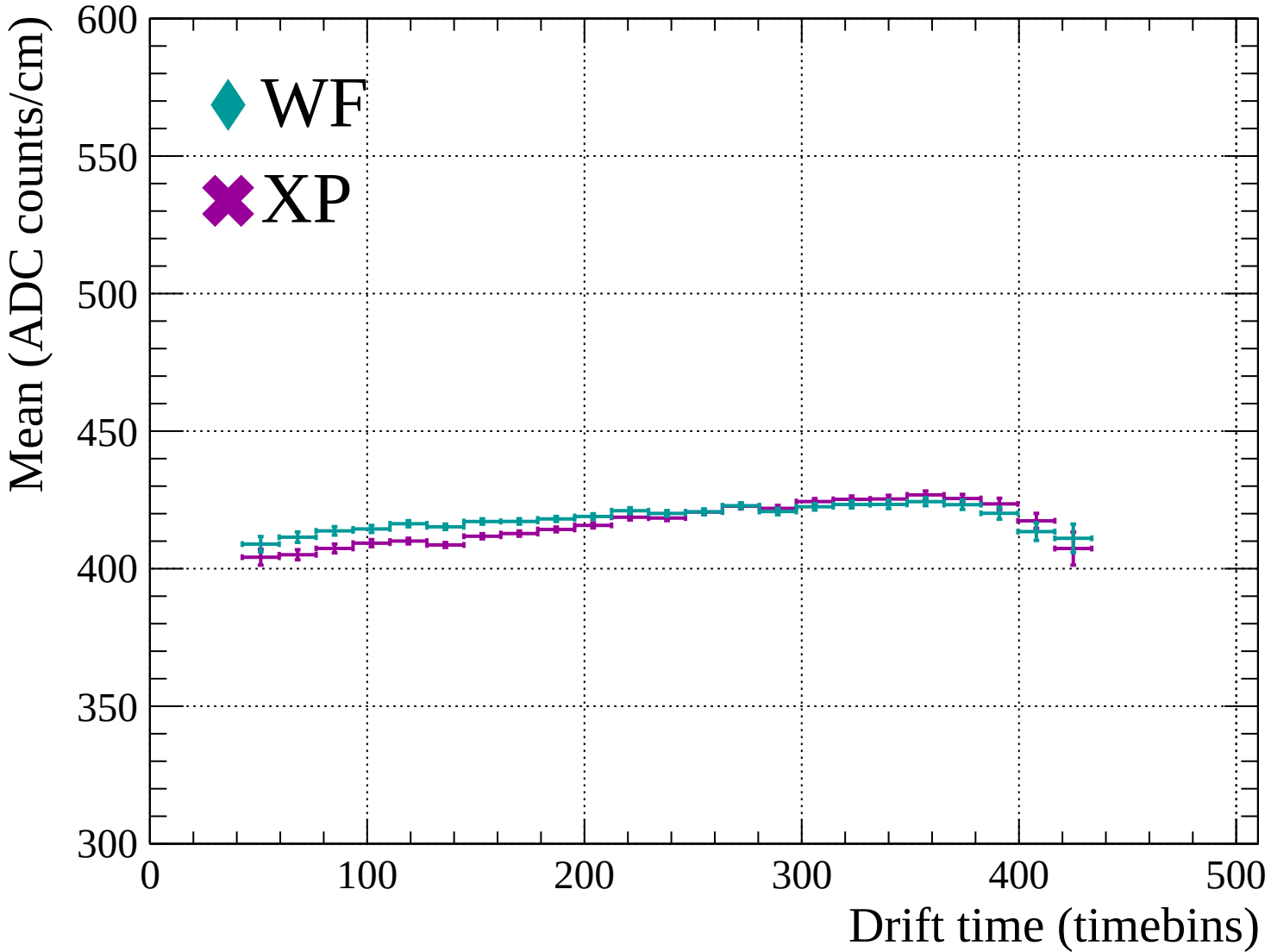


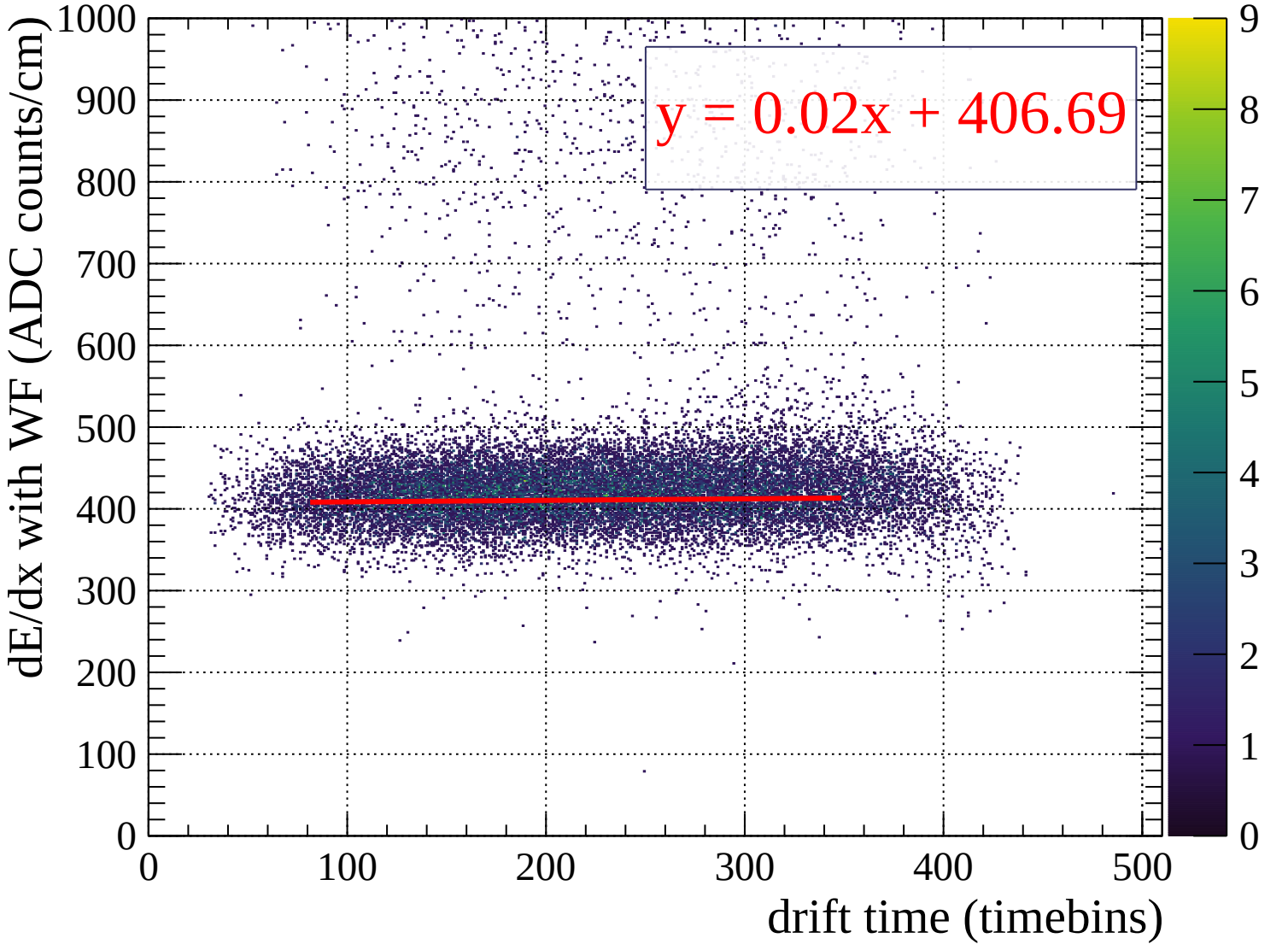


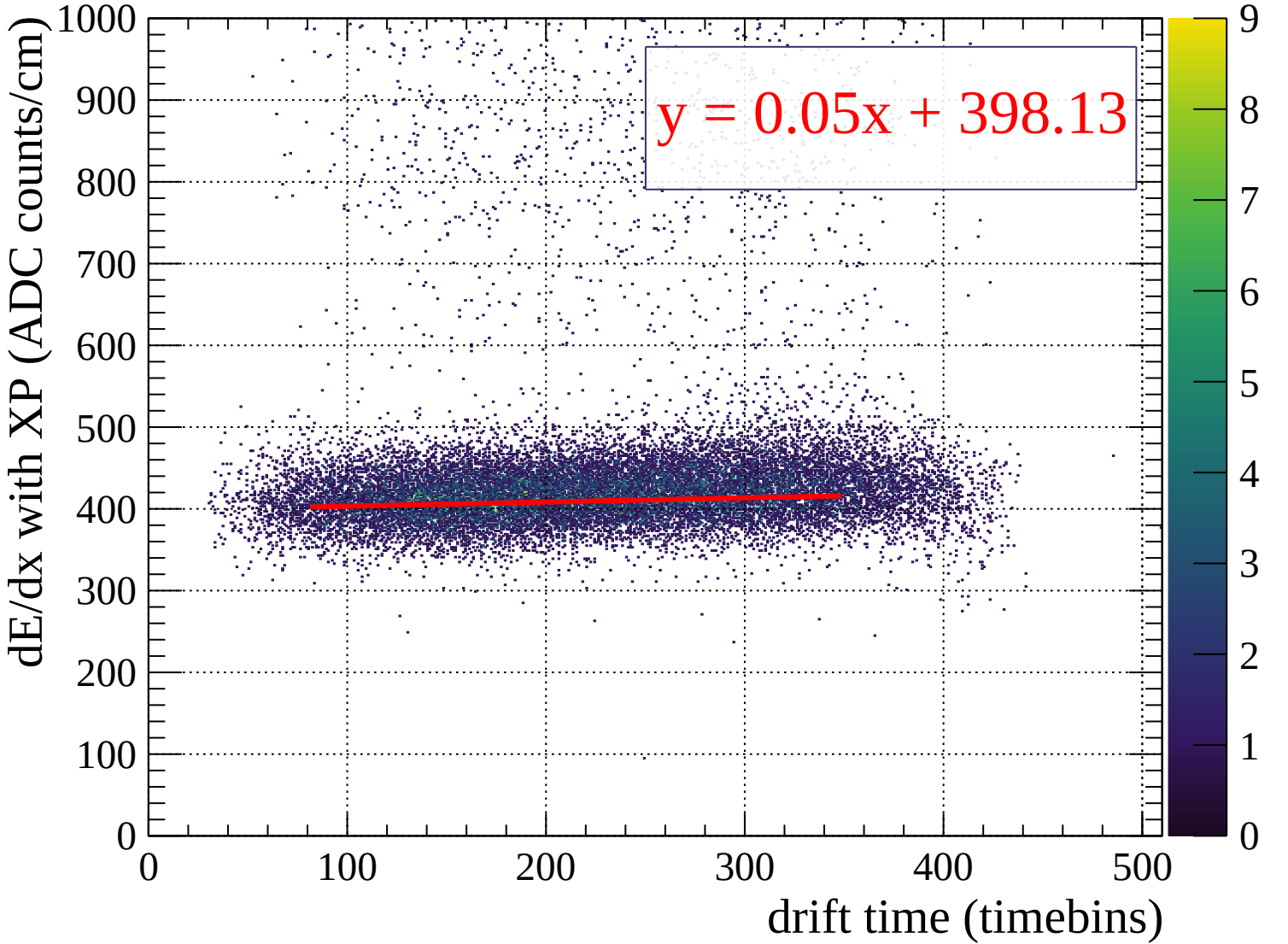


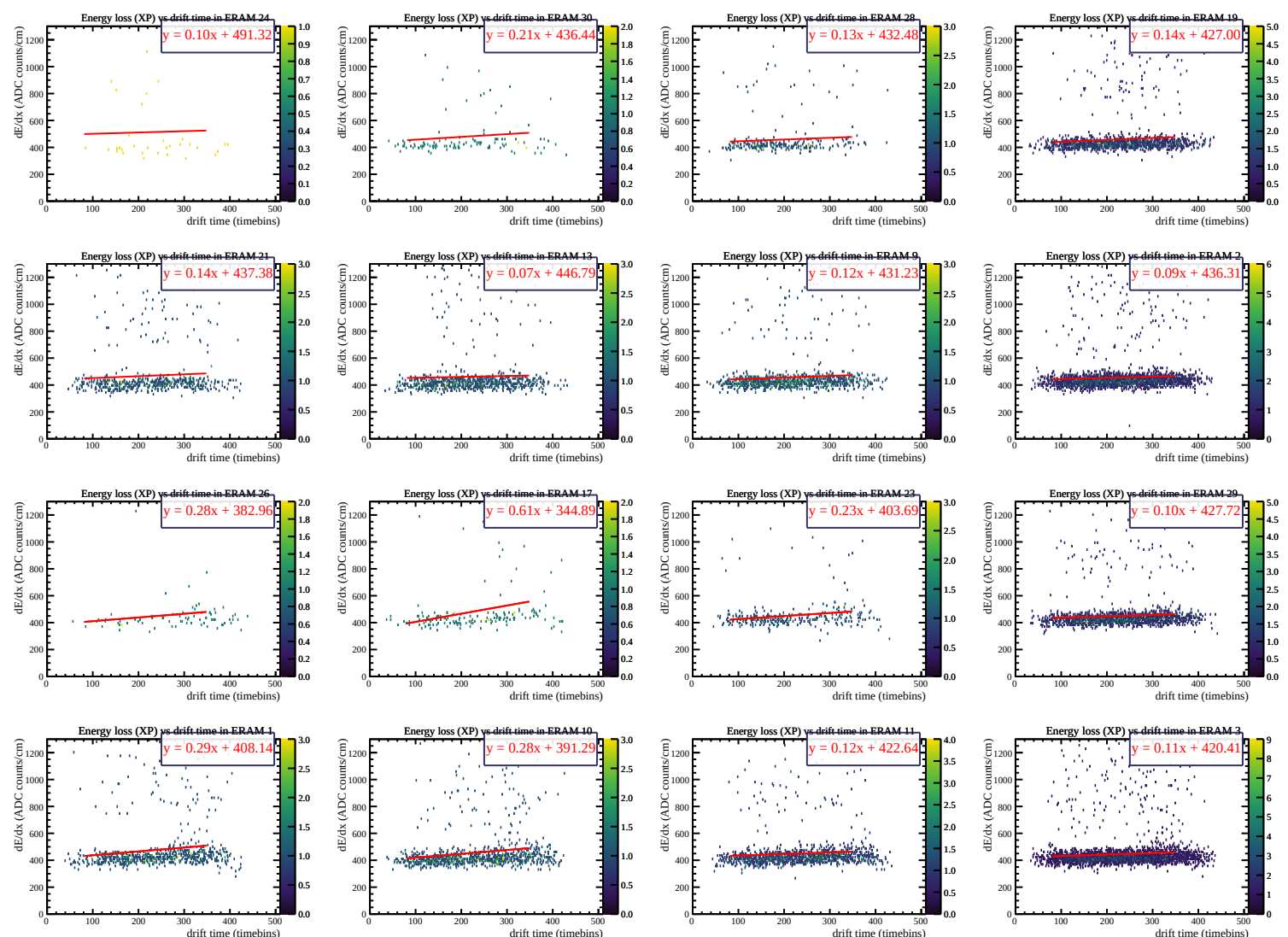


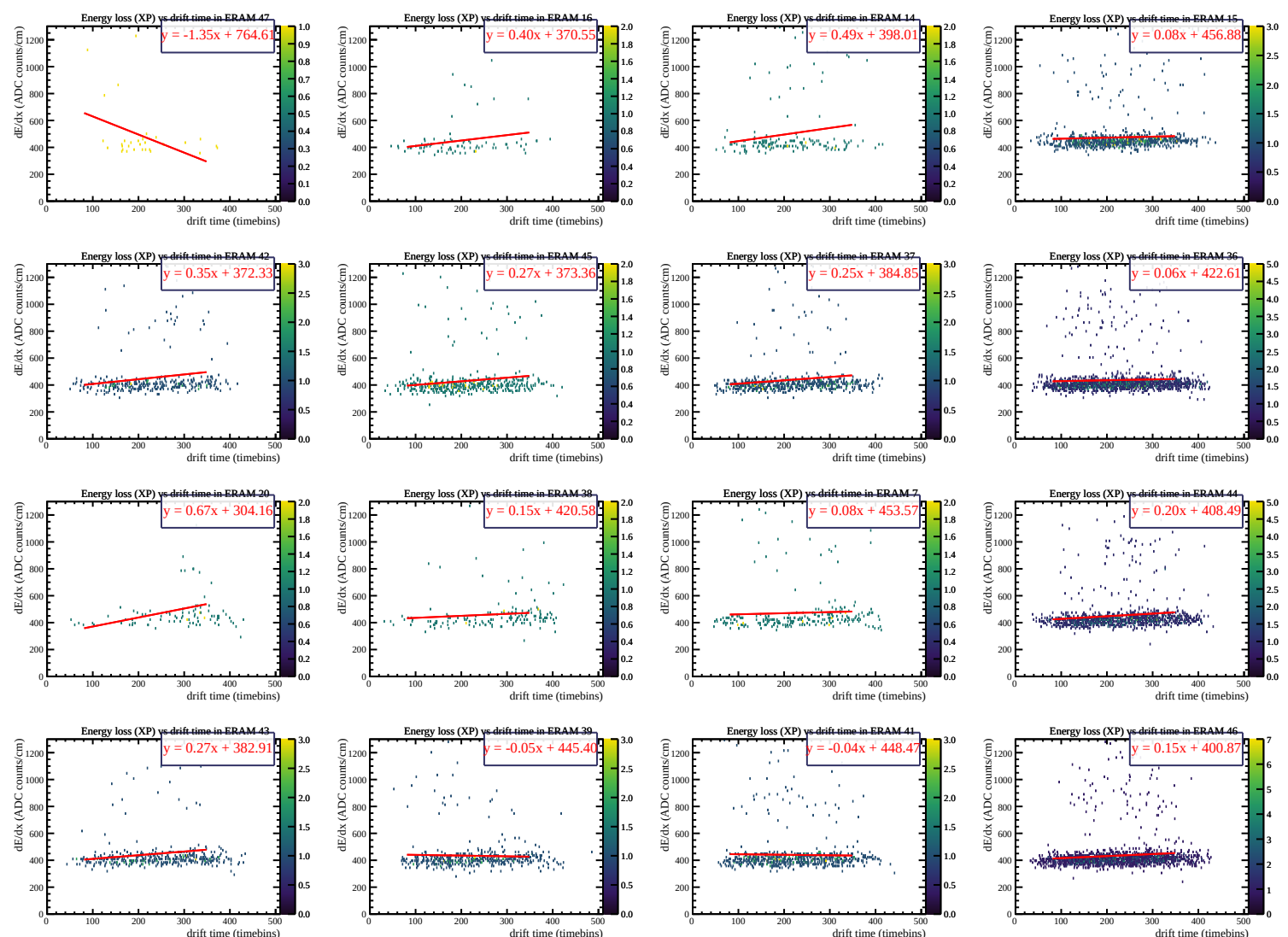


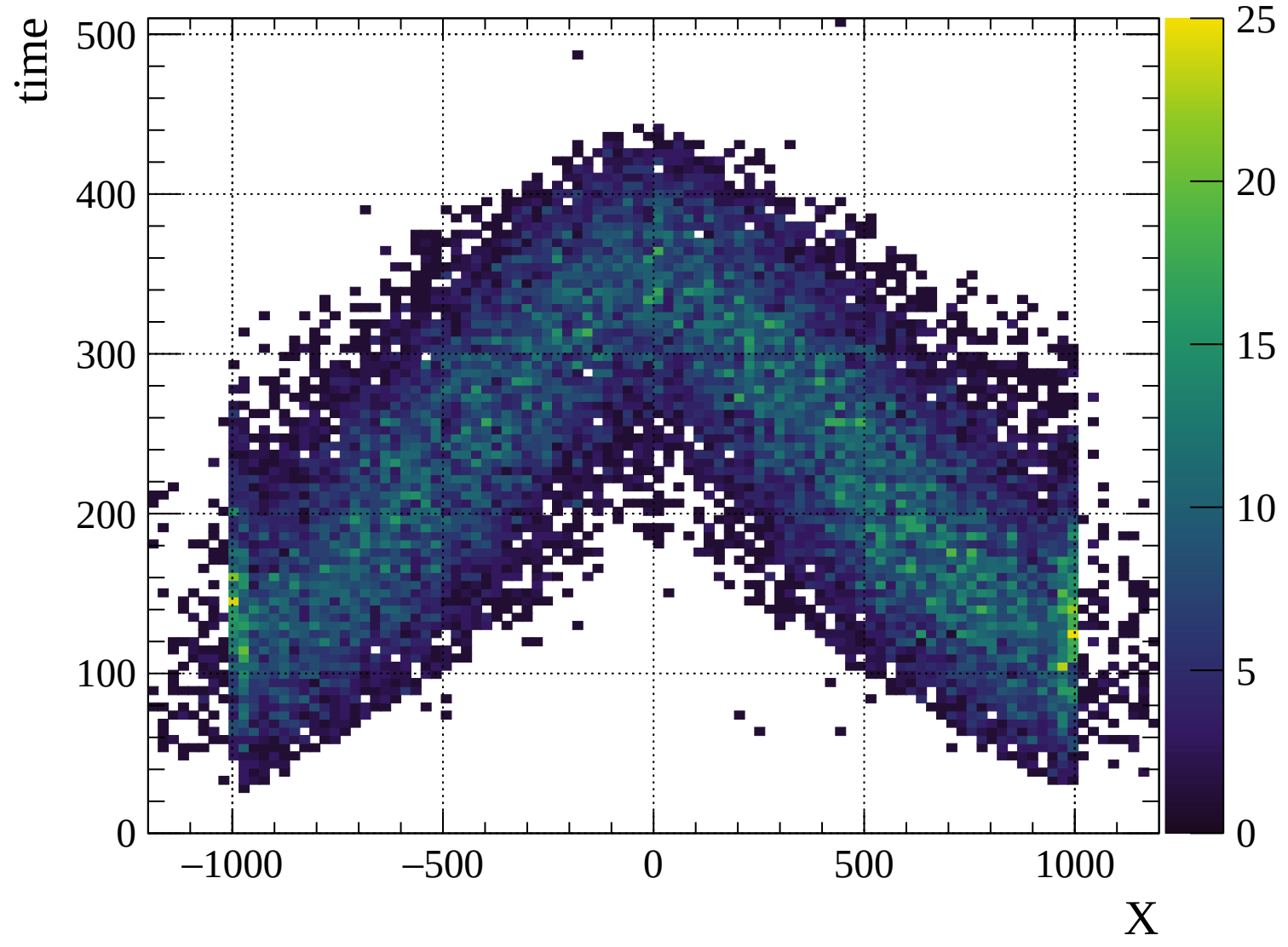


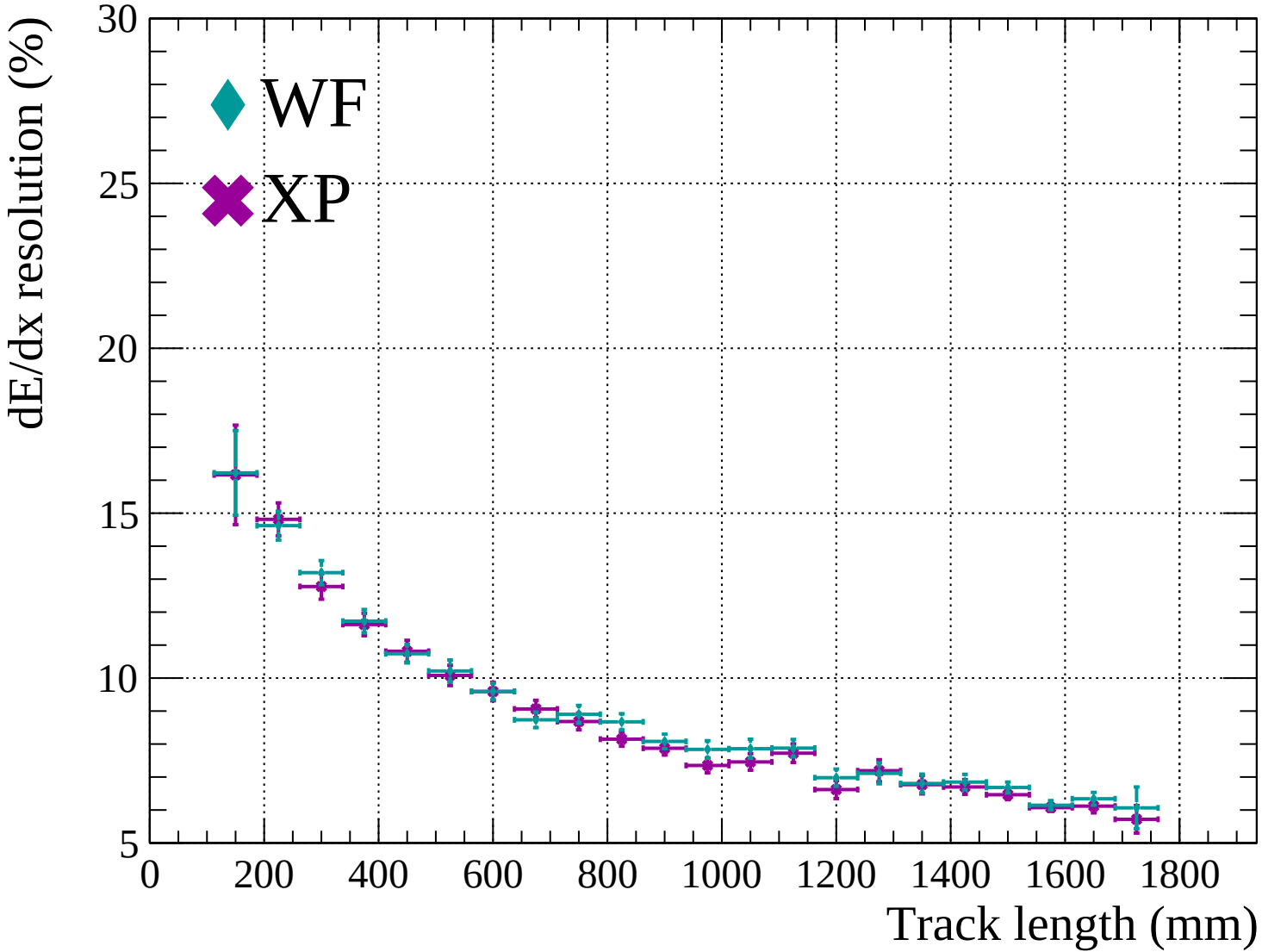


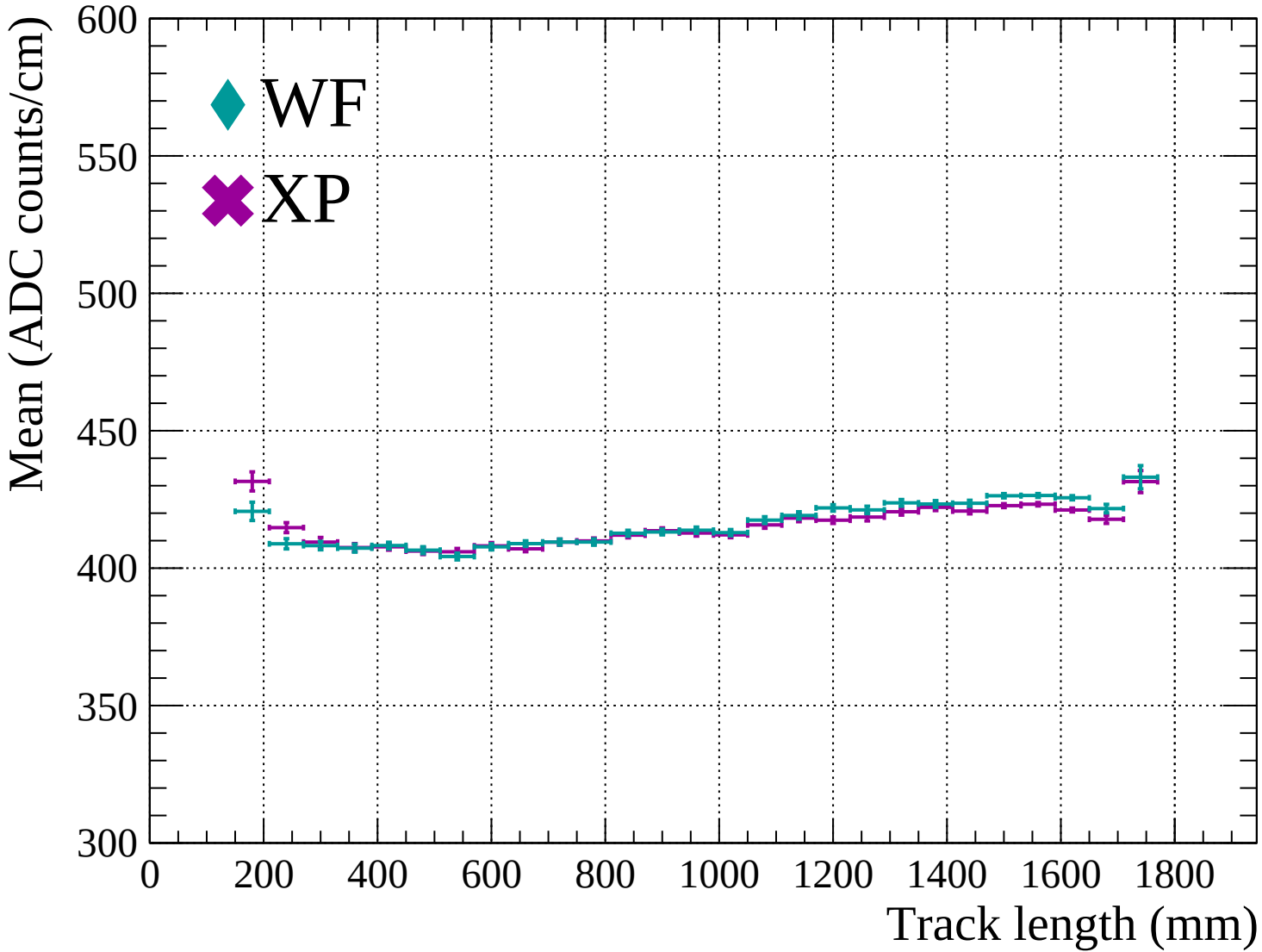


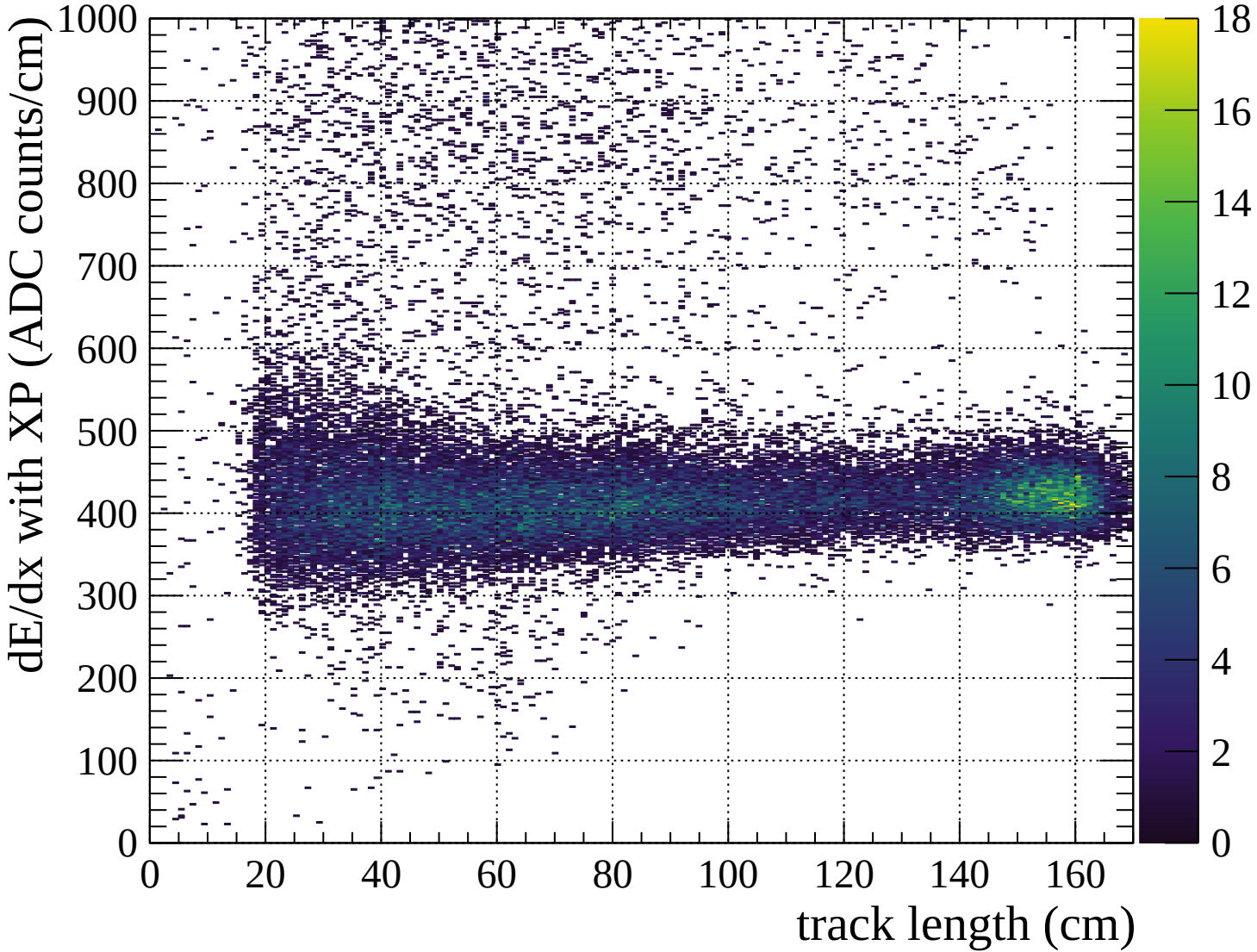


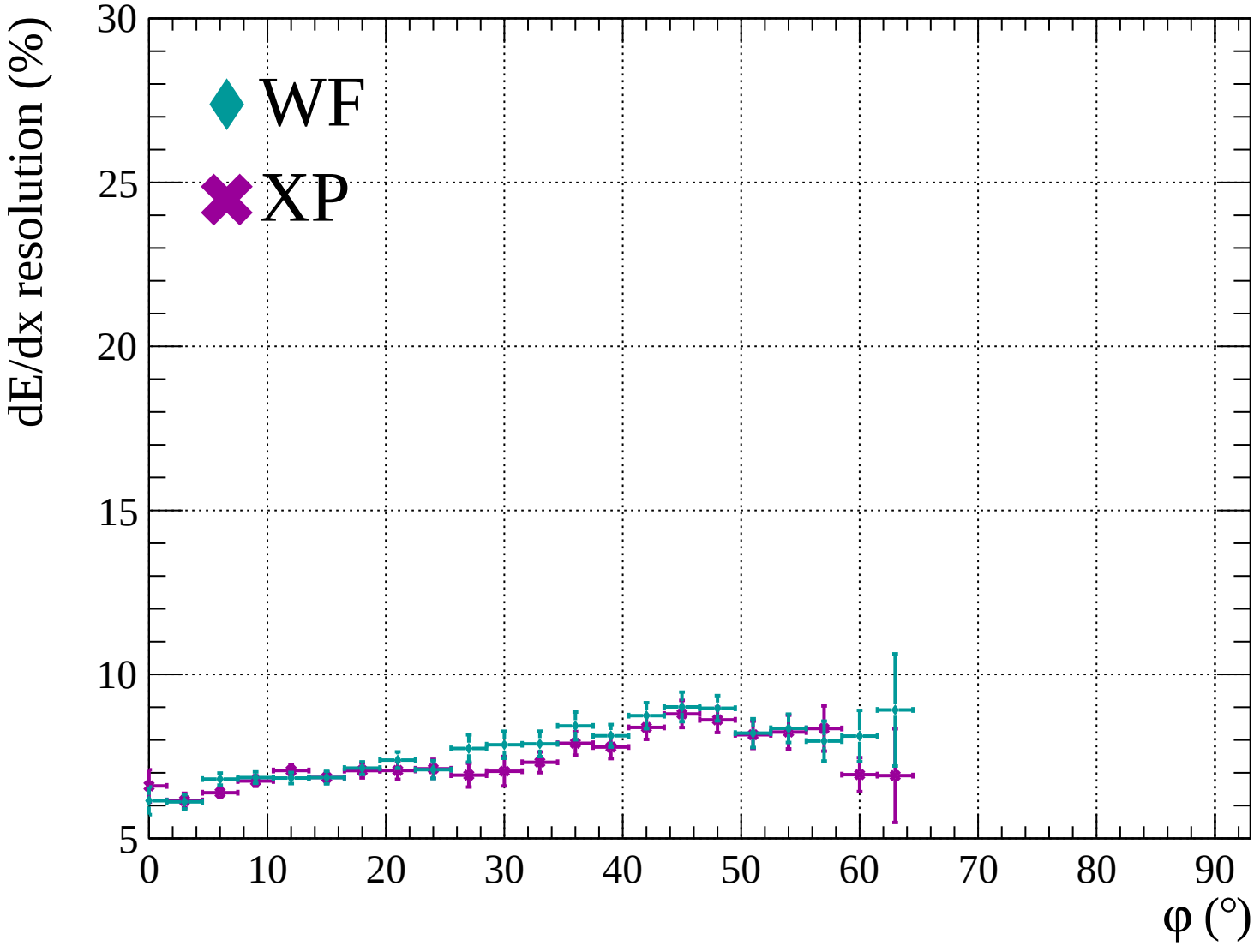


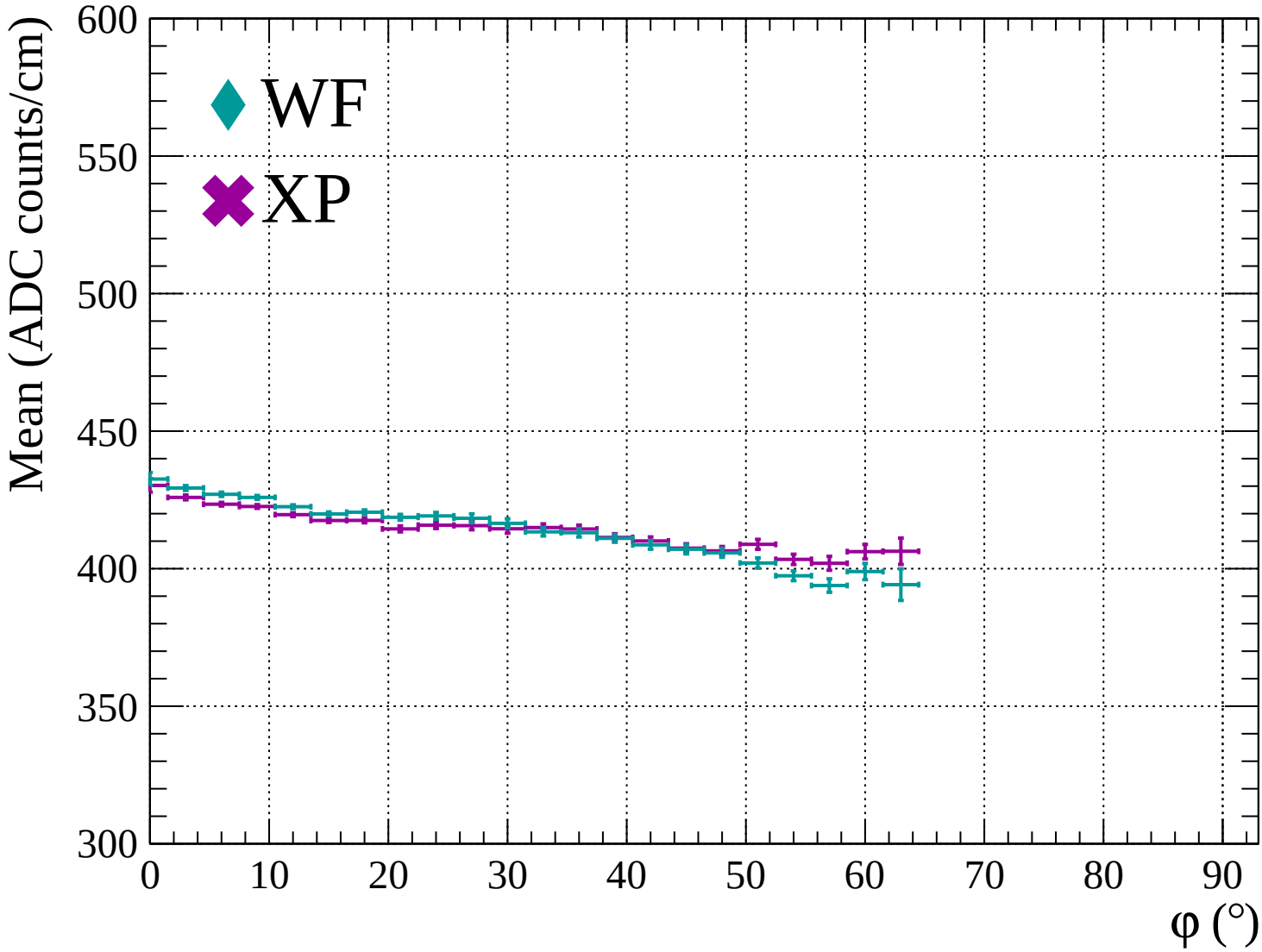


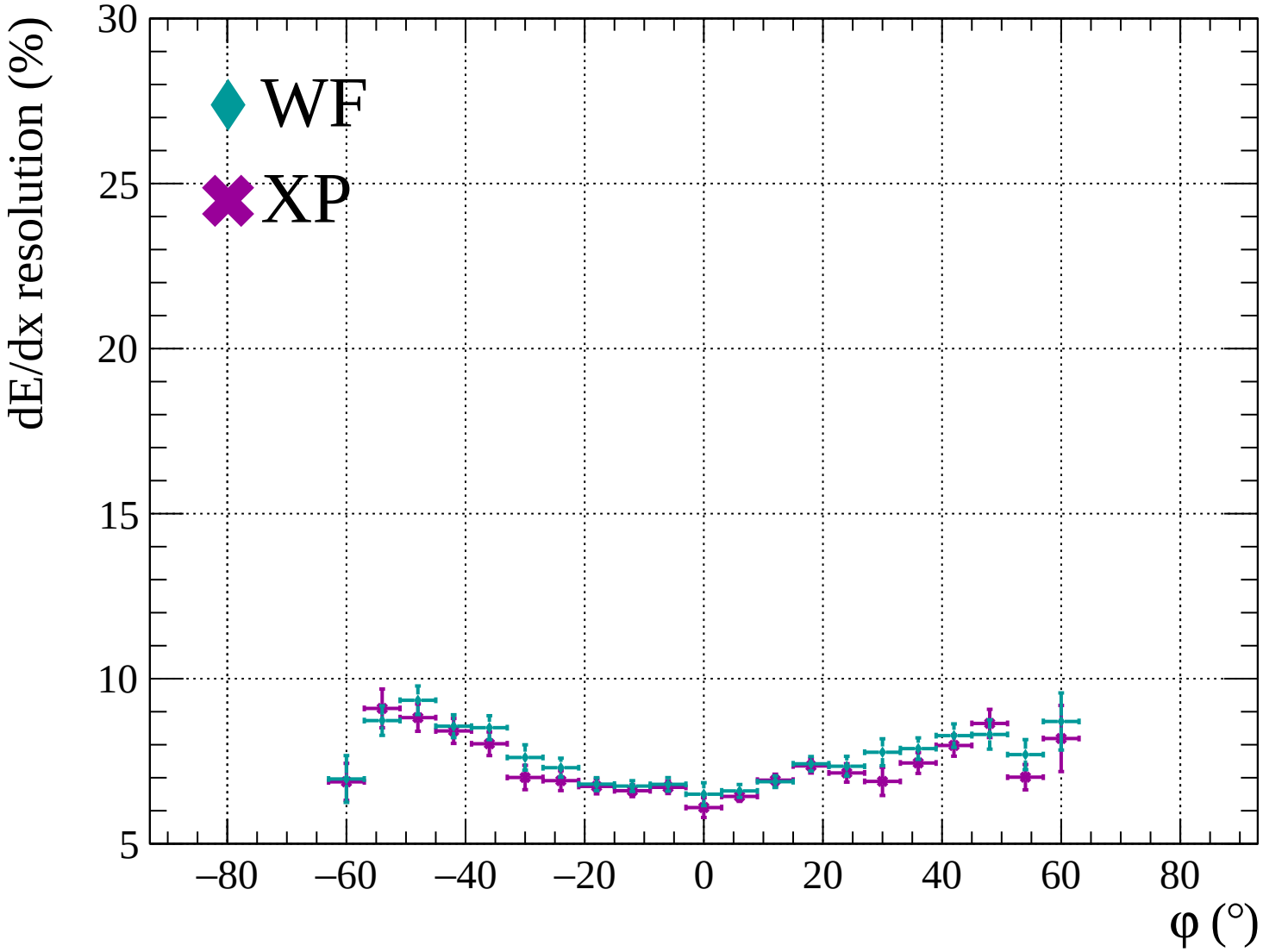


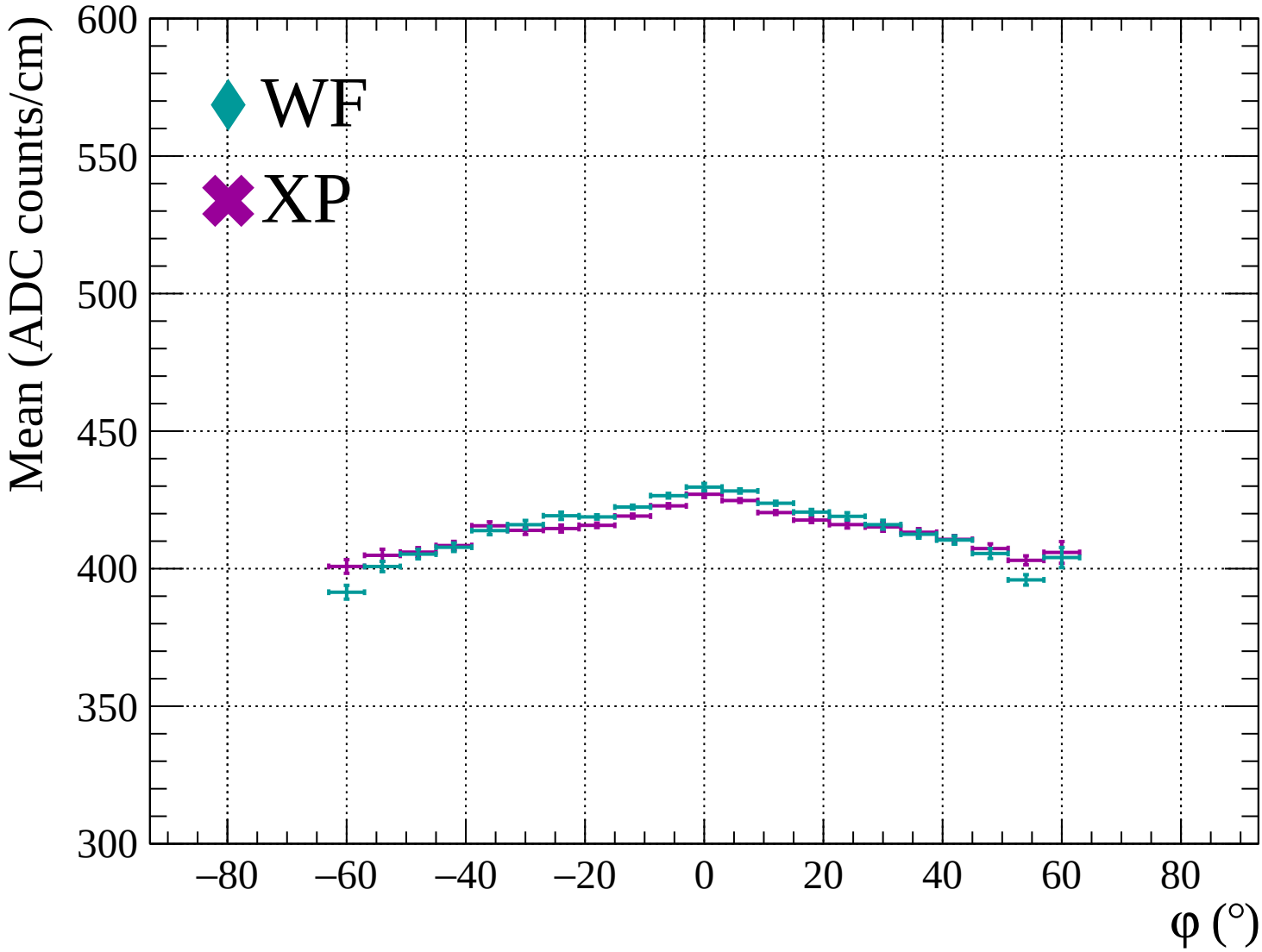


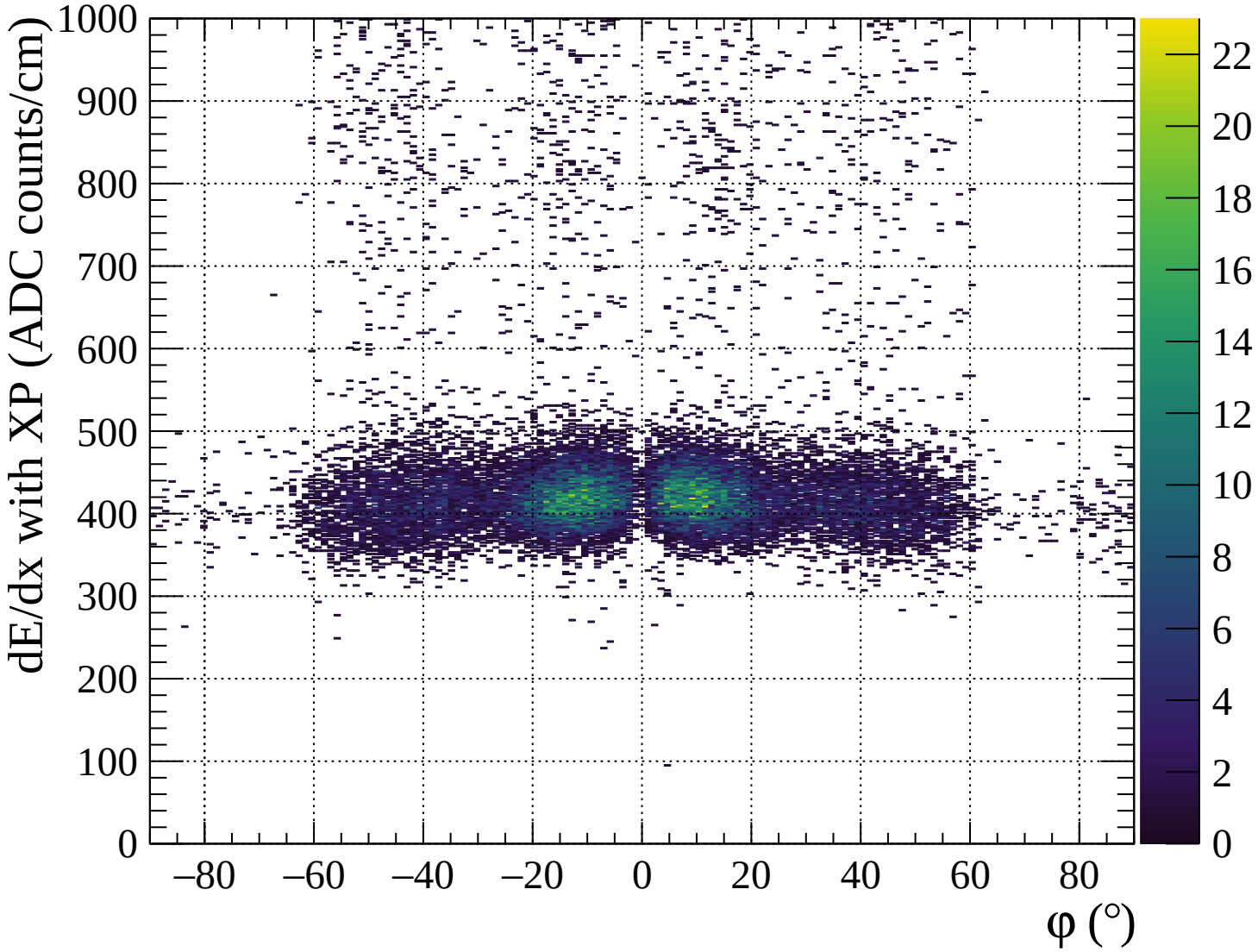


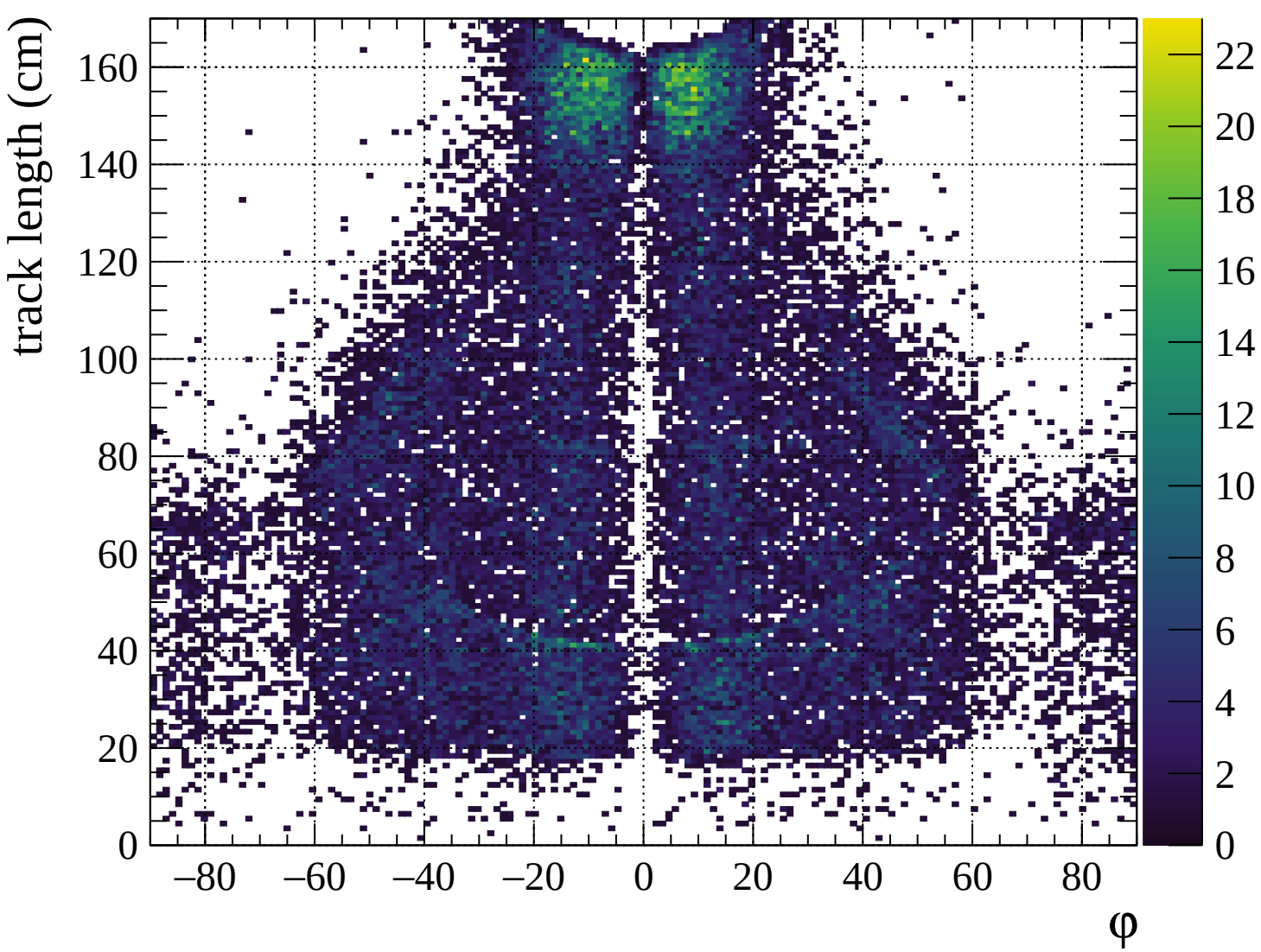


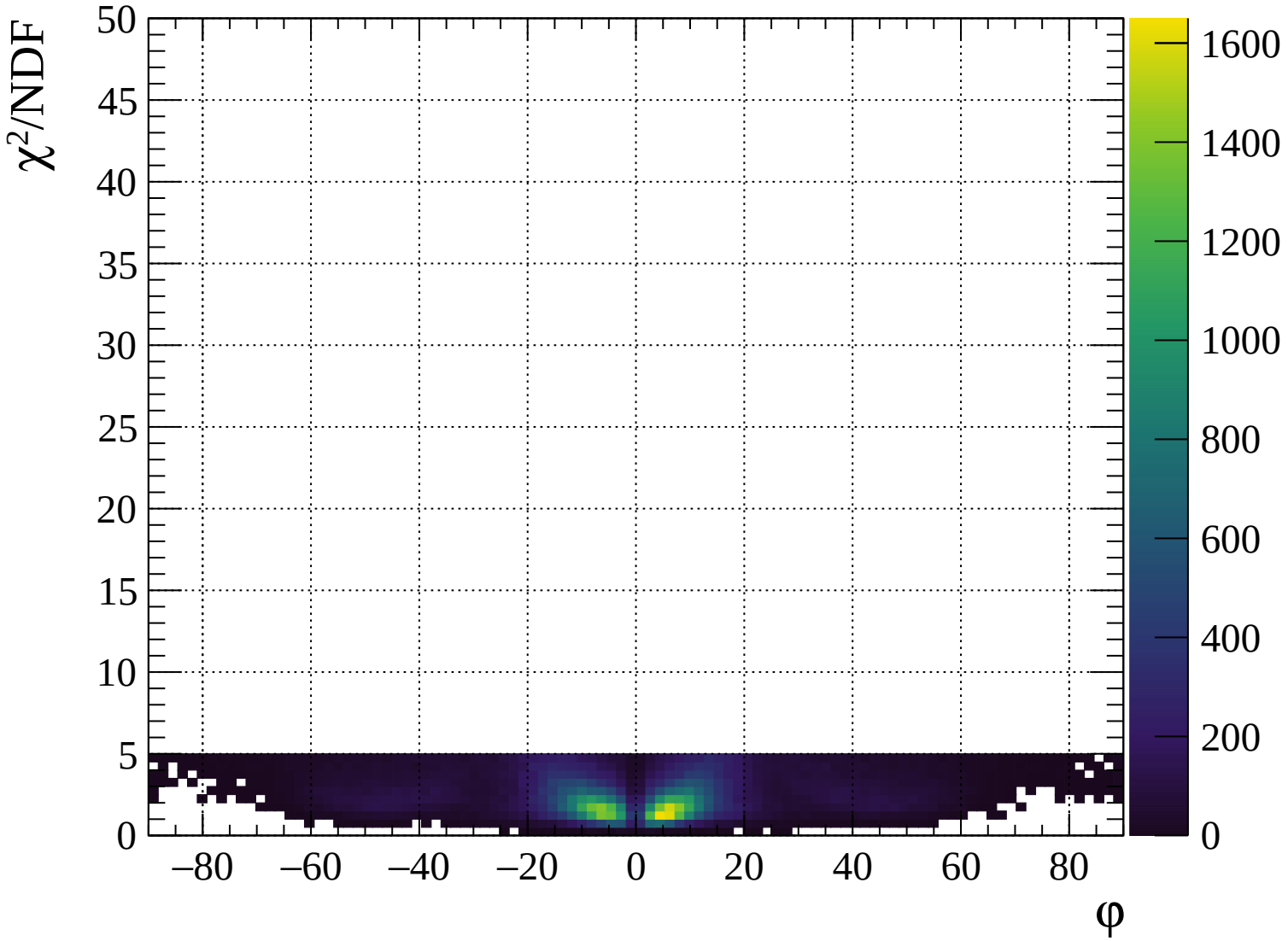


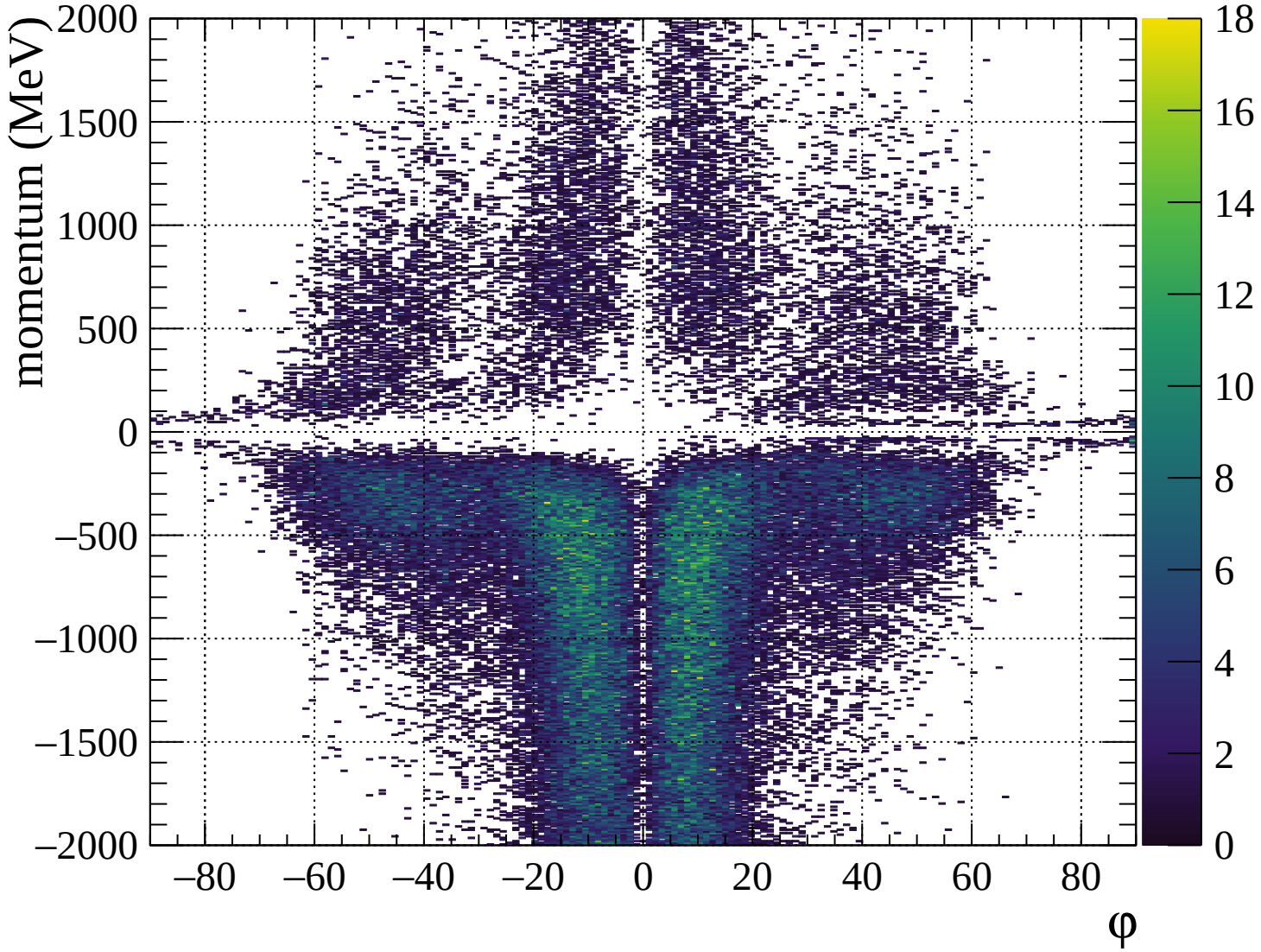


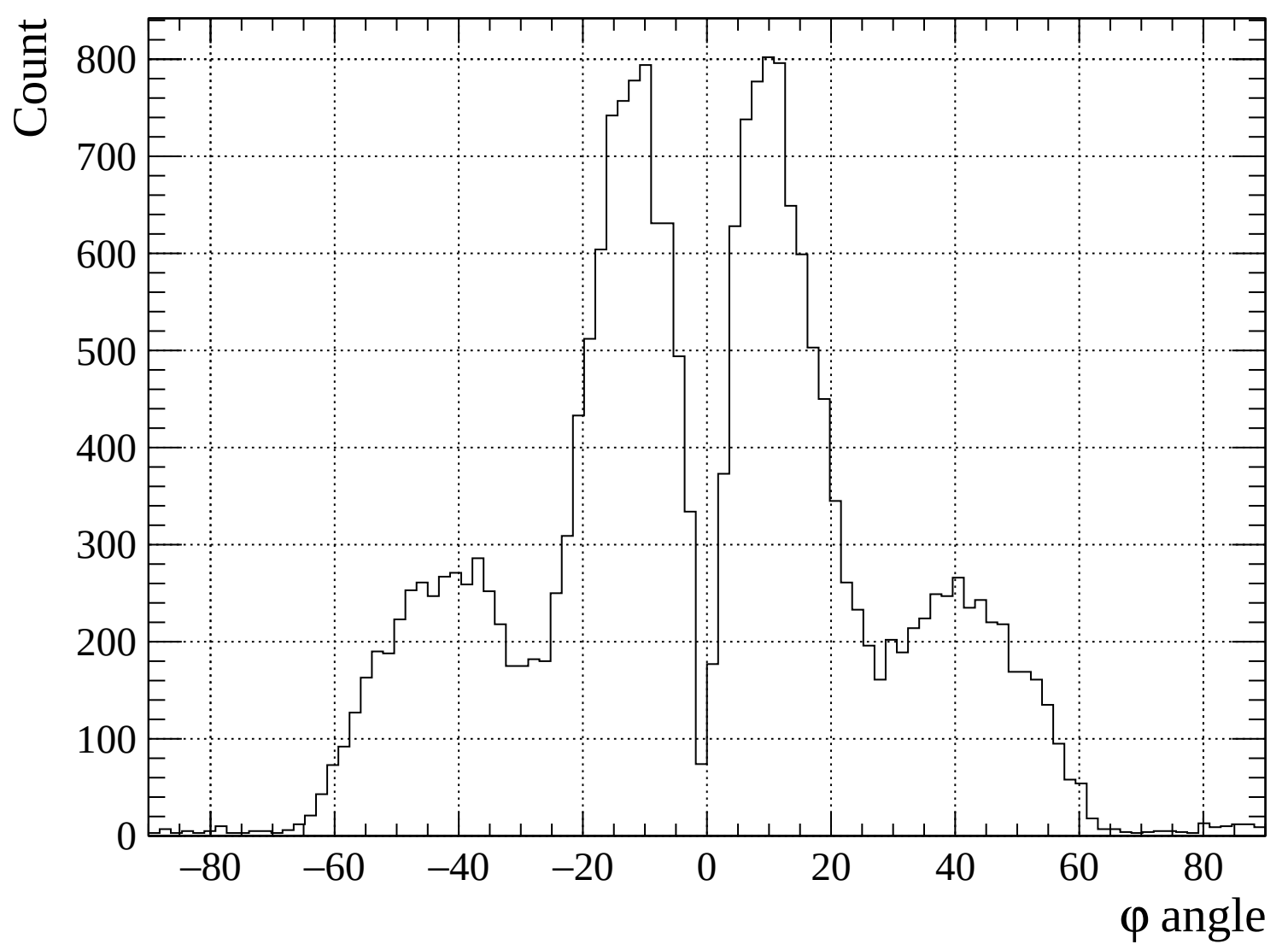


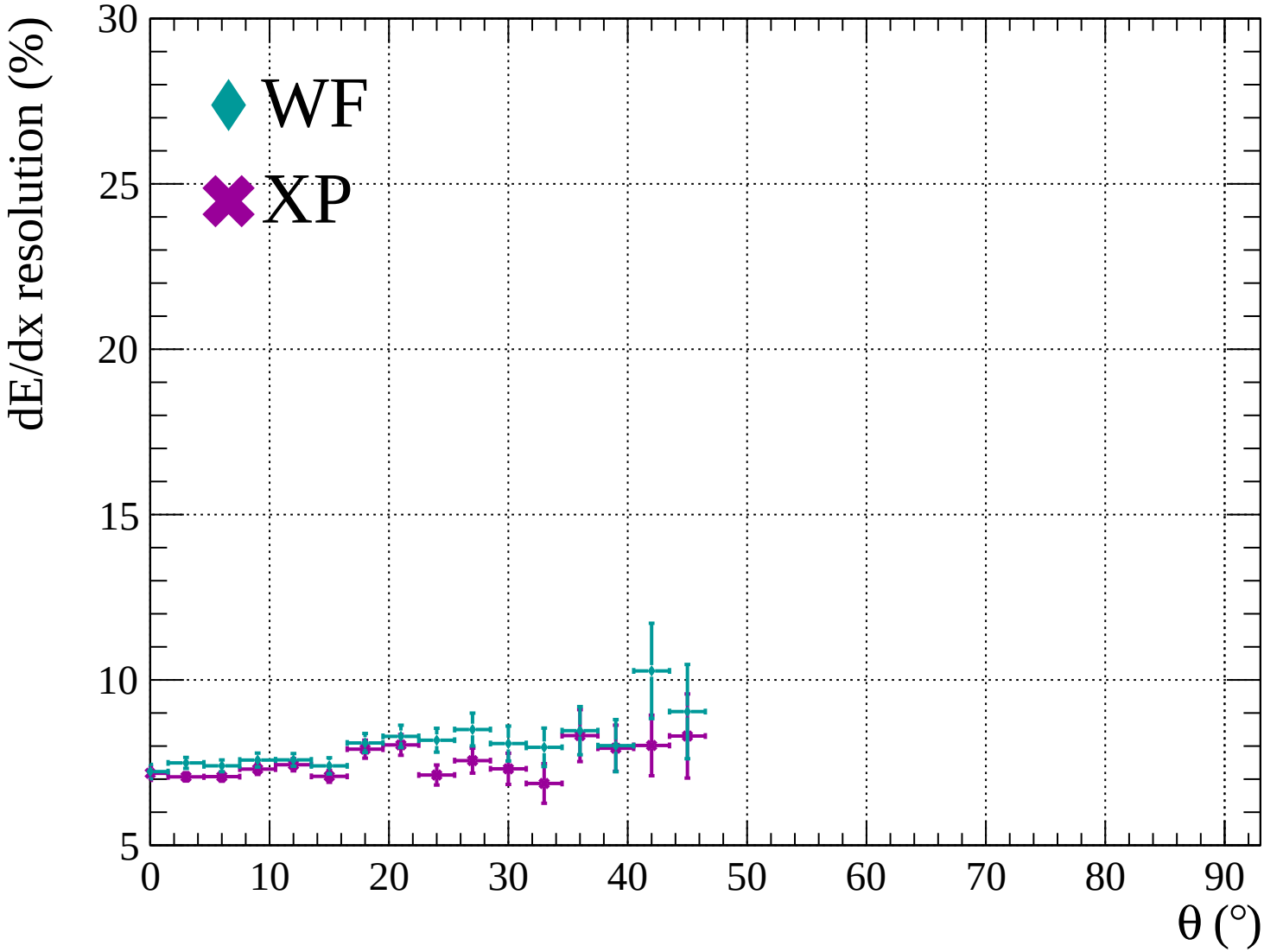


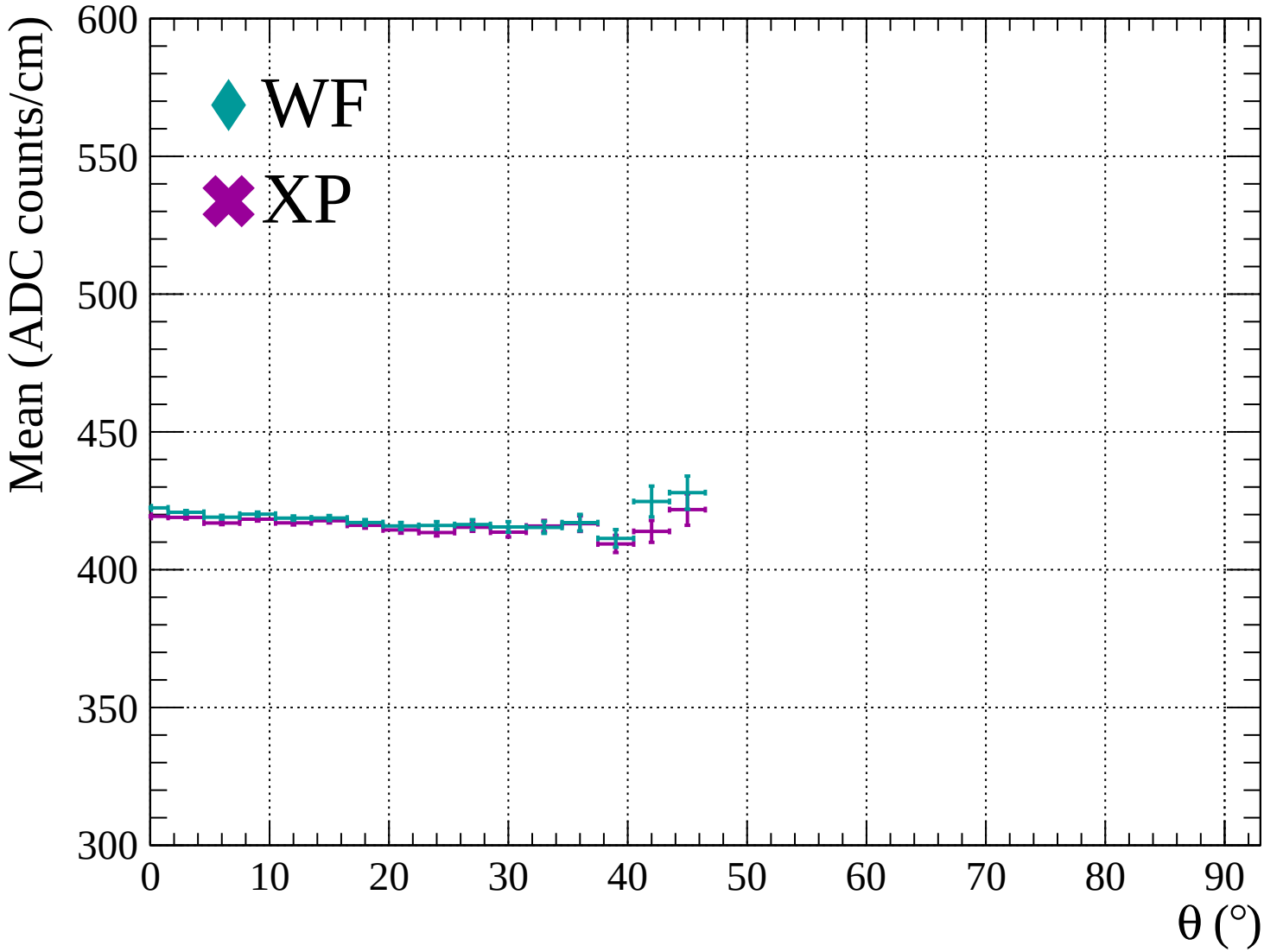


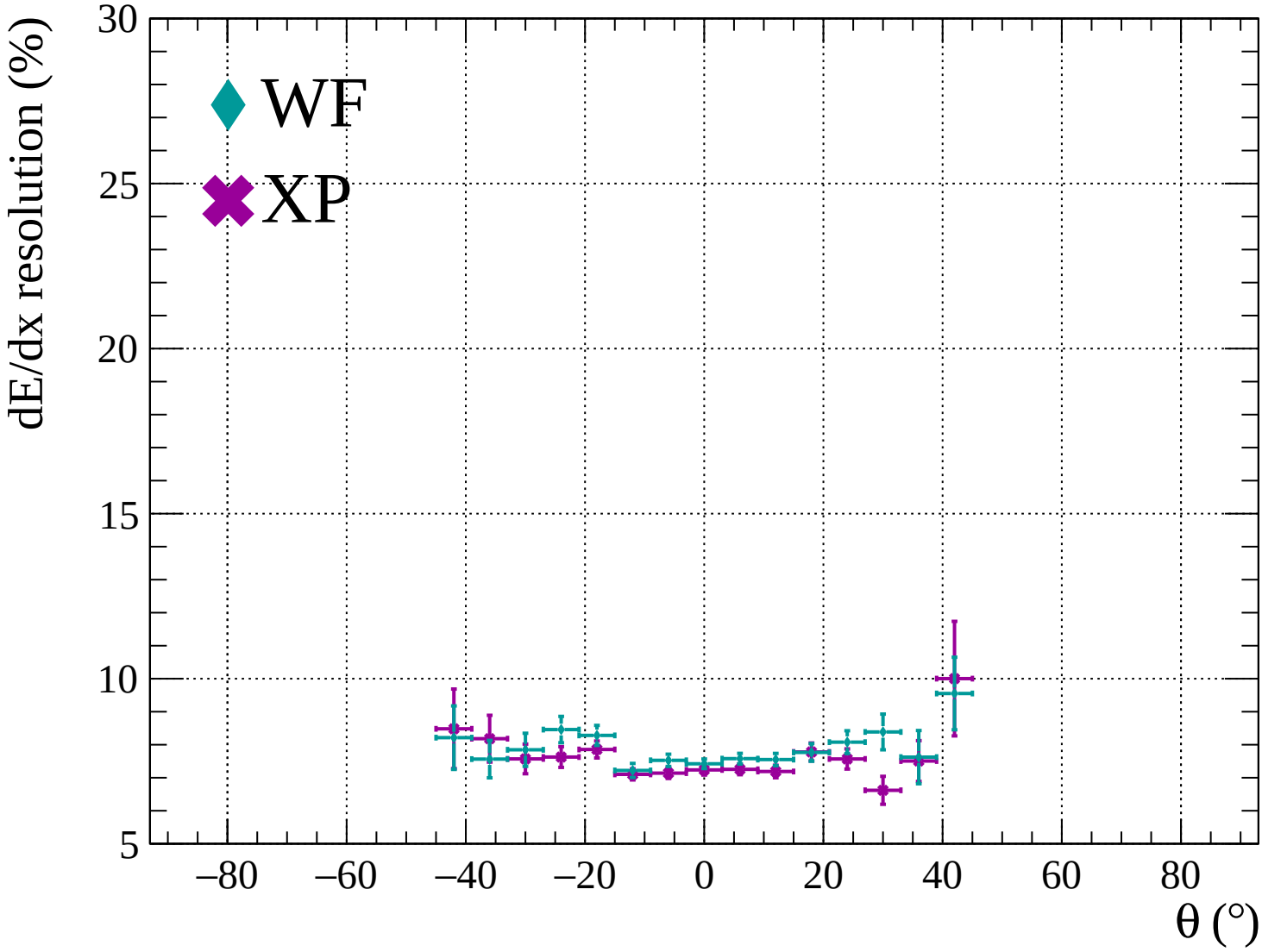


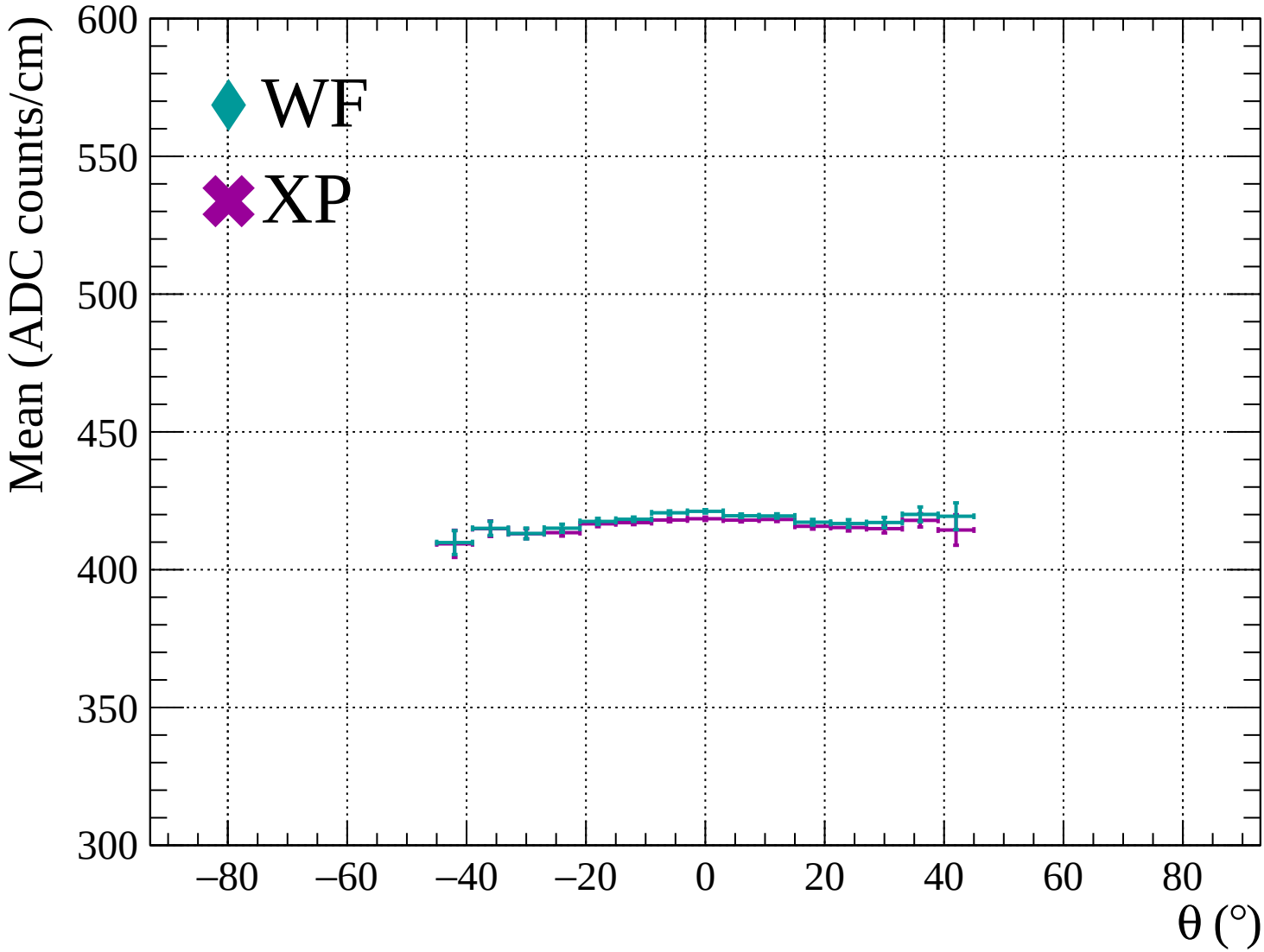


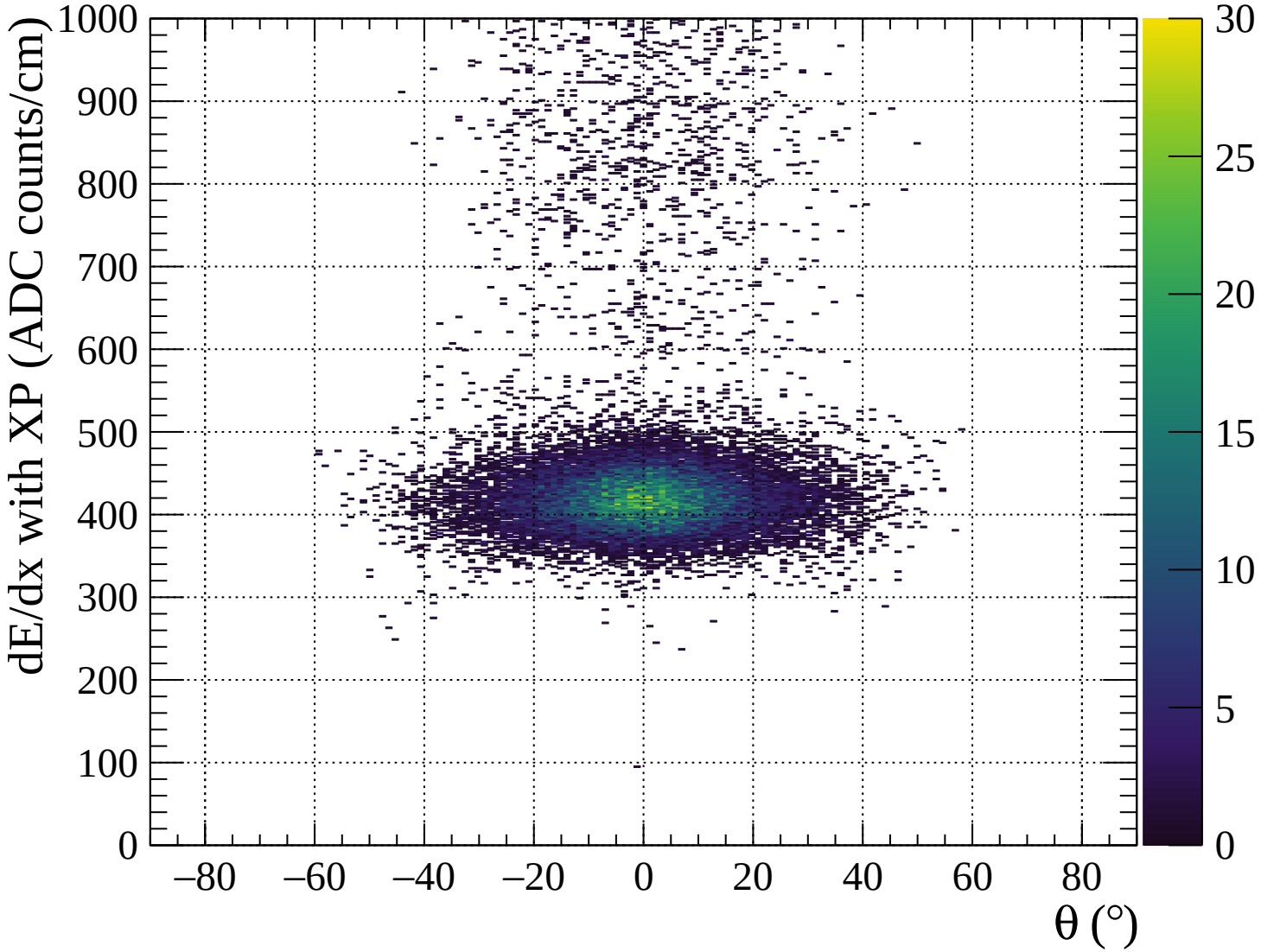


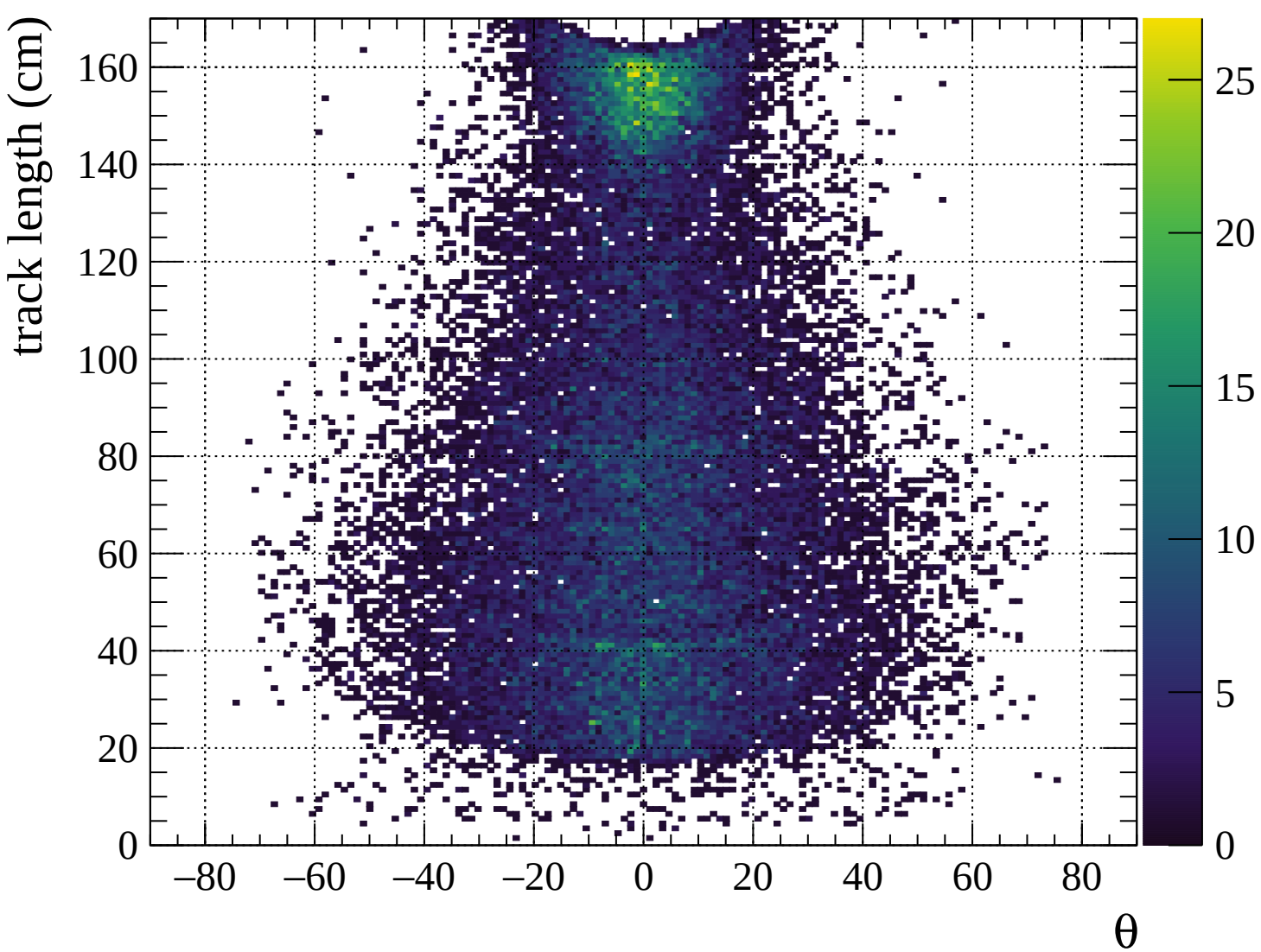


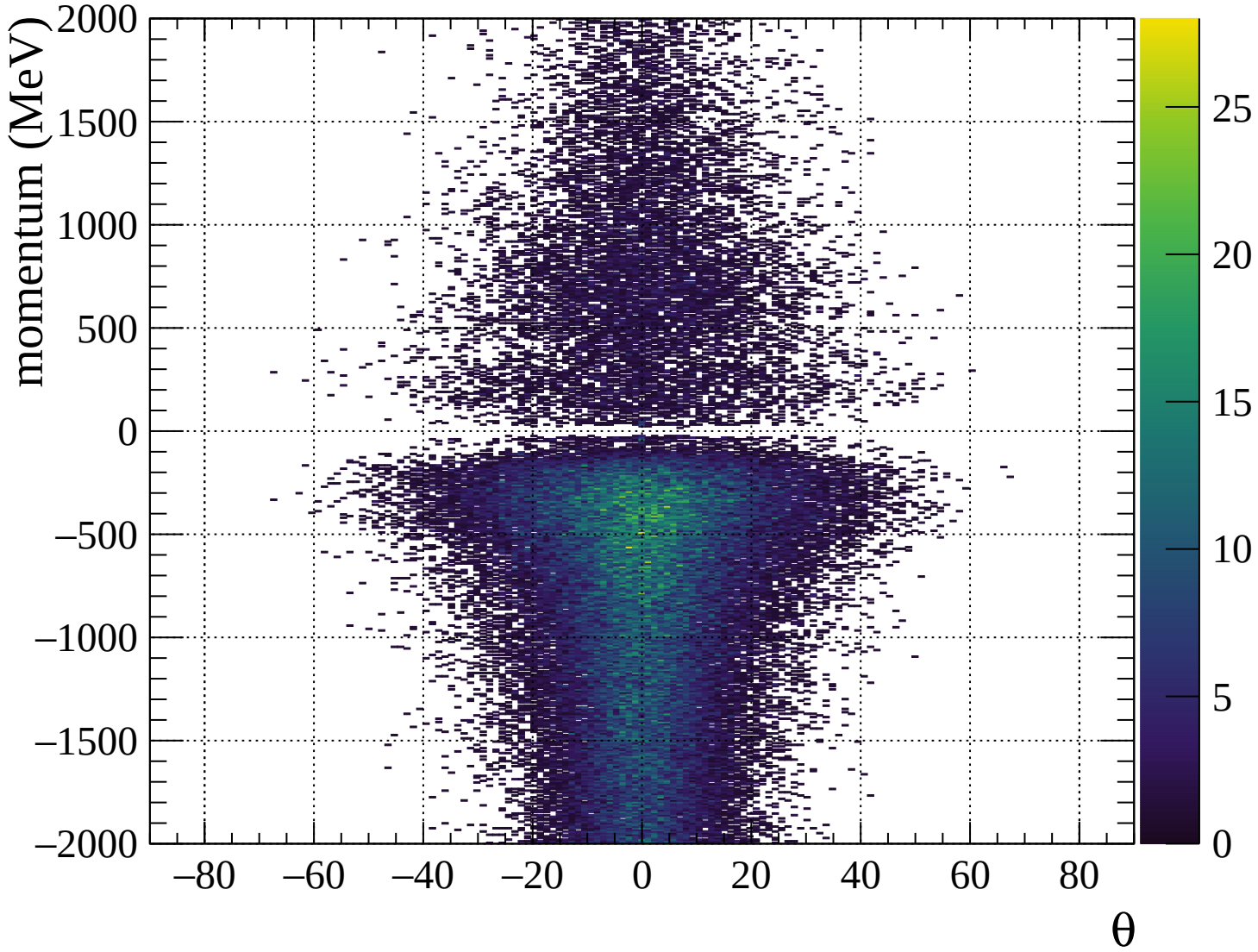


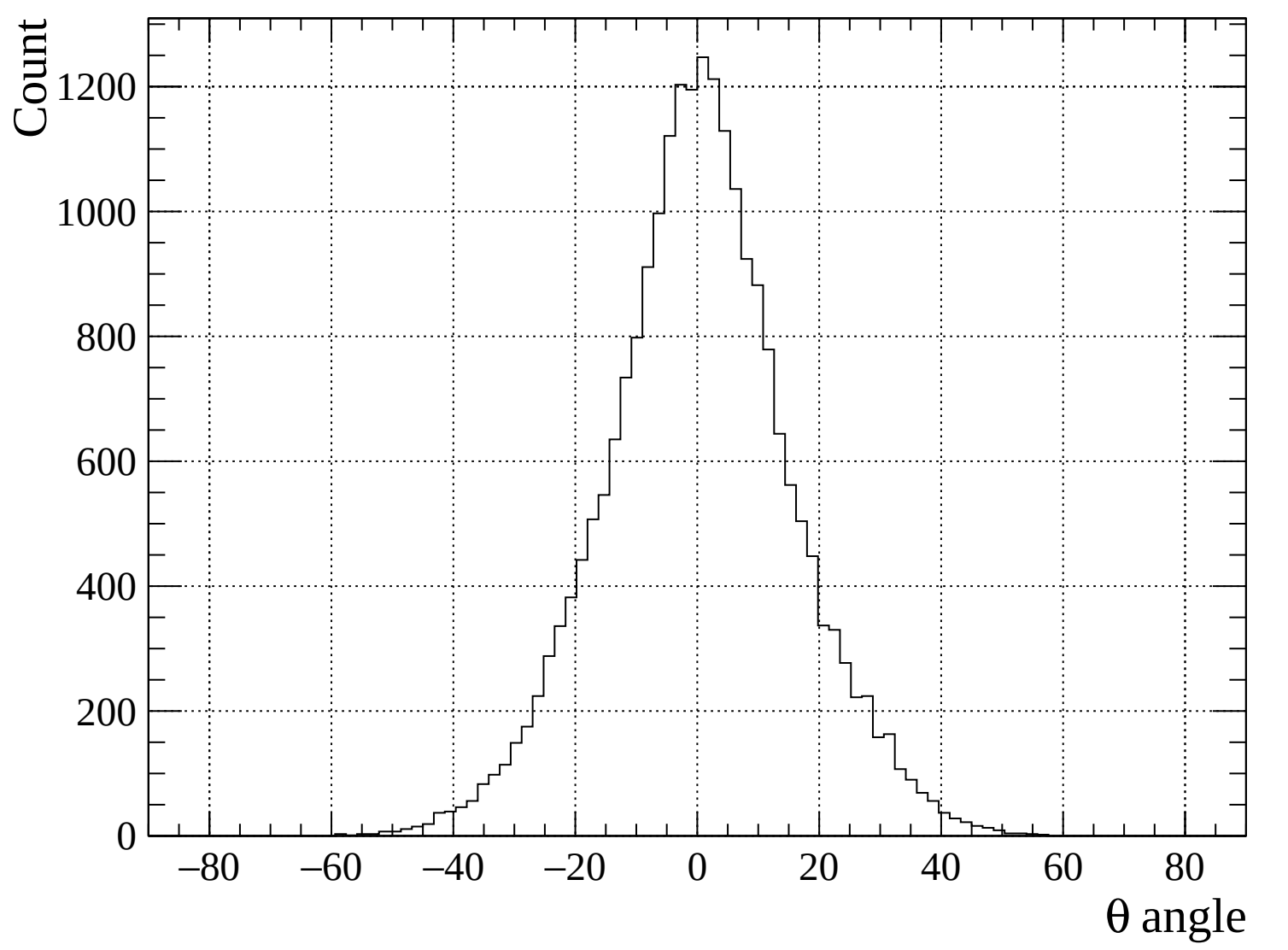


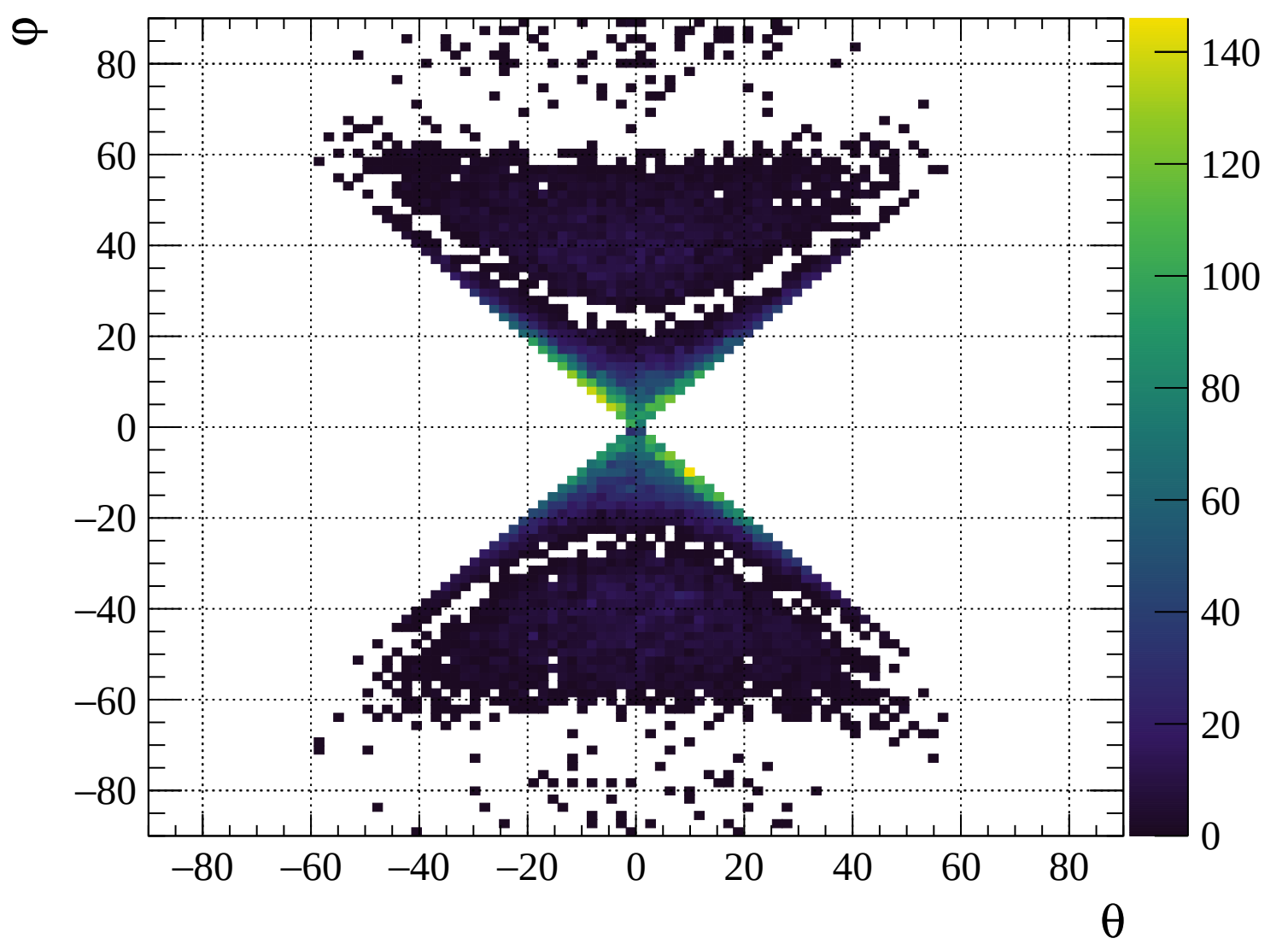


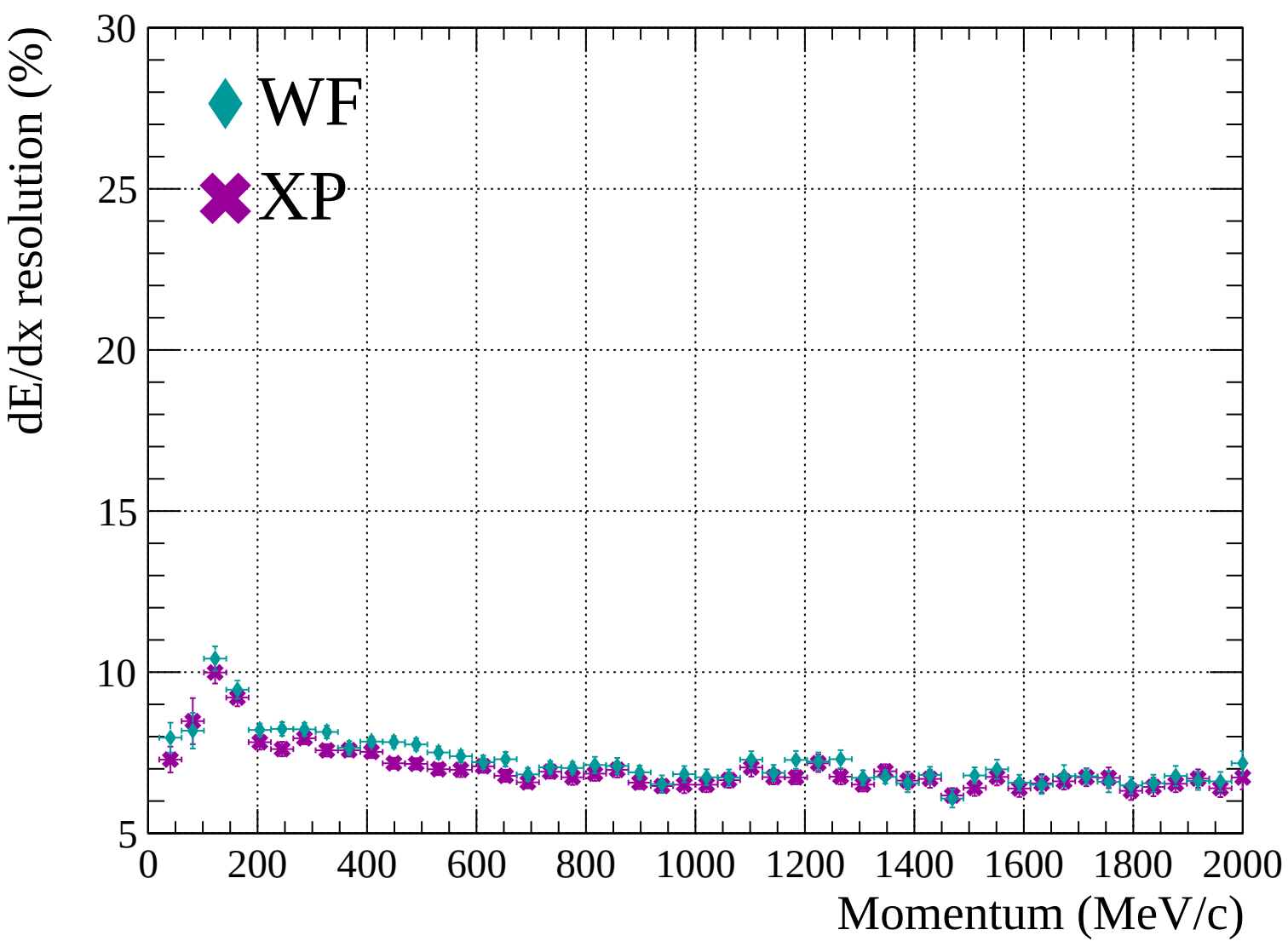


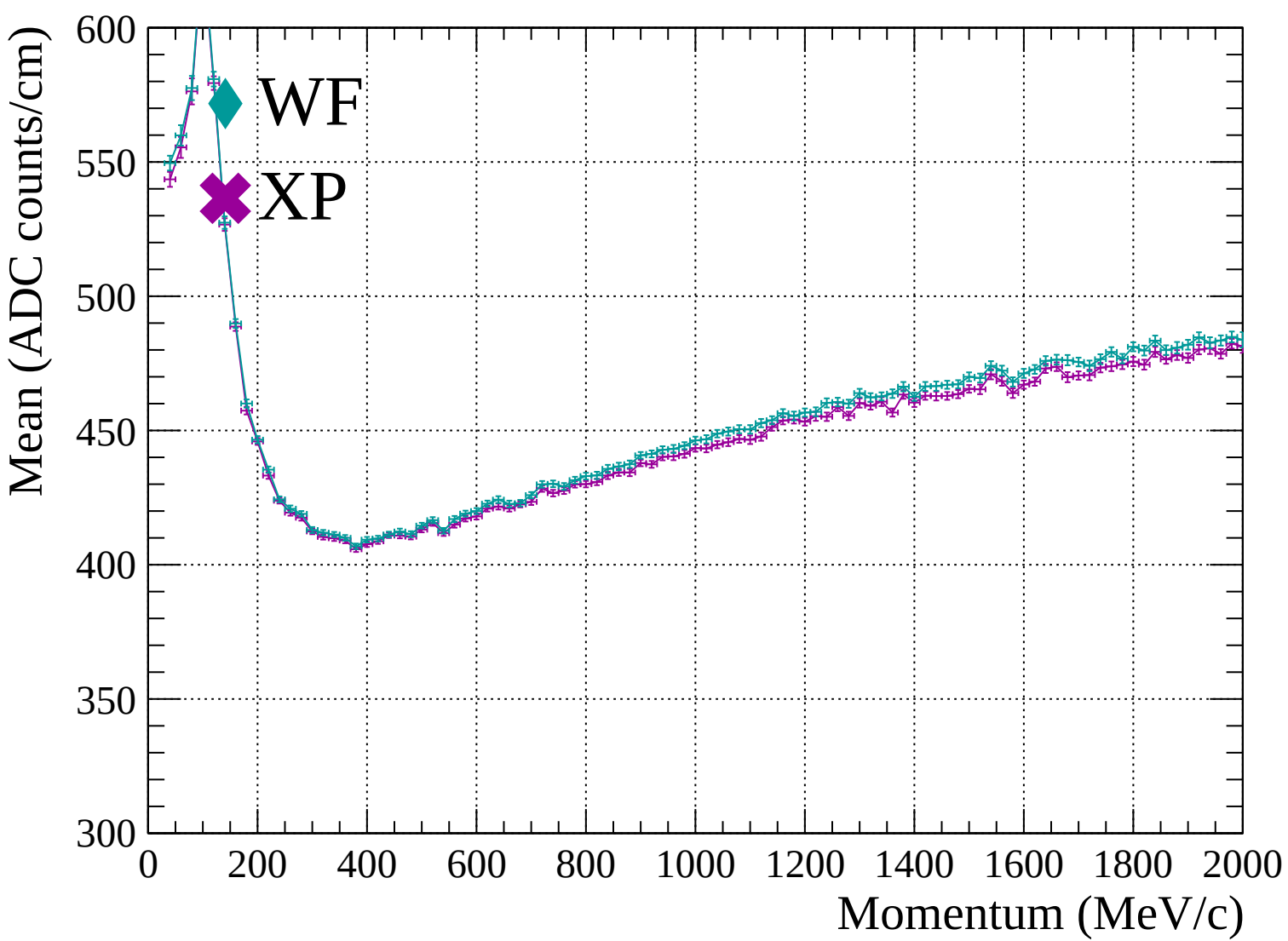


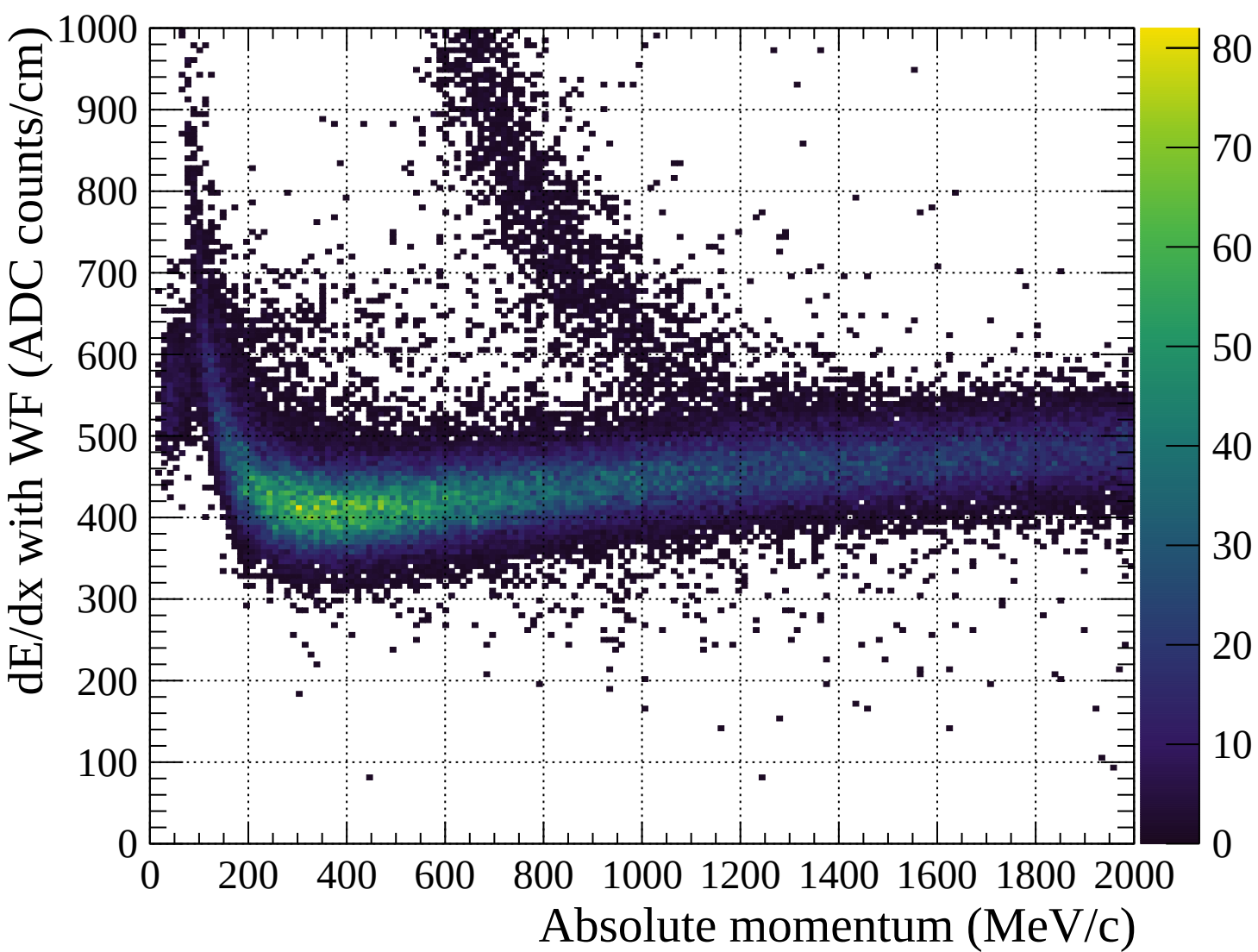


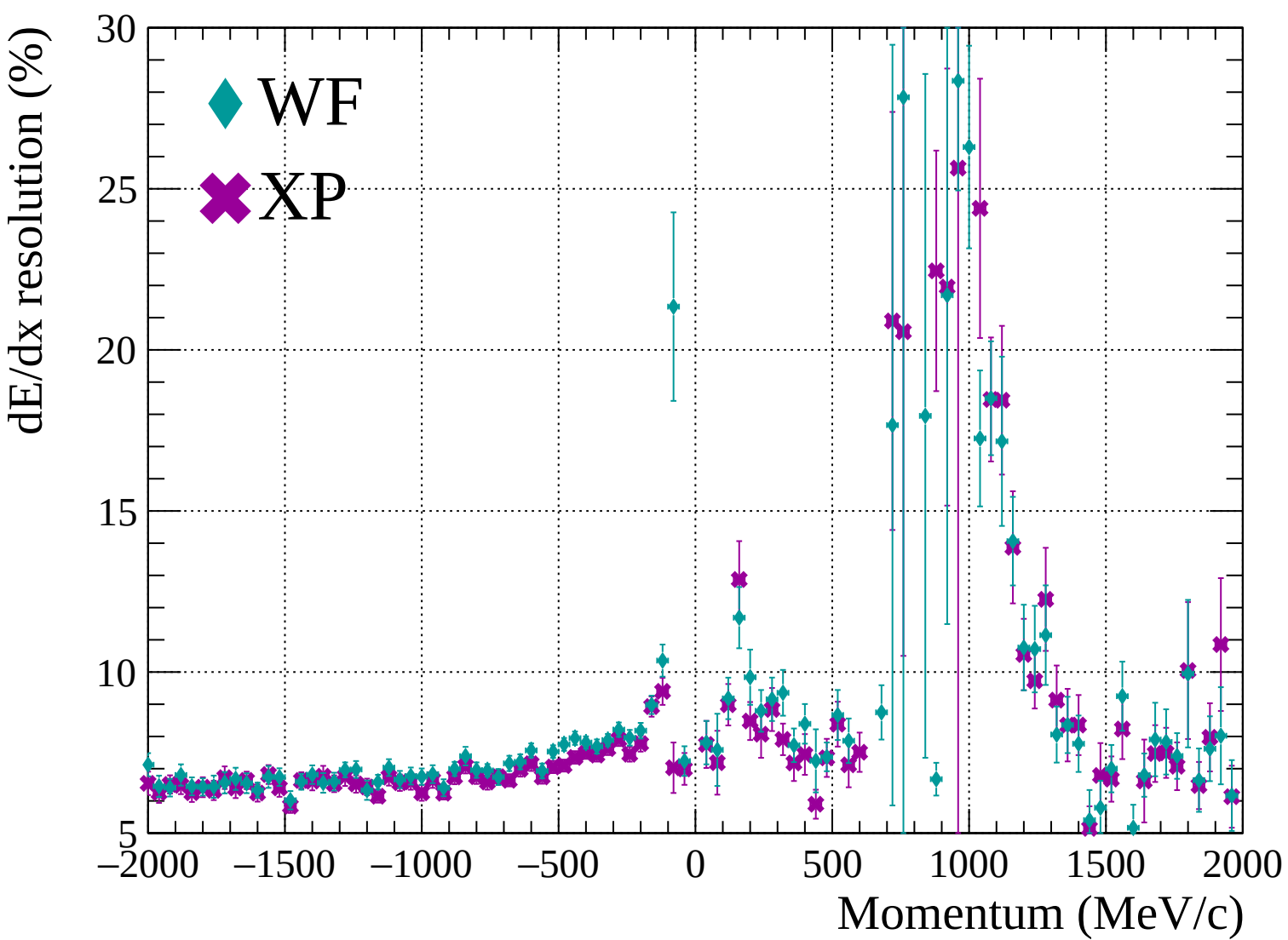


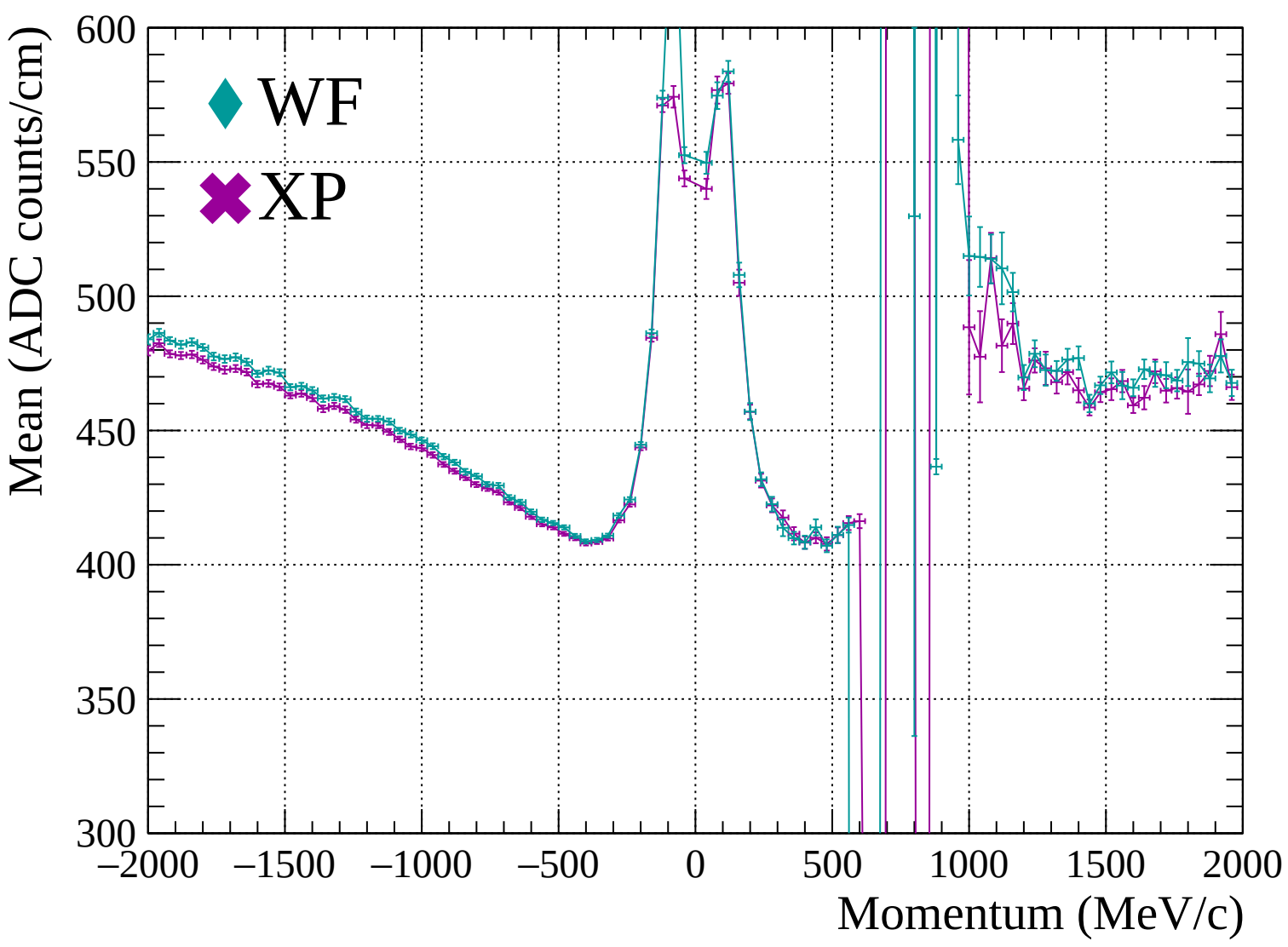


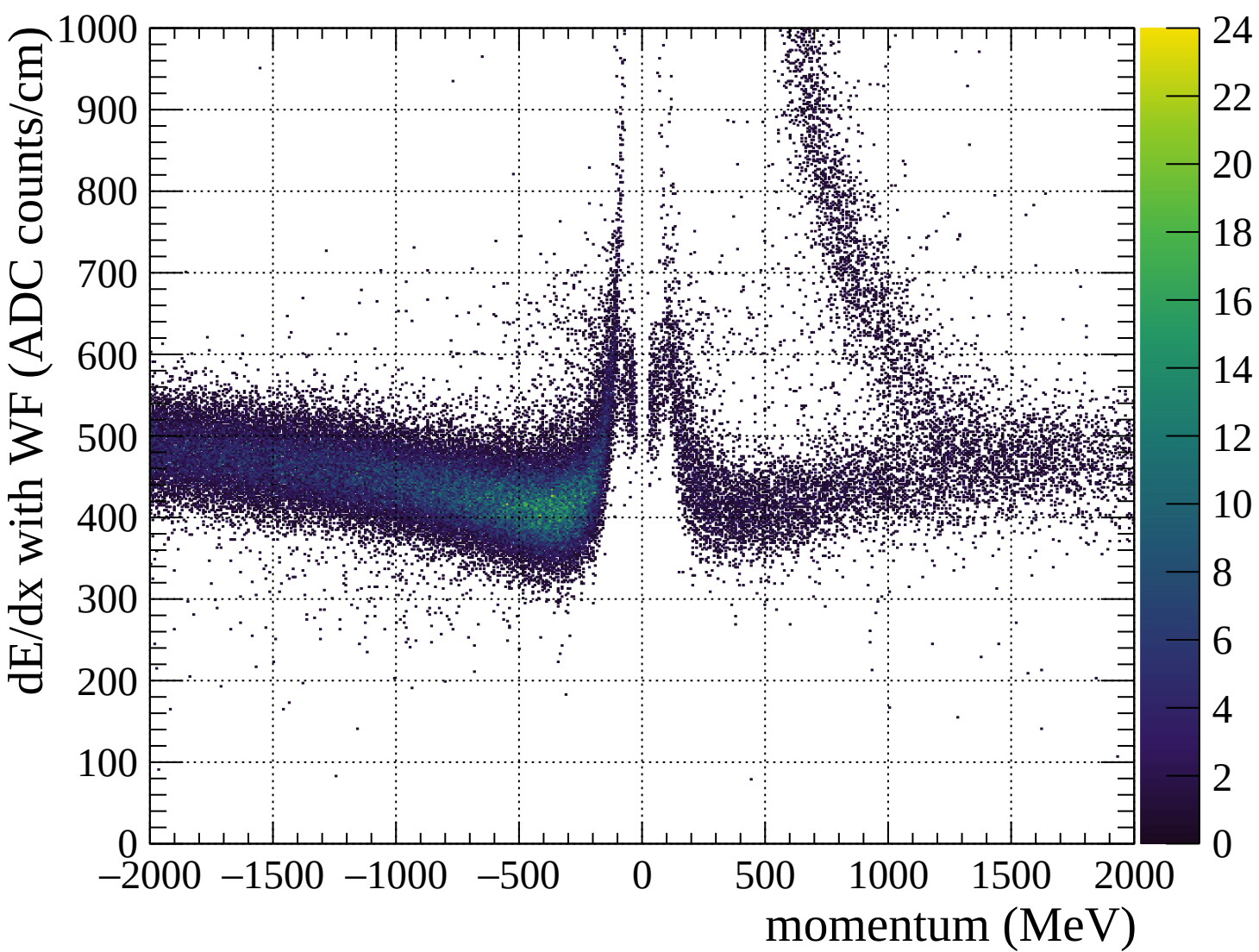


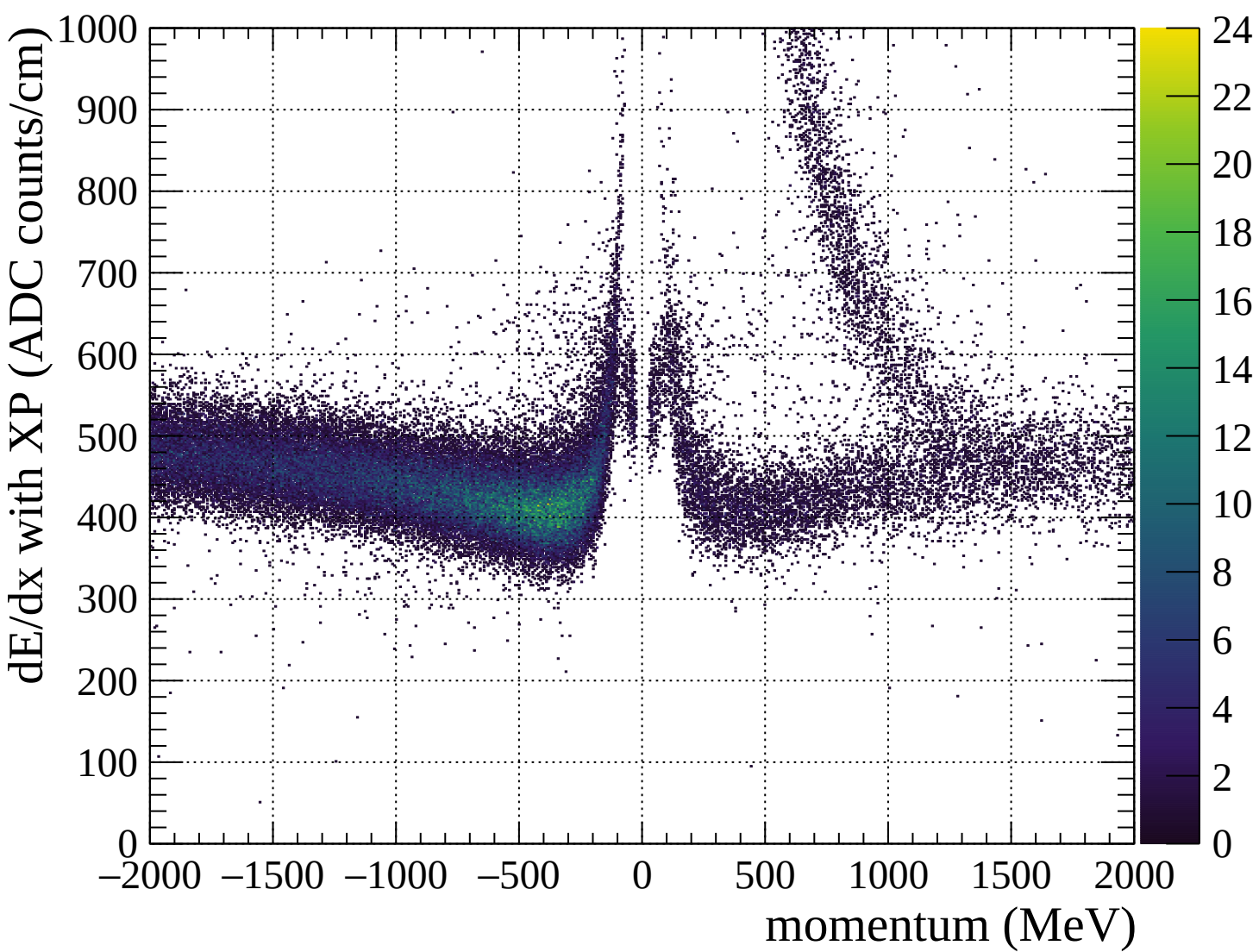


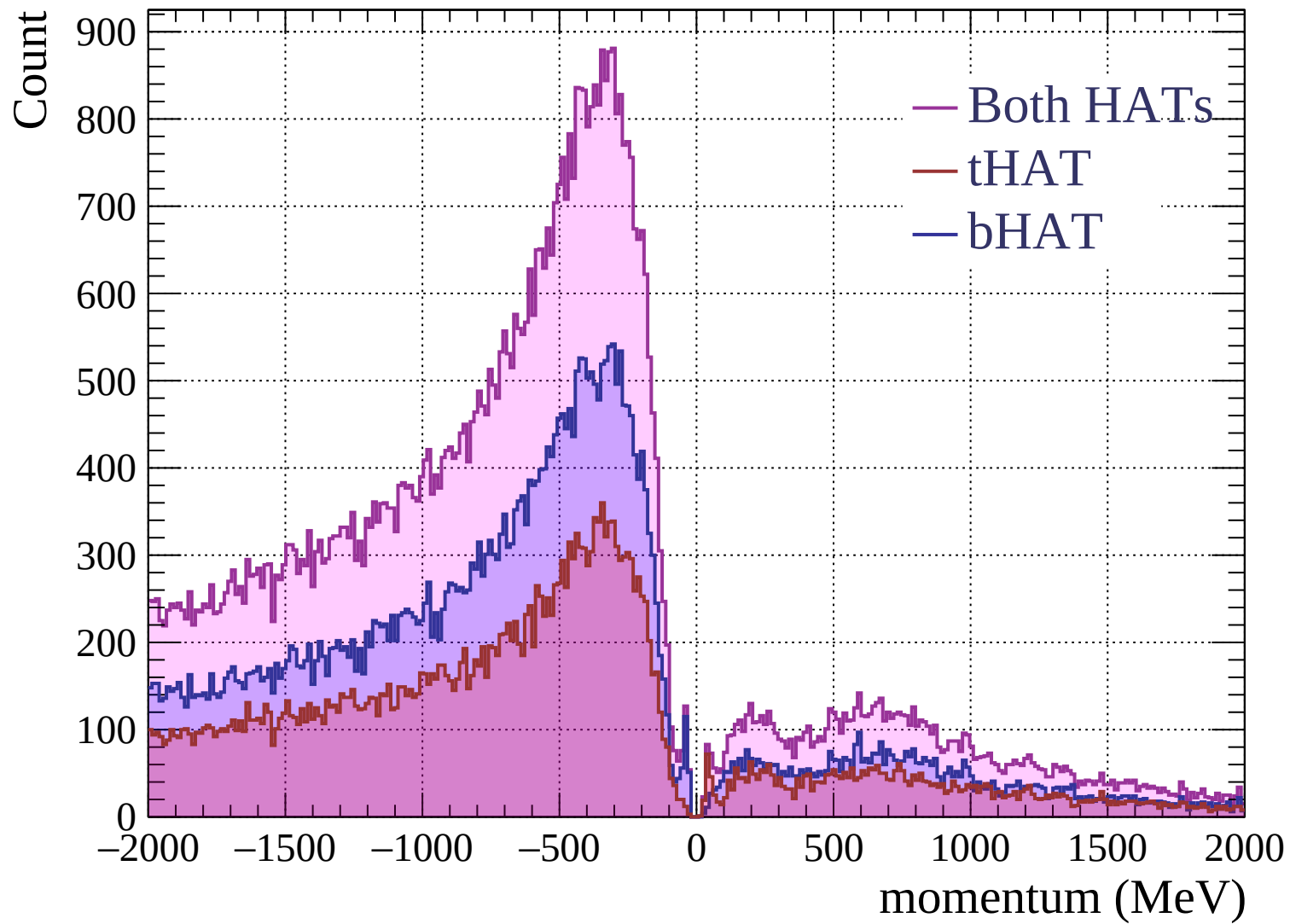


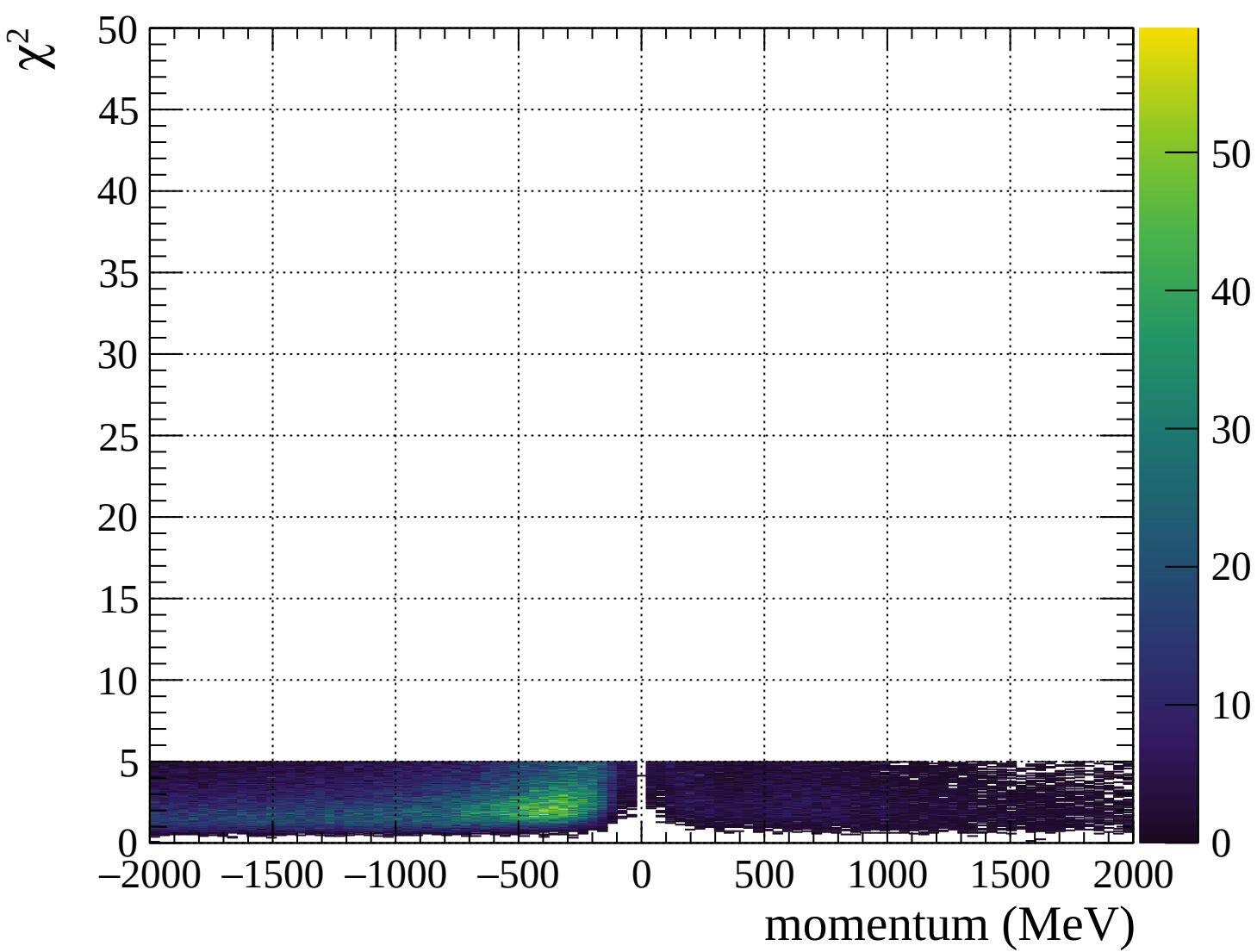




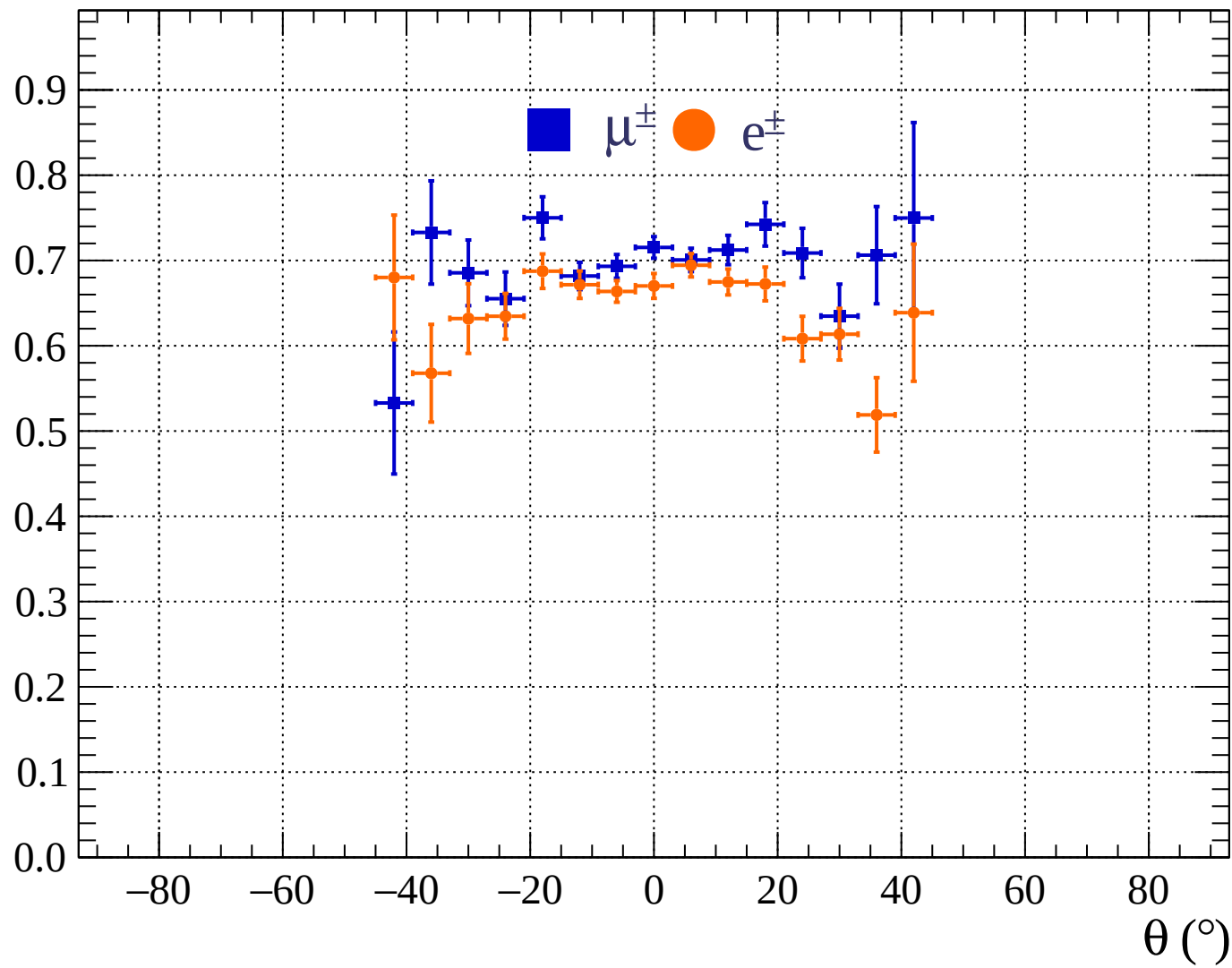




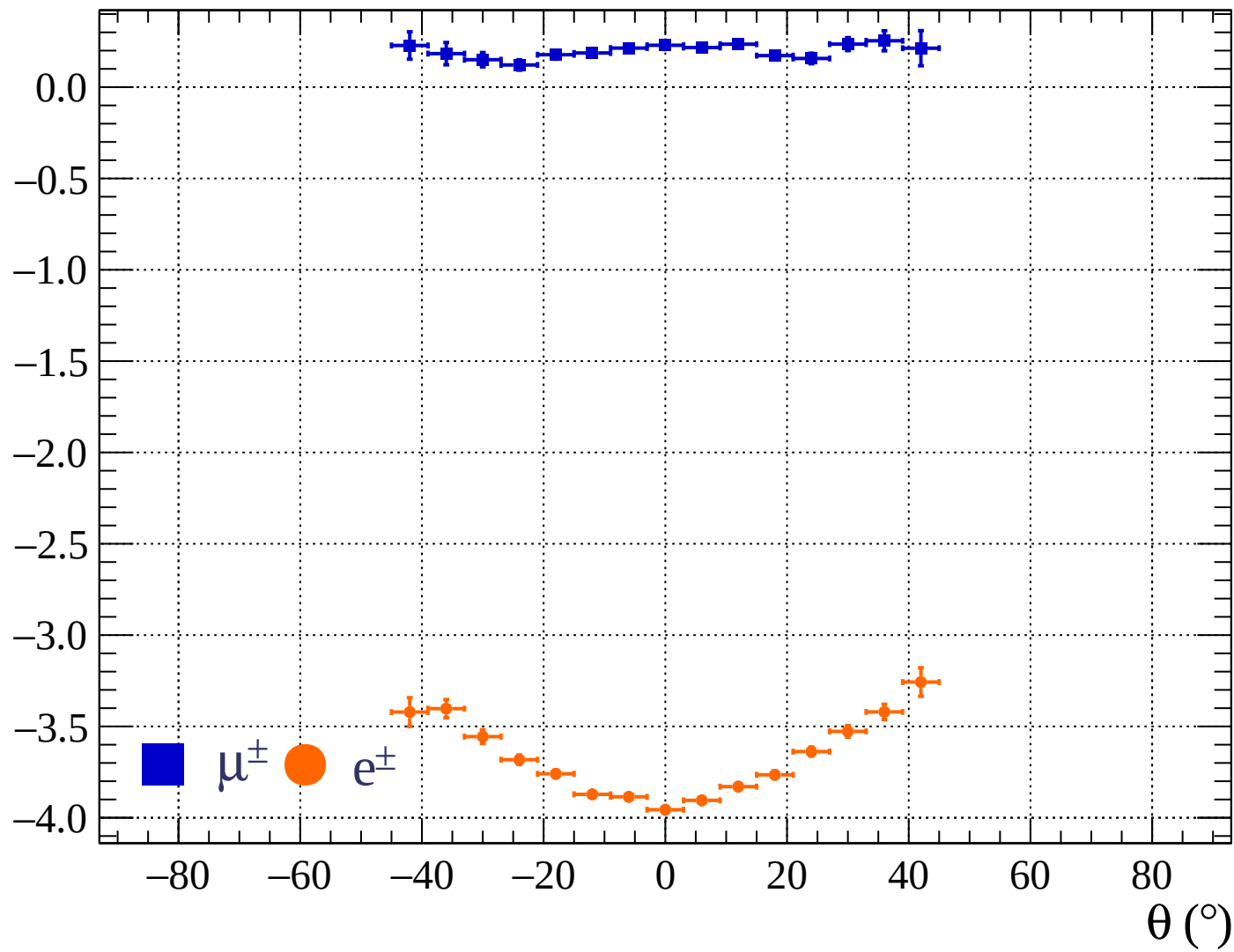




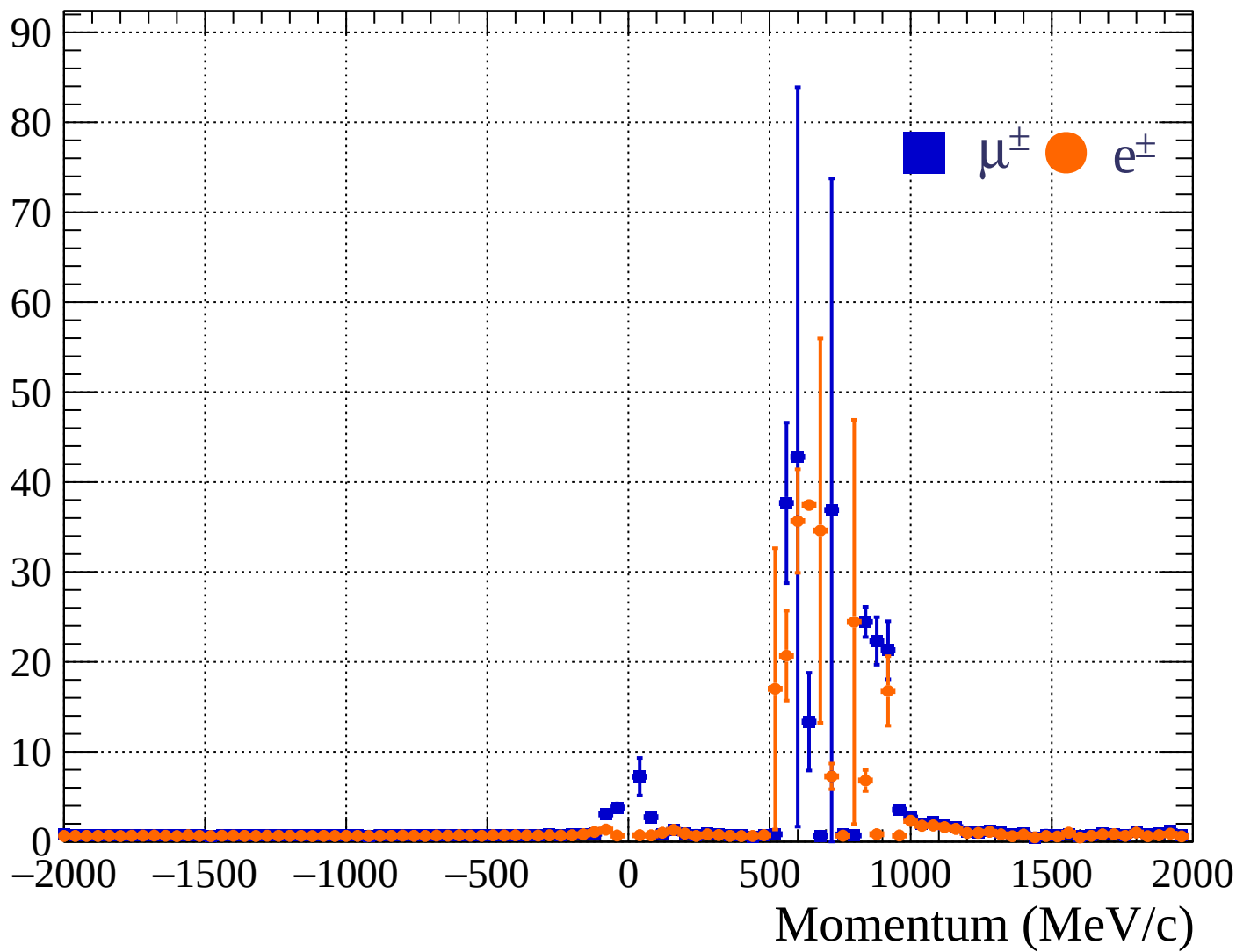
Pulls standard deviation

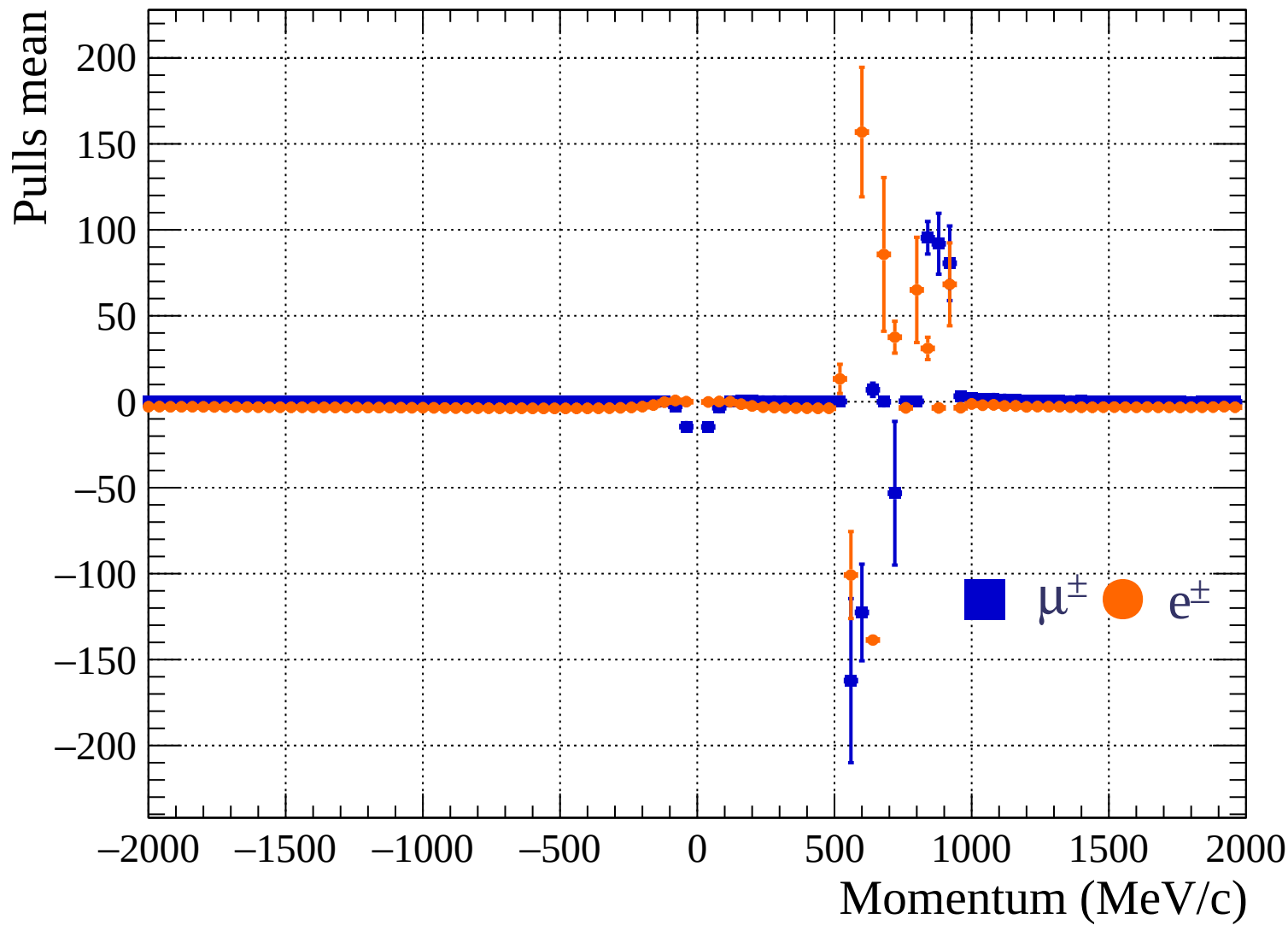


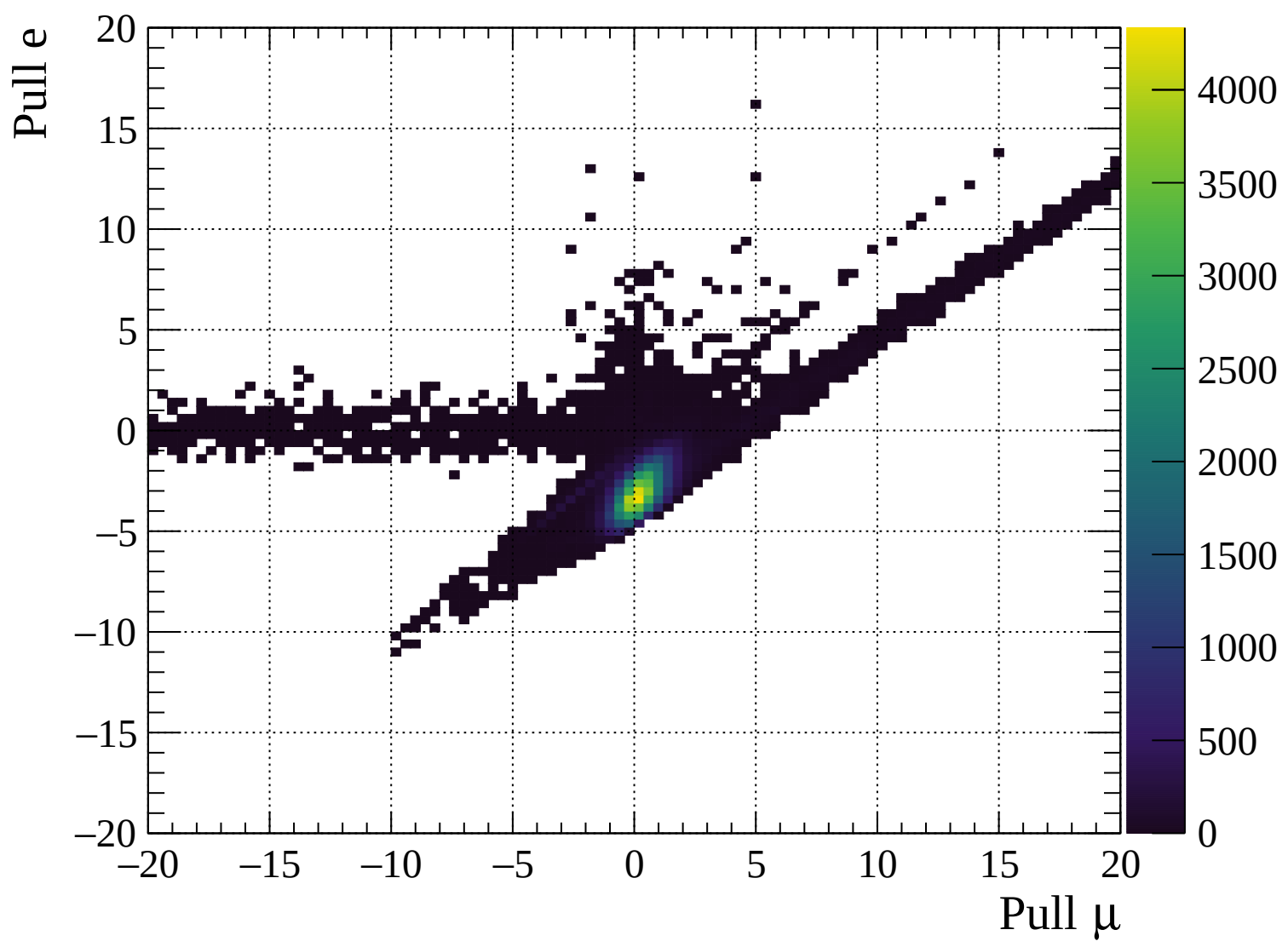
Pulls mean

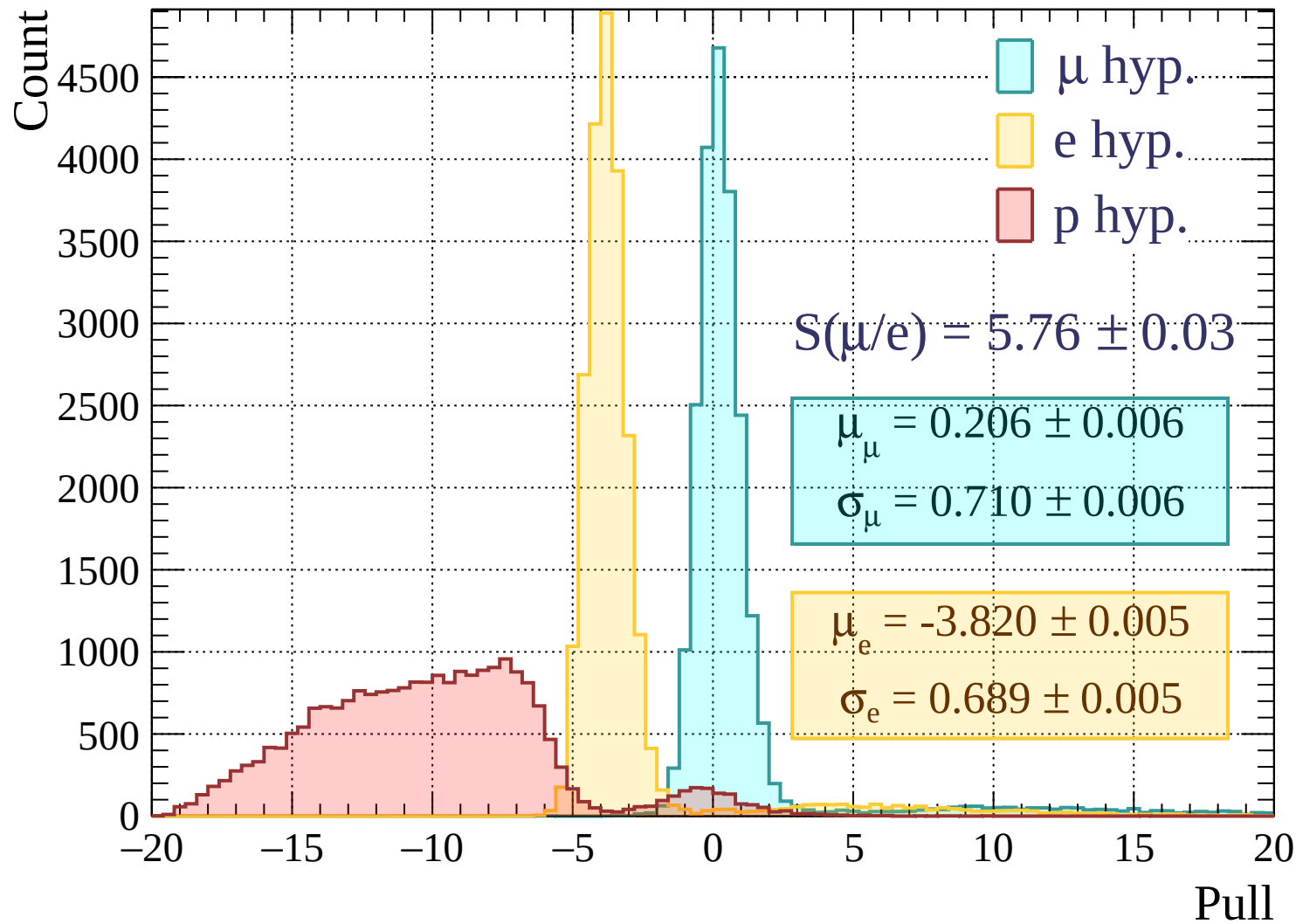


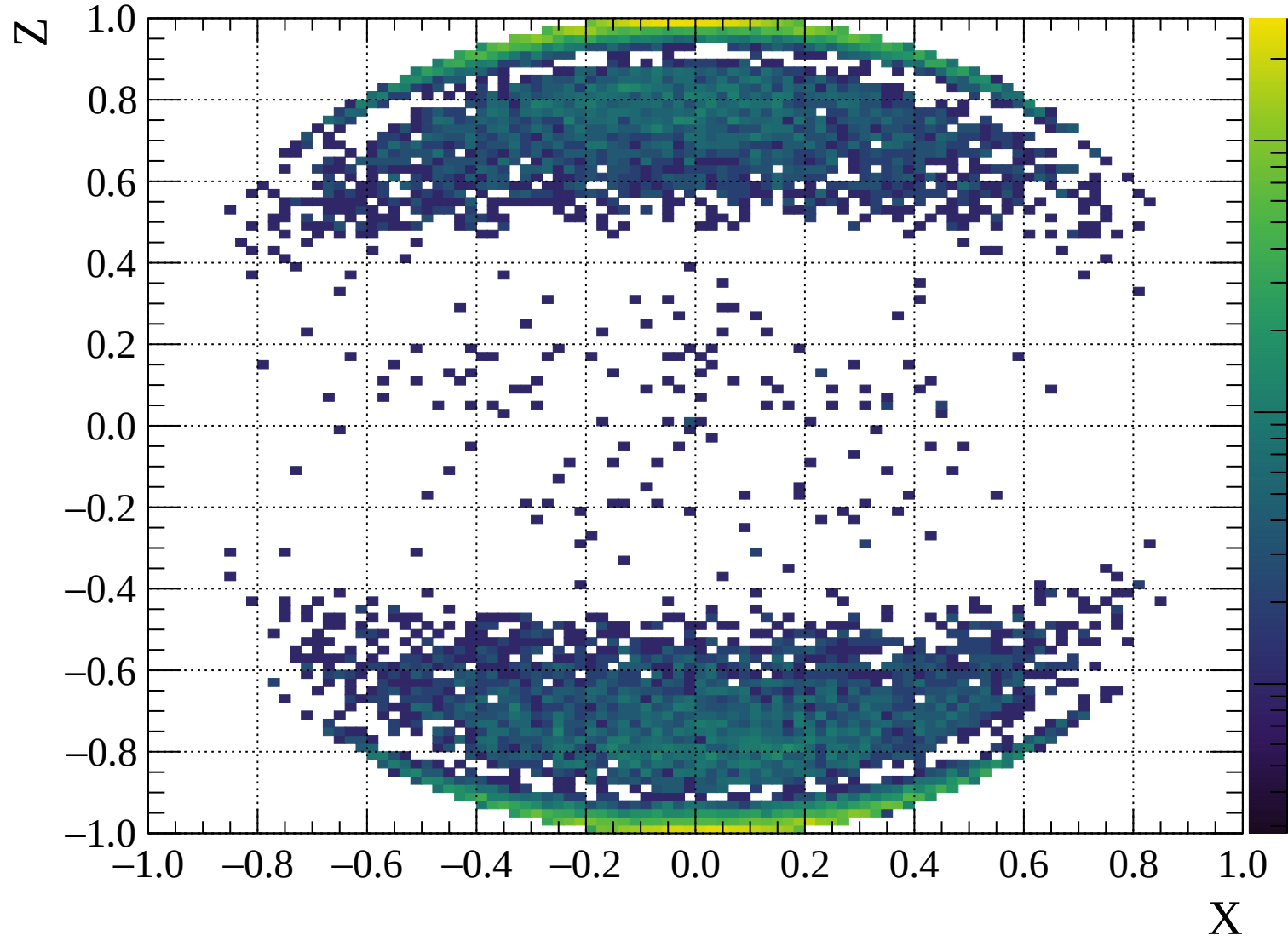
Pulls standard deviation

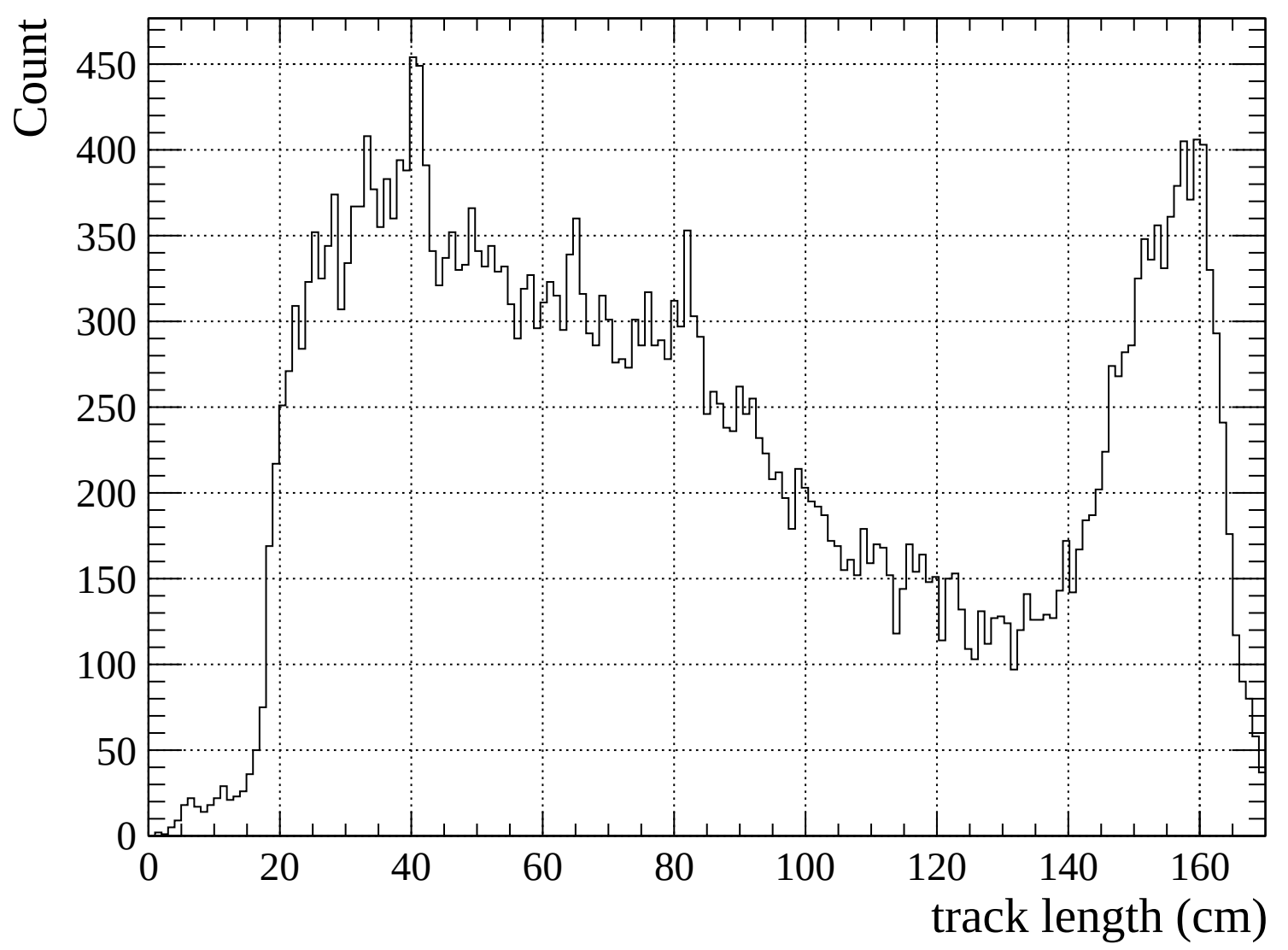


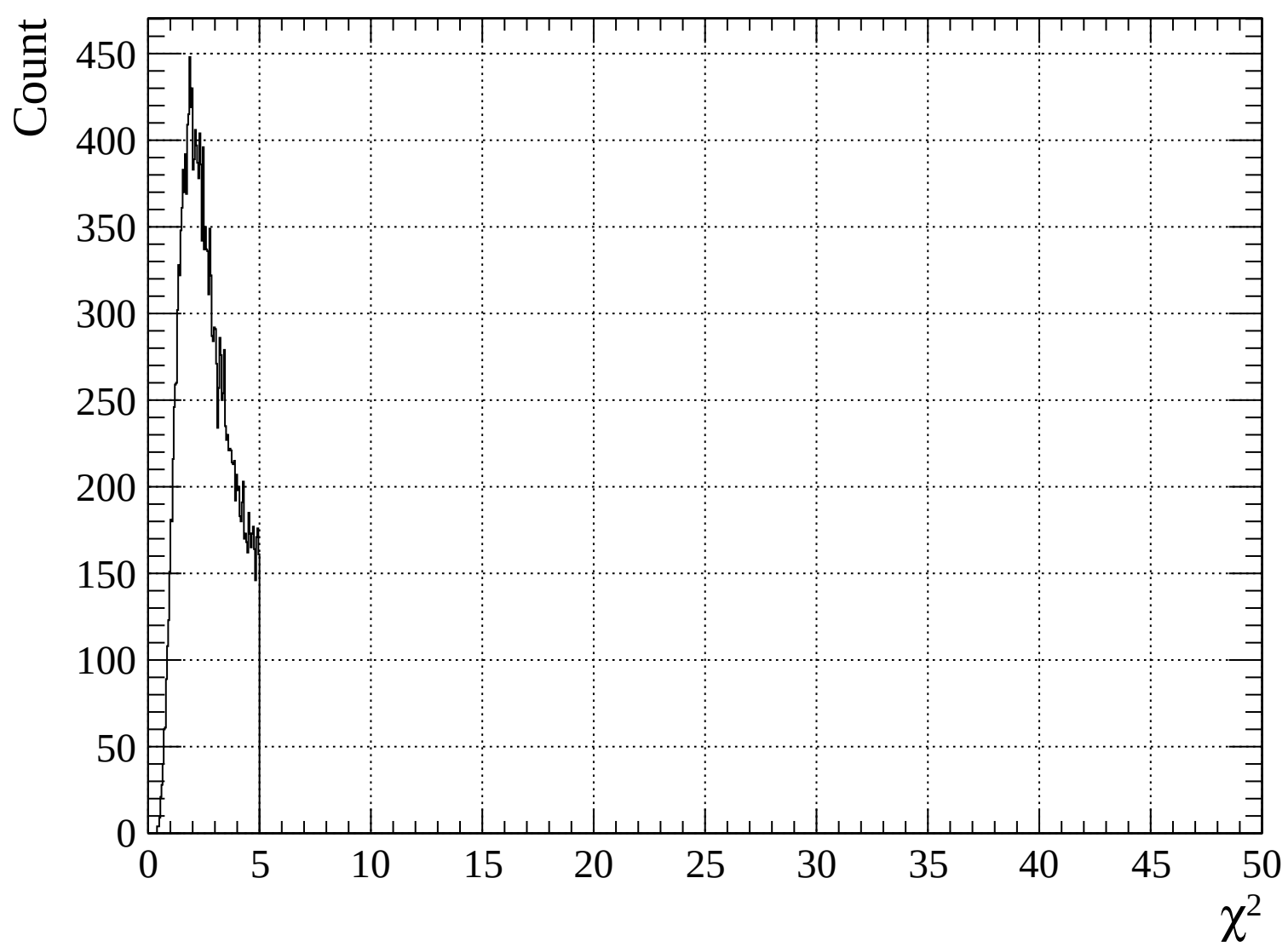


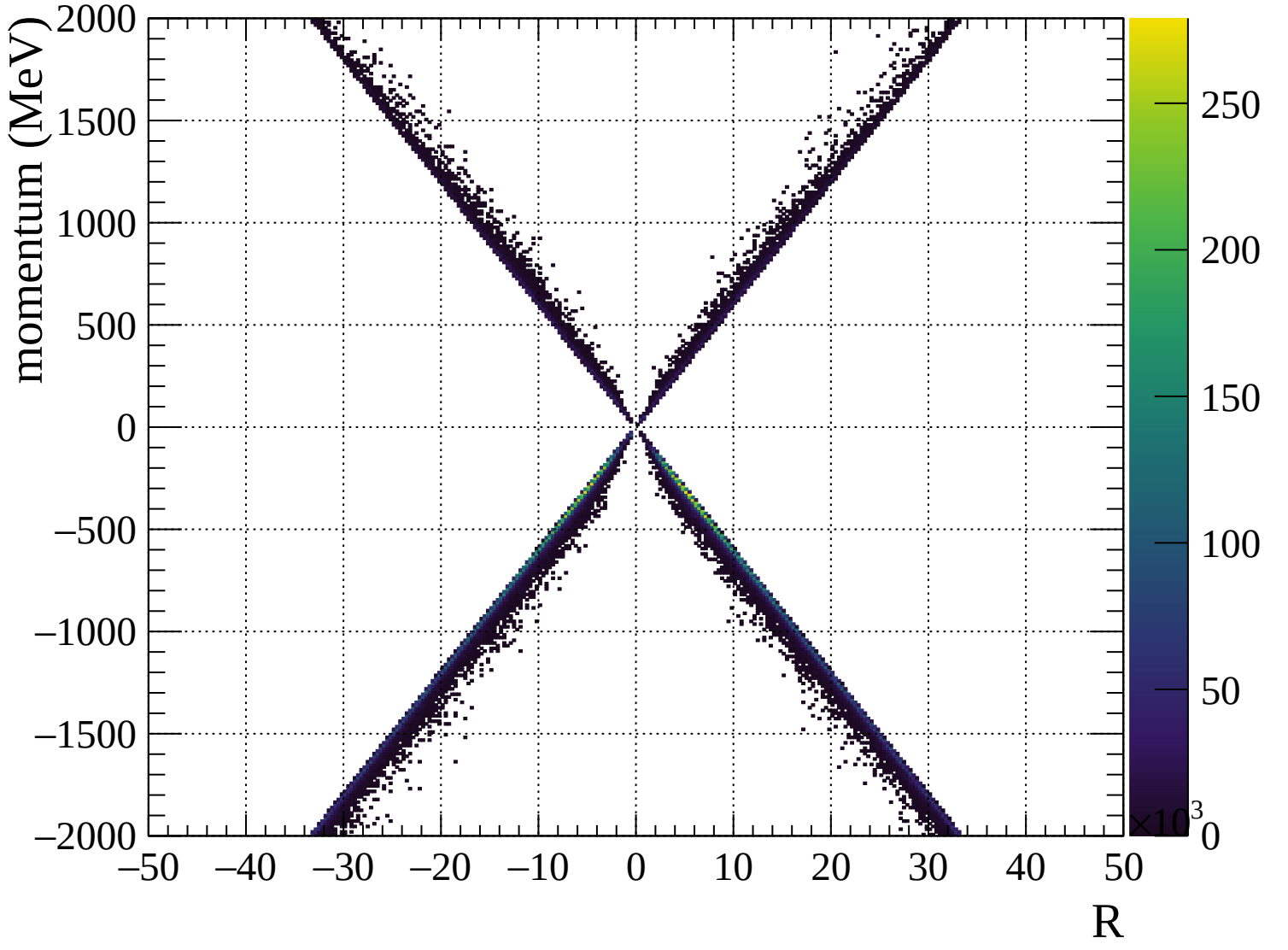


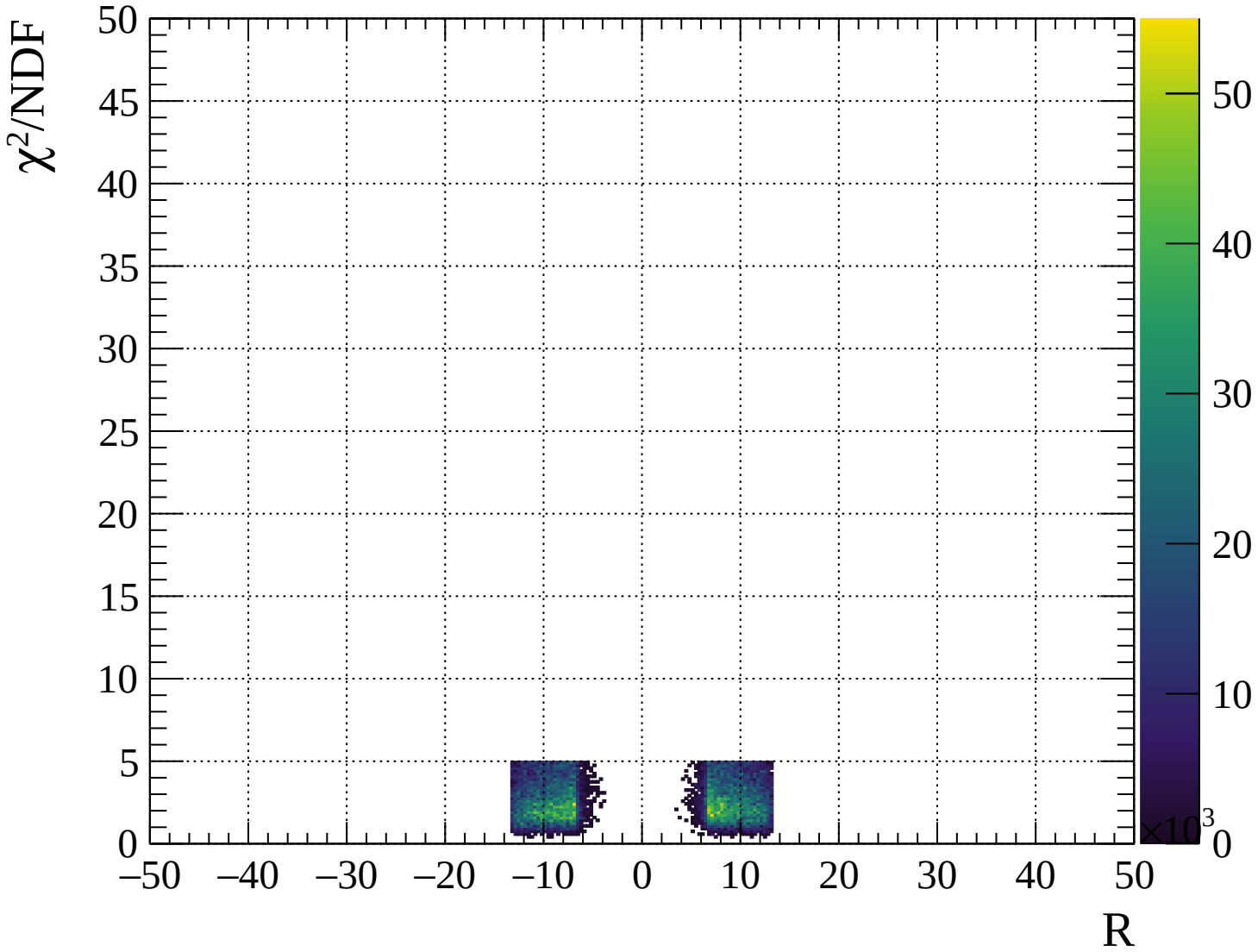






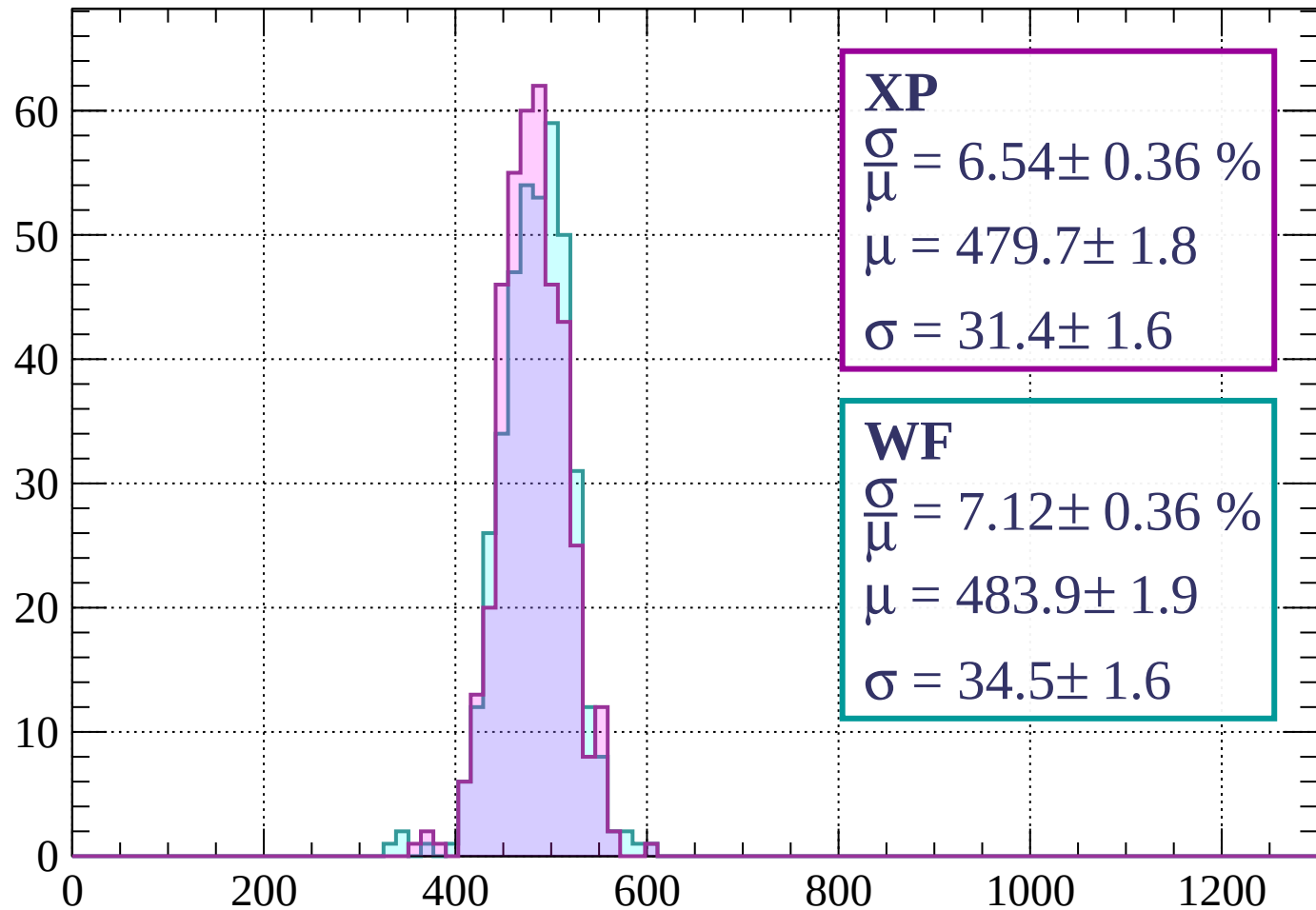






Energy loss | $-2000 < p < -1960$

Count



XP

$$\frac{\sigma}{\mu} = 6.54 \pm 0.36 \%$$

$$\mu = 479.7 \pm 1.8$$

$$\sigma = 31.4 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 7.12 \pm 0.36 \%$$

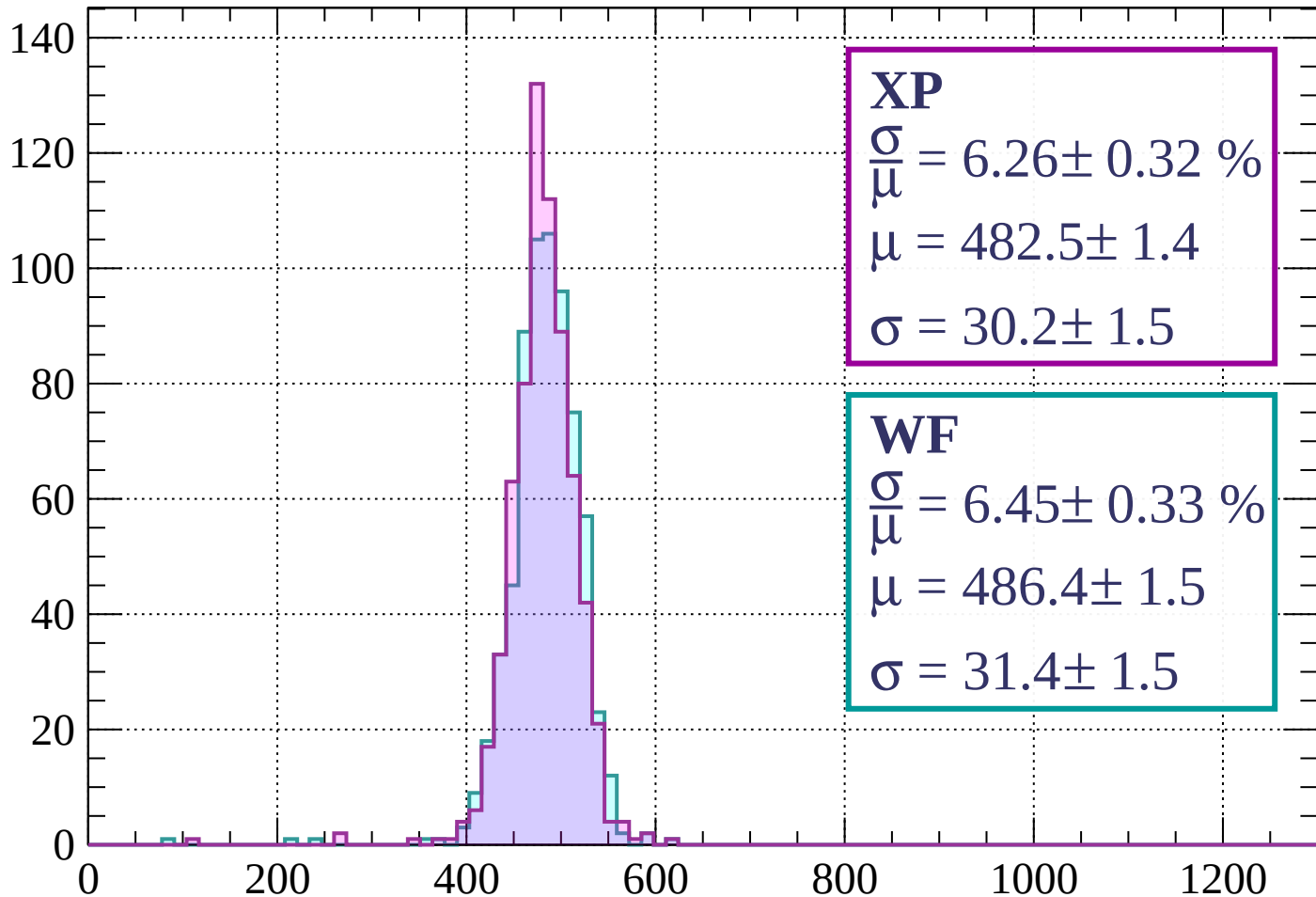
$$\mu = 483.9 \pm 1.9$$

$$\sigma = 34.5 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | -1960 < p < -1920

Count



XP

$$\frac{\sigma}{\mu} = 6.26 \pm 0.32 \%$$

$$\mu = 482.5 \pm 1.4$$

$$\sigma = 30.2 \pm 1.5$$

WF

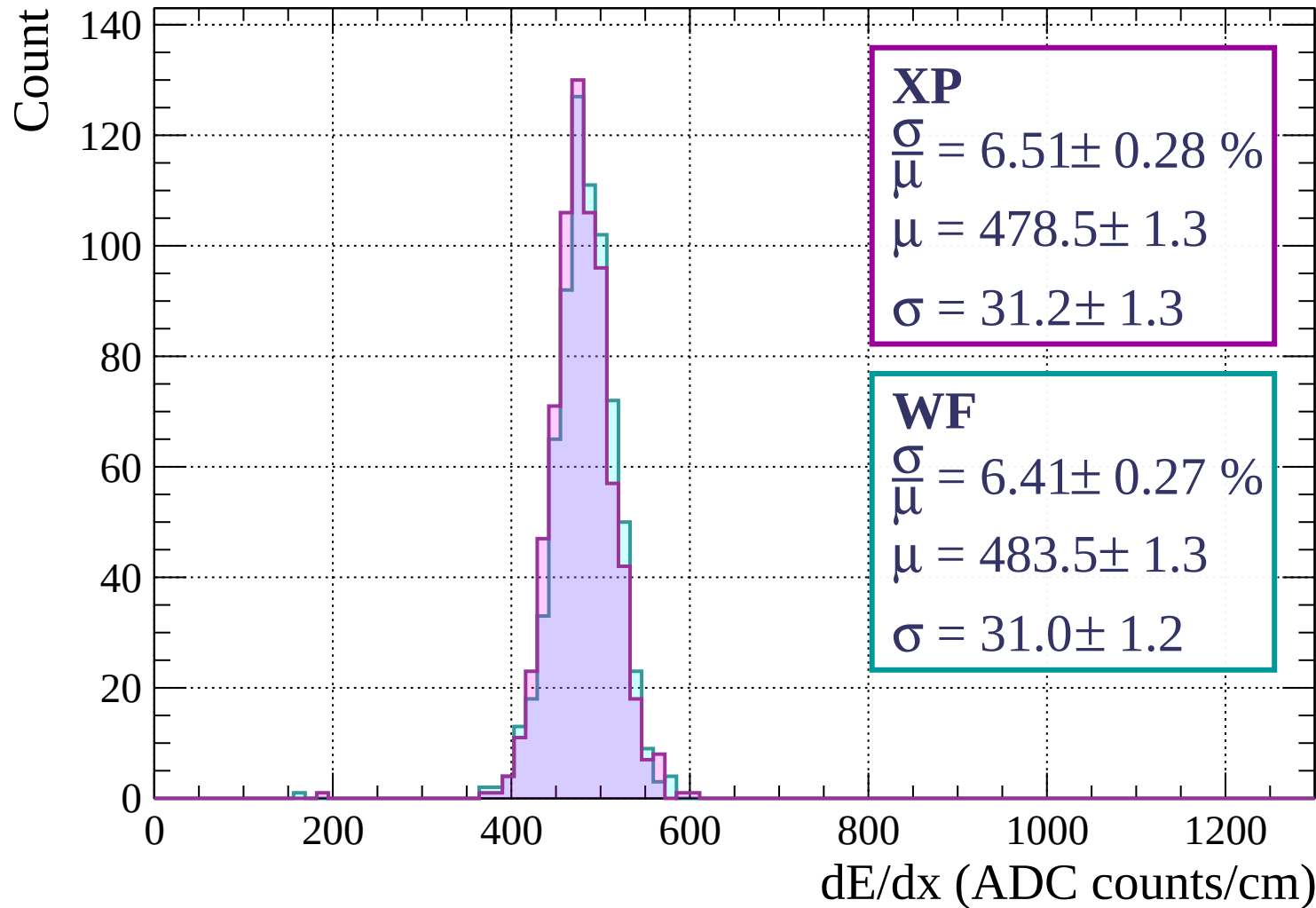
$$\frac{\sigma}{\mu} = 6.45 \pm 0.33 \%$$

$$\mu = 486.4 \pm 1.5$$

$$\sigma = 31.4 \pm 1.5$$

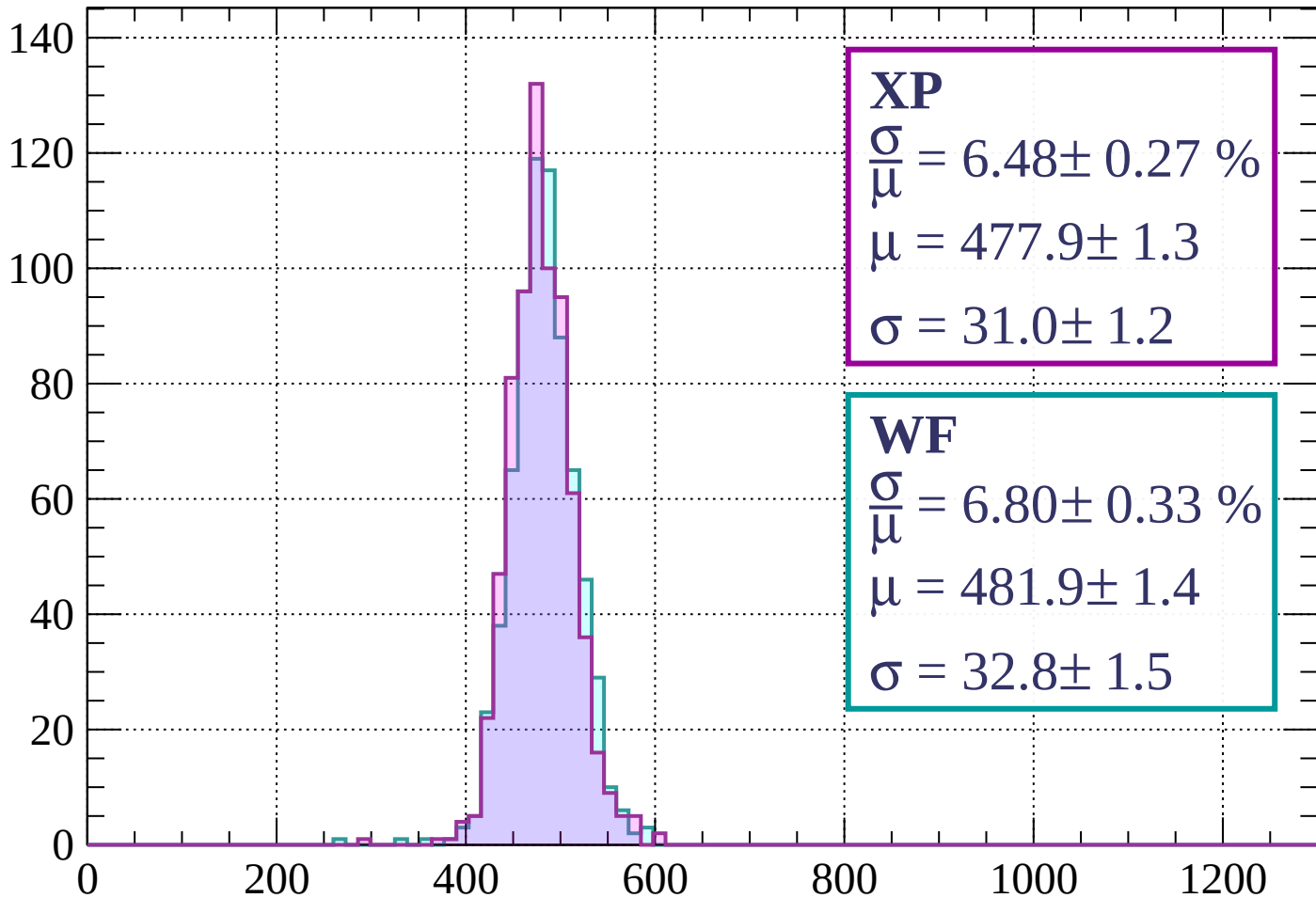
dE/dx (ADC counts/cm)

Energy loss | -1920 < p < -1880



Energy loss | $-1880 < p < -1840$

Count



XP

$$\frac{\sigma}{\mu} = 6.48 \pm 0.27 \%$$

$$\mu = 477.9 \pm 1.3$$

$$\sigma = 31.0 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.80 \pm 0.33 \%$$

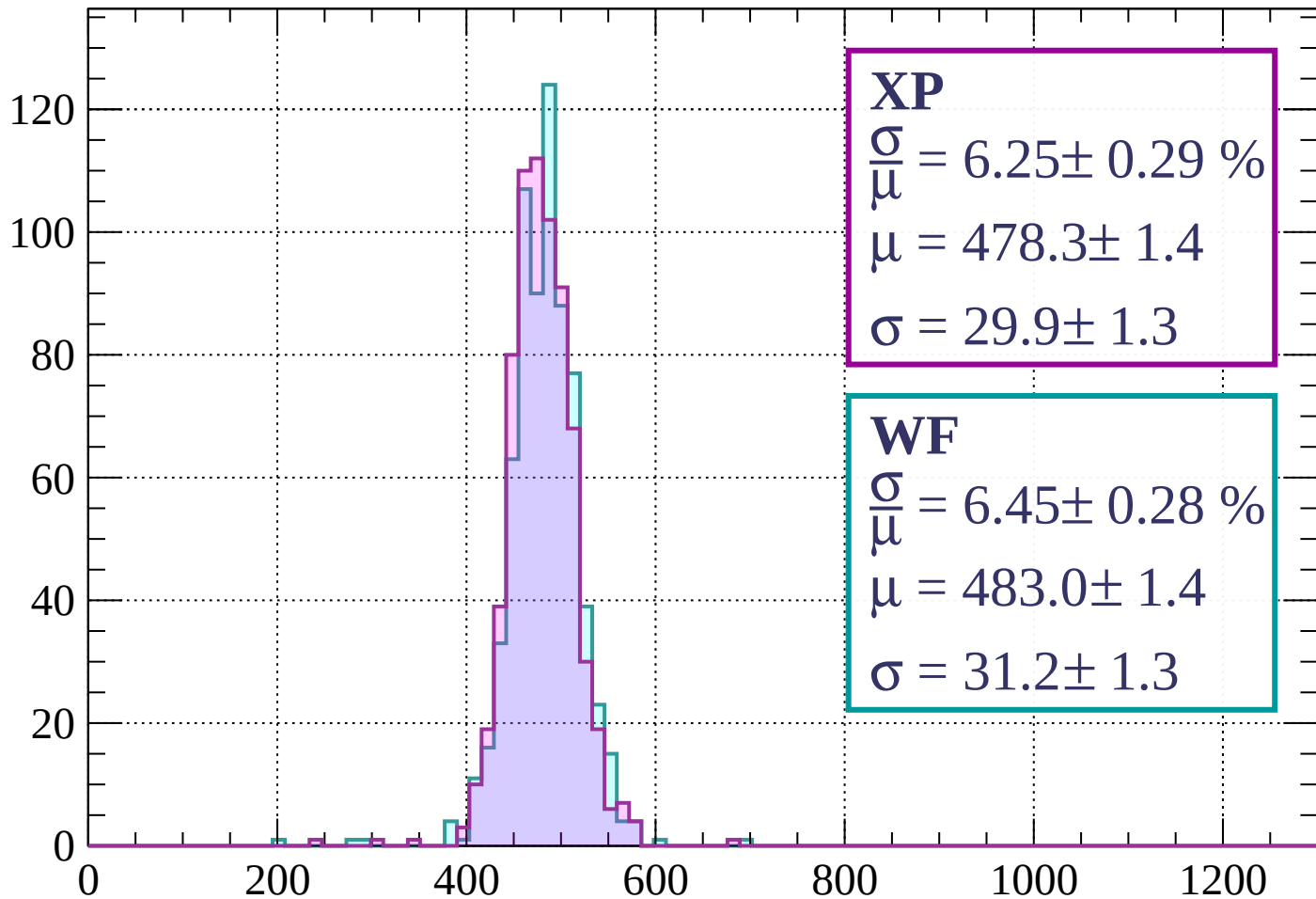
$$\mu = 481.9 \pm 1.4$$

$$\sigma = 32.8 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1840 < p < -1800$

Count



XP

$$\frac{\sigma}{\mu} = 6.25 \pm 0.29 \%$$

$$\mu = 478.3 \pm 1.4$$

$$\sigma = 29.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.45 \pm 0.28 \%$$

$$\mu = 483.0 \pm 1.4$$

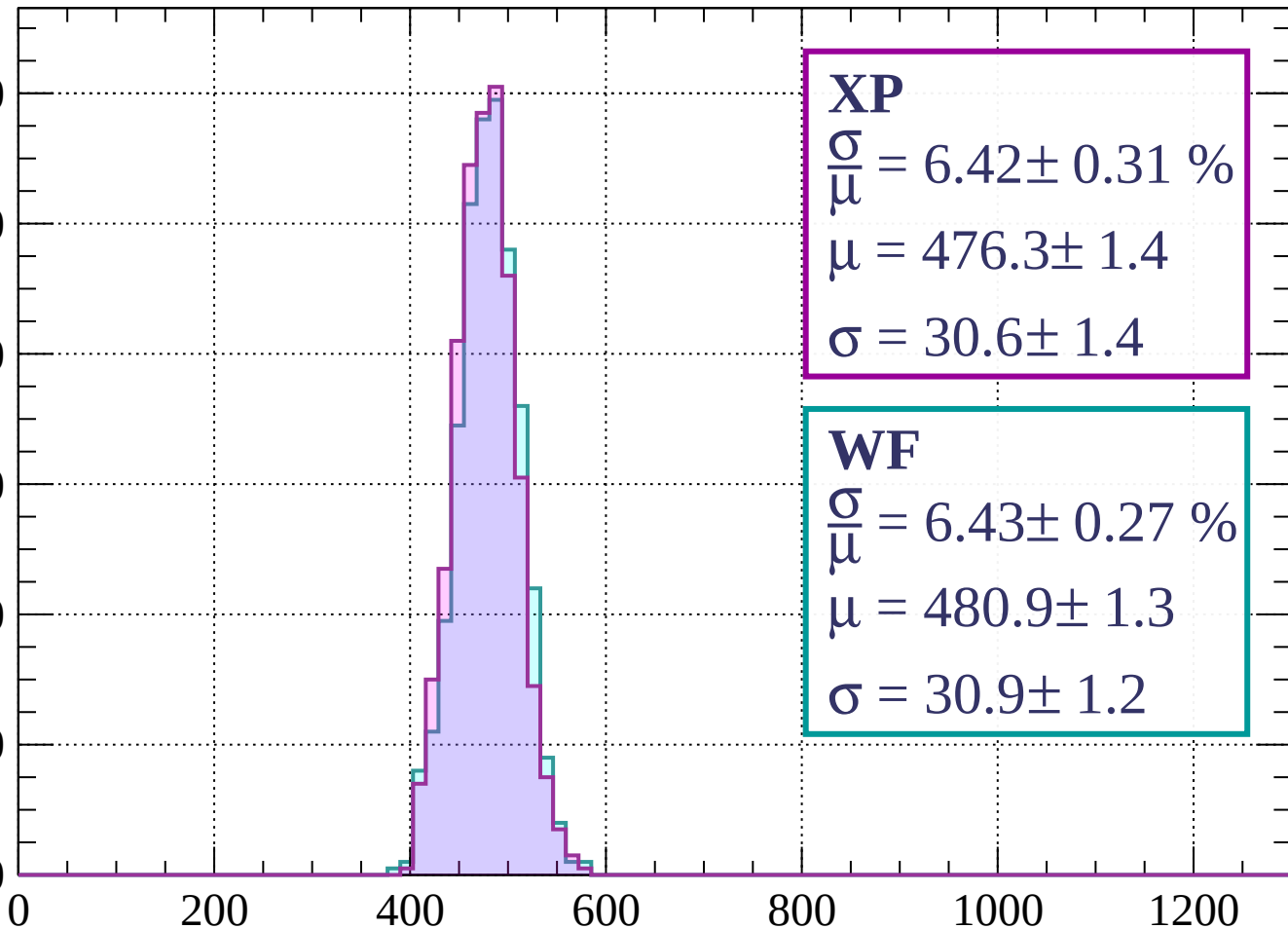
$$\sigma = 31.2 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1760$

Count

120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 6.42 \pm 0.31 \%$$

$$\mu = 476.3 \pm 1.4$$

$$\sigma = 30.6 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.43 \pm 0.27 \%$$

$$\mu = 480.9 \pm 1.3$$

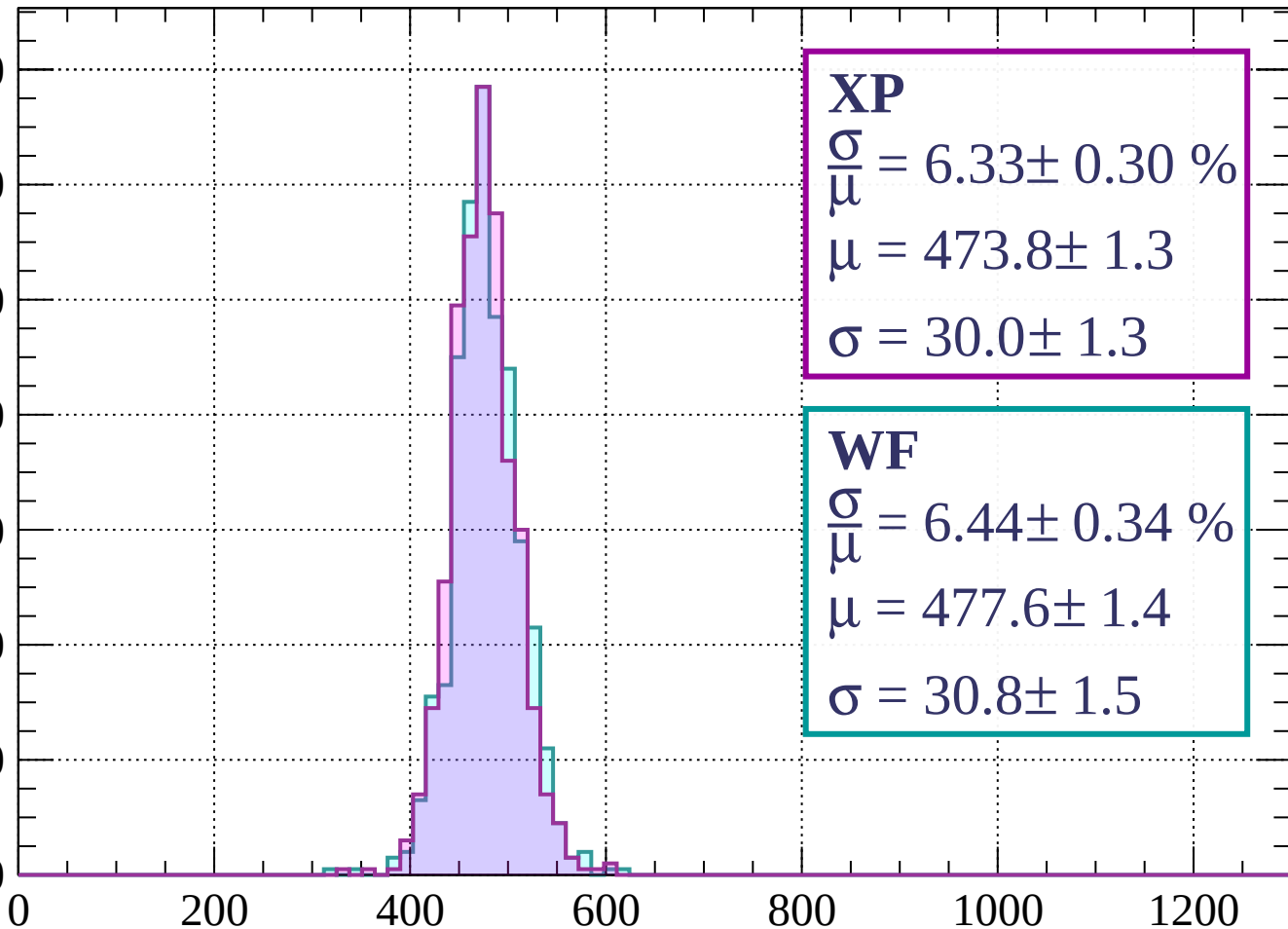
$$\sigma = 30.9 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1760 < p < -1720$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 6.33 \pm 0.30 \%$$

$$\mu = 473.8 \pm 1.3$$

$$\sigma = 30.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.44 \pm 0.34 \%$$

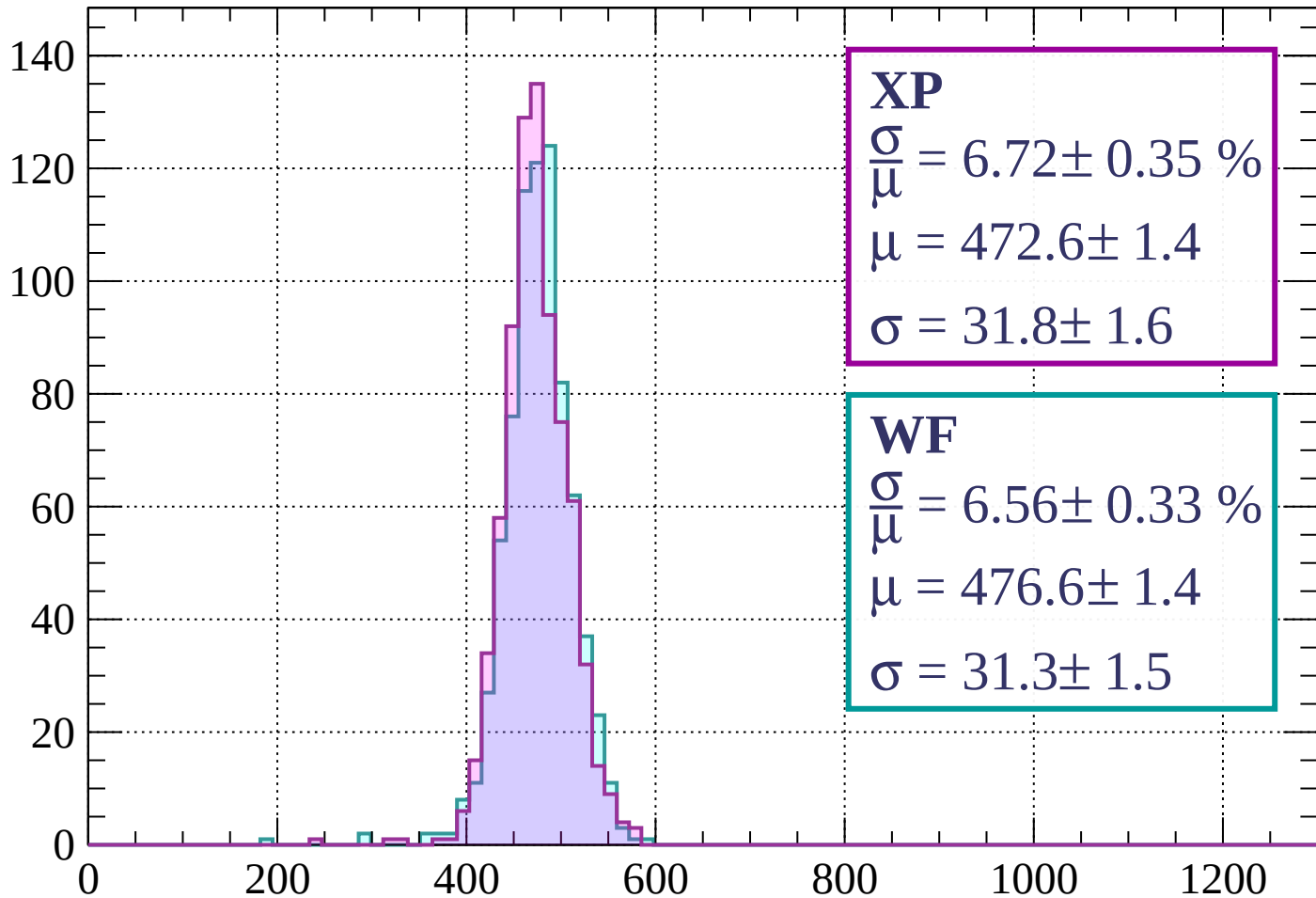
$$\mu = 477.6 \pm 1.4$$

$$\sigma = 30.8 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 6.72 \pm 0.35 \%$$

$$\mu = 472.6 \pm 1.4$$

$$\sigma = 31.8 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 6.56 \pm 0.33 \%$$

$$\mu = 476.6 \pm 1.4$$

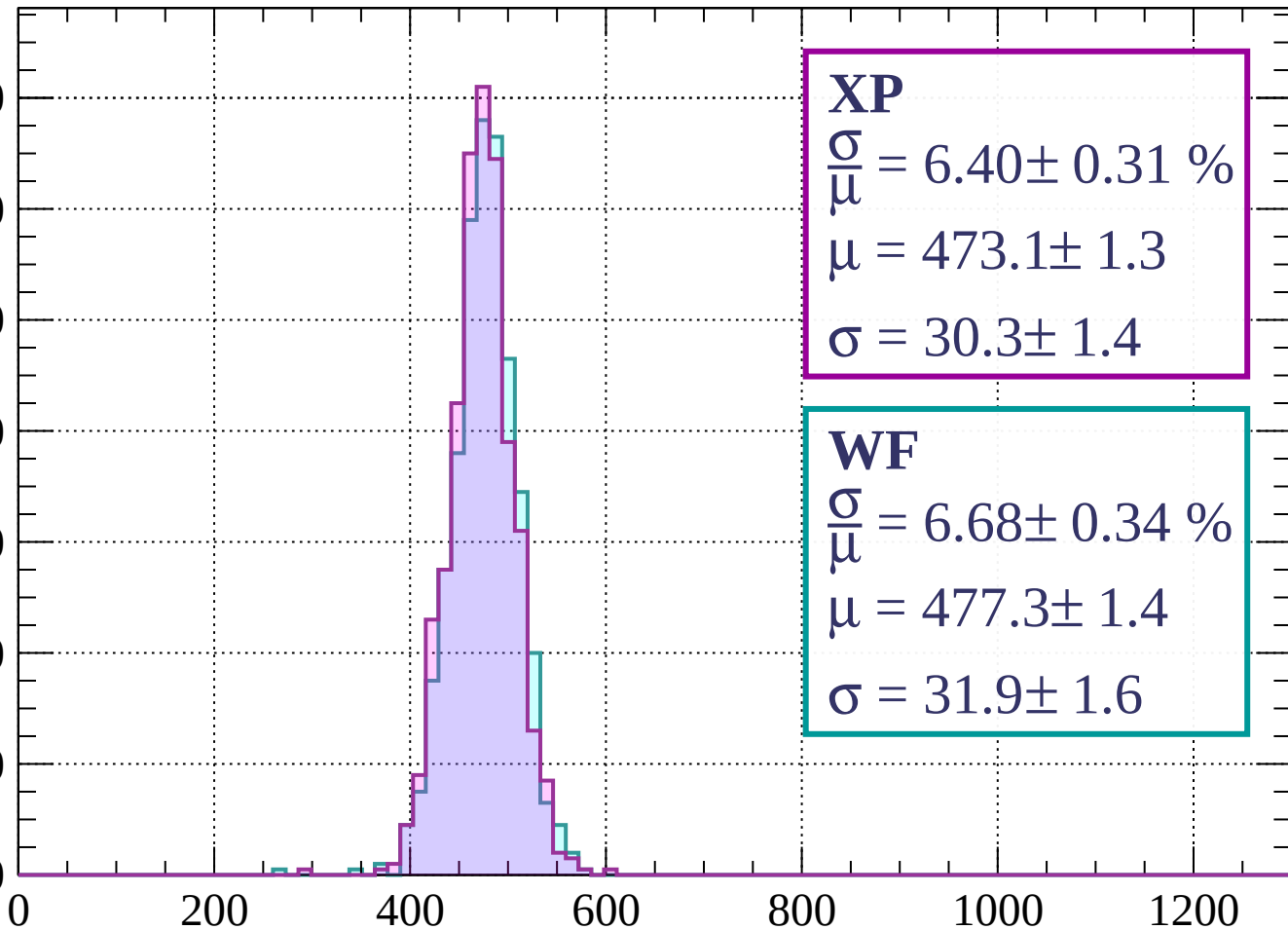
$$\sigma = 31.3 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 6.40 \pm 0.31 \%$$

$$\mu = 473.1 \pm 1.3$$

$$\sigma = 30.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.68 \pm 0.34 \%$$

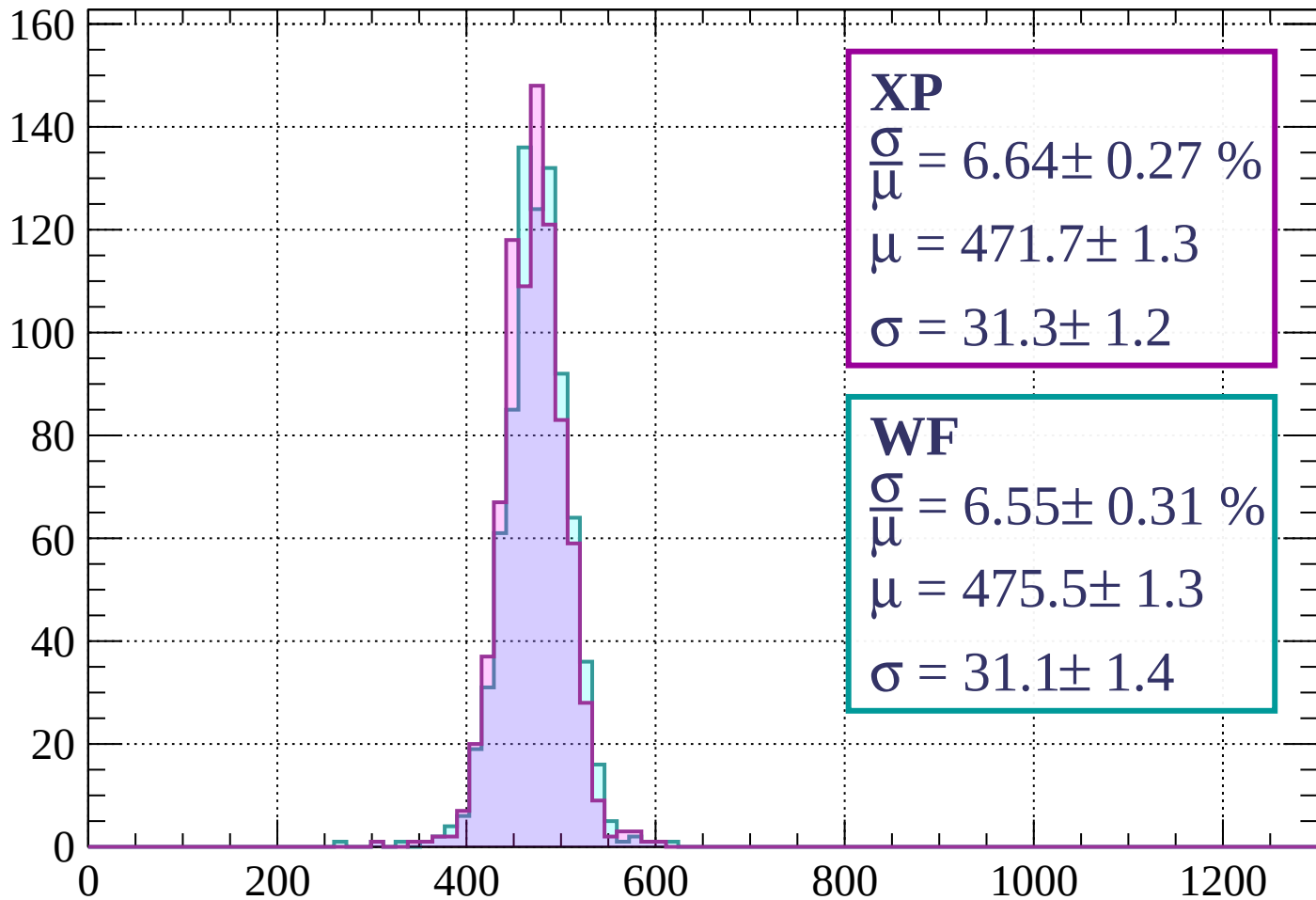
$$\mu = 477.3 \pm 1.4$$

$$\sigma = 31.9 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$

Count



XP

$$\frac{\sigma}{\mu} = 6.64 \pm 0.27 \%$$

$$\mu = 471.7 \pm 1.3$$

$$\sigma = 31.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.55 \pm 0.31 \%$$

$$\mu = 475.5 \pm 1.3$$

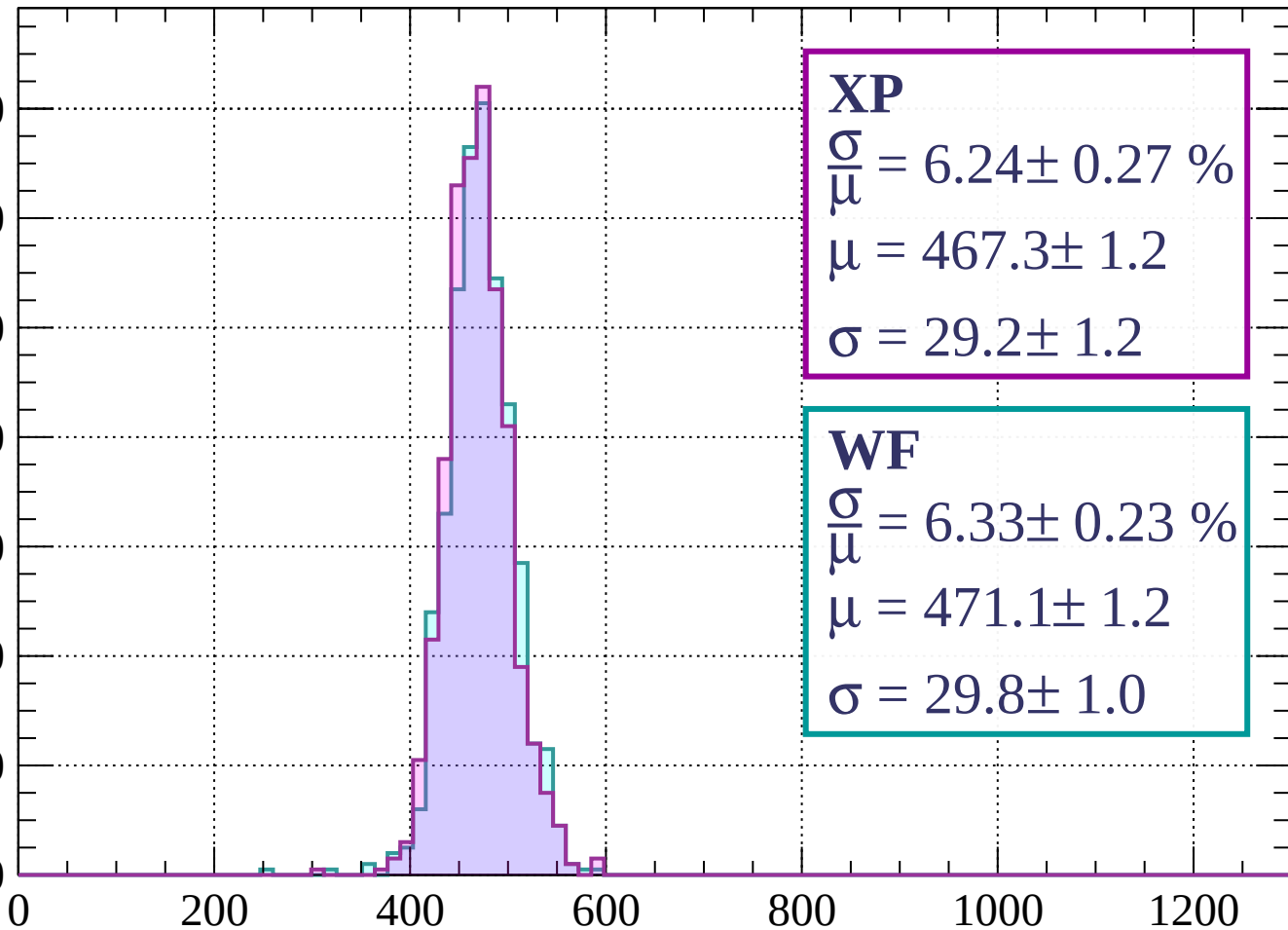
$$\sigma = 31.1 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-1600 < p < -1560$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 6.24 \pm 0.27 \%$$

$$\mu = 467.3 \pm 1.2$$

$$\sigma = 29.2 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.33 \pm 0.23 \%$$

$$\mu = 471.1 \pm 1.2$$

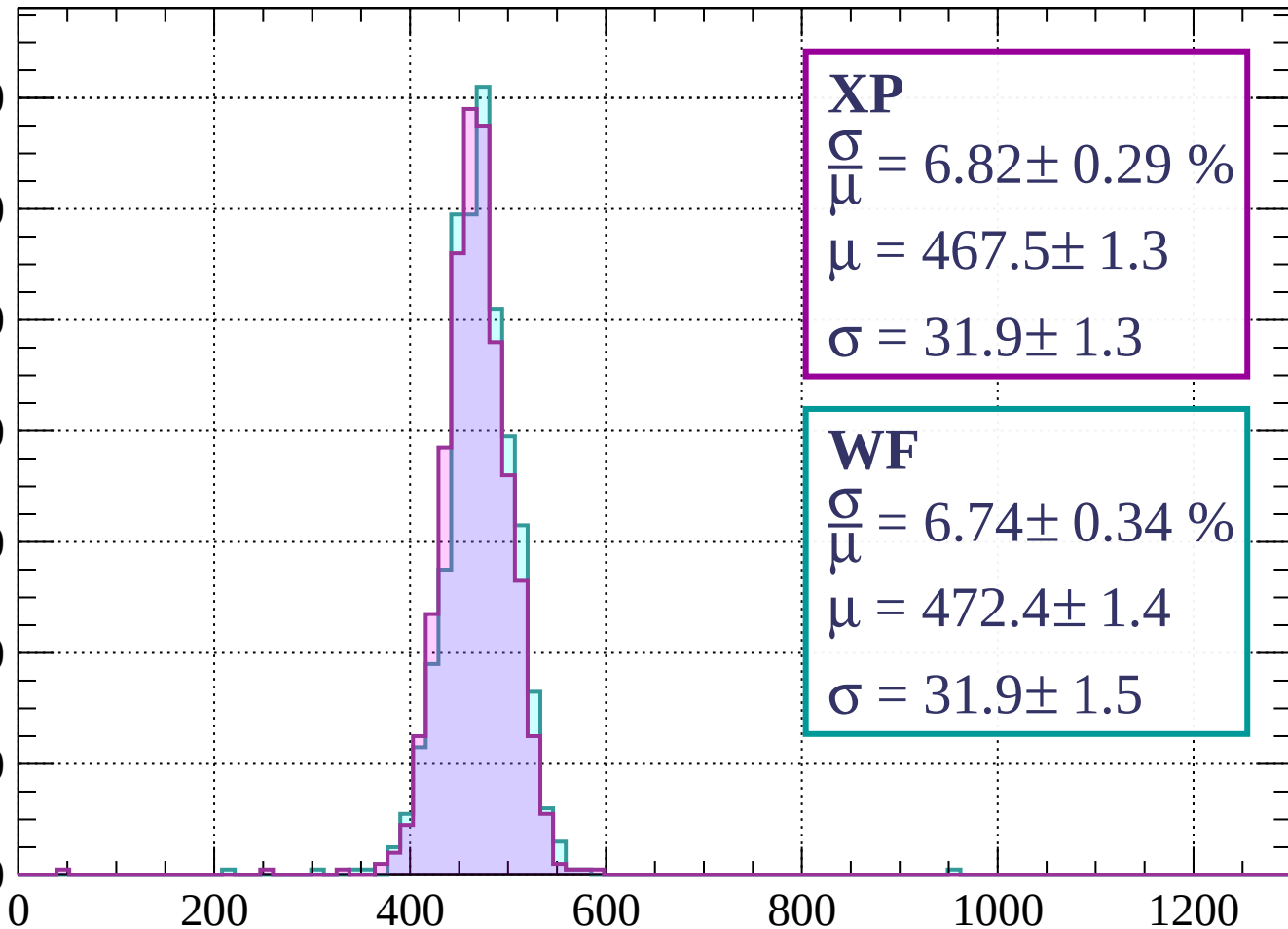
$$\sigma = 29.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1560 < p < -1520$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 6.82 \pm 0.29 \%$$

$$\mu = 467.5 \pm 1.3$$

$$\sigma = 31.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.74 \pm 0.34 \%$$

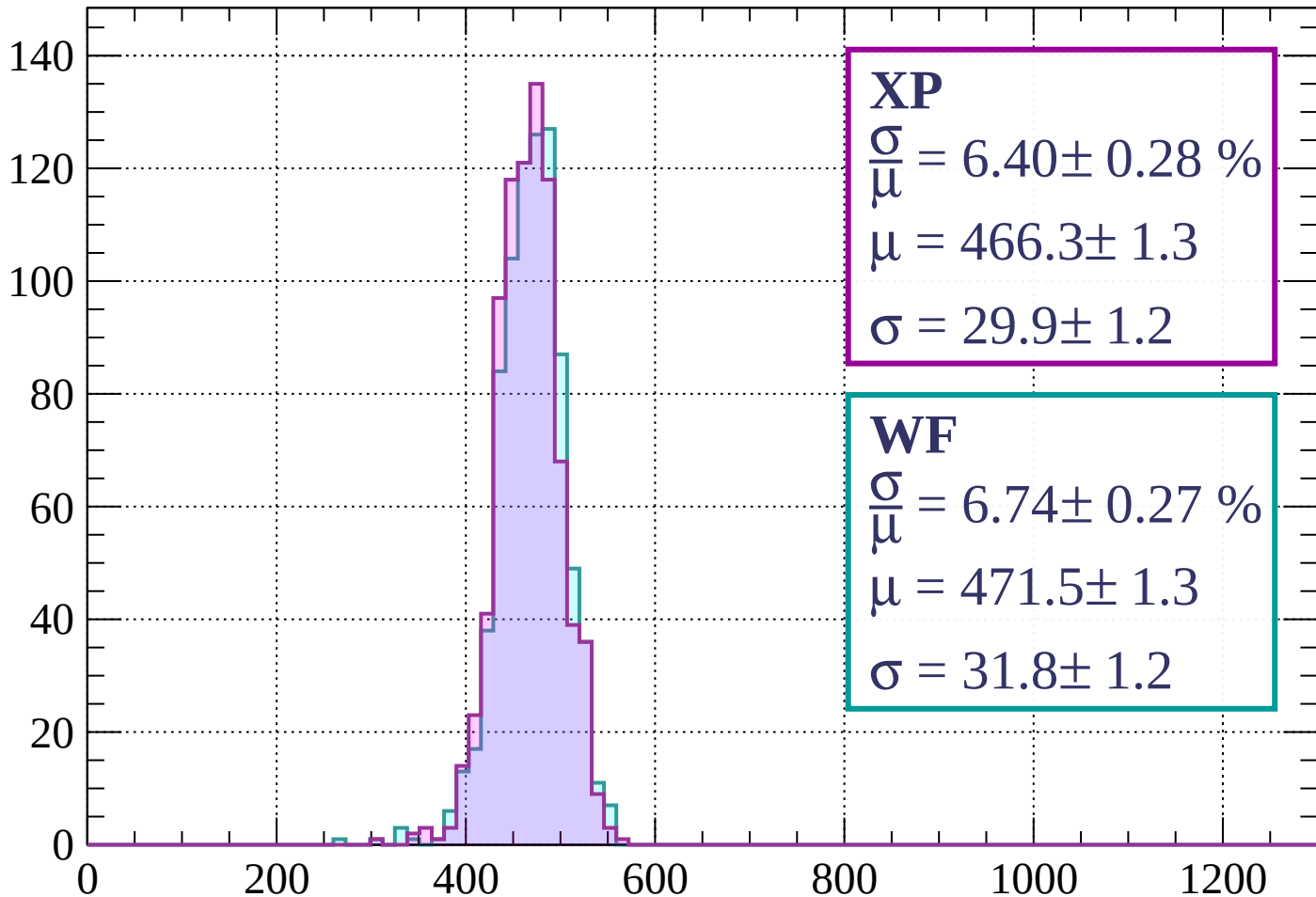
$$\mu = 472.4 \pm 1.4$$

$$\sigma = 31.9 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1520 < p < -1480$

Count



XP

$$\frac{\sigma}{\mu} = 6.40 \pm 0.28 \%$$

$$\mu = 466.3 \pm 1.3$$

$$\sigma = 29.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.74 \pm 0.27 \%$$

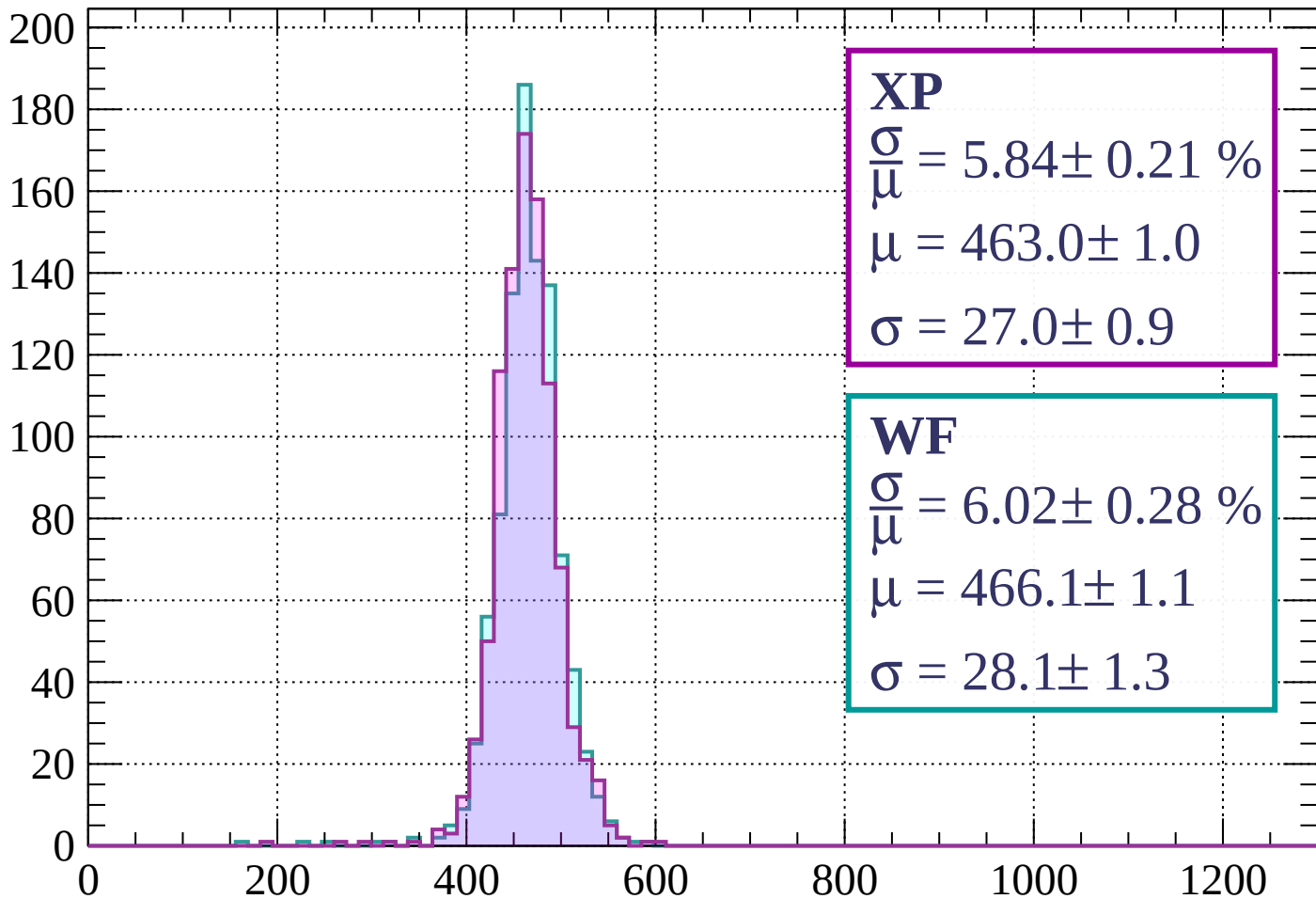
$$\mu = 471.5 \pm 1.3$$

$$\sigma = 31.8 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1480 < p < -1440$

Count



XP

$$\frac{\sigma}{\mu} = 5.84 \pm 0.21 \%$$

$$\mu = 463.0 \pm 1.0$$

$$\sigma = 27.0 \pm 0.9$$

WF

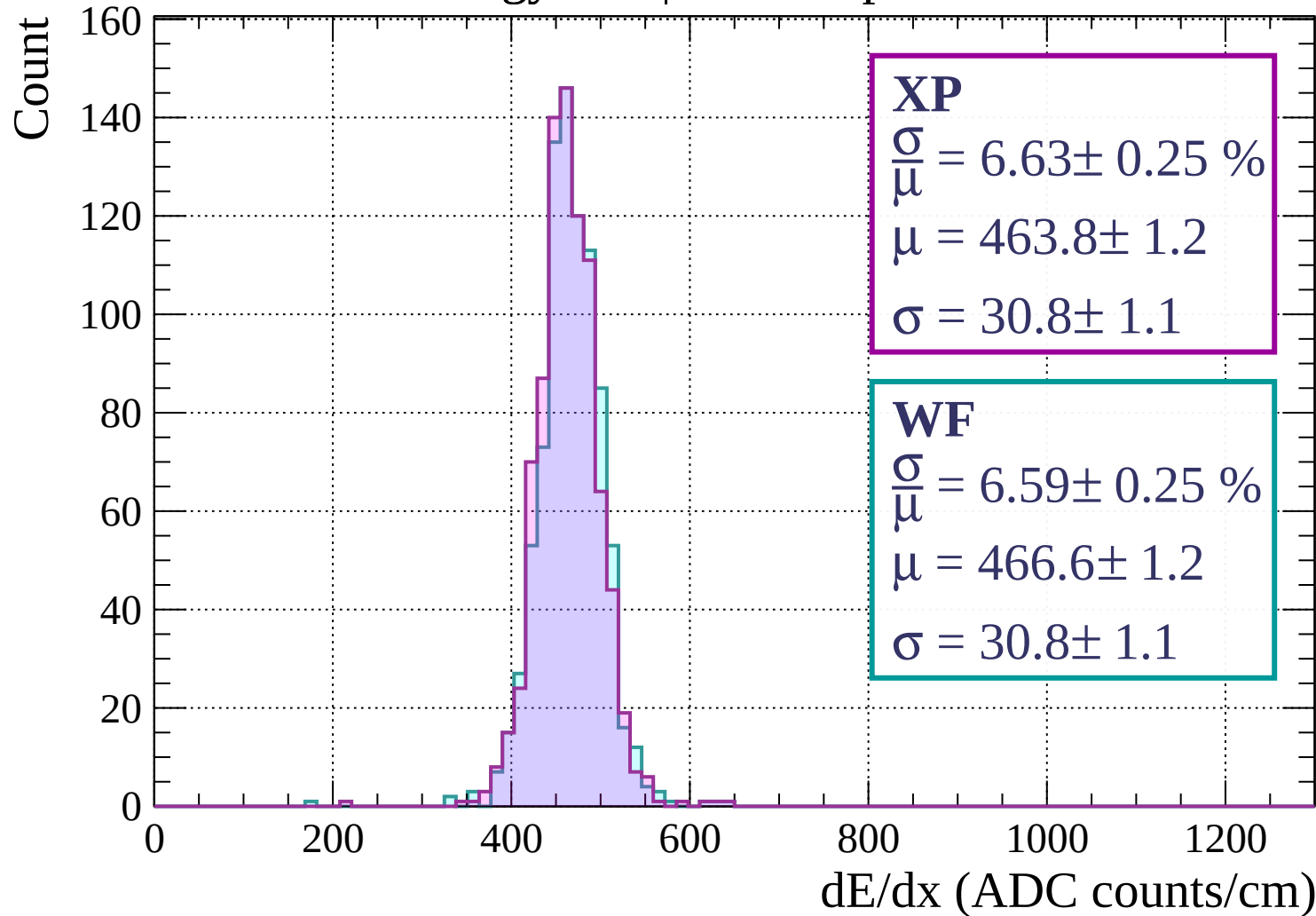
$$\frac{\sigma}{\mu} = 6.02 \pm 0.28 \%$$

$$\mu = 466.1 \pm 1.1$$

$$\sigma = 28.1 \pm 1.3$$

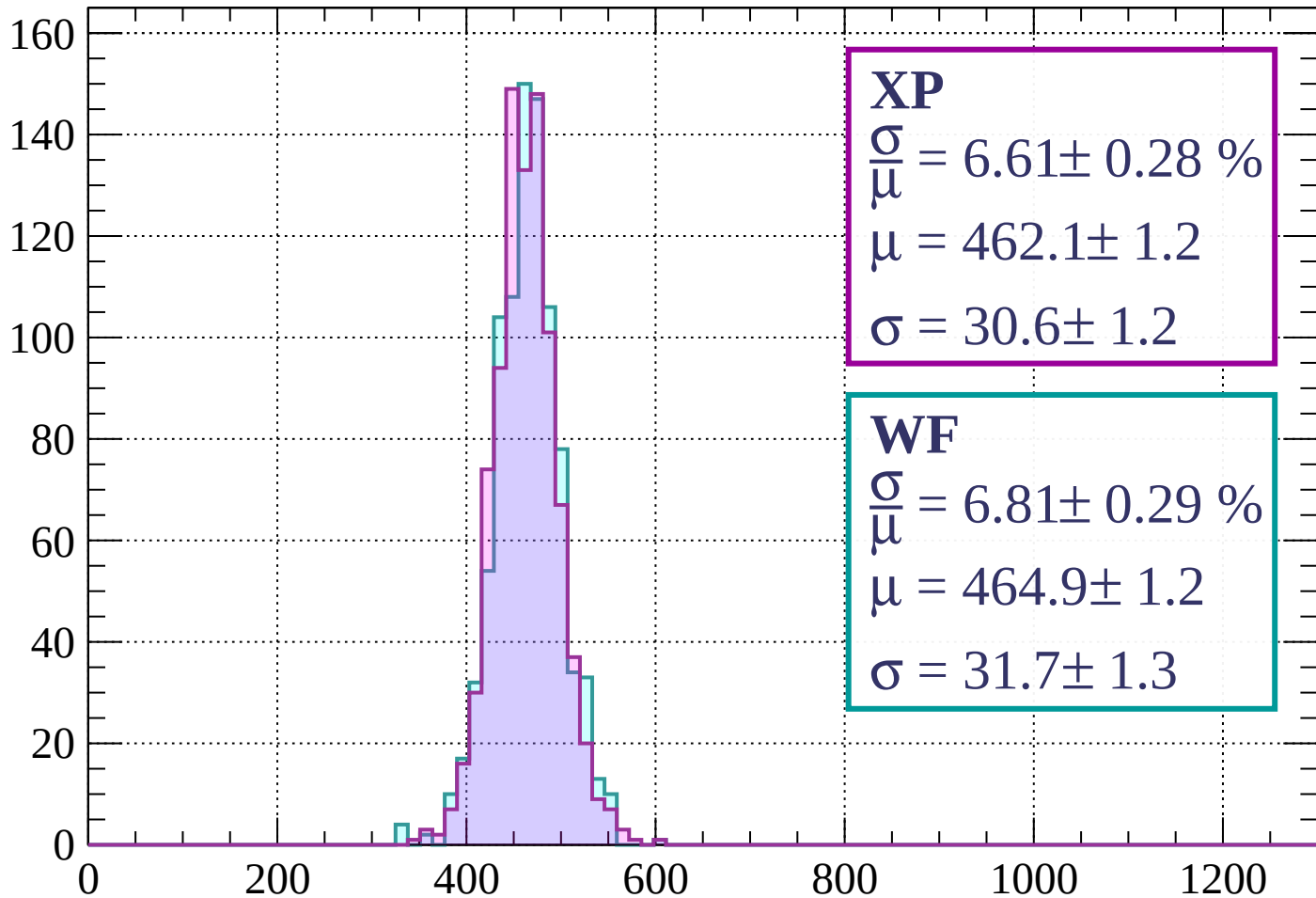
dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1400$



Energy loss | $-1400 < p < -1360$

Count



XP

$$\frac{\sigma}{\mu} = 6.61 \pm 0.28 \%$$

$$\mu = 462.1 \pm 1.2$$

$$\sigma = 30.6 \pm 1.2$$

WF

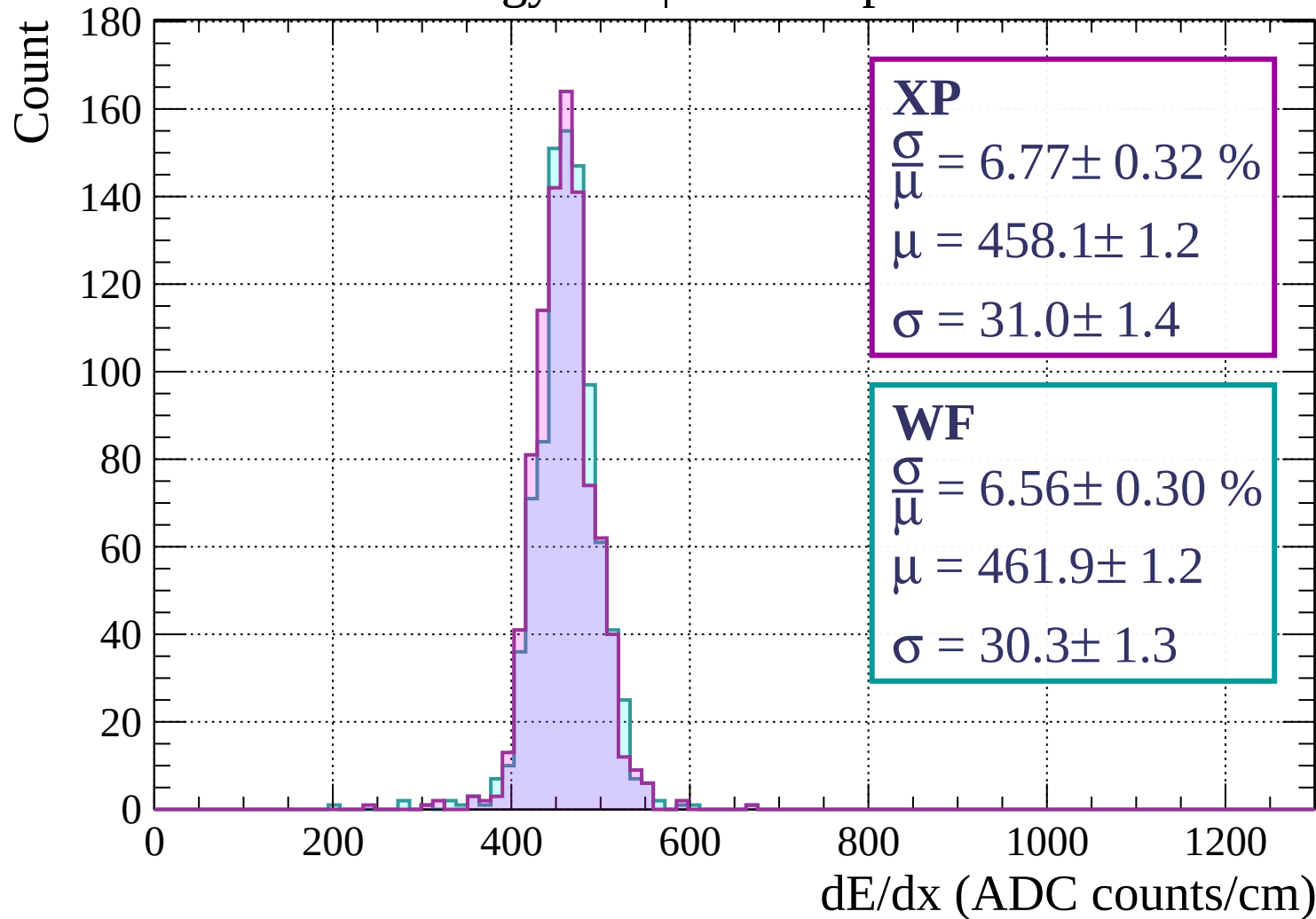
$$\frac{\sigma}{\mu} = 6.81 \pm 0.29 \%$$

$$\mu = 464.9 \pm 1.2$$

$$\sigma = 31.7 \pm 1.3$$

dE/dx (ADC counts/cm)

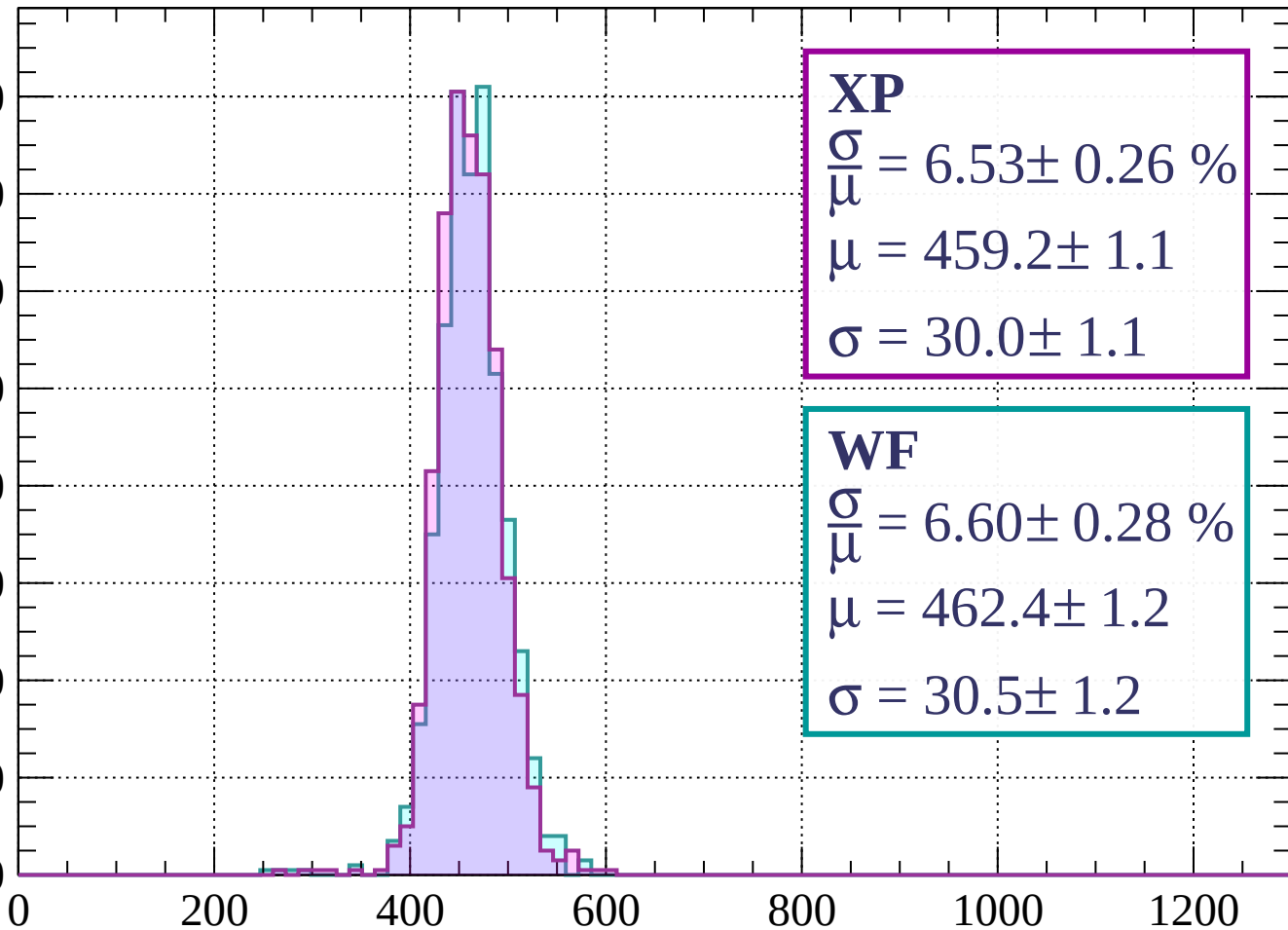
Energy loss | $-1360 < p < -1320$



Energy loss | $-1320 < p < -1280$

Count

160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 6.53 \pm 0.26 \%$$

$$\mu = 459.2 \pm 1.1$$

$$\sigma = 30.0 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.60 \pm 0.28 \%$$

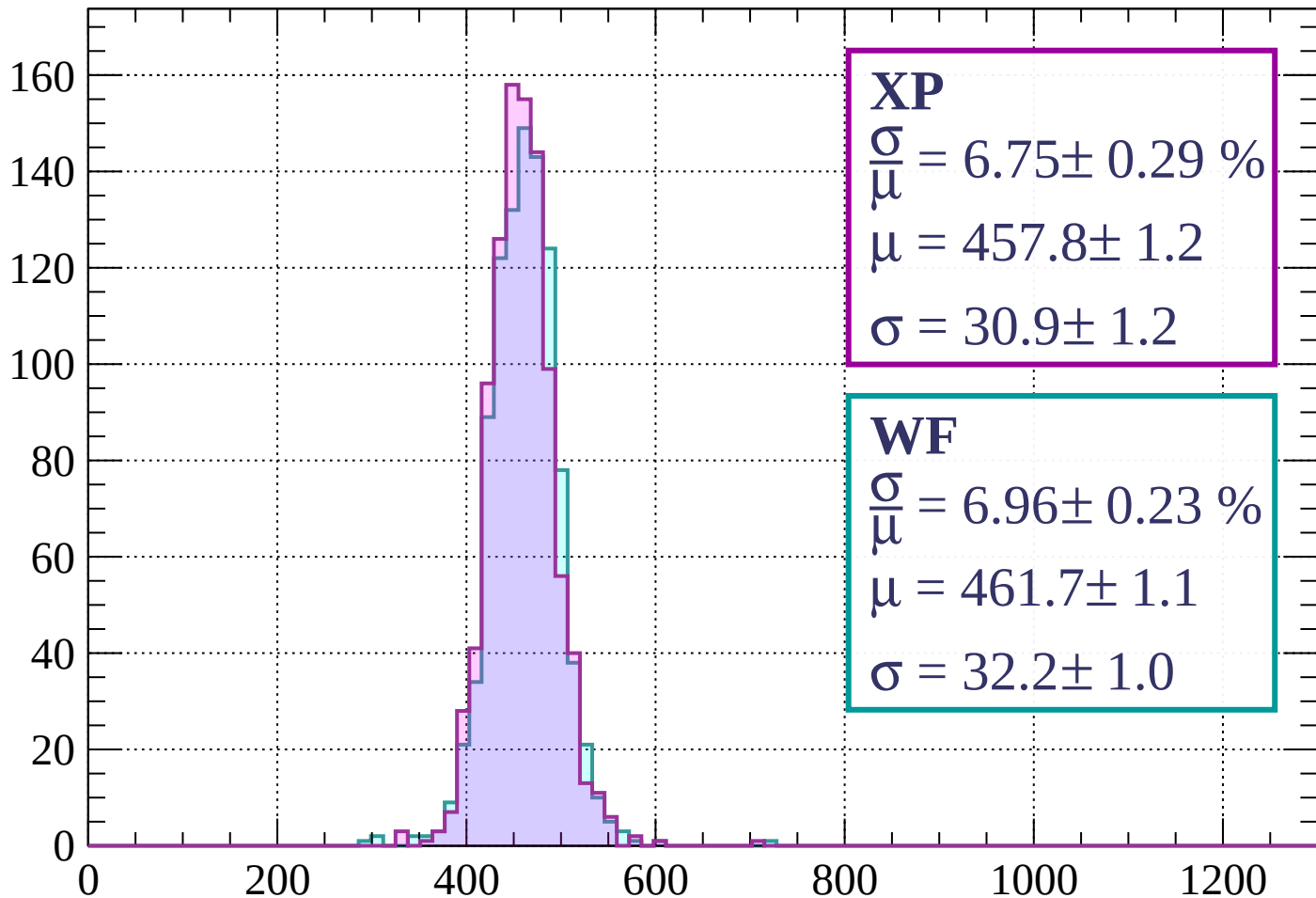
$$\mu = 462.4 \pm 1.2$$

$$\sigma = 30.5 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 6.75 \pm 0.29 \%$$

$$\mu = 457.8 \pm 1.2$$

$$\sigma = 30.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 6.96 \pm 0.23 \%$$

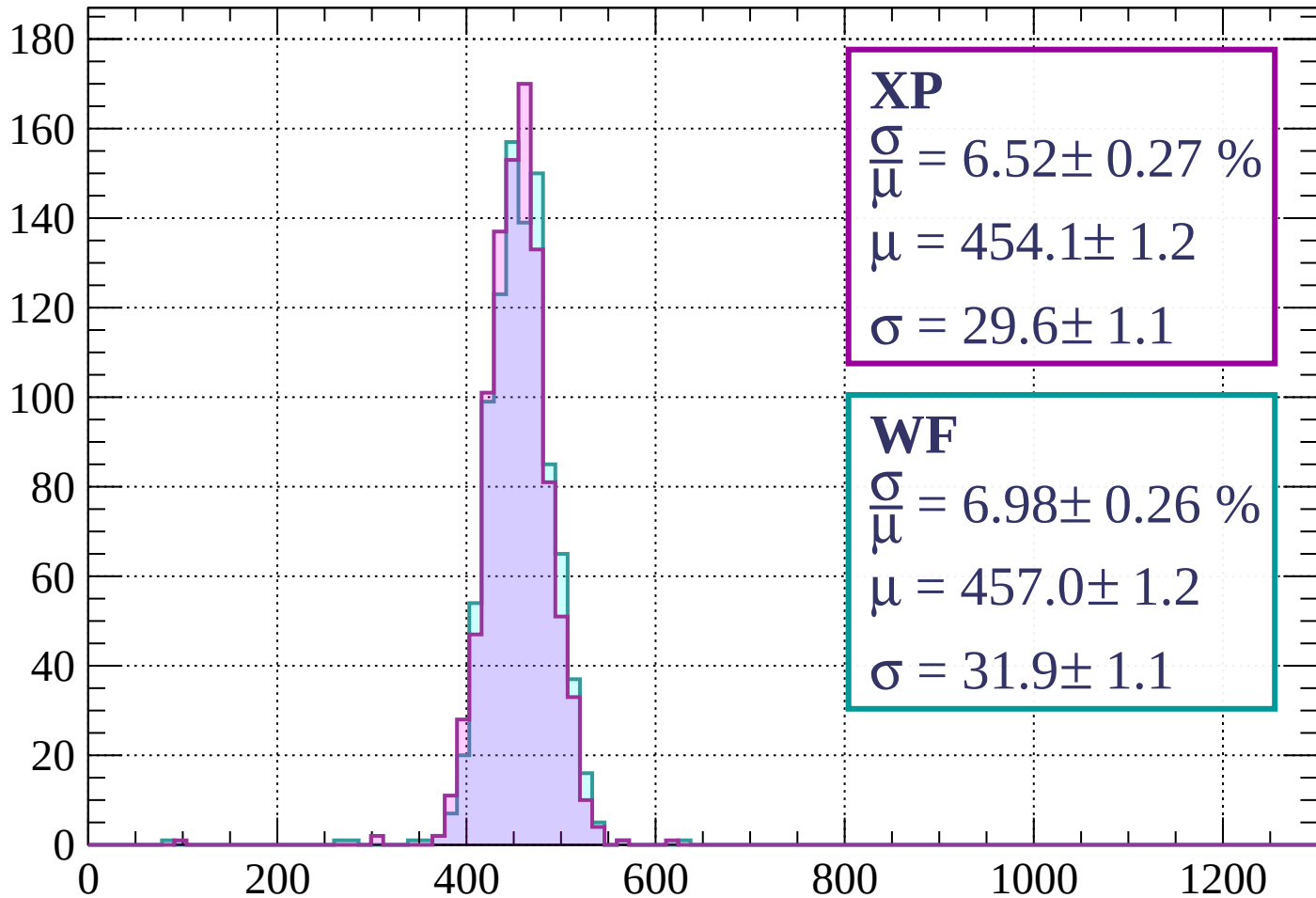
$$\mu = 461.7 \pm 1.1$$

$$\sigma = 32.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 6.52 \pm 0.27 \%$$

$$\mu = 454.1 \pm 1.2$$

$$\sigma = 29.6 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.98 \pm 0.26 \%$$

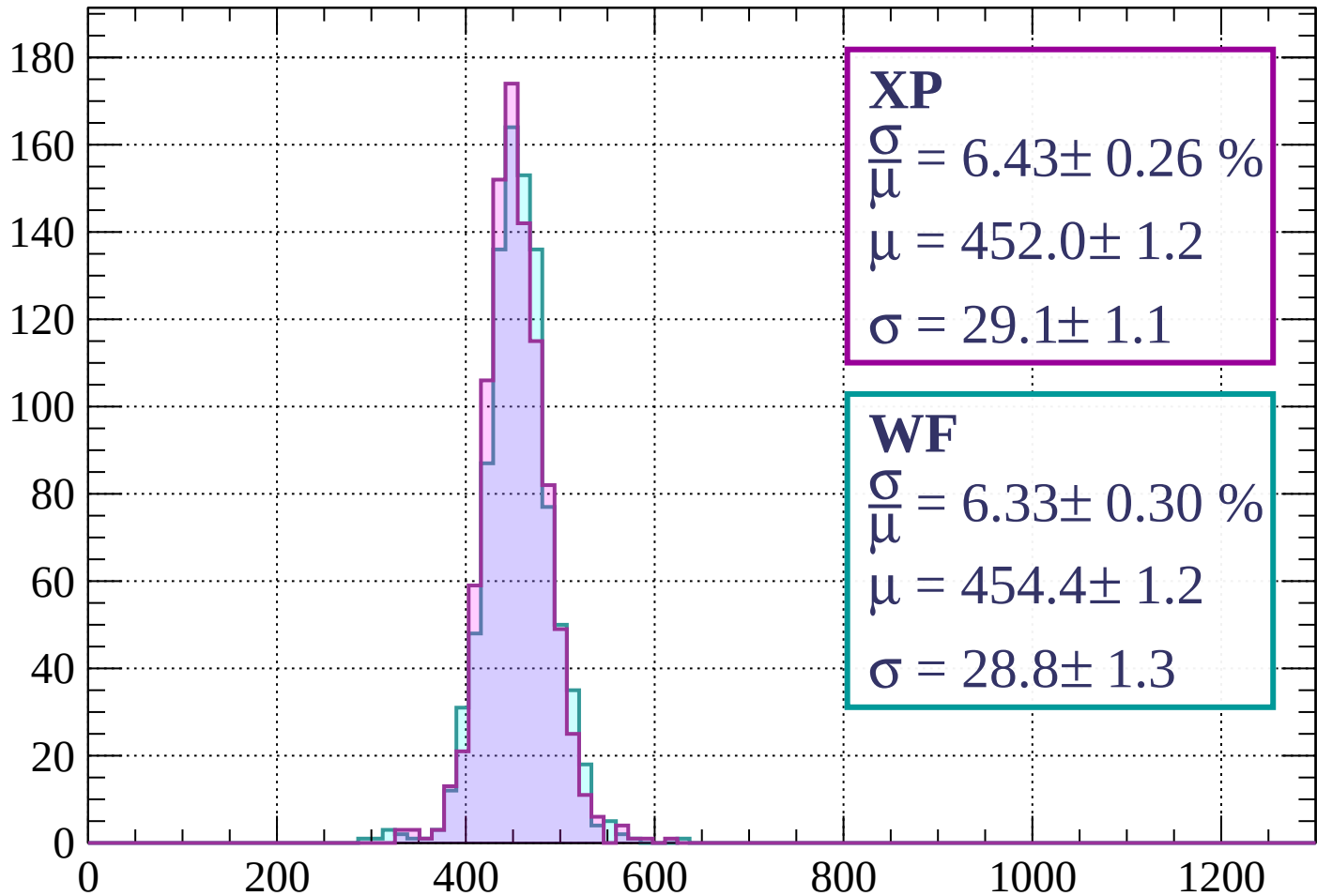
$$\mu = 457.0 \pm 1.2$$

$$\sigma = 31.9 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1200 < p < -1160$

Count



XP

$$\frac{\sigma}{\mu} = 6.43 \pm 0.26 \%$$

$$\mu = 452.0 \pm 1.2$$

$$\sigma = 29.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.33 \pm 0.30 \%$$

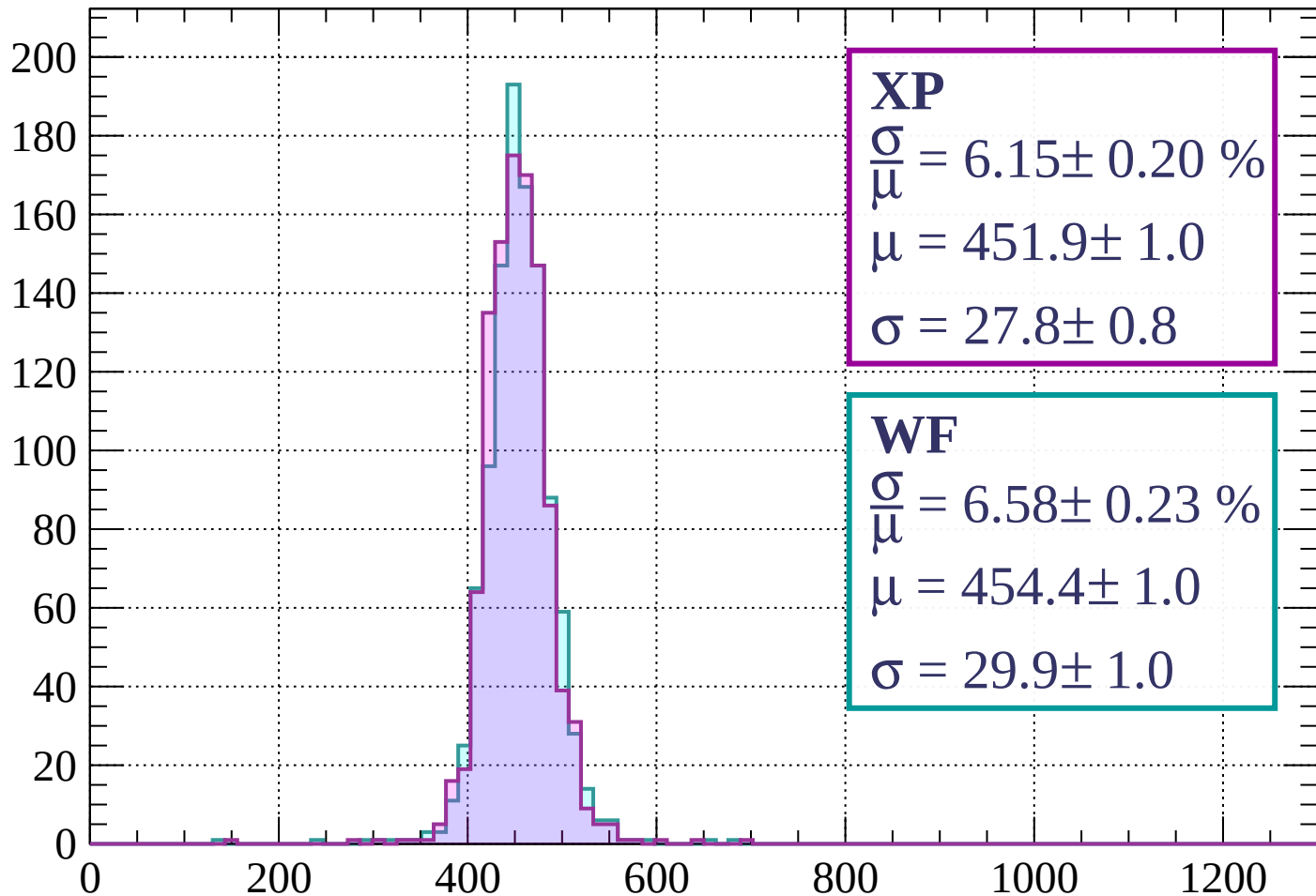
$$\mu = 454.4 \pm 1.2$$

$$\sigma = 28.8 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-1160 < p < -1120$

Count



XP

$$\frac{\sigma}{\mu} = 6.15 \pm 0.20 \%$$

$$\mu = 451.9 \pm 1.0$$

$$\sigma = 27.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.58 \pm 0.23 \%$$

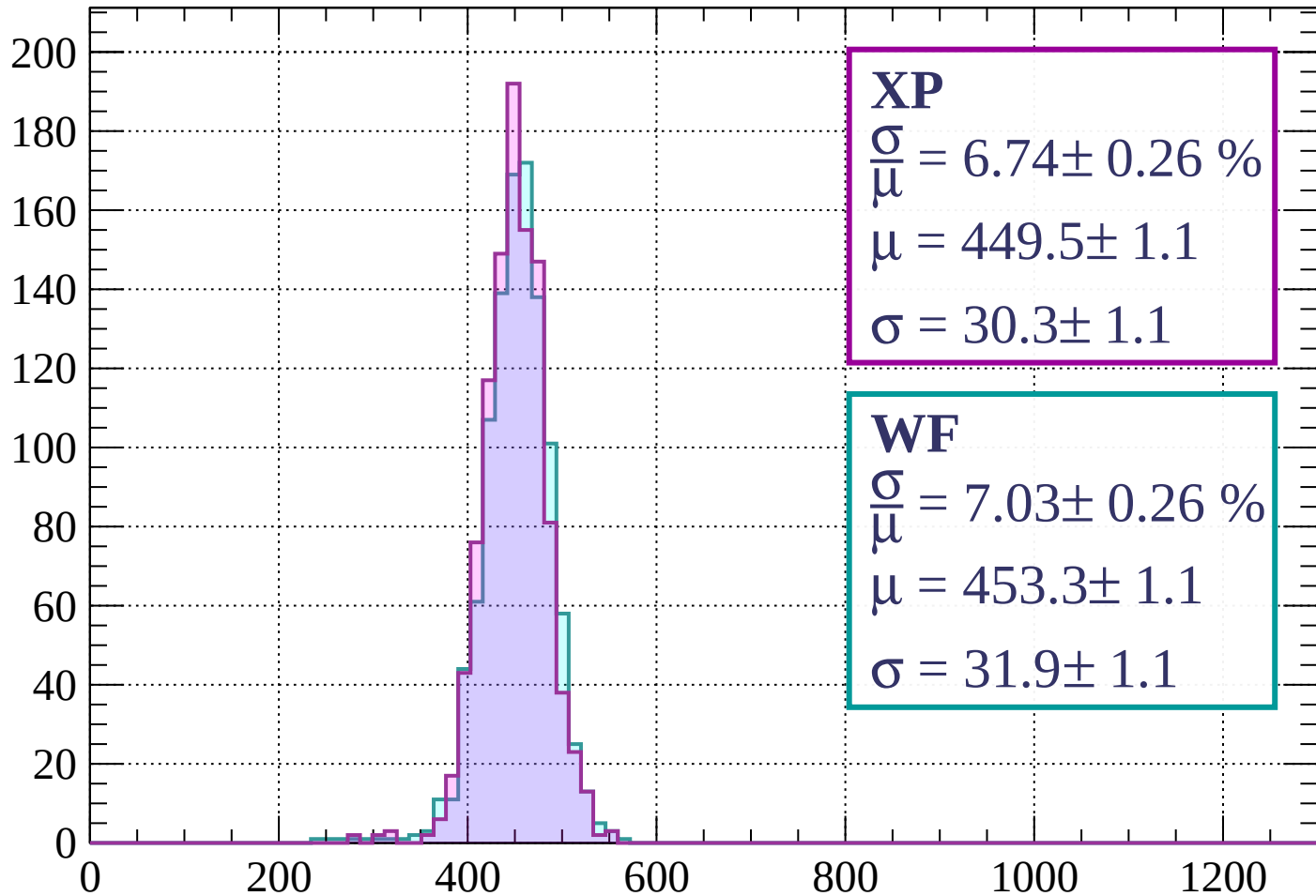
$$\mu = 454.4 \pm 1.0$$

$$\sigma = 29.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-1120 < p < -1080$

Count



XP

$$\frac{\sigma}{\mu} = 6.74 \pm 0.26 \%$$

$$\mu = 449.5 \pm 1.1$$

$$\sigma = 30.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.03 \pm 0.26 \%$$

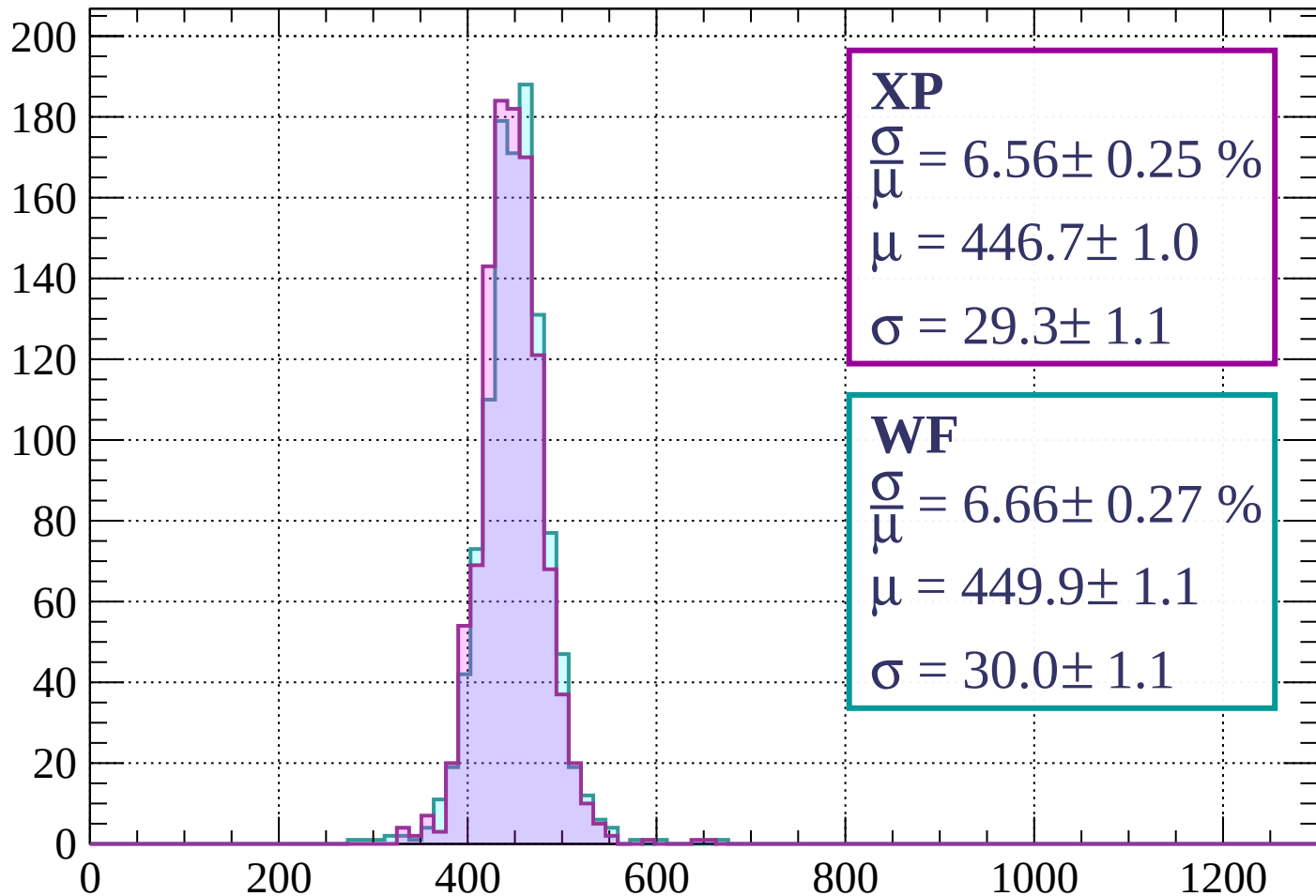
$$\mu = 453.3 \pm 1.1$$

$$\sigma = 31.9 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1040$

Count



XP

$$\frac{\sigma}{\mu} = 6.56 \pm 0.25 \%$$

$$\mu = 446.7 \pm 1.0$$

$$\sigma = 29.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 6.66 \pm 0.27 \%$$

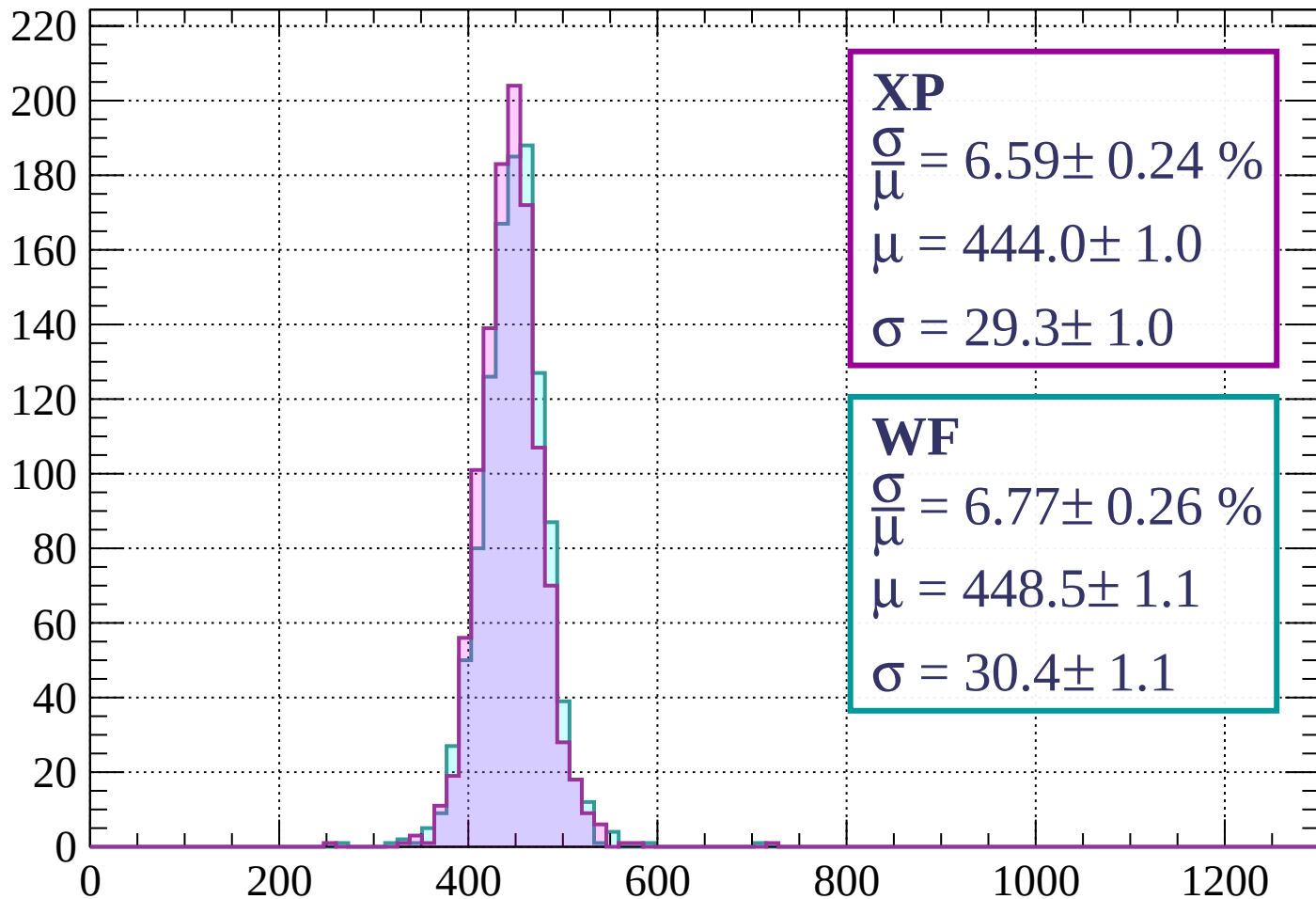
$$\mu = 449.9 \pm 1.1$$

$$\sigma = 30.0 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\mu} = 6.59 \pm 0.24 \%$$

$$\mu = 444.0 \pm 1.0$$

$$\sigma = 29.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.77 \pm 0.26 \%$$

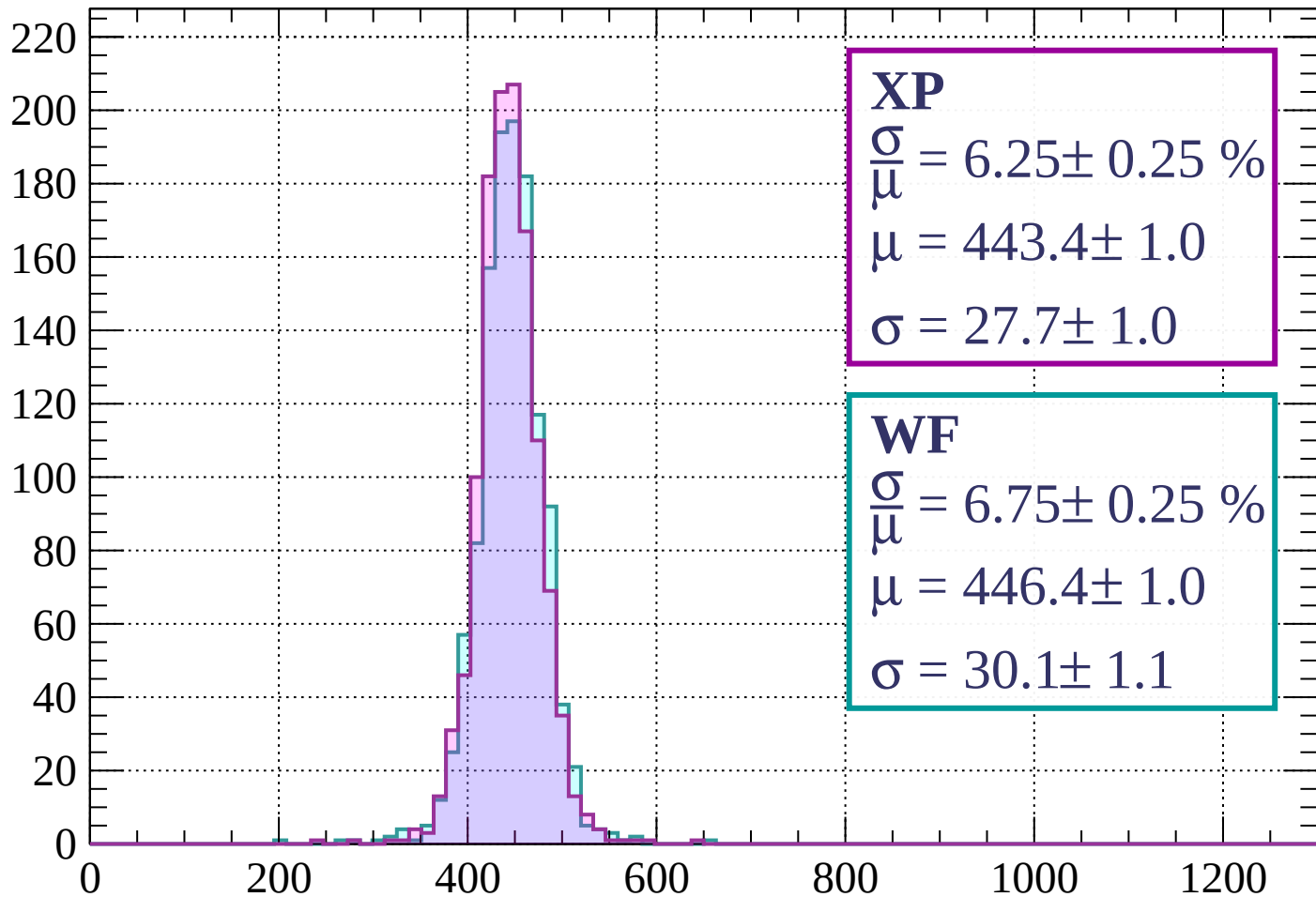
$$\mu = 448.5 \pm 1.1$$

$$\sigma = 30.4 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 6.25 \pm 0.25 \%$$

$$\mu = 443.4 \pm 1.0$$

$$\sigma = 27.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.75 \pm 0.25 \%$$

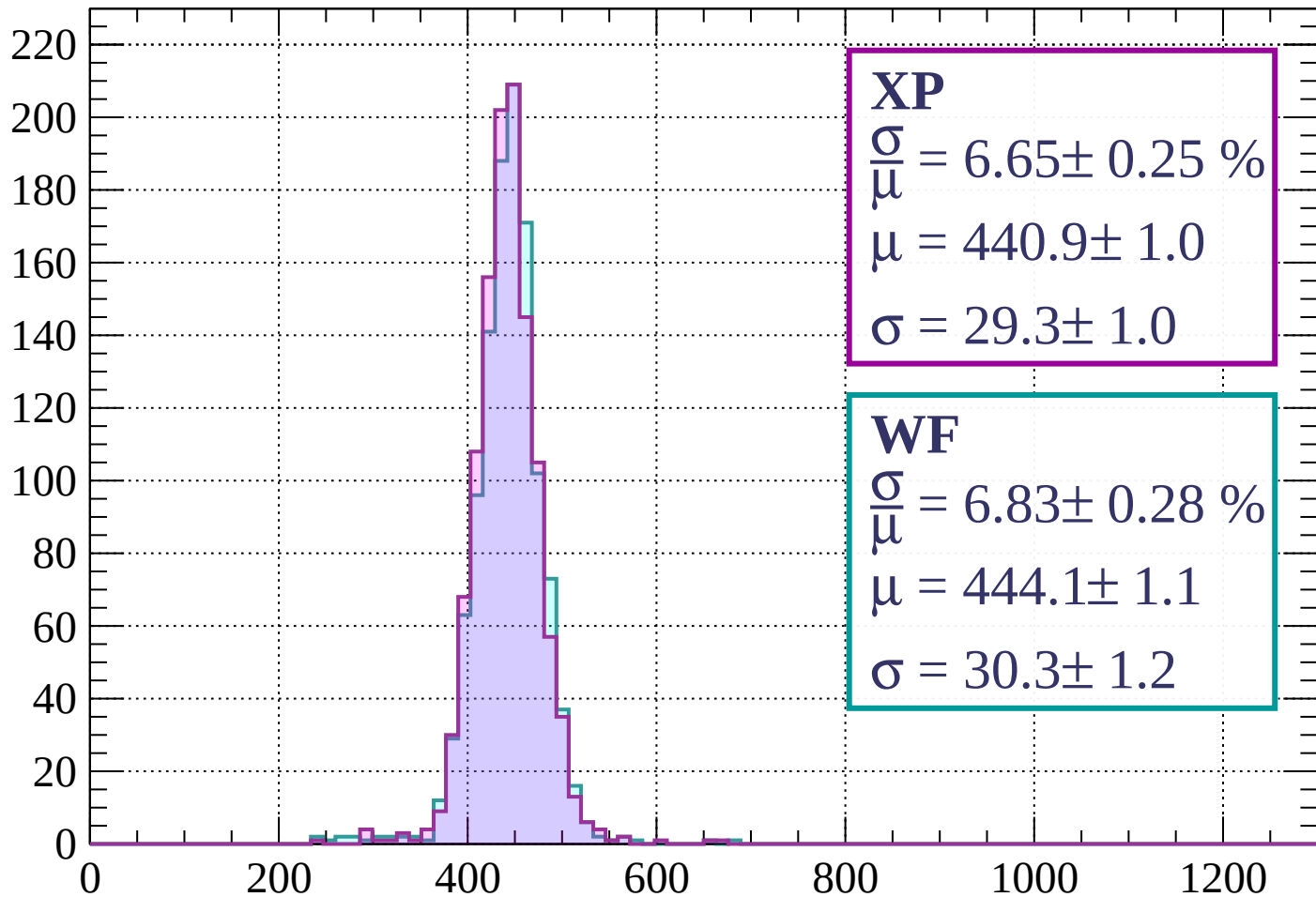
$$\mu = 446.4 \pm 1.0$$

$$\sigma = 30.1 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 6.65 \pm 0.25 \%$$

$$\mu = 440.9 \pm 1.0$$

$$\sigma = 29.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.83 \pm 0.28 \%$$

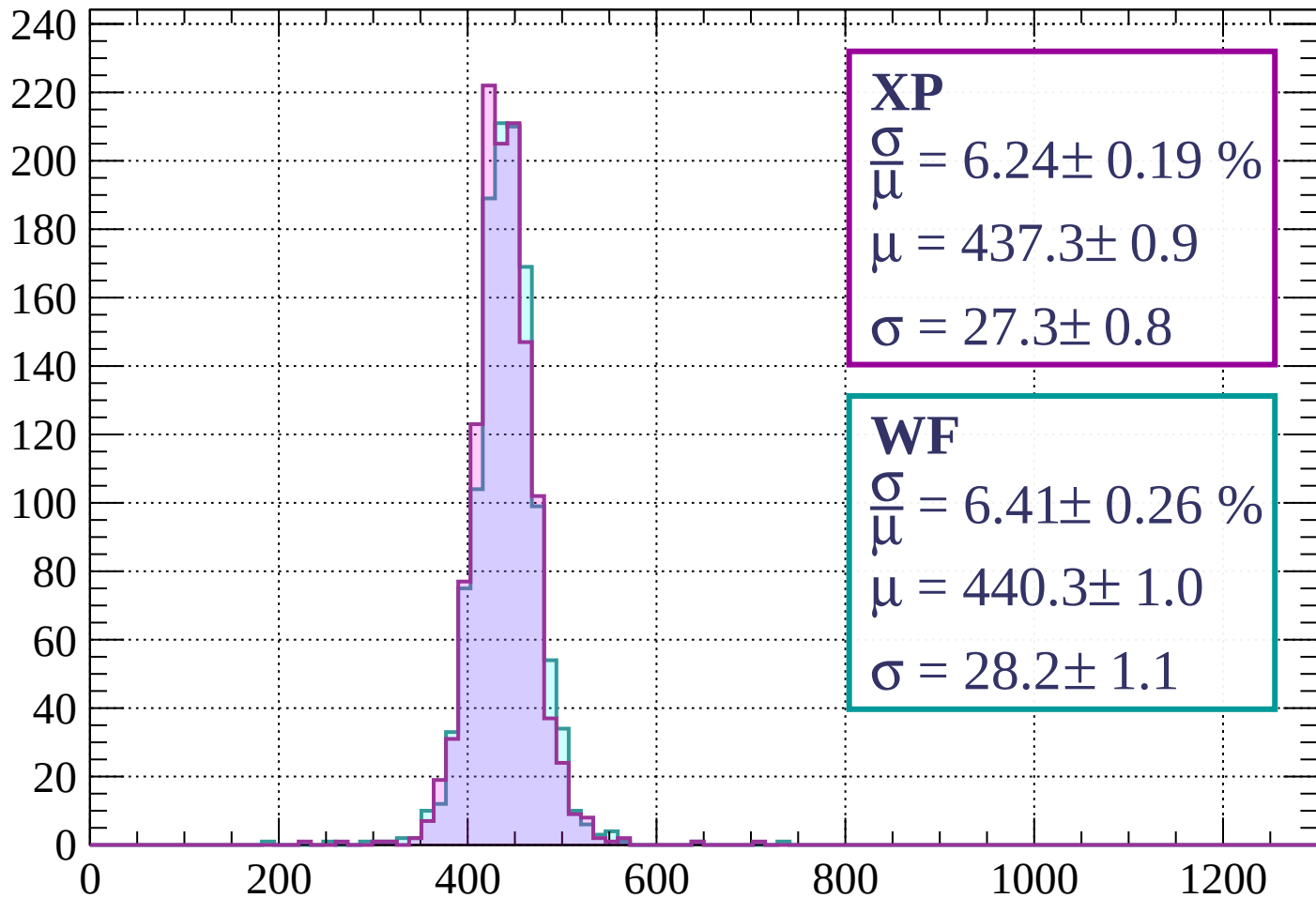
$$\mu = 444.1 \pm 1.1$$

$$\sigma = 30.3 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 6.24 \pm 0.19 \%$$

$$\mu = 437.3 \pm 0.9$$

$$\sigma = 27.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.41 \pm 0.26 \%$$

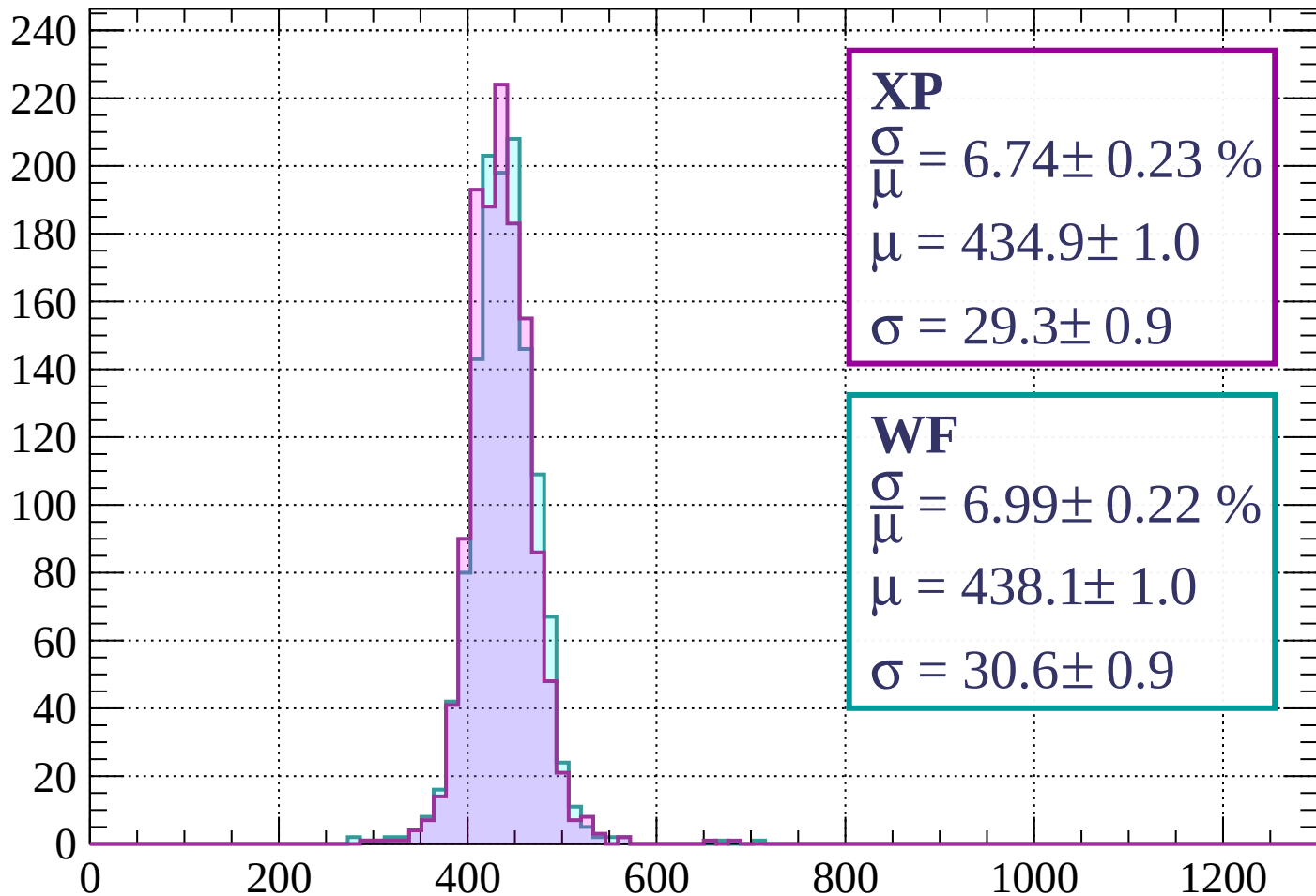
$$\mu = 440.3 \pm 1.0$$

$$\sigma = 28.2 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 6.74 \pm 0.23 \%$$

$$\mu = 434.9 \pm 1.0$$

$$\sigma = 29.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.99 \pm 0.22 \%$$

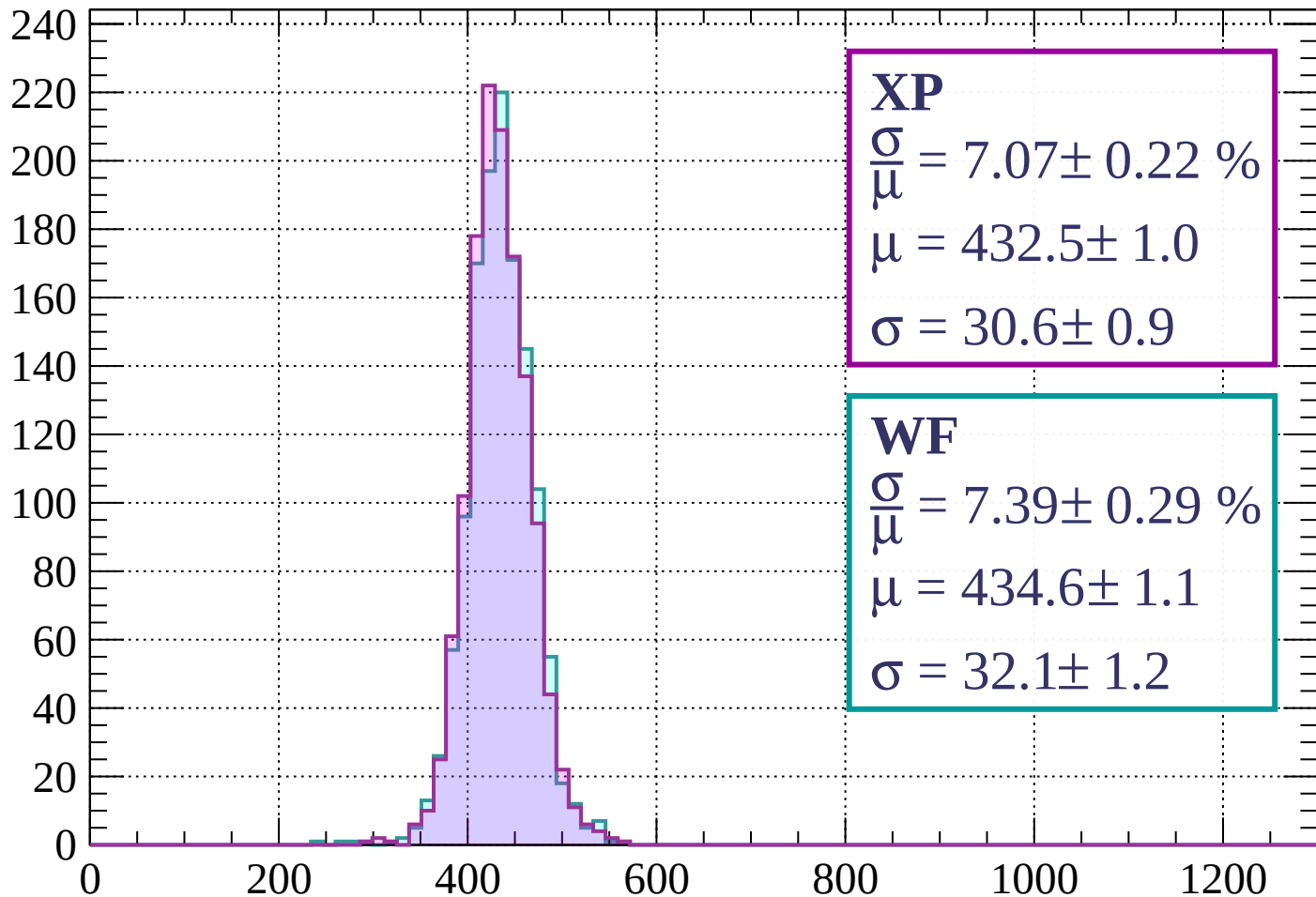
$$\mu = 438.1 \pm 1.0$$

$$\sigma = 30.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 7.07 \pm 0.22 \%$$

$$\mu = 432.5 \pm 1.0$$

$$\sigma = 30.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.39 \pm 0.29 \%$$

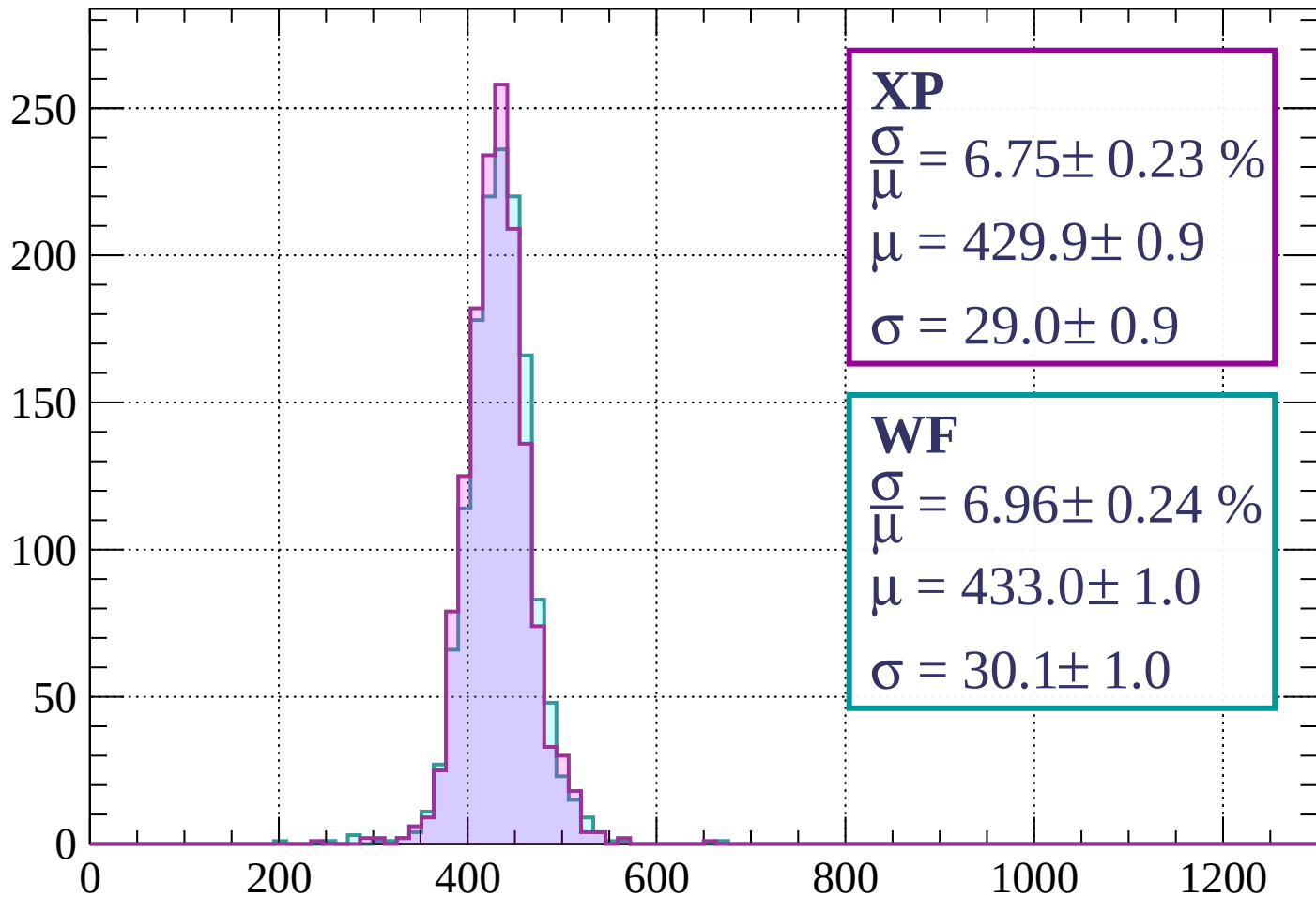
$$\mu = 434.6 \pm 1.1$$

$$\sigma = 32.1 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 6.75 \pm 0.23 \%$$

$$\mu = 429.9 \pm 0.9$$

$$\sigma = 29.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.96 \pm 0.24 \%$$

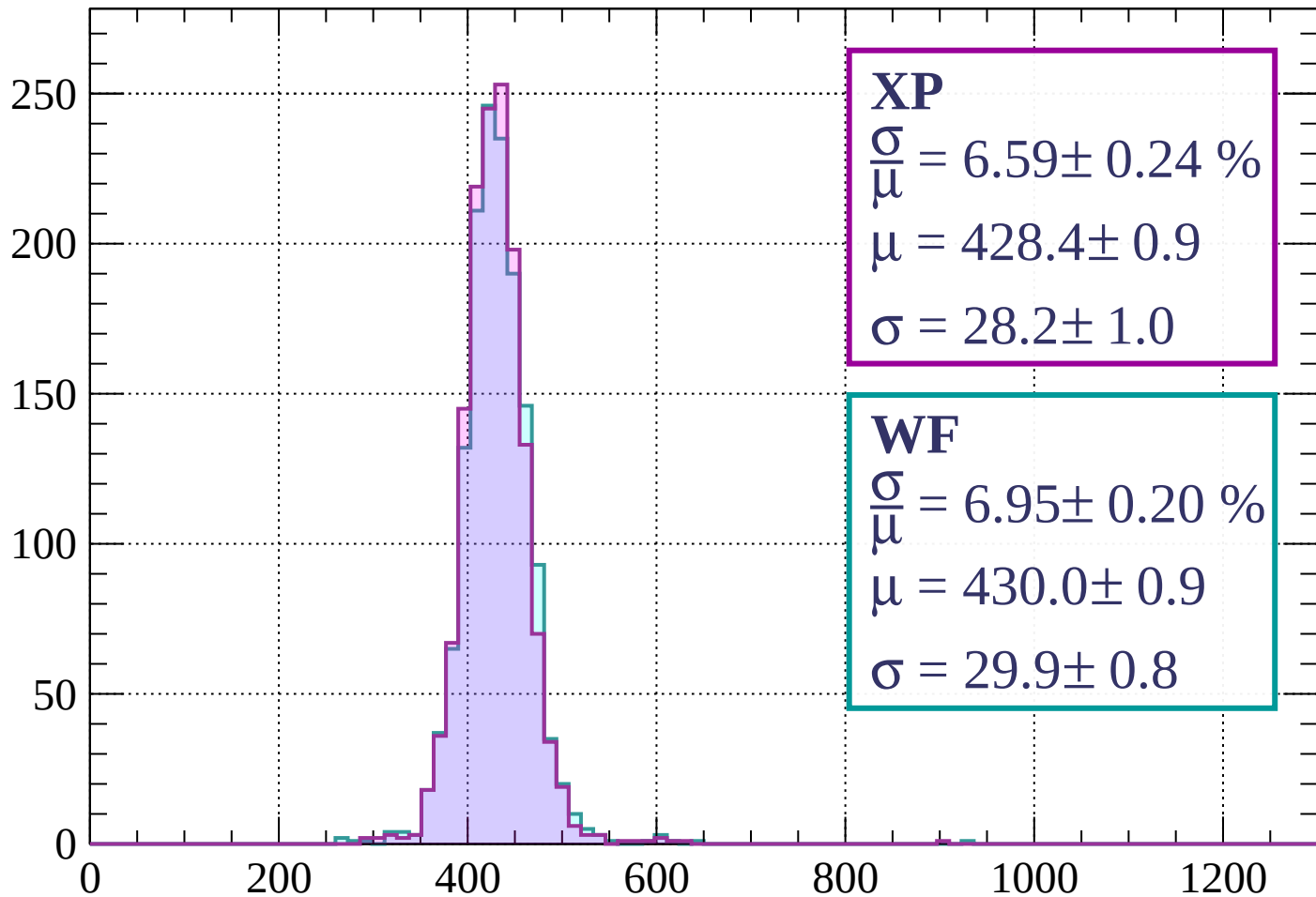
$$\mu = 433.0 \pm 1.0$$

$$\sigma = 30.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 6.59 \pm 0.24 \%$$

$$\mu = 428.4 \pm 0.9$$

$$\sigma = 28.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.95 \pm 0.20 \%$$

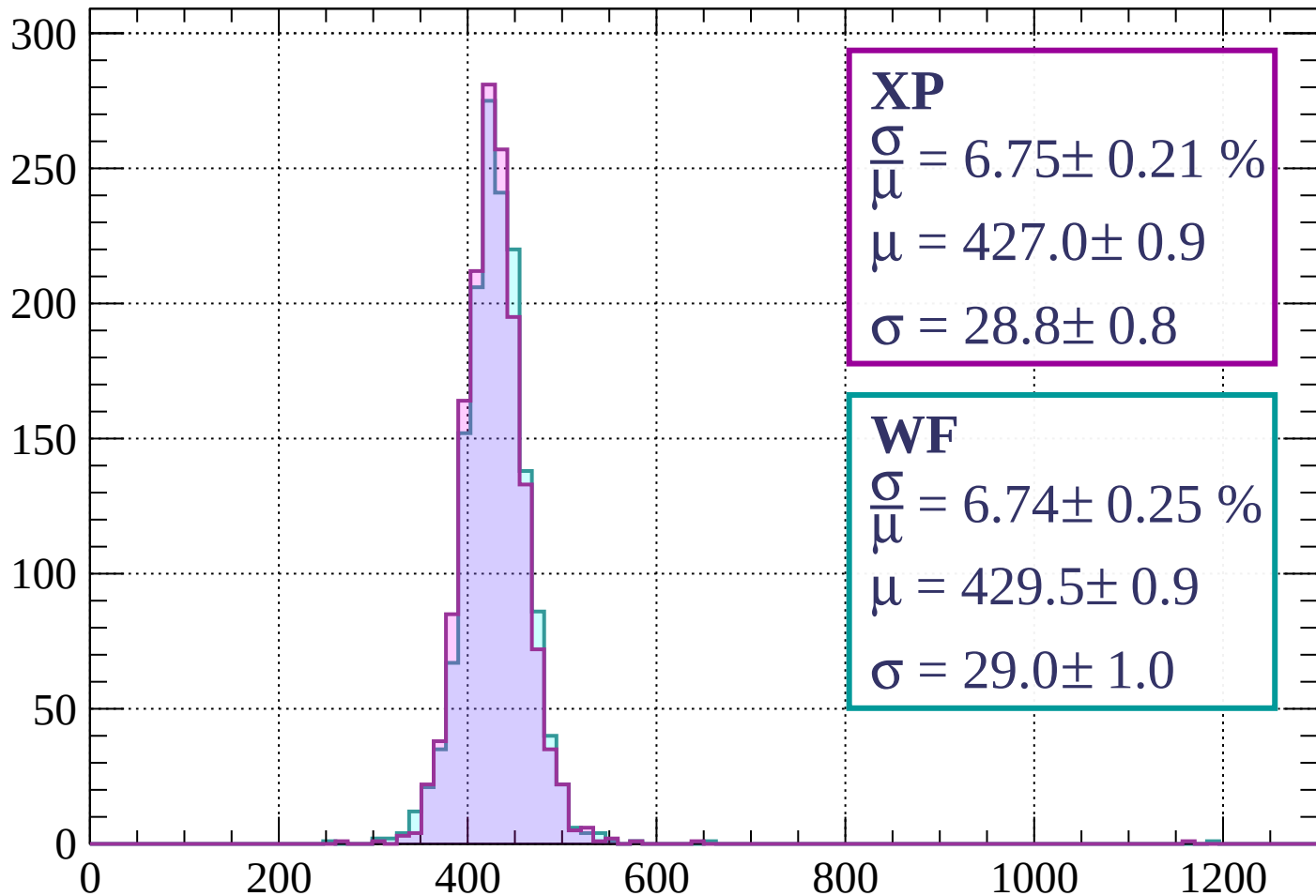
$$\mu = 430.0 \pm 0.9$$

$$\sigma = 29.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 6.75 \pm 0.21 \%$$

$$\mu = 427.0 \pm 0.9$$

$$\sigma = 28.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.74 \pm 0.25 \%$$

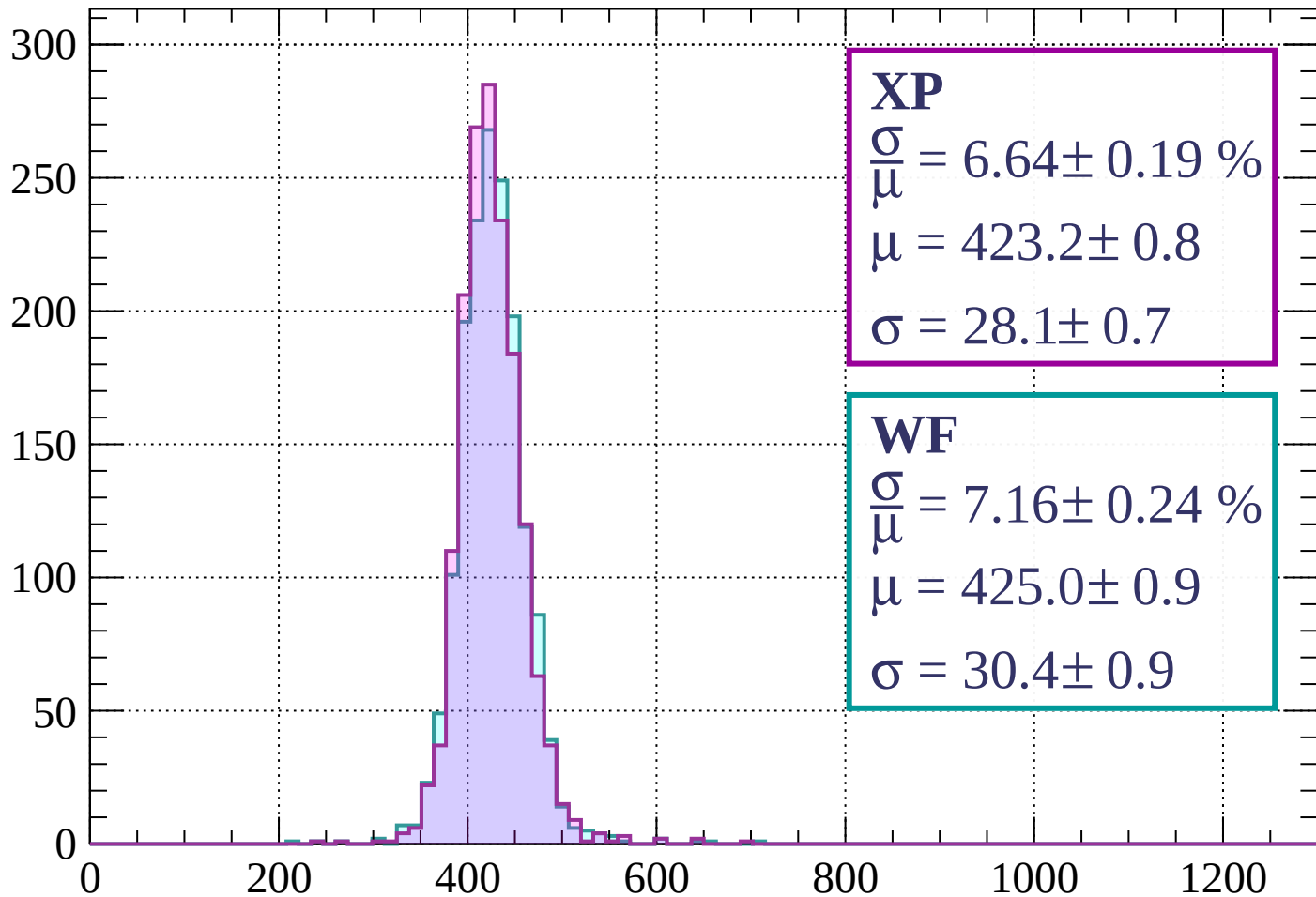
$$\mu = 429.5 \pm 0.9$$

$$\sigma = 29.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 6.64 \pm 0.19 \%$$

$$\mu = 423.2 \pm 0.8$$

$$\sigma = 28.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.16 \pm 0.24 \%$$

$$\mu = 425.0 \pm 0.9$$

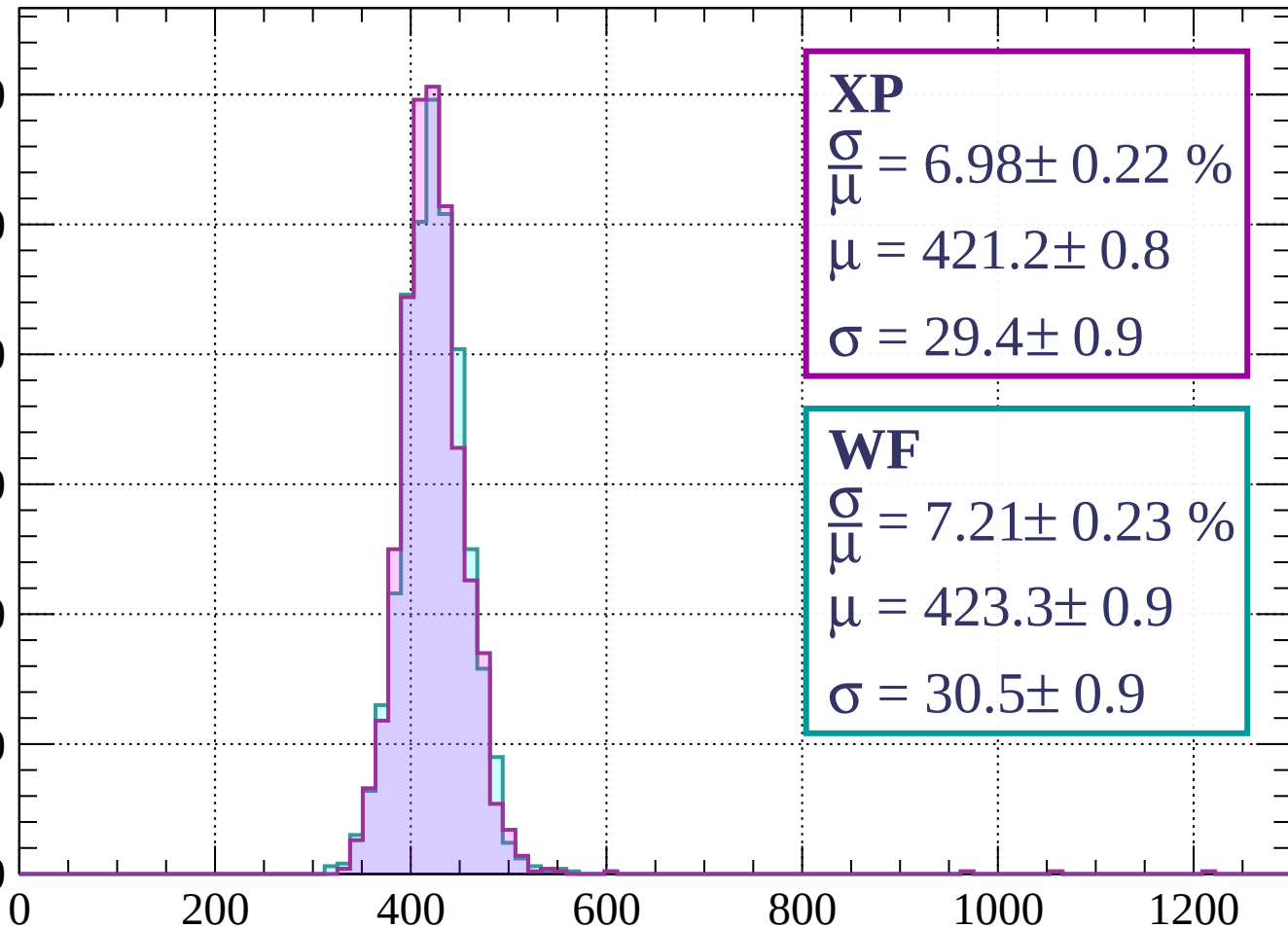
$$\sigma = 30.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 6.98 \pm 0.22 \%$$

$$\mu = 421.2 \pm 0.8$$

$$\sigma = 29.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.21 \pm 0.23 \%$$

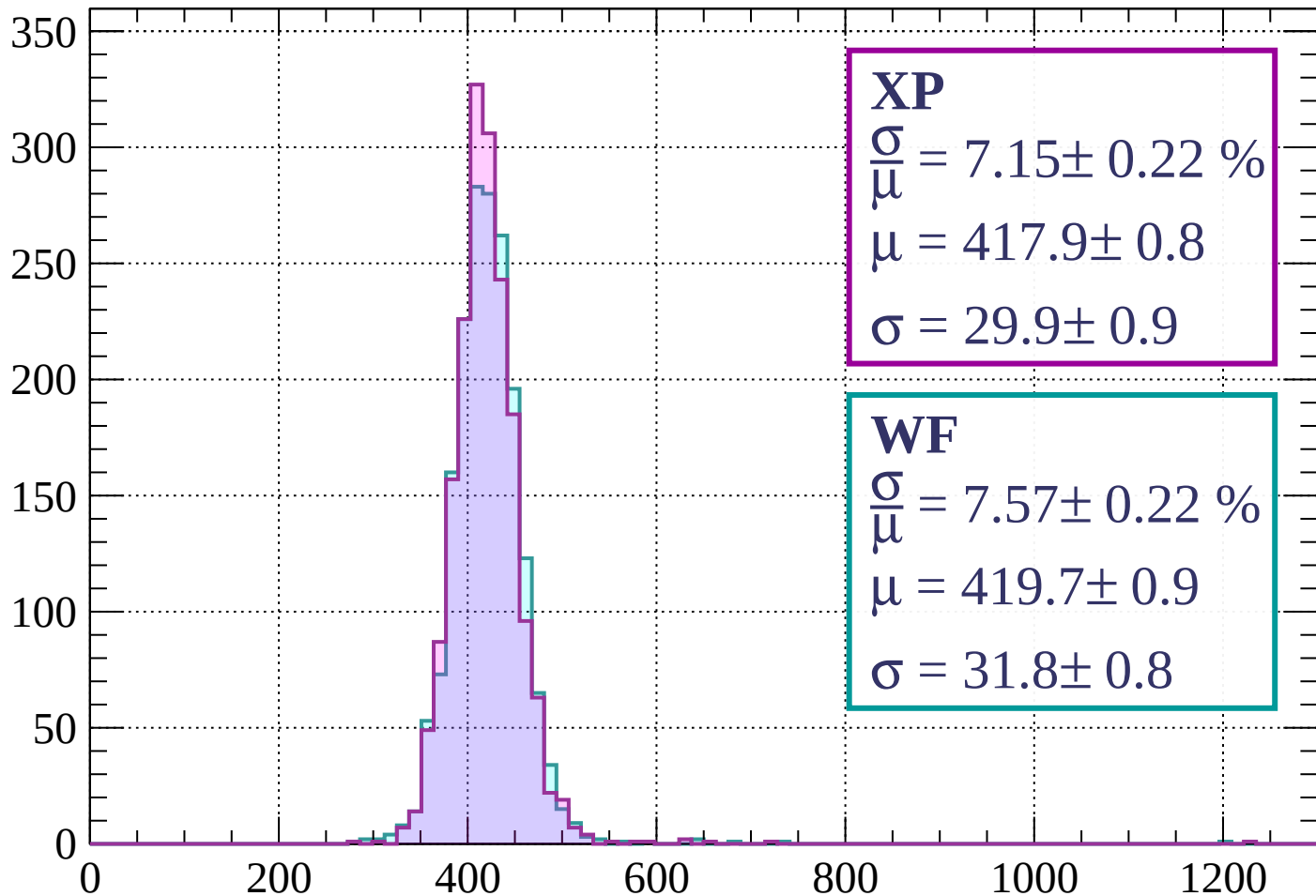
$$\mu = 423.3 \pm 0.9$$

$$\sigma = 30.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.15 \pm 0.22 \%$$

$$\mu = 417.9 \pm 0.8$$

$$\sigma = 29.9 \pm 0.9$$

WF

$$\frac{\sigma}{\bar{\mu}} = 7.57 \pm 0.22 \%$$

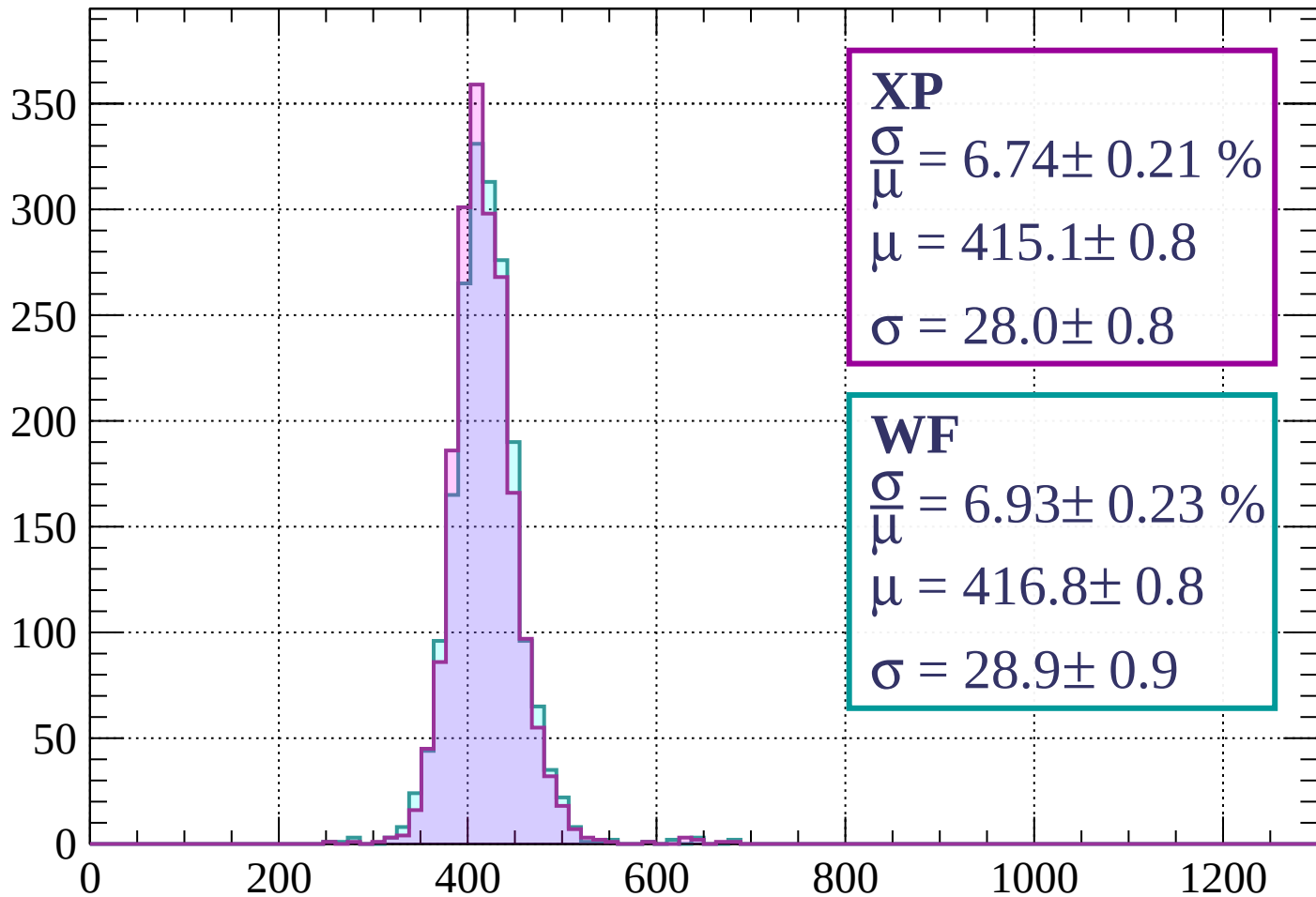
$$\mu = 419.7 \pm 0.9$$

$$\sigma = 31.8 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 6.74 \pm 0.21 \%$$

$$\mu = 415.1 \pm 0.8$$

$$\sigma = 28.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.93 \pm 0.23 \%$$

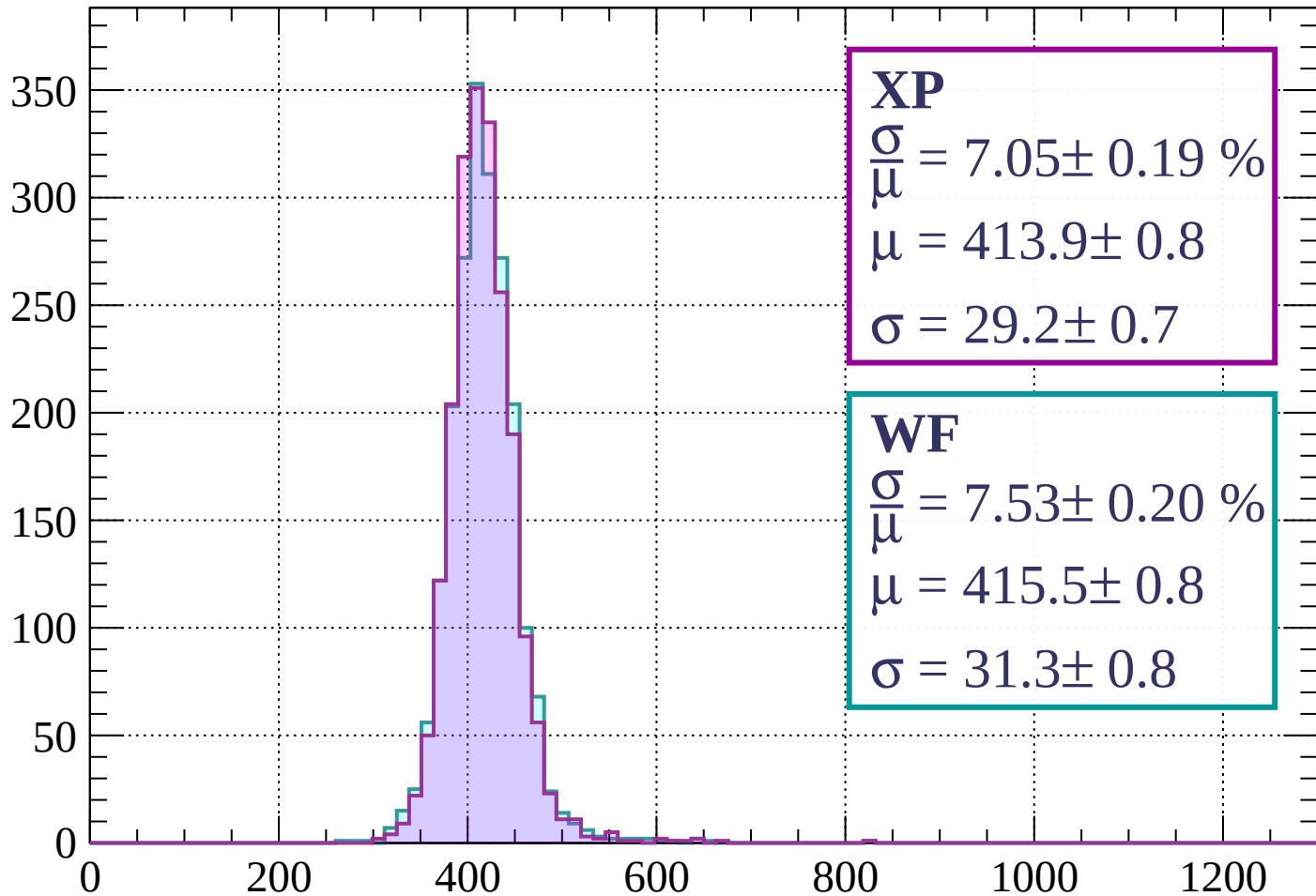
$$\mu = 416.8 \pm 0.8$$

$$\sigma = 28.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 7.05 \pm 0.19 \%$$

$$\mu = 413.9 \pm 0.8$$

$$\sigma = 29.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.53 \pm 0.20 \%$$

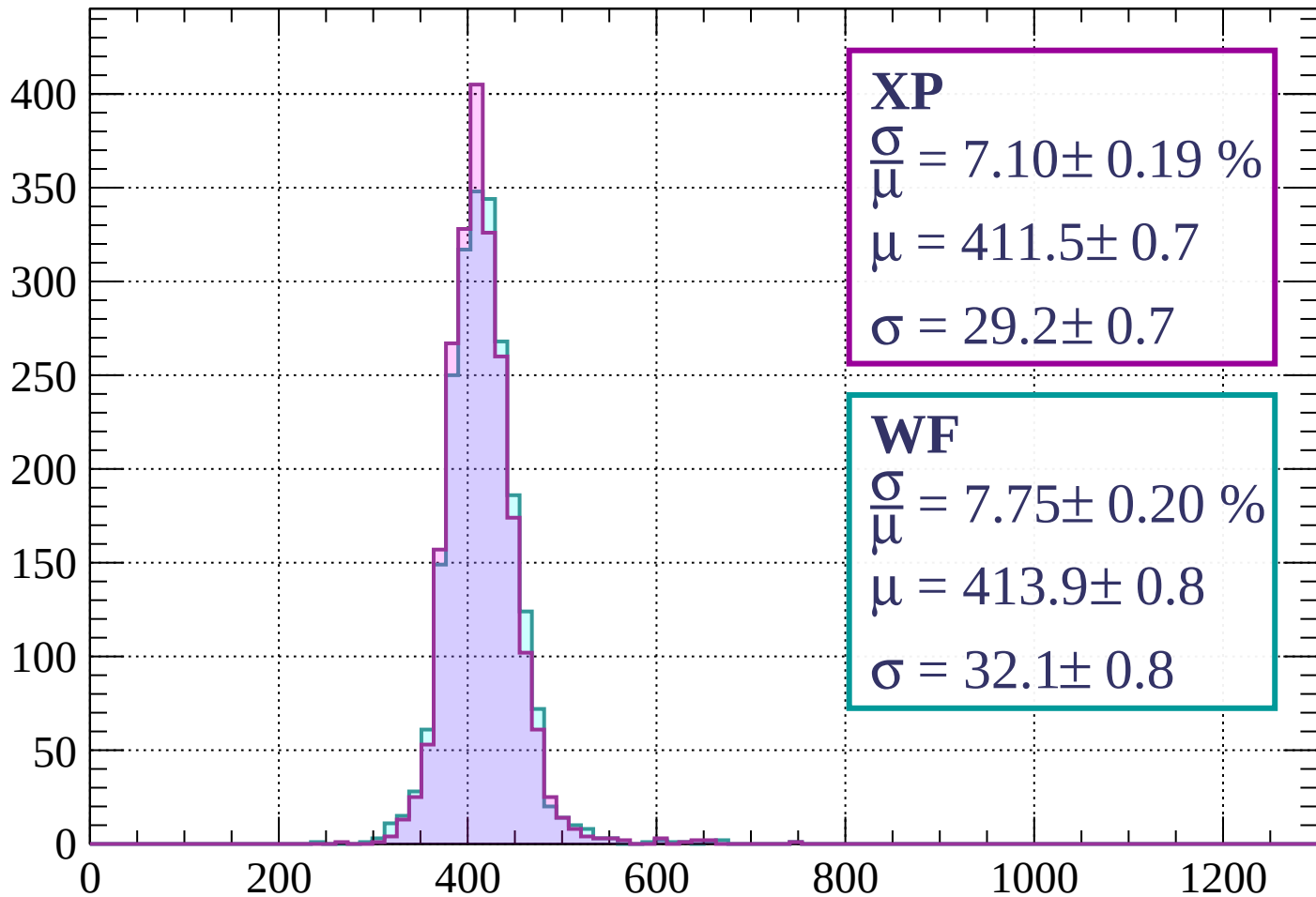
$$\mu = 415.5 \pm 0.8$$

$$\sigma = 31.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 7.10 \pm 0.19 \%$$

$$\mu = 411.5 \pm 0.7$$

$$\sigma = 29.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.75 \pm 0.20 \%$$

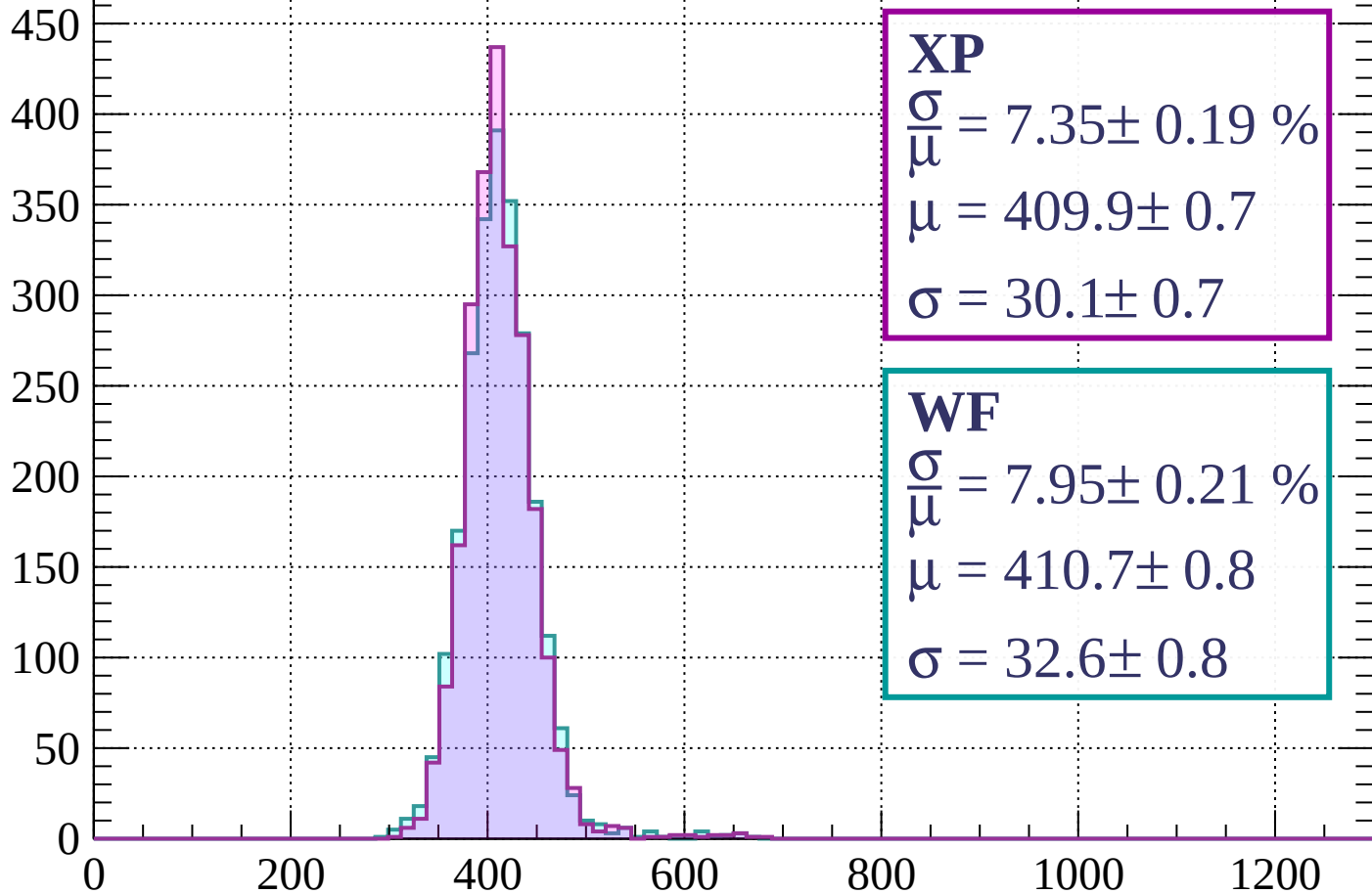
$$\mu = 413.9 \pm 0.8$$

$$\sigma = 32.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count



XP

$$\frac{\sigma}{\mu} = 7.35 \pm 0.19 \%$$

$$\mu = 409.9 \pm 0.7$$

$$\sigma = 30.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.95 \pm 0.21 \%$$

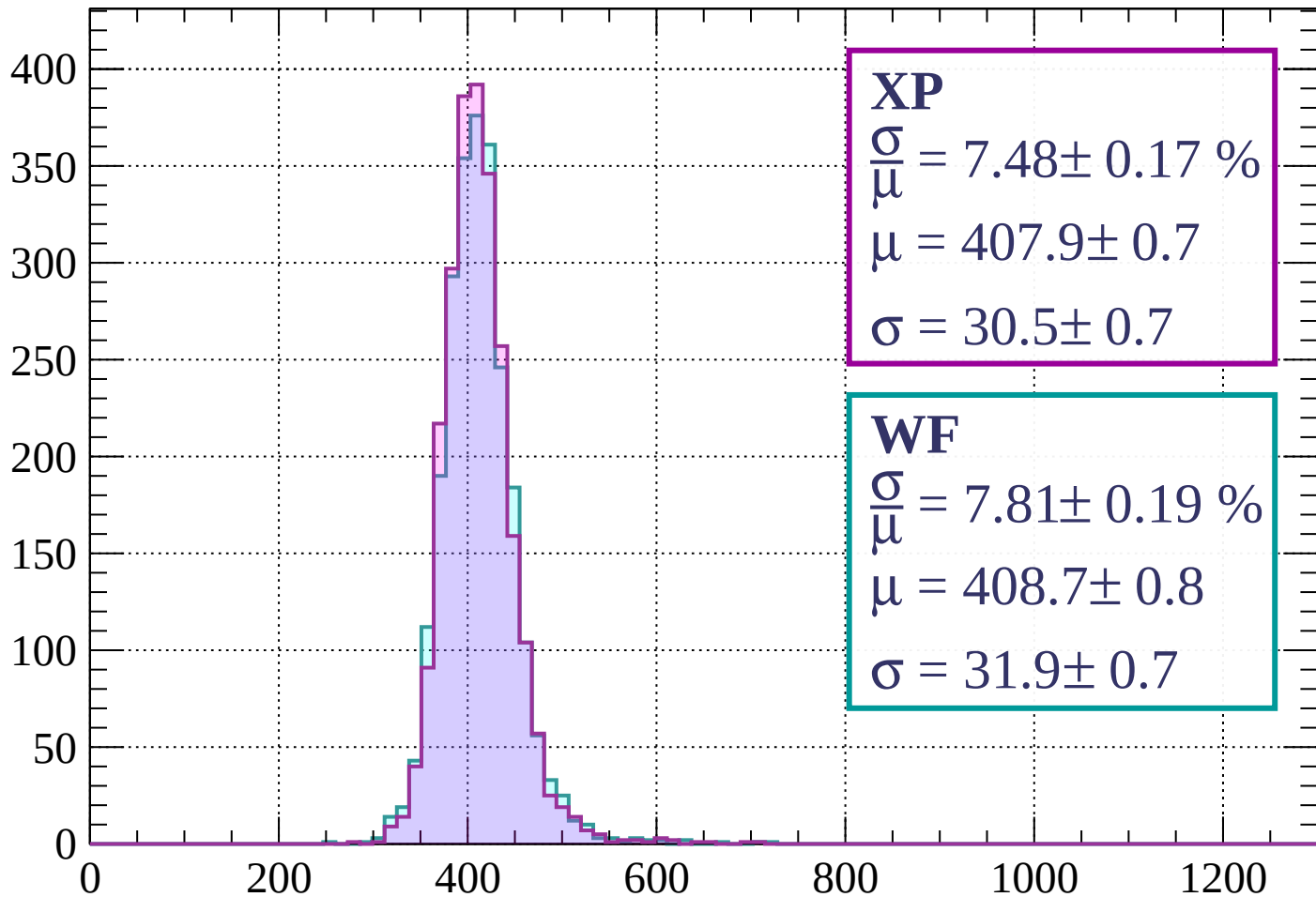
$$\mu = 410.7 \pm 0.8$$

$$\sigma = 32.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 7.48 \pm 0.17 \%$$

$$\mu = 407.9 \pm 0.7$$

$$\sigma = 30.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.81 \pm 0.19 \%$$

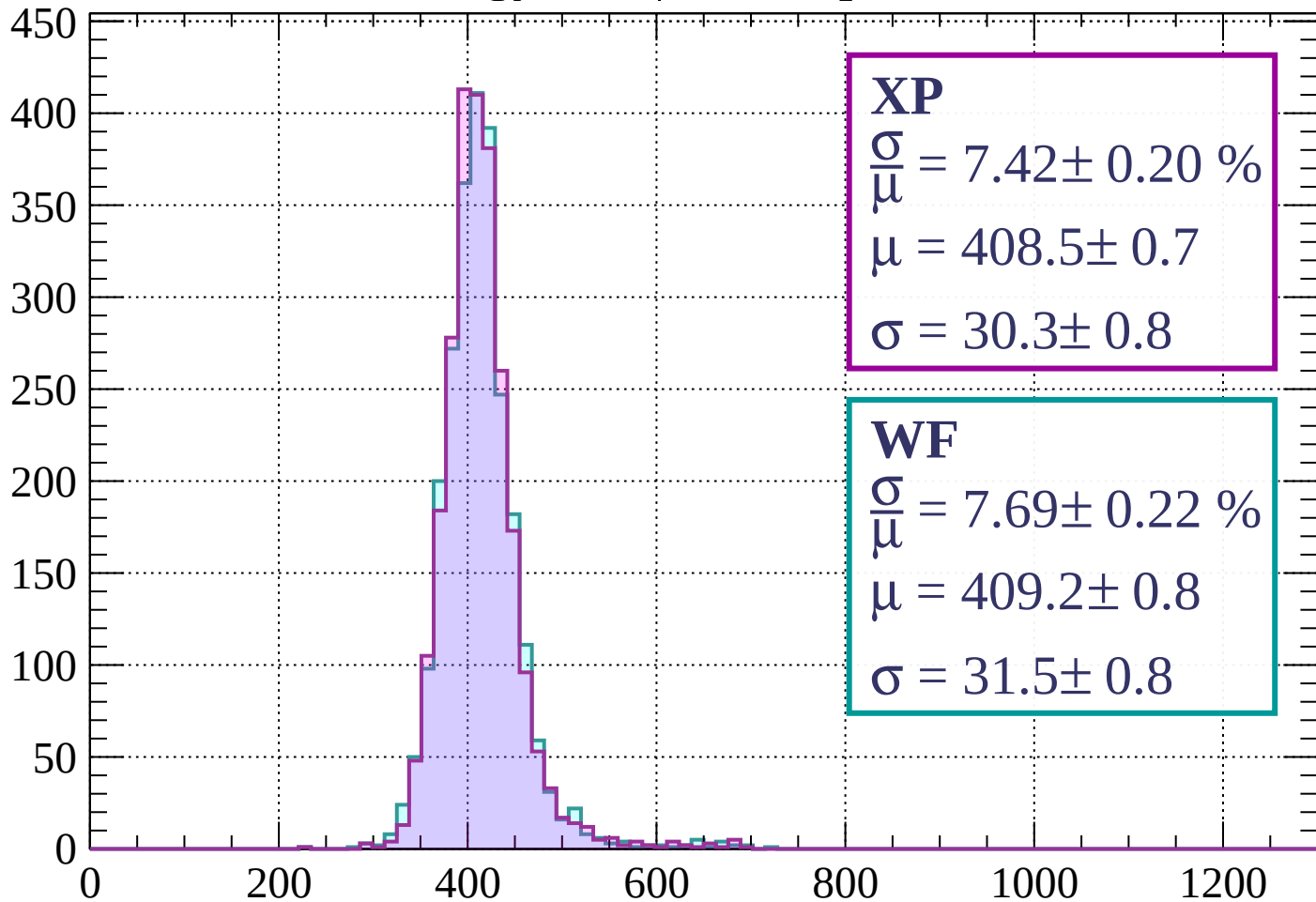
$$\mu = 408.7 \pm 0.8$$

$$\sigma = 31.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 7.42 \pm 0.20 \%$$

$$\mu = 408.5 \pm 0.7$$

$$\sigma = 30.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.69 \pm 0.22 \%$$

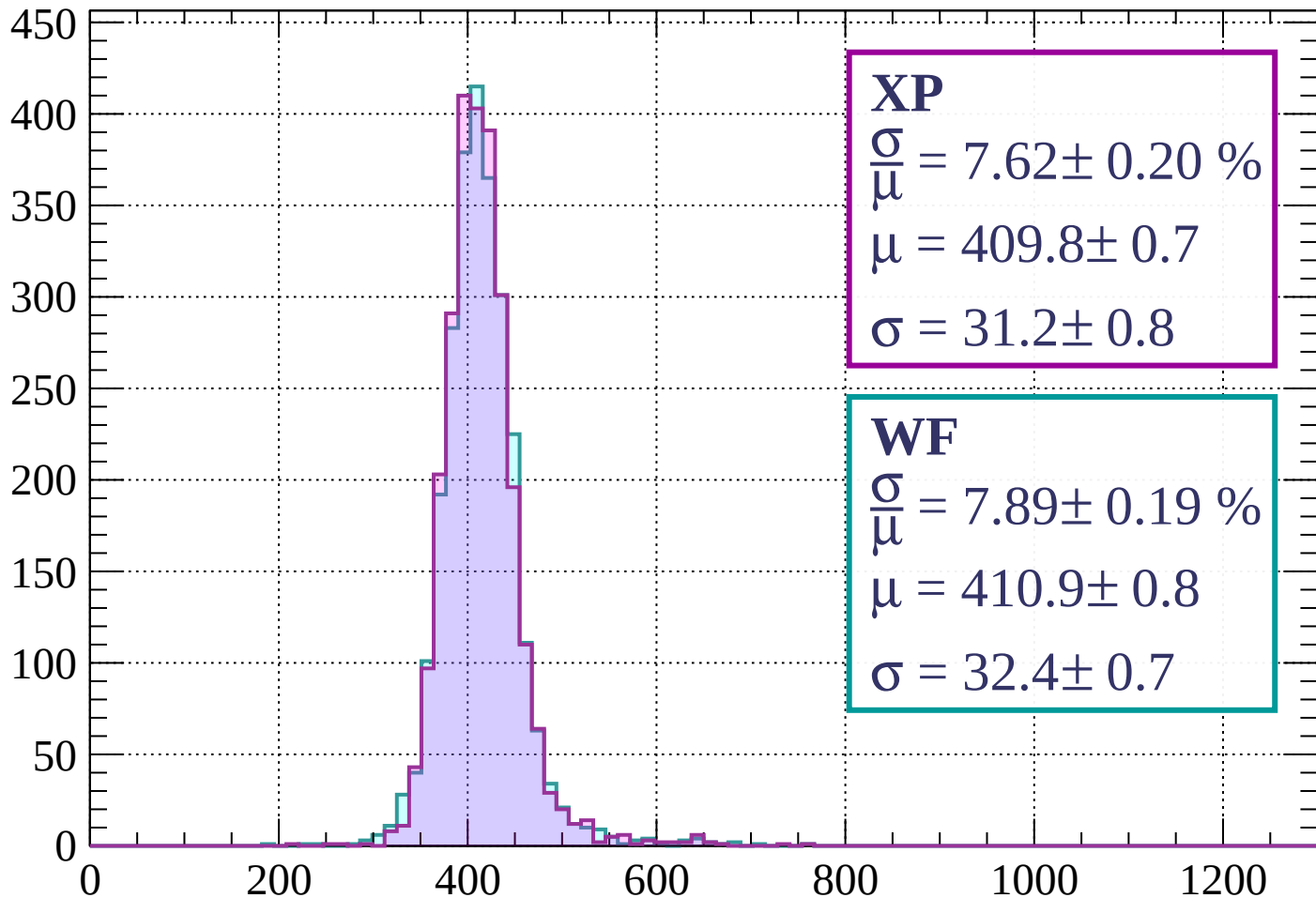
$$\mu = 409.2 \pm 0.8$$

$$\sigma = 31.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 7.62 \pm 0.20 \%$$

$$\mu = 409.8 \pm 0.7$$

$$\sigma = 31.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 7.89 \pm 0.19 \%$$

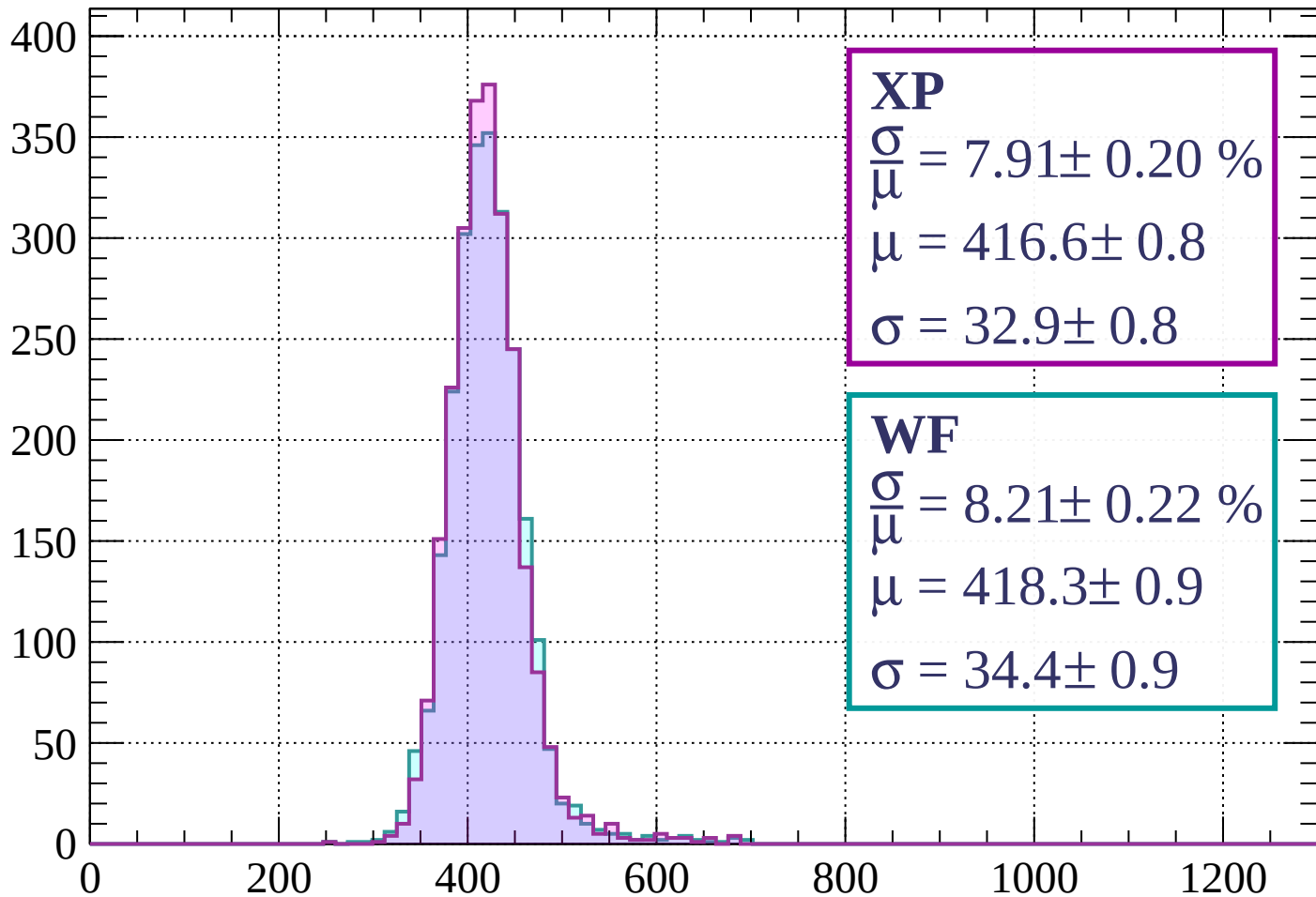
$$\mu = 410.9 \pm 0.8$$

$$\sigma = 32.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 7.91 \pm 0.20 \%$$

$$\mu = 416.6 \pm 0.8$$

$$\sigma = 32.9 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.21 \pm 0.22 \%$$

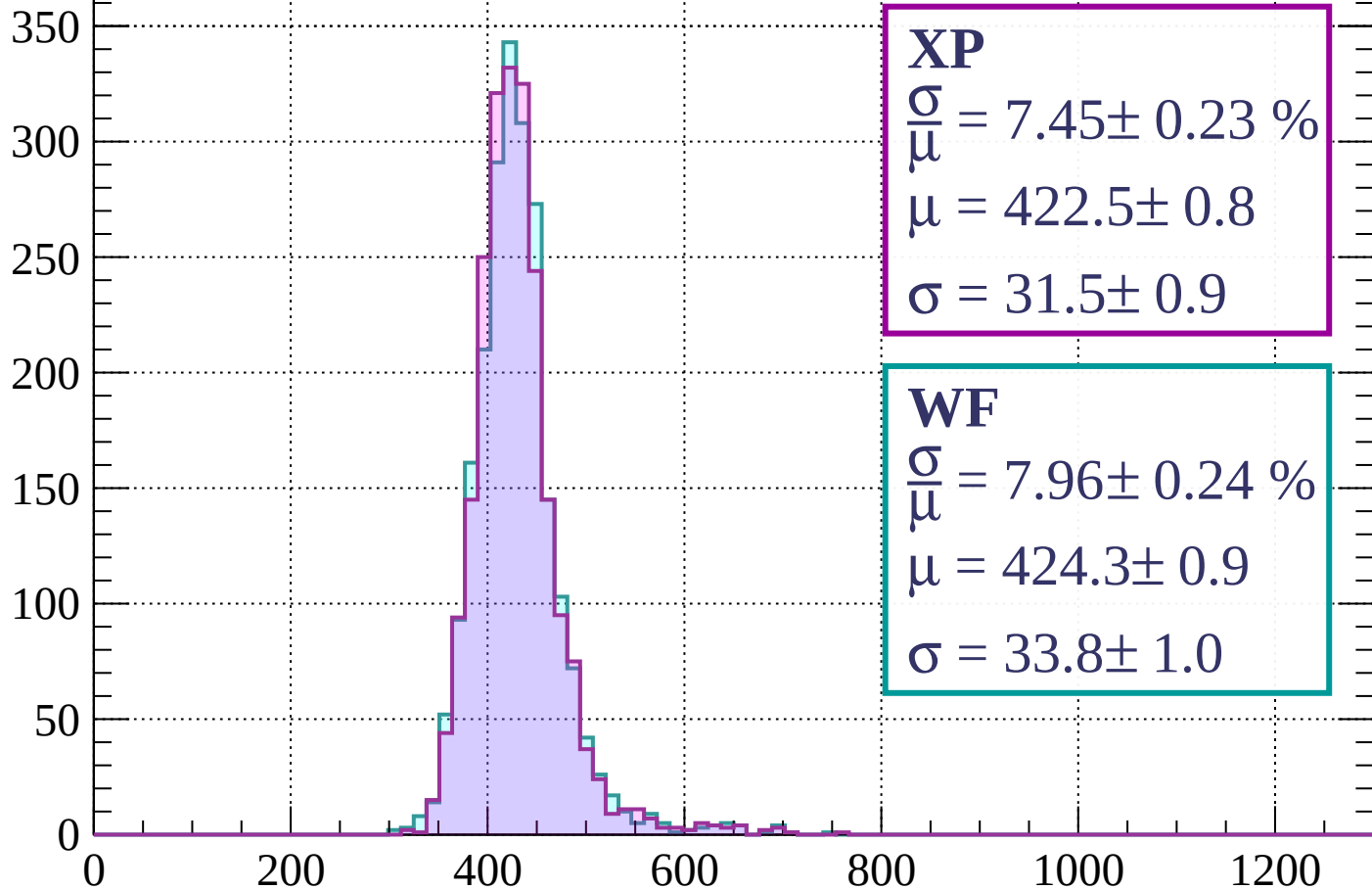
$$\mu = 418.3 \pm 0.9$$

$$\sigma = 34.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 7.45 \pm 0.23 \%$$

$$\mu = 422.5 \pm 0.8$$

$$\sigma = 31.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 7.96 \pm 0.24 \%$$

$$\mu = 424.3 \pm 0.9$$

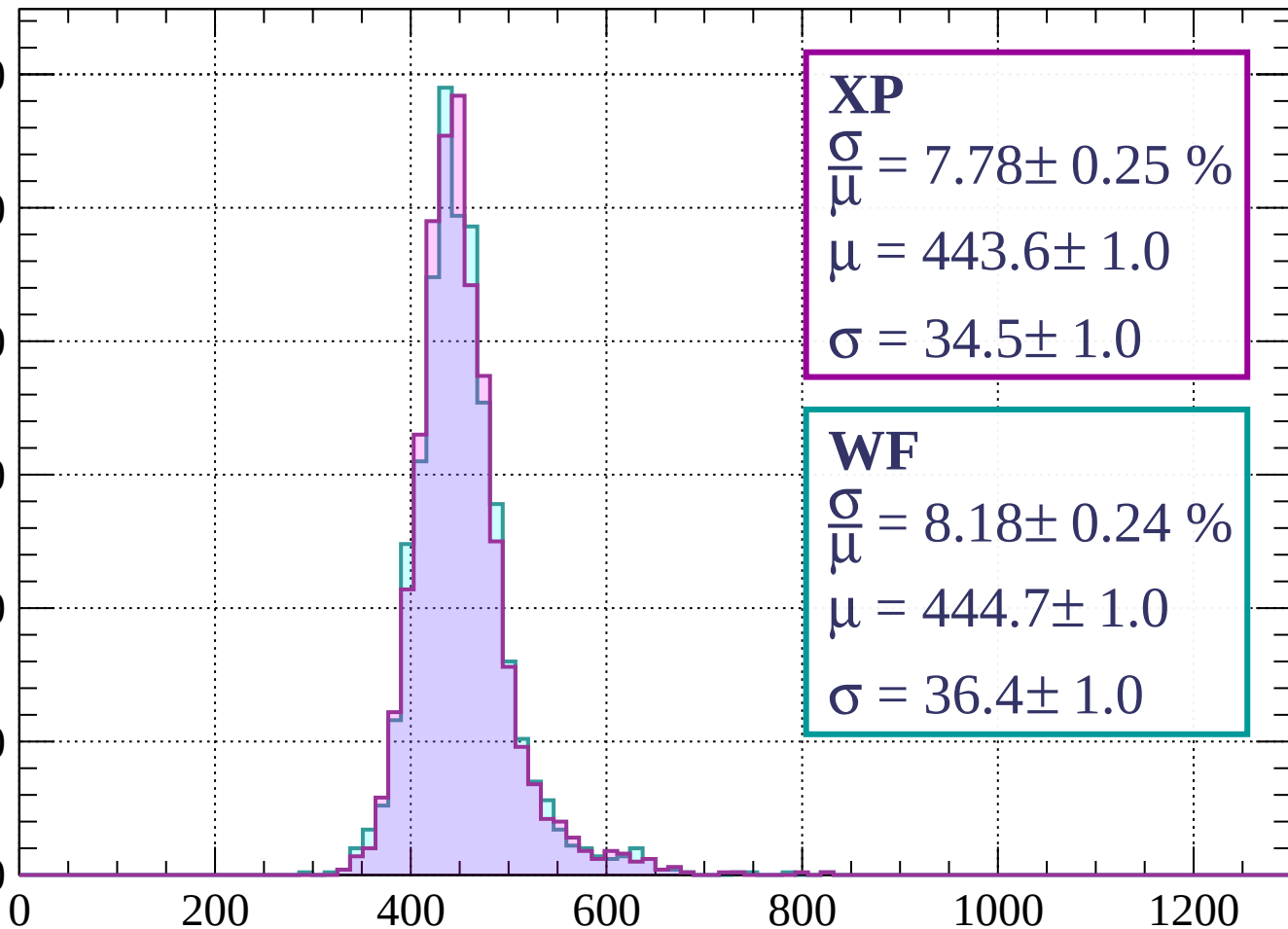
$$\sigma = 33.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 7.78 \pm 0.25 \%$$

$$\mu = 443.6 \pm 1.0$$

$$\sigma = 34.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.18 \pm 0.24 \%$$

$$\mu = 444.7 \pm 1.0$$

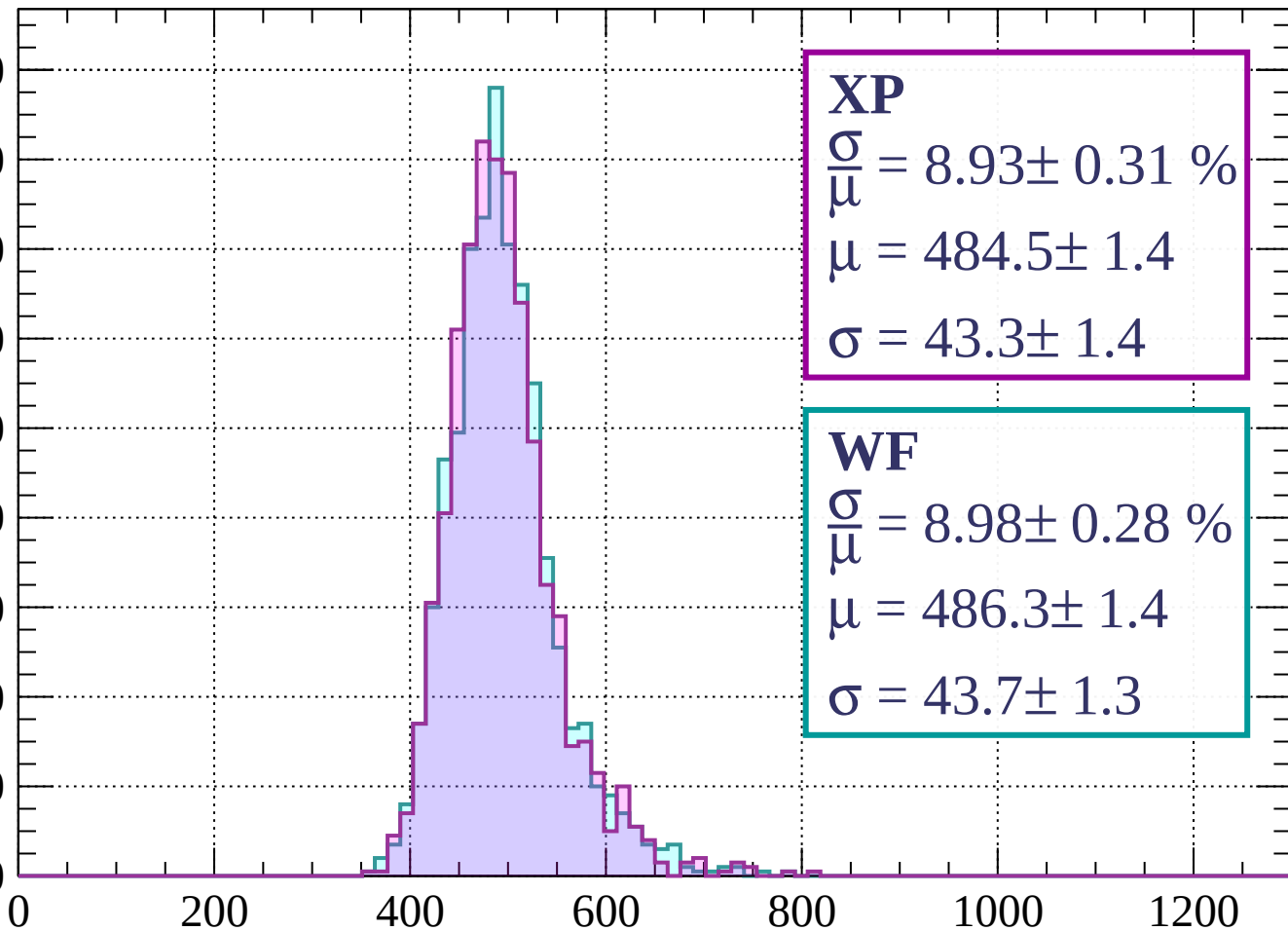
$$\sigma = 36.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-160 < p < -120$

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.93 \pm 0.31 \%$$

$$\mu = 484.5 \pm 1.4$$

$$\sigma = 43.3 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 8.98 \pm 0.28 \%$$

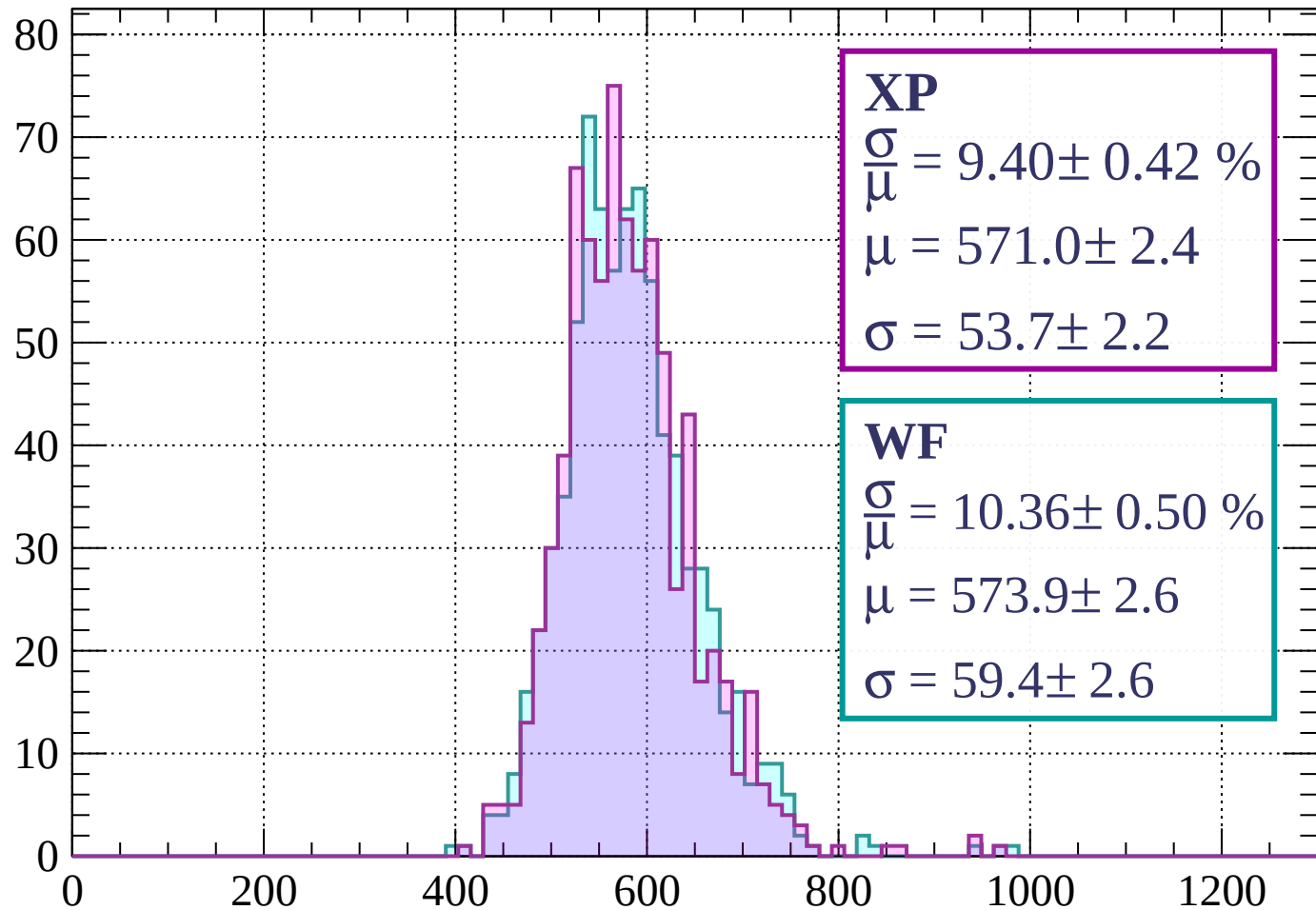
$$\mu = 486.3 \pm 1.4$$

$$\sigma = 43.7 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -80$

Count



XP

$$\frac{\sigma}{\mu} = 9.40 \pm 0.42 \%$$

$$\mu = 571.0 \pm 2.4$$

$$\sigma = 53.7 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 10.36 \pm 0.50 \%$$

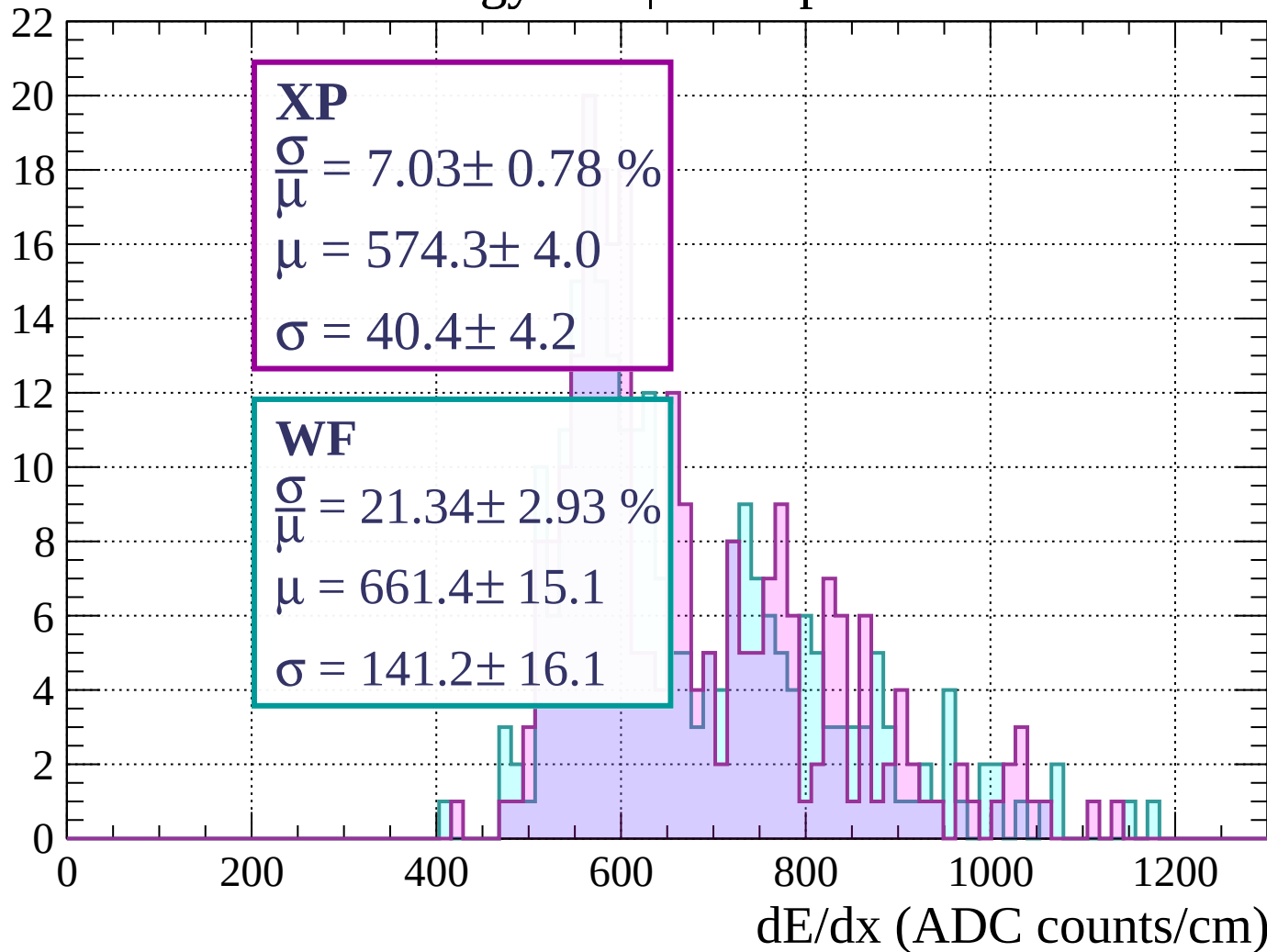
$$\mu = 573.9 \pm 2.6$$

$$\sigma = 59.4 \pm 2.6$$

dE/dx (ADC counts/cm)

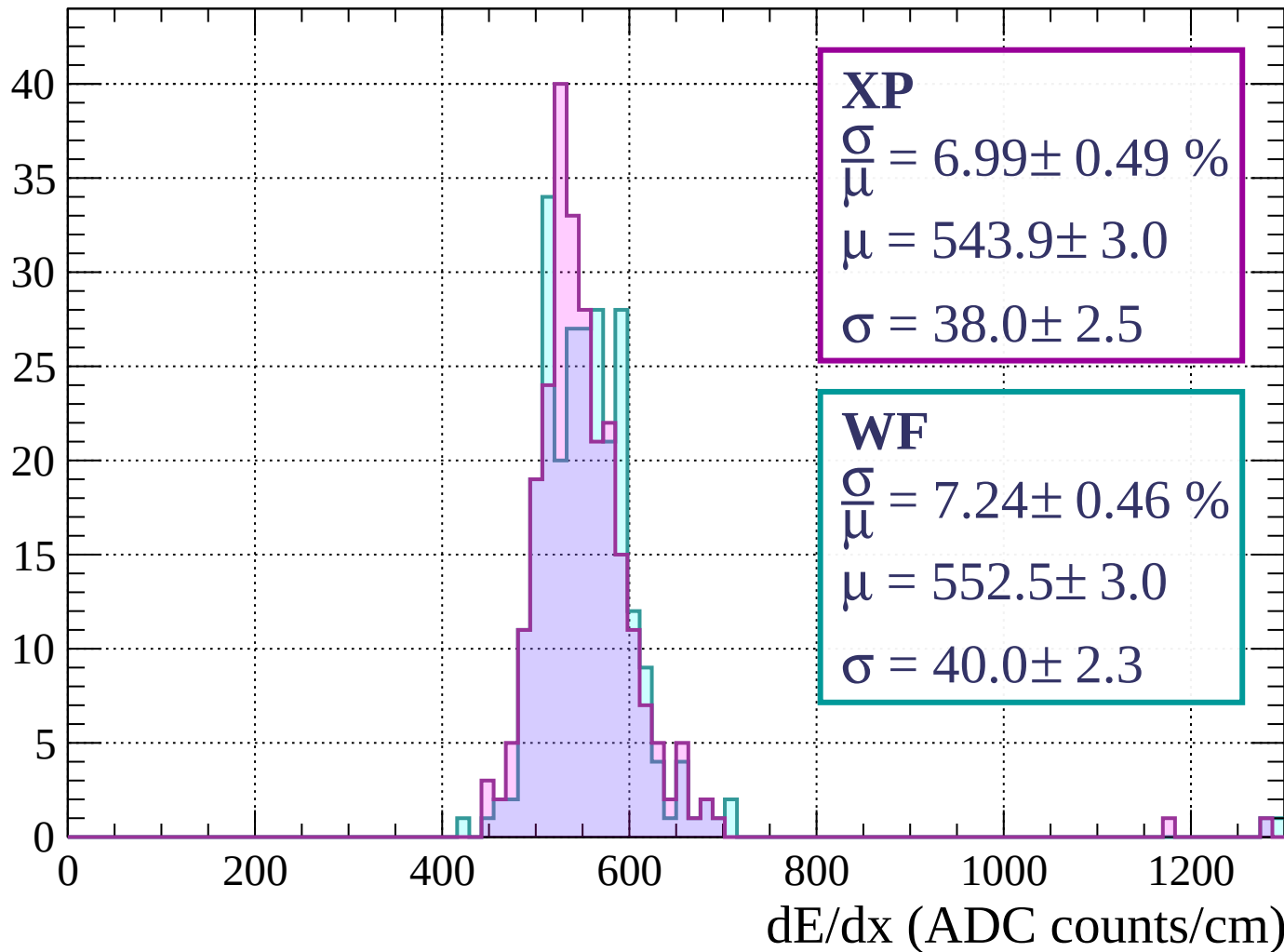
Energy loss | $-80 < p < -40$

Count



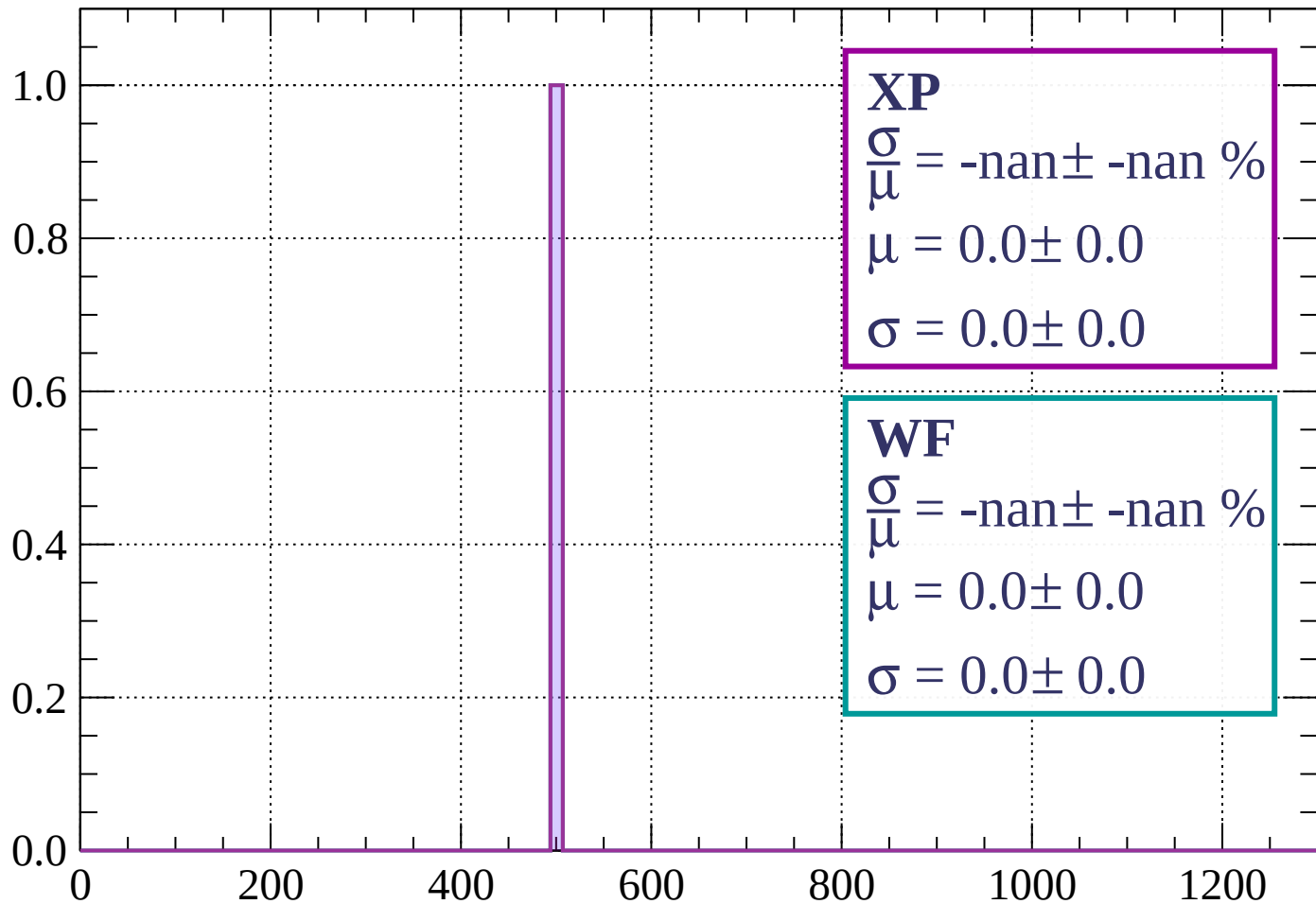
Energy loss | $-40 < p < 0$

Count



Energy loss | $0 < p < 40$

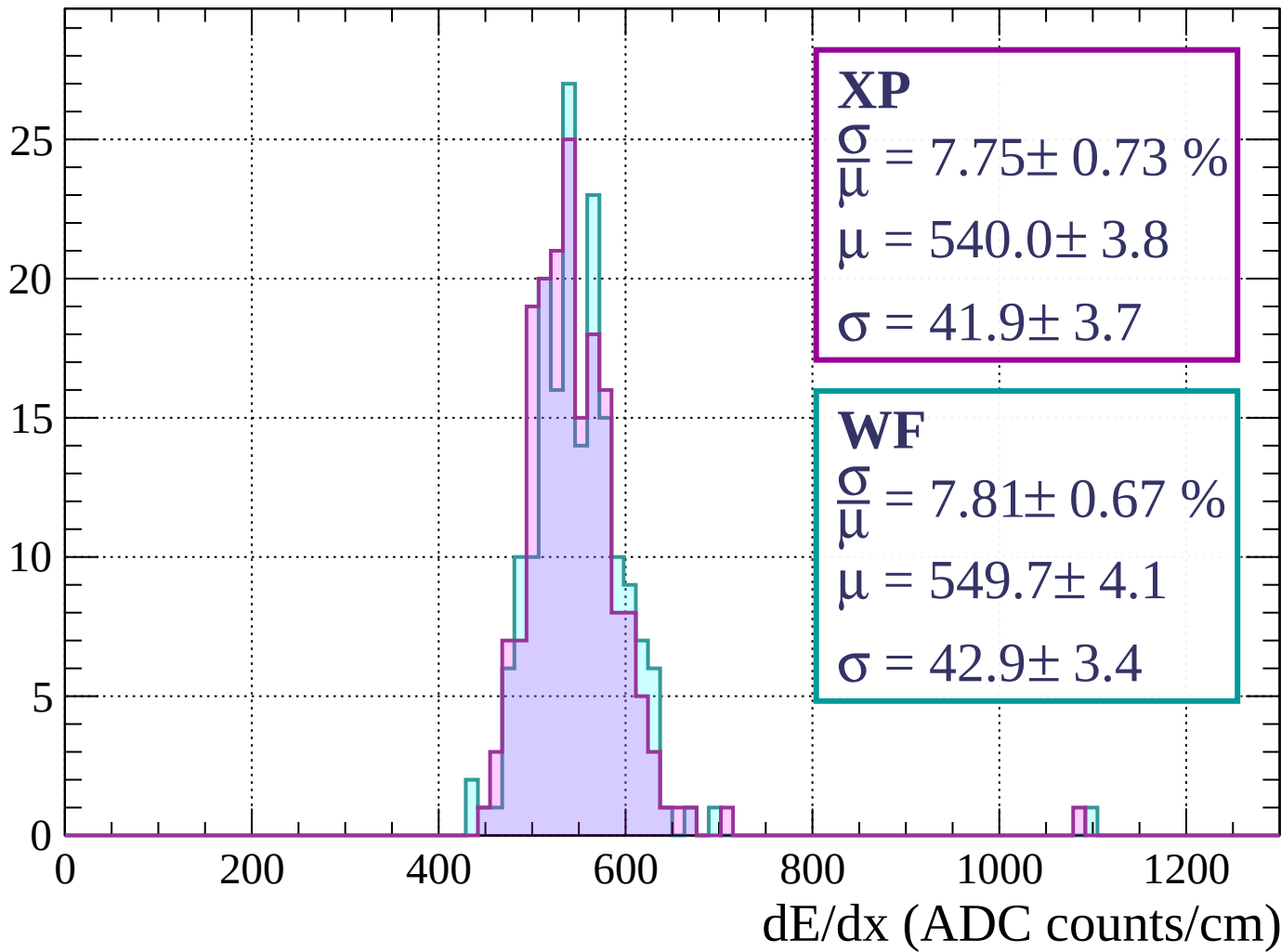
Count



dE/dx (ADC counts/cm)

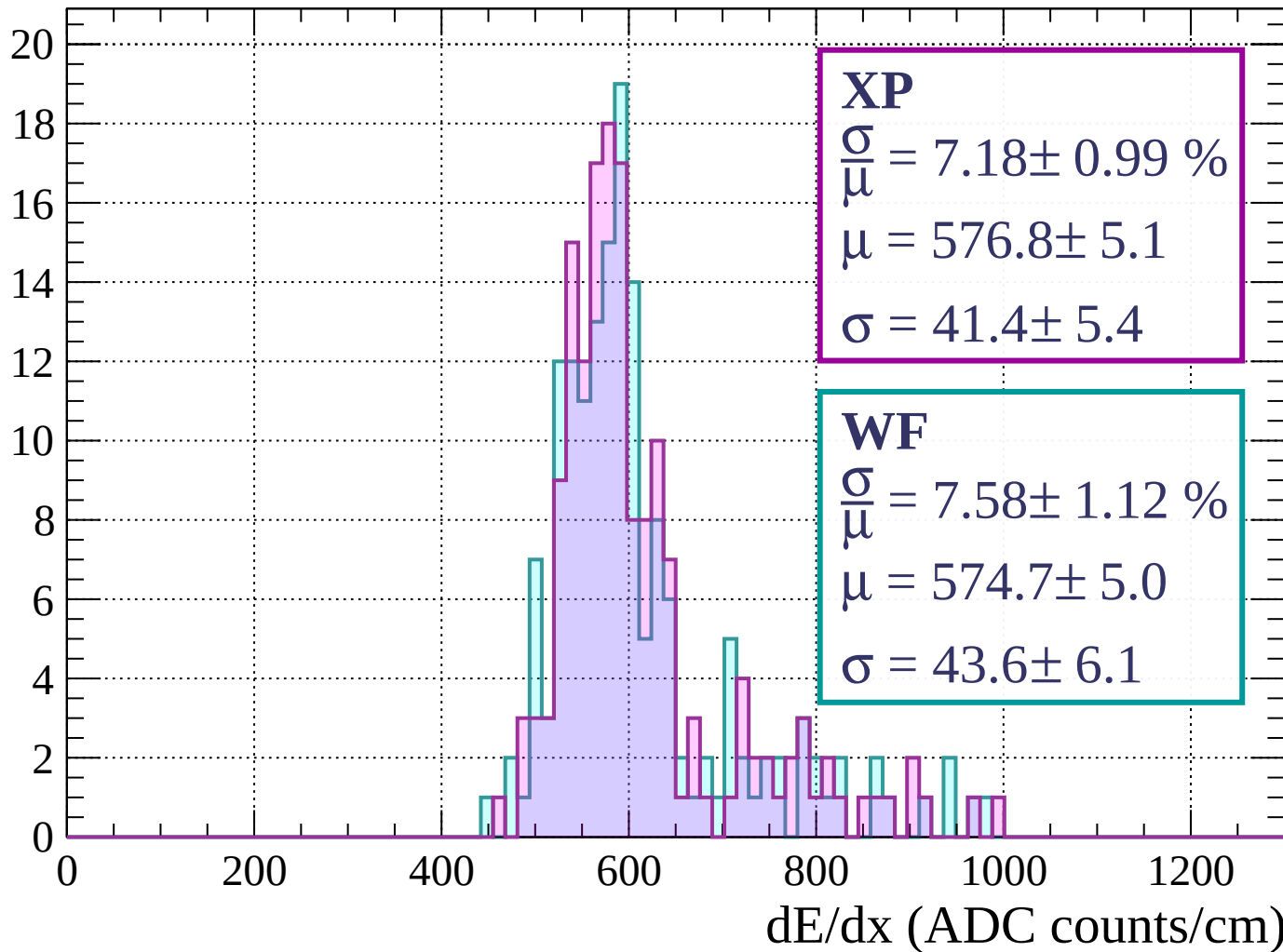
Energy loss | $40 < p < 80$

Count



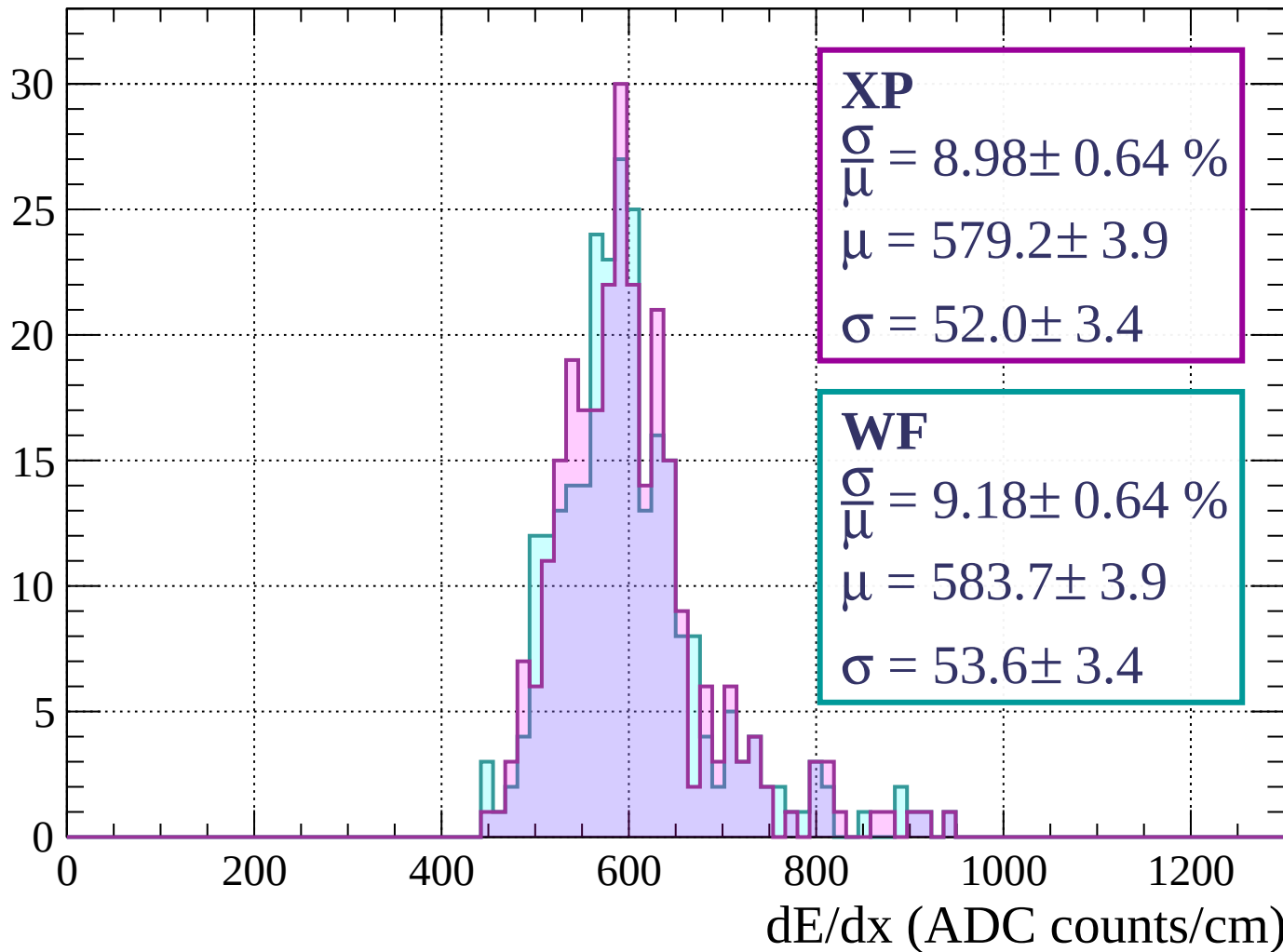
Energy loss | $80 < p < 120$

Count



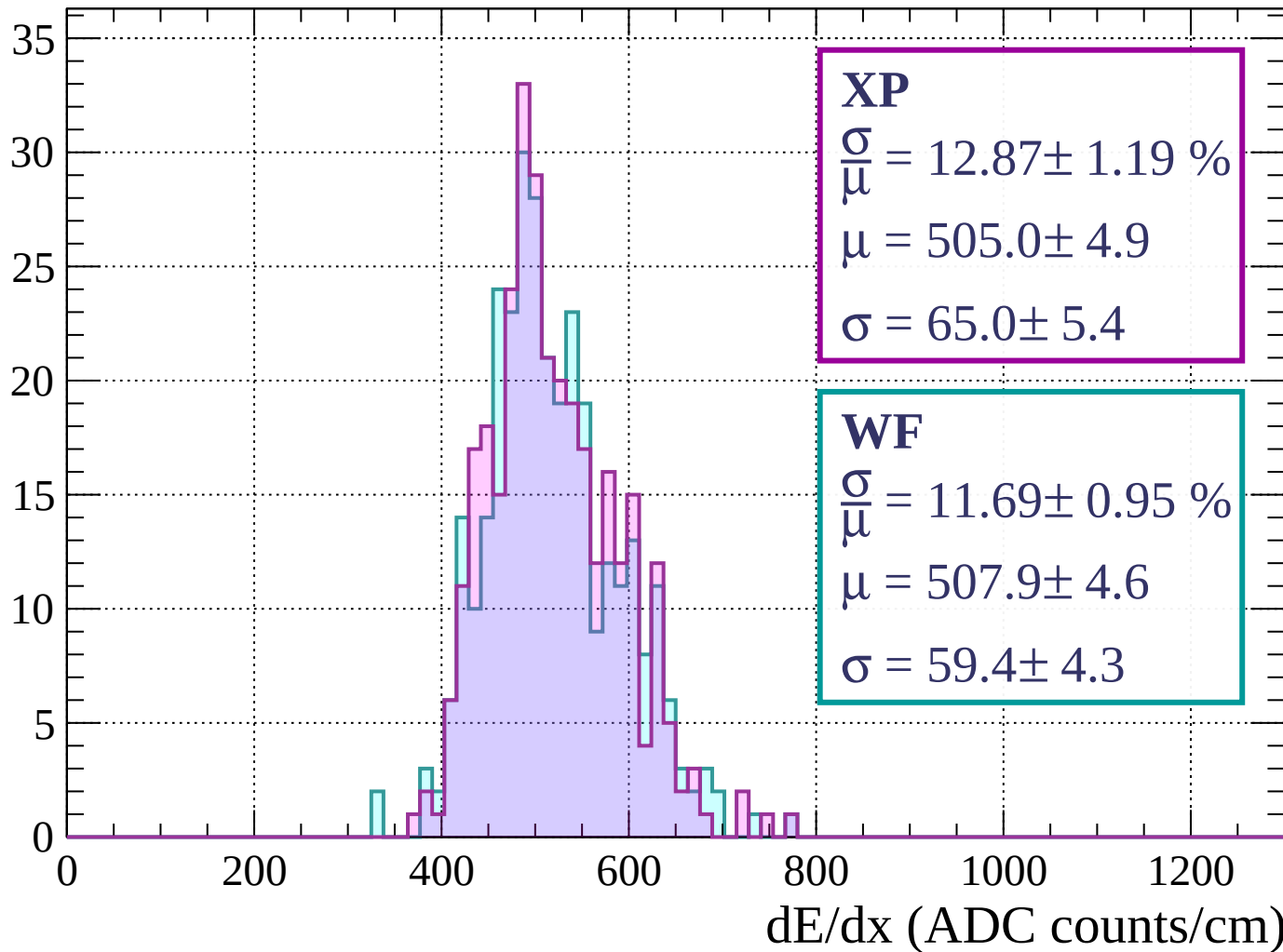
Energy loss | $120 < p < 160$

Count



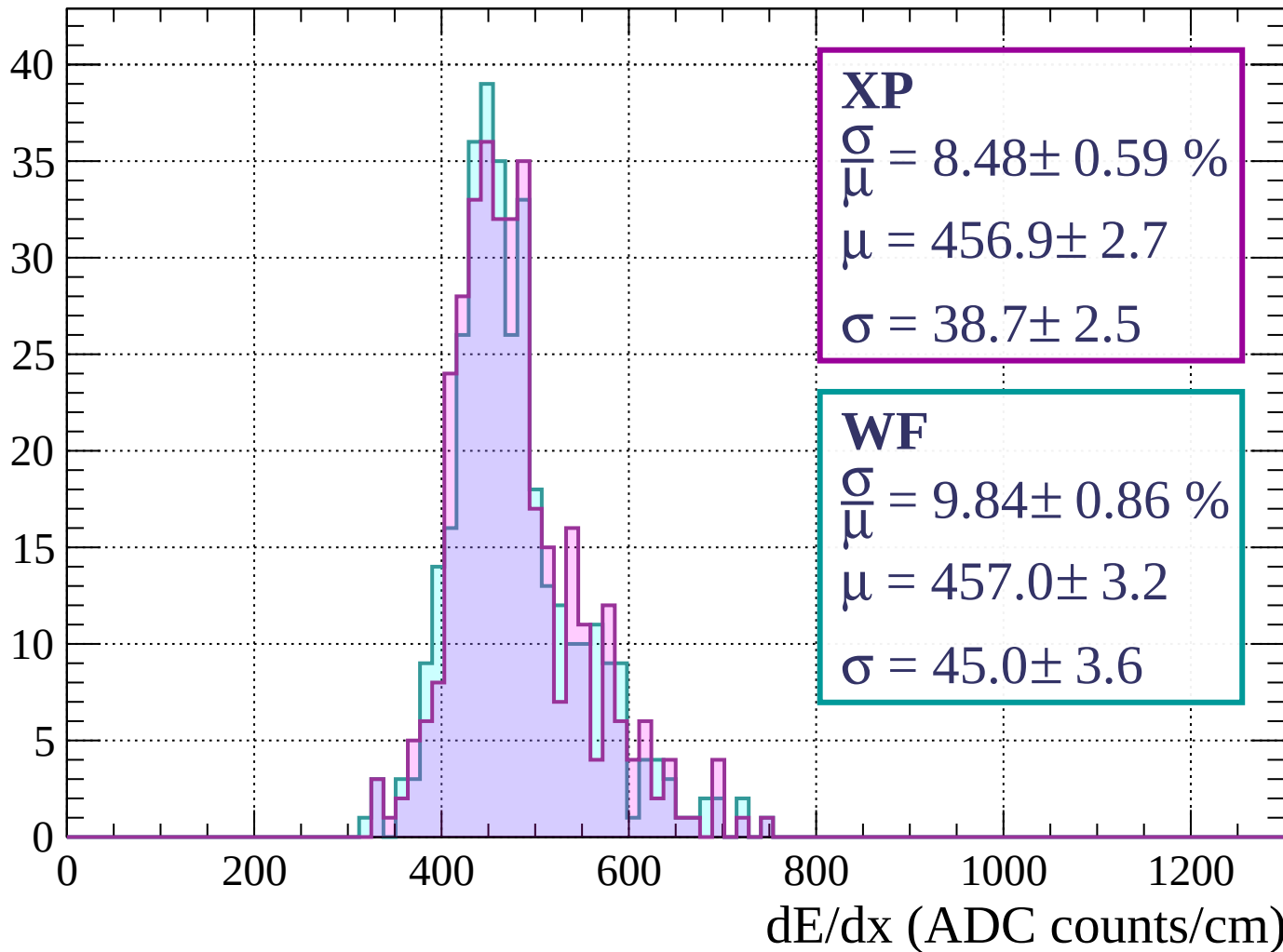
Energy loss | $160 < p < 200$

Count



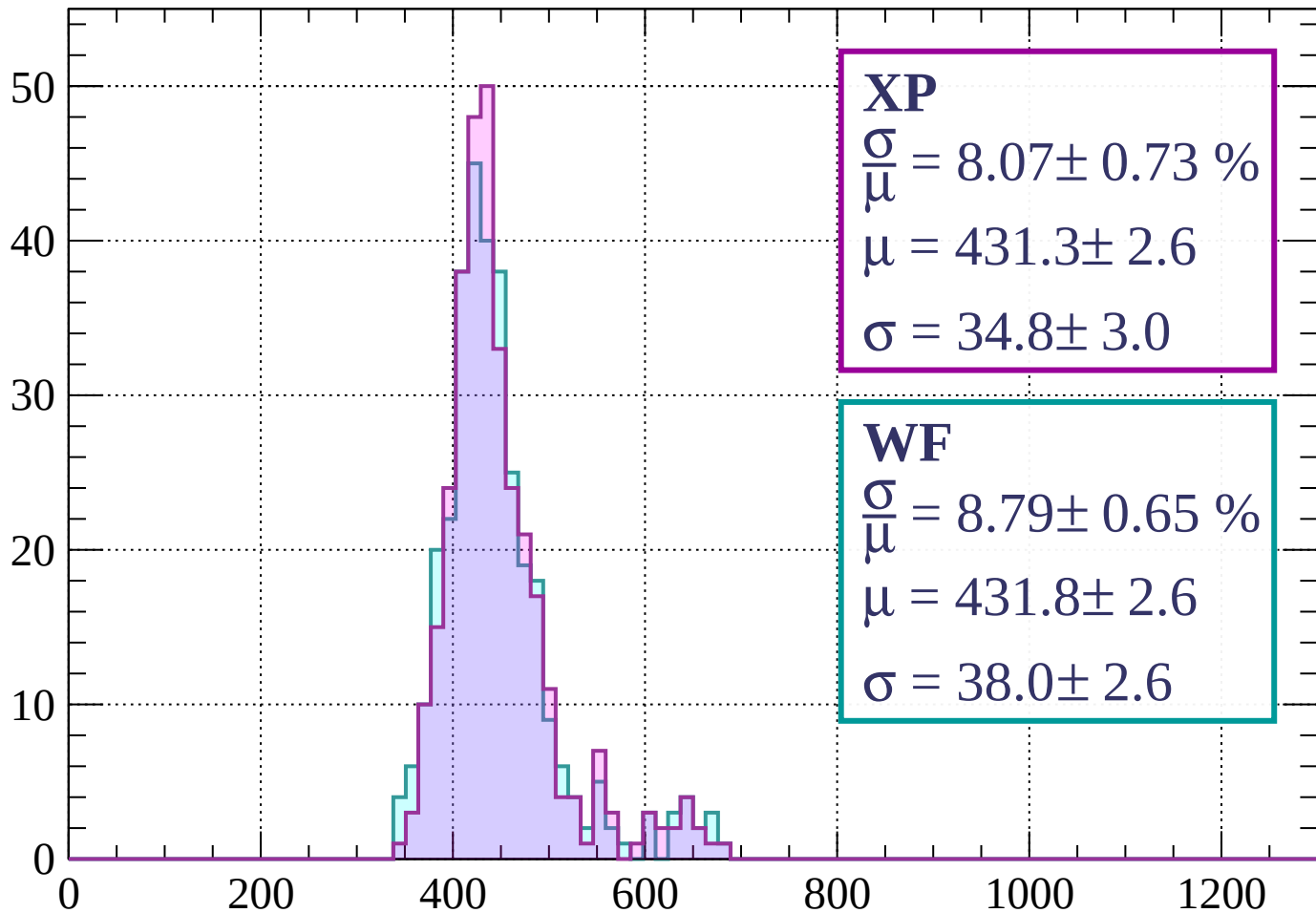
Energy loss | $200 < p < 240$

Count



Energy loss | $240 < p < 280$

Count



XP

$$\frac{\sigma}{\mu} = 8.07 \pm 0.73 \%$$

$$\mu = 431.3 \pm 2.6$$

$$\sigma = 34.8 \pm 3.0$$

WF

$$\frac{\sigma}{\mu} = 8.79 \pm 0.65 \%$$

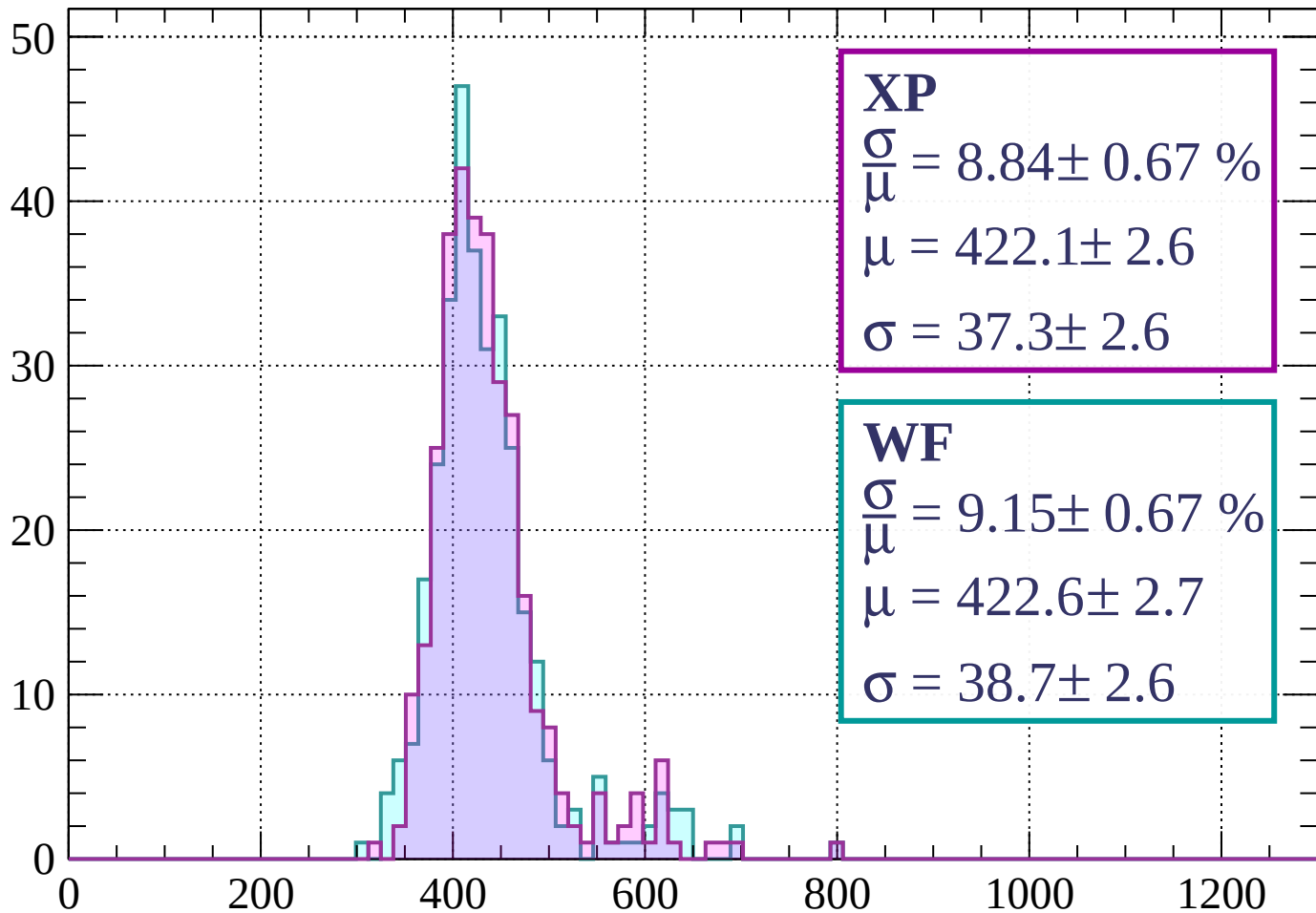
$$\mu = 431.8 \pm 2.6$$

$$\sigma = 38.0 \pm 2.6$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 8.84 \pm 0.67 \%$$

$$\mu = 422.1 \pm 2.6$$

$$\sigma = 37.3 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 9.15 \pm 0.67 \%$$

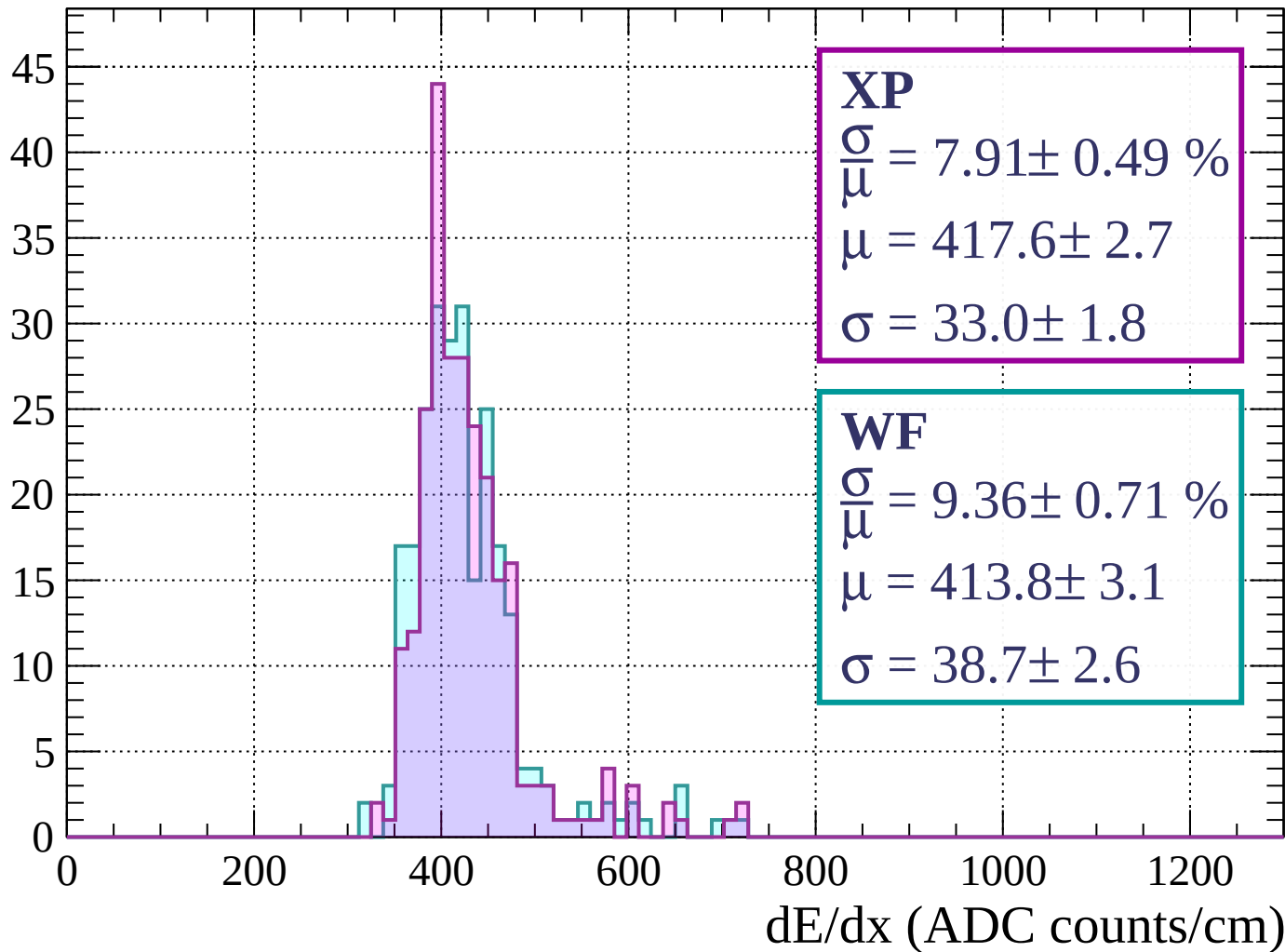
$$\mu = 422.6 \pm 2.7$$

$$\sigma = 38.7 \pm 2.6$$

dE/dx (ADC counts/cm)

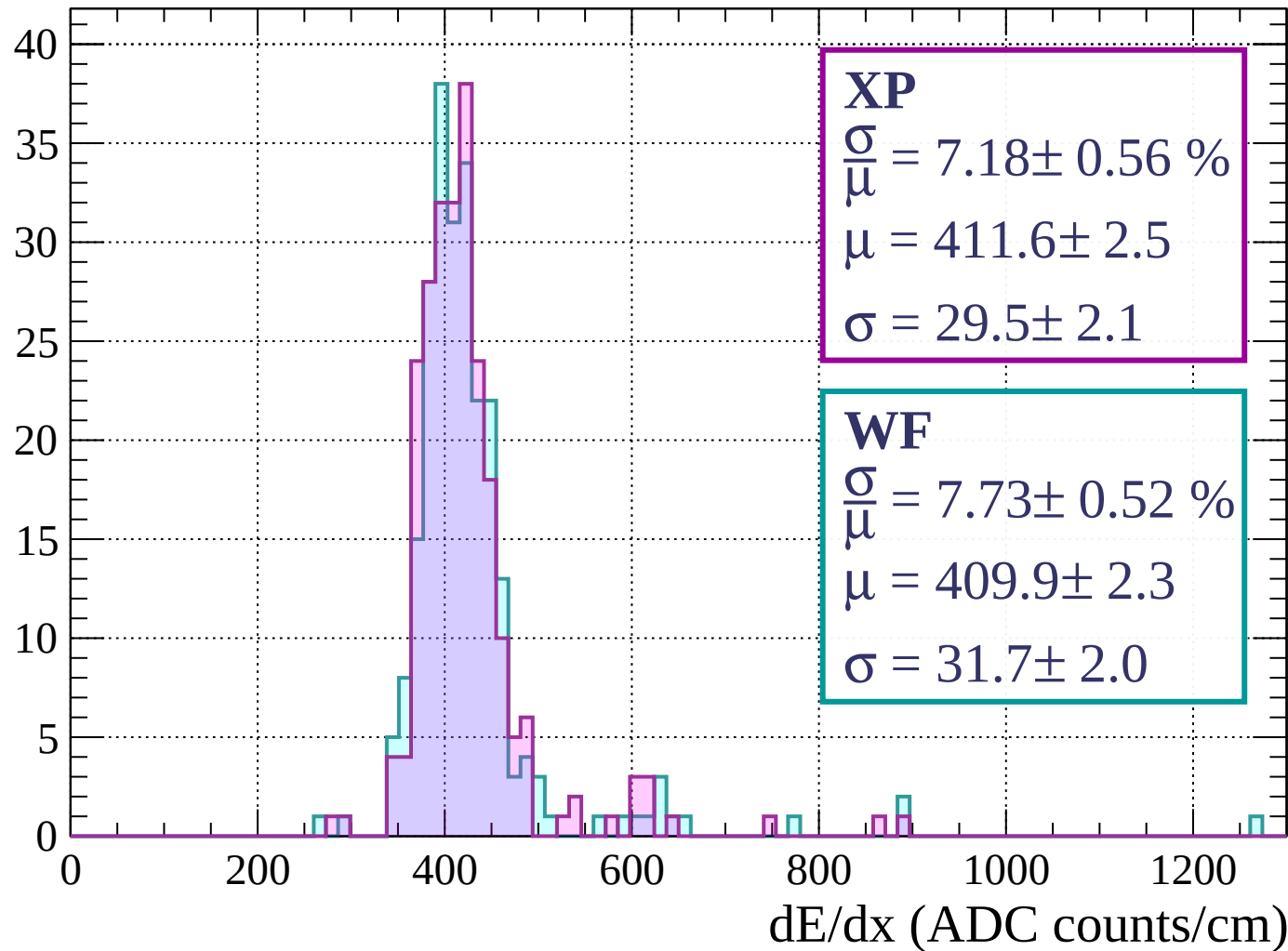
Energy loss | $320 < p < 360$

Count



Energy loss | $360 < p < 400$

Count



Energy loss | $400 < p < 440$

Count

XP

$$\frac{\sigma}{\mu} = 7.44 \pm 0.63 \%$$

$$\mu = 408.2 \pm 2.3$$

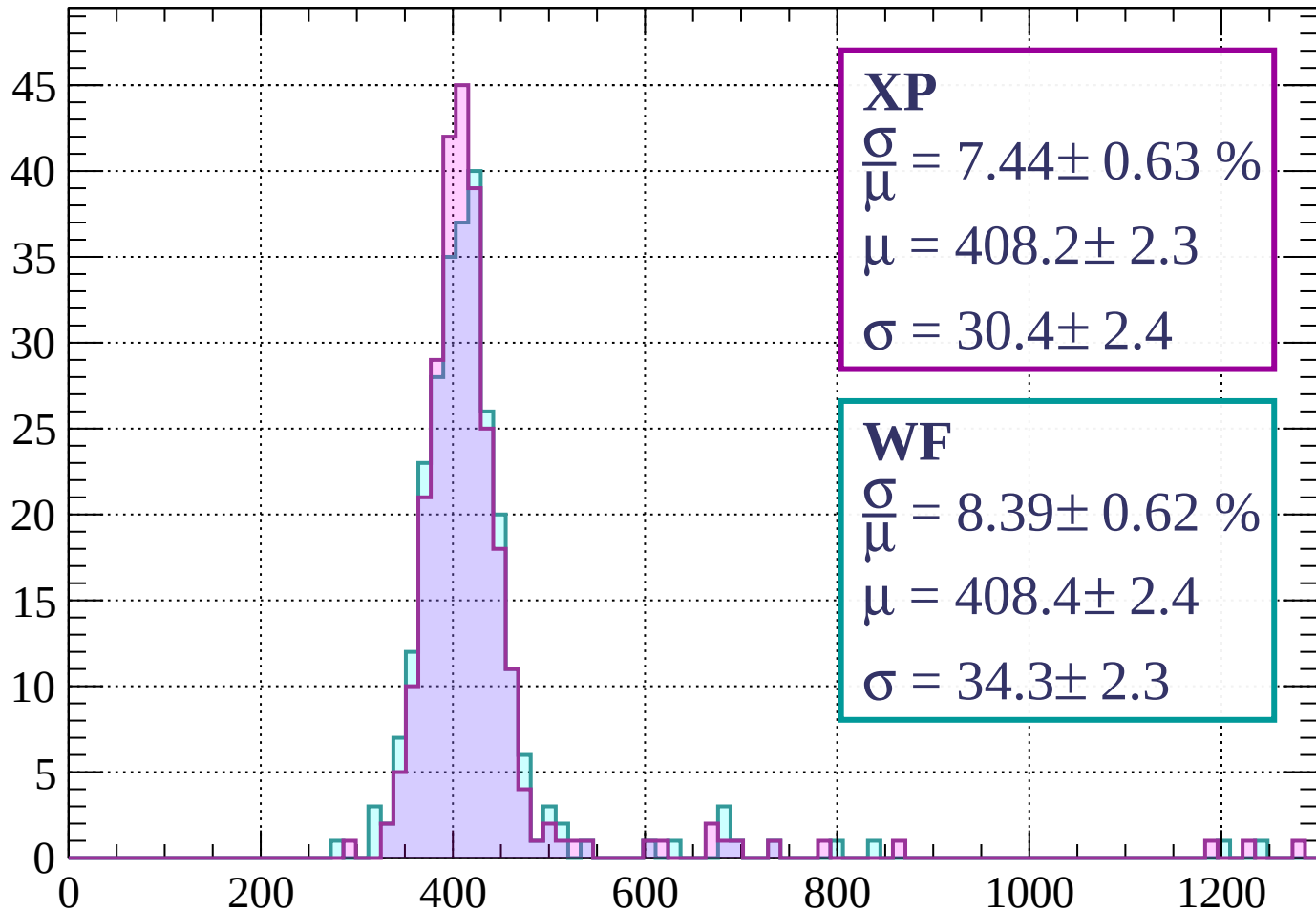
$$\sigma = 30.4 \pm 2.4$$

WF

$$\frac{\sigma}{\mu} = 8.39 \pm 0.62 \%$$

$$\mu = 408.4 \pm 2.4$$

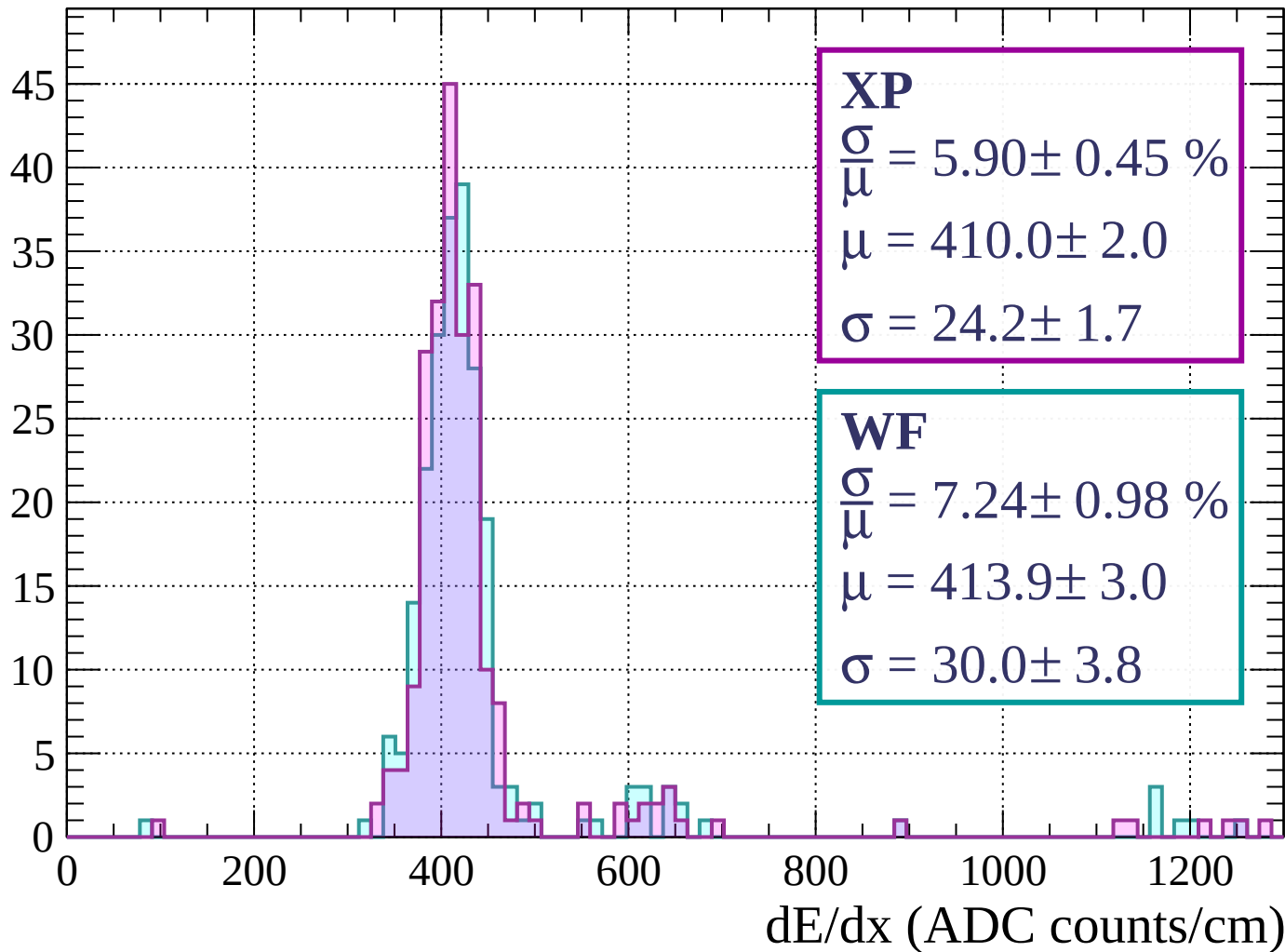
$$\sigma = 34.3 \pm 2.3$$



dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count



Energy loss | $480 < p < 520$

Count

XP

$$\frac{\sigma}{\mu} = 7.34 \pm 0.59 \%$$

$$\mu = 407.8 \pm 2.5$$

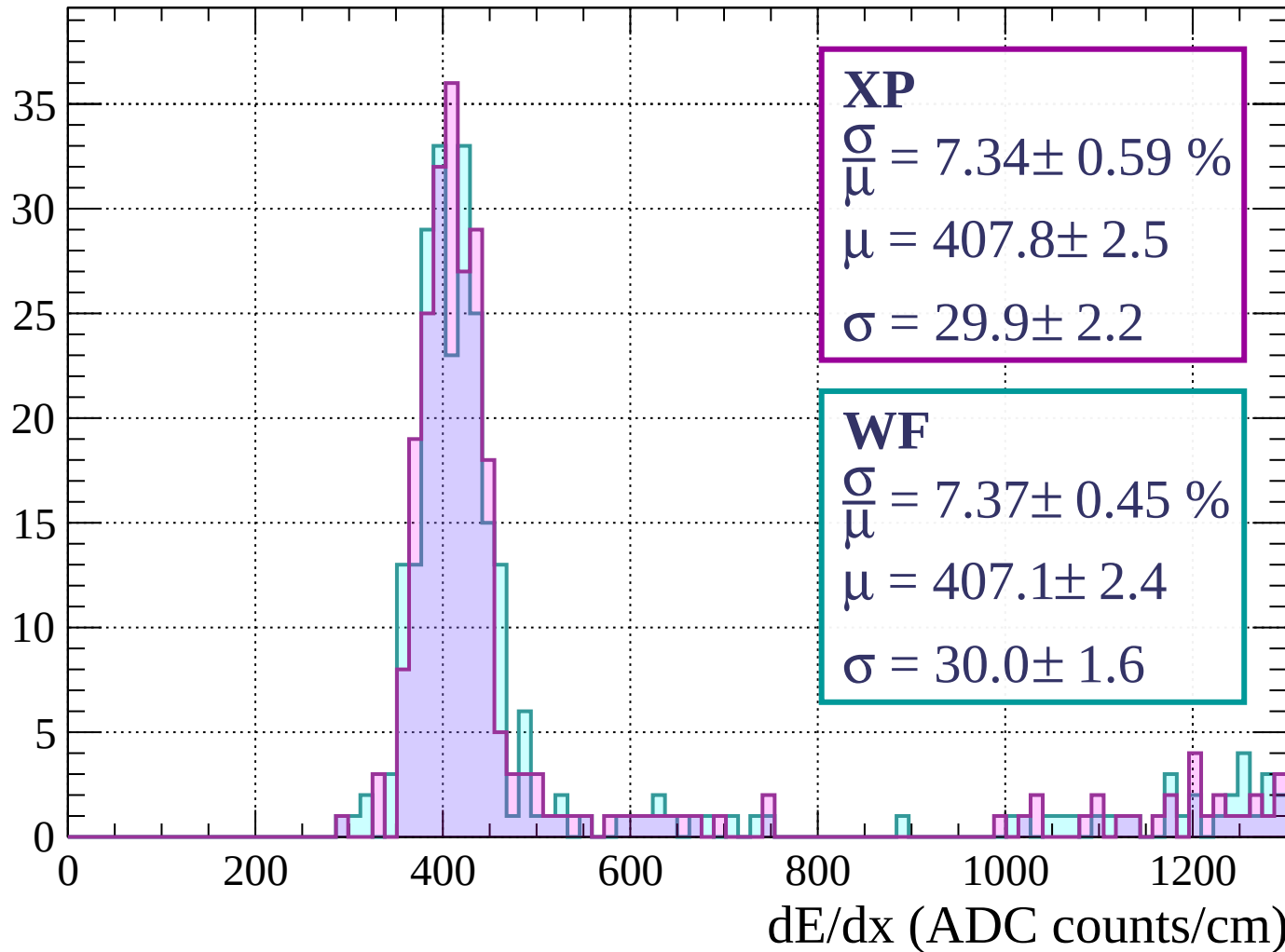
$$\sigma = 29.9 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 7.37 \pm 0.45 \%$$

$$\mu = 407.1 \pm 2.4$$

$$\sigma = 30.0 \pm 1.6$$



Energy loss | $520 < p < 560$

Count

XP

$$\frac{\sigma}{\mu} = 8.38 \pm 0.70 \%$$

$$\mu = 411.0 \pm 2.9$$

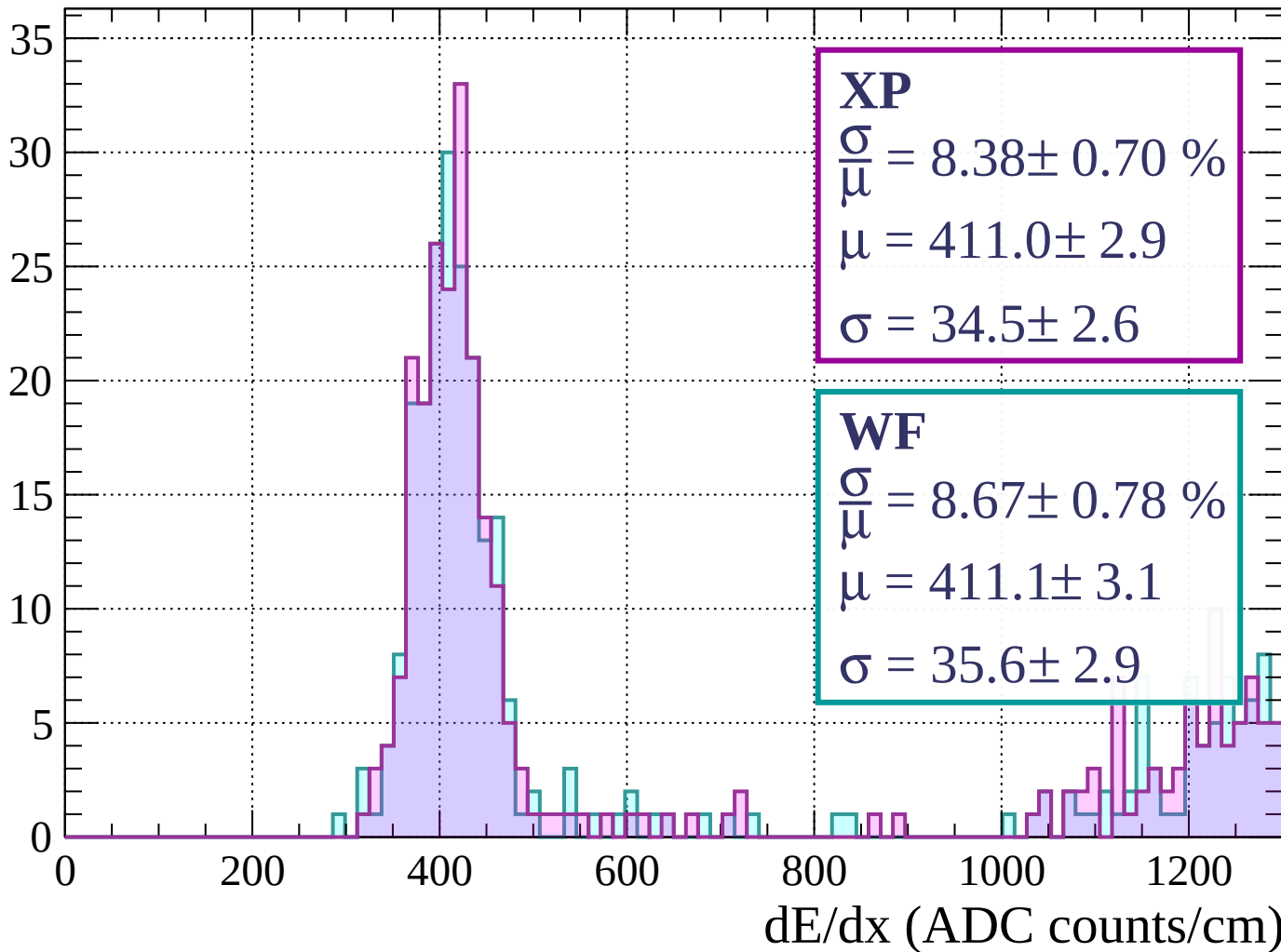
$$\sigma = 34.5 \pm 2.6$$

WF

$$\frac{\sigma}{\mu} = 8.67 \pm 0.78 \%$$

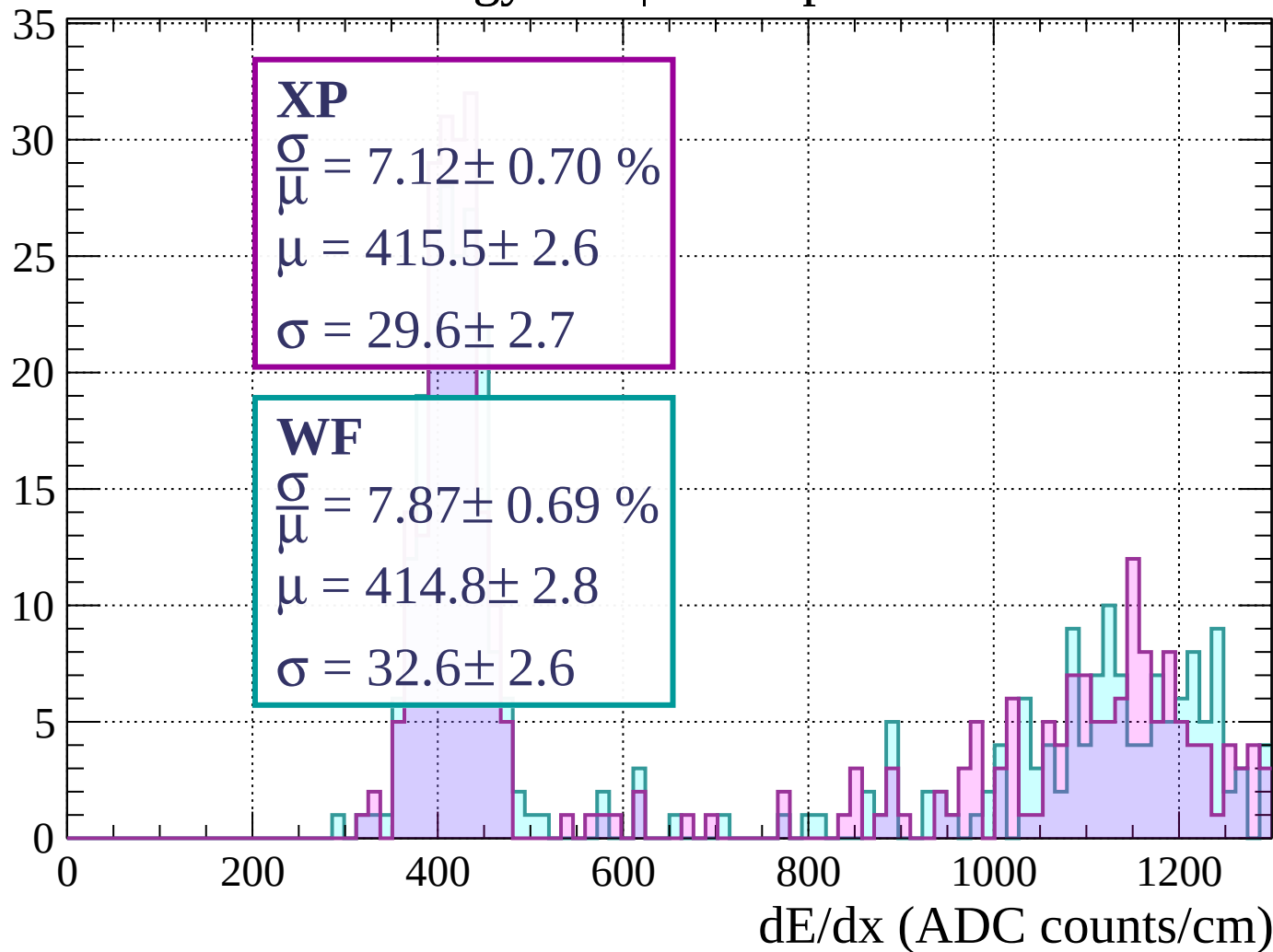
$$\mu = 411.1 \pm 3.1$$

$$\sigma = 35.6 \pm 2.9$$



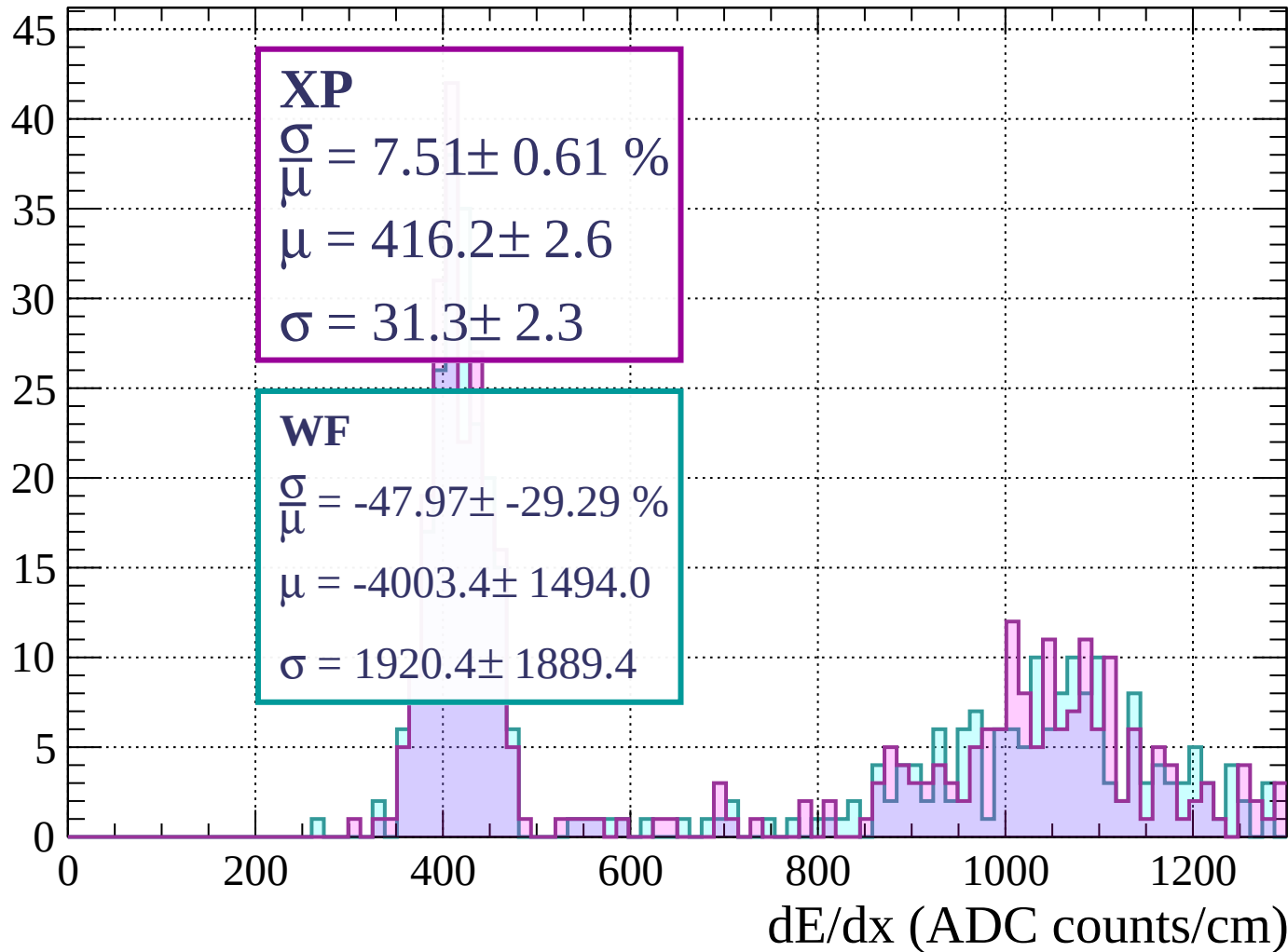
Energy loss | $560 < p < 600$

Count



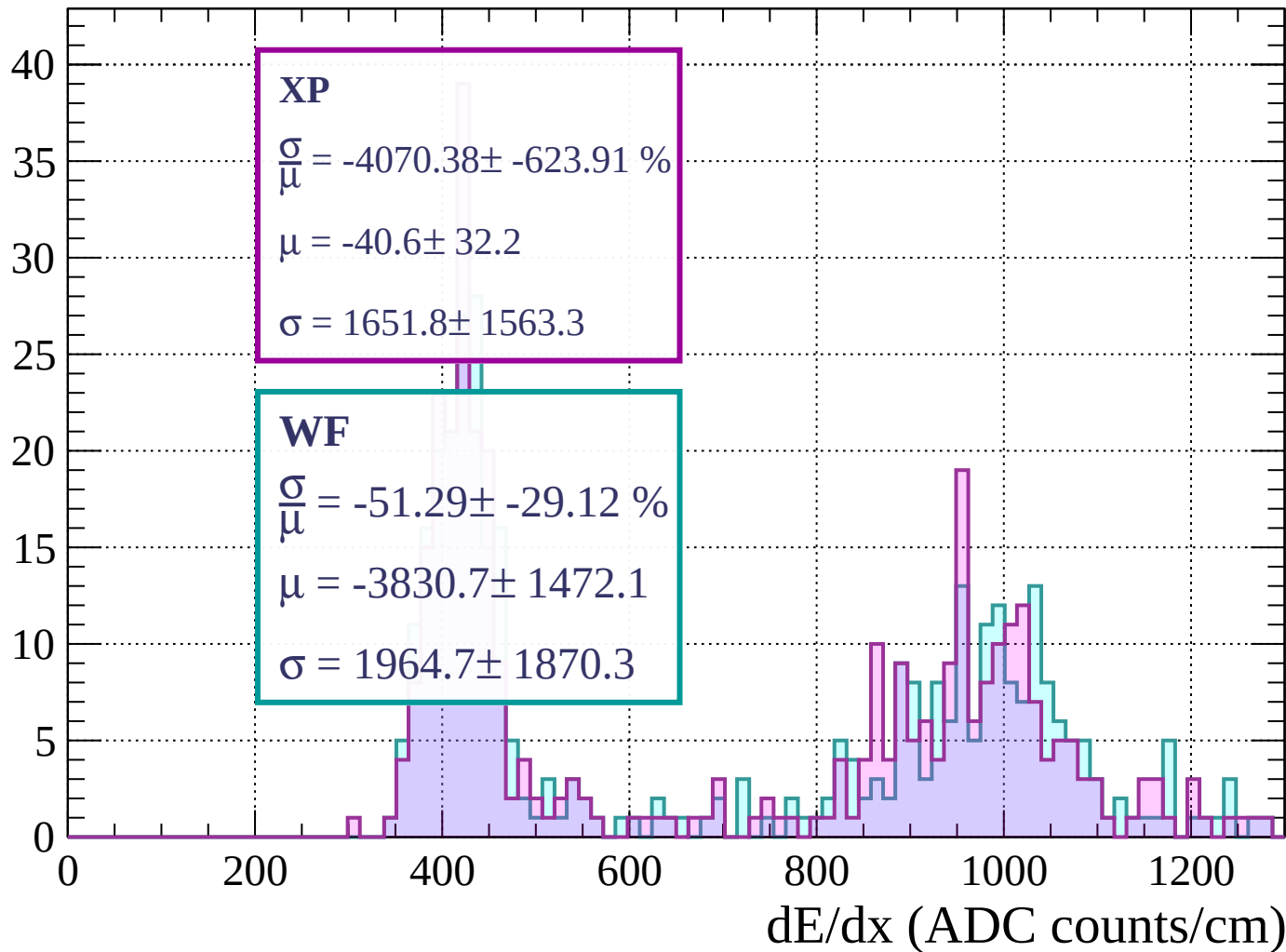
Energy loss | $600 < p < 640$

Count



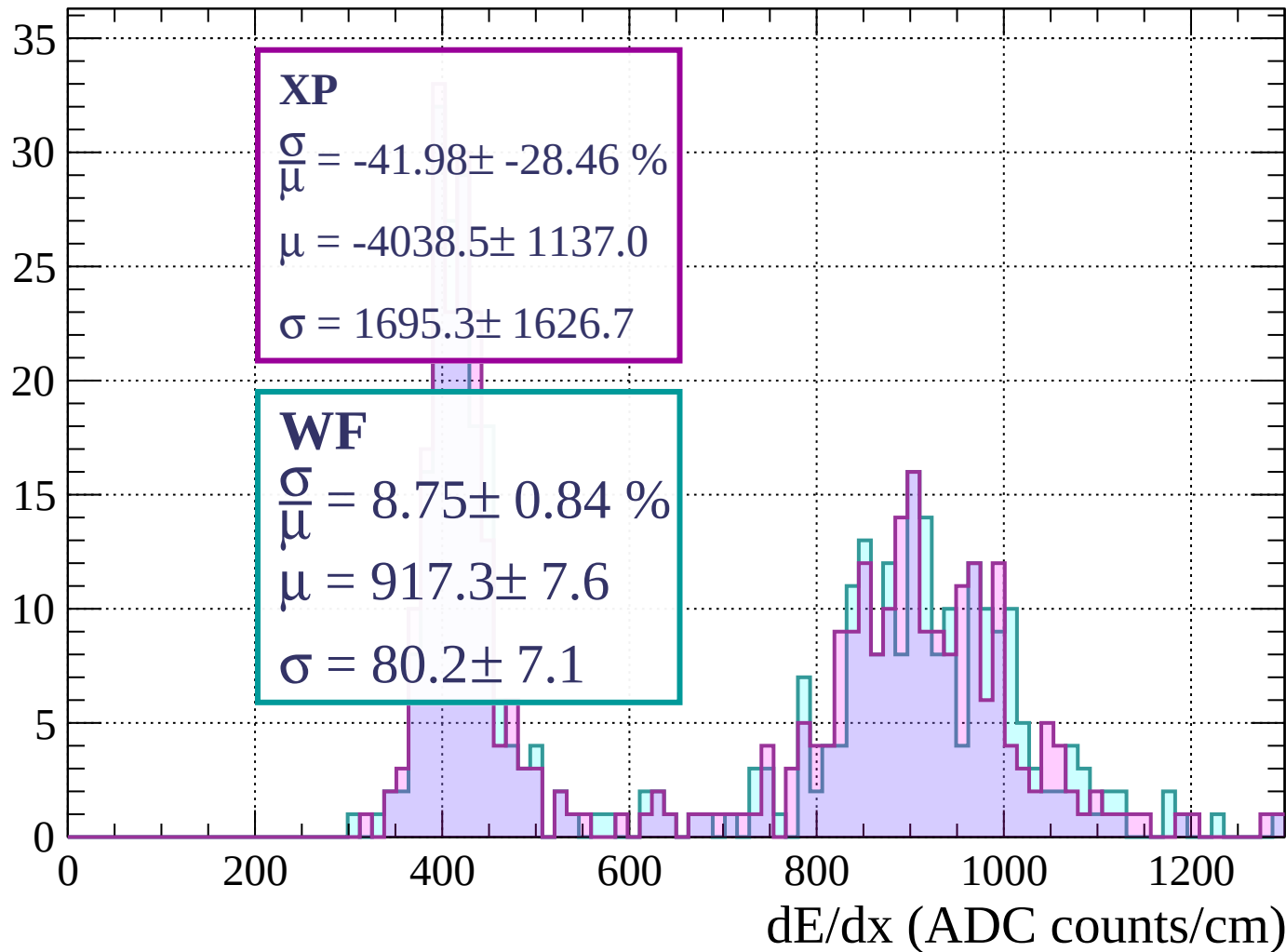
Energy loss | $640 < p < 680$

Count



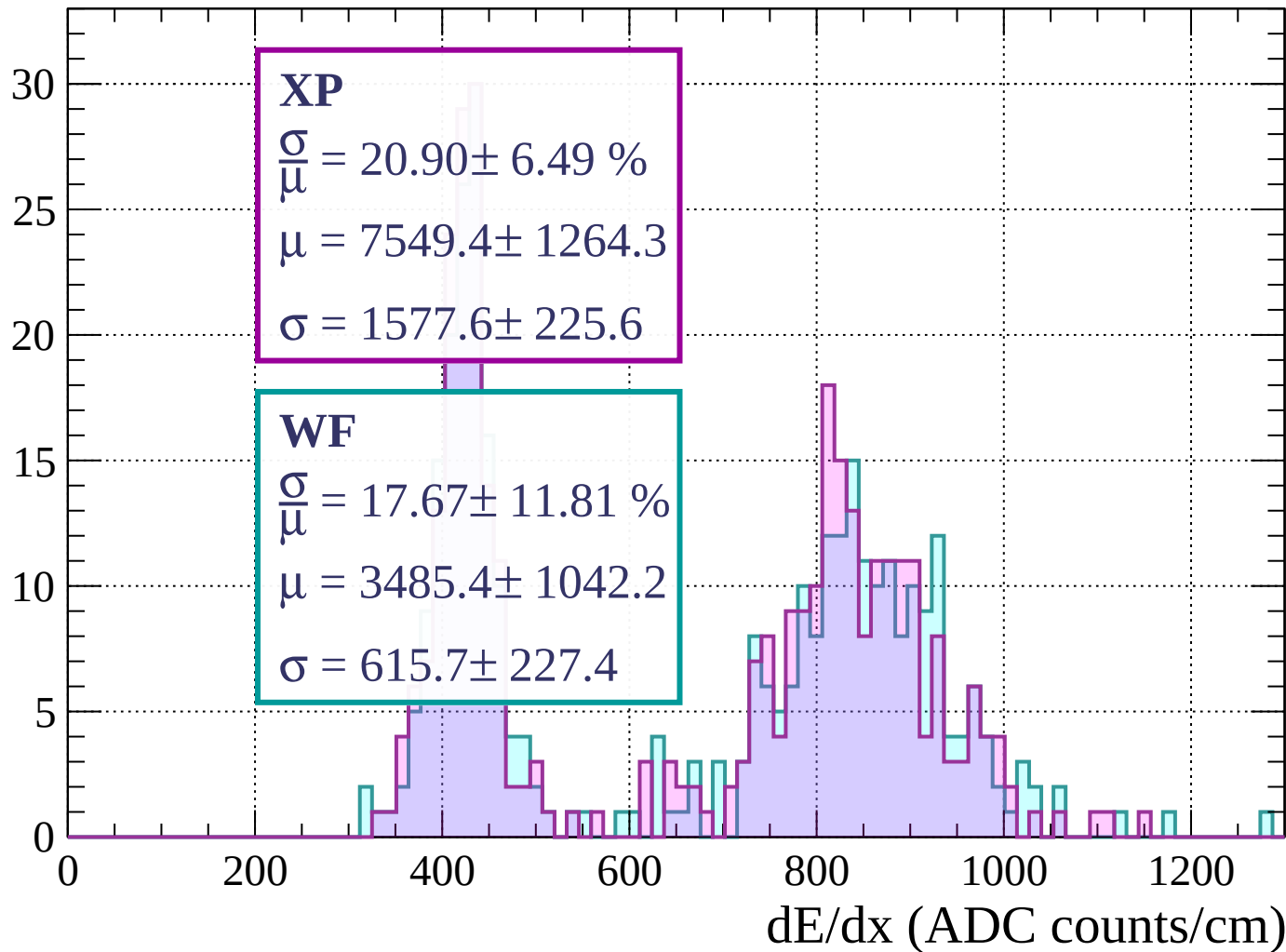
Energy loss | $680 < p < 720$

Count



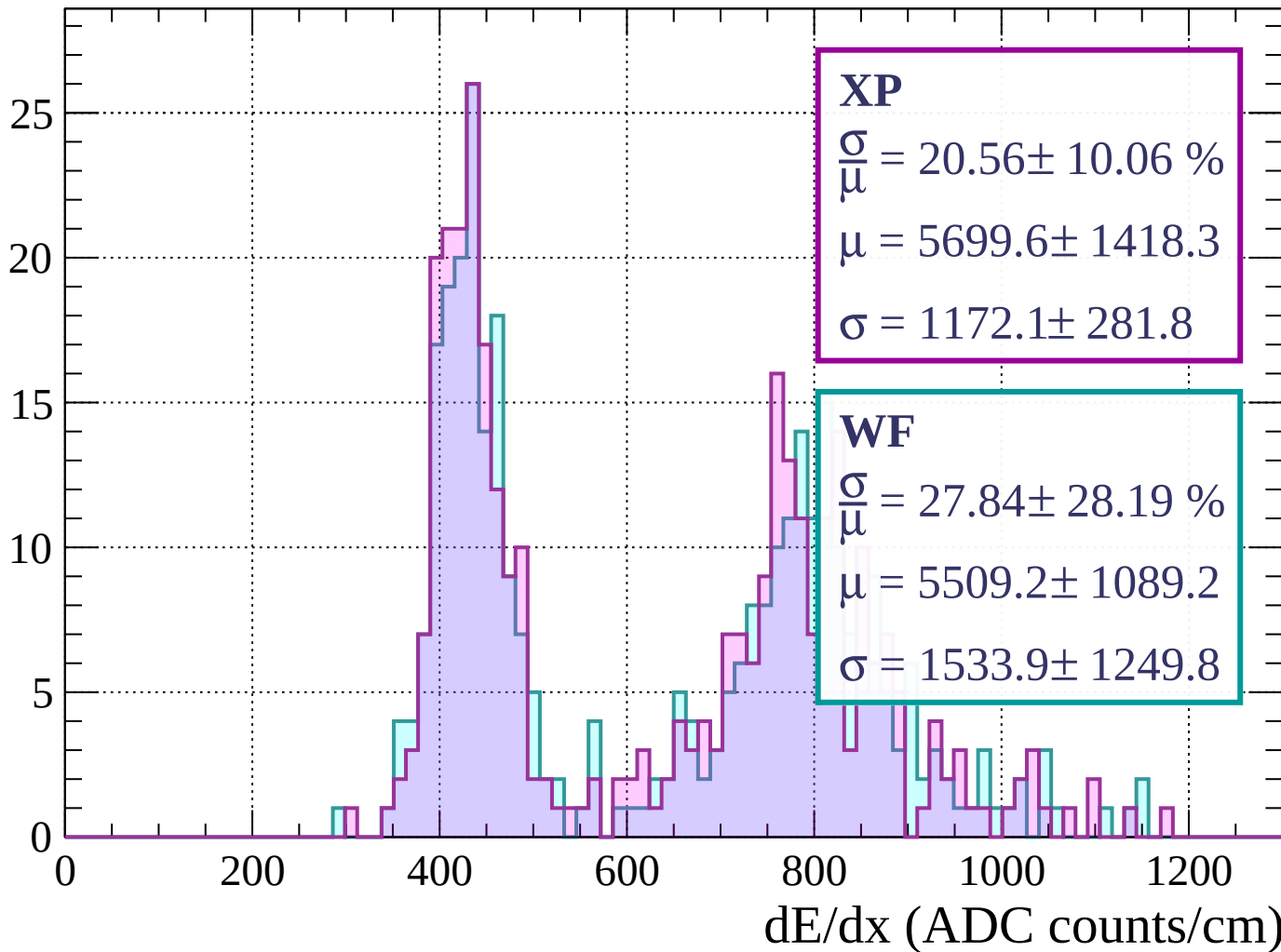
Energy loss | $720 < p < 760$

Count



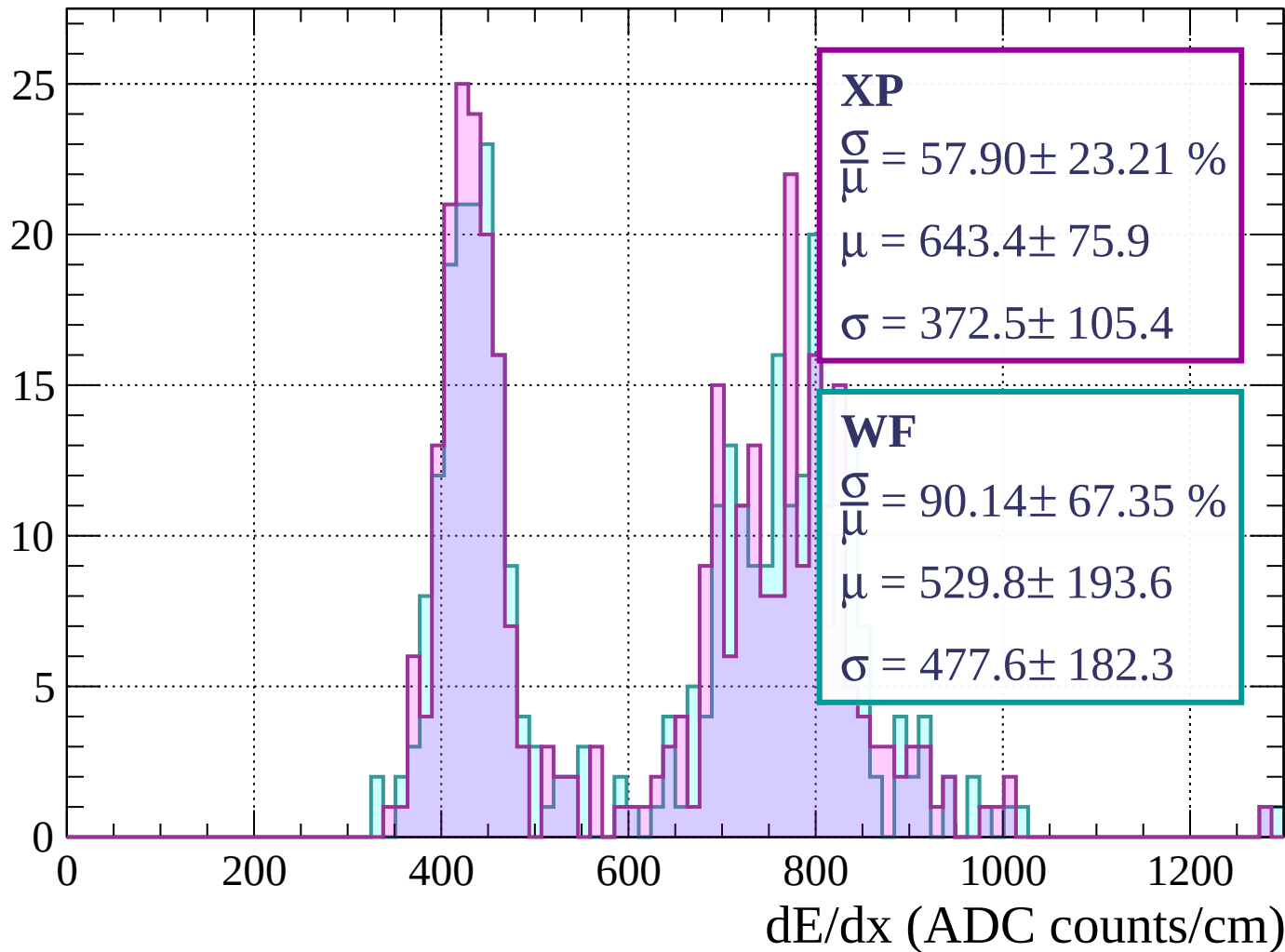
Energy loss | $760 < p < 800$

Count



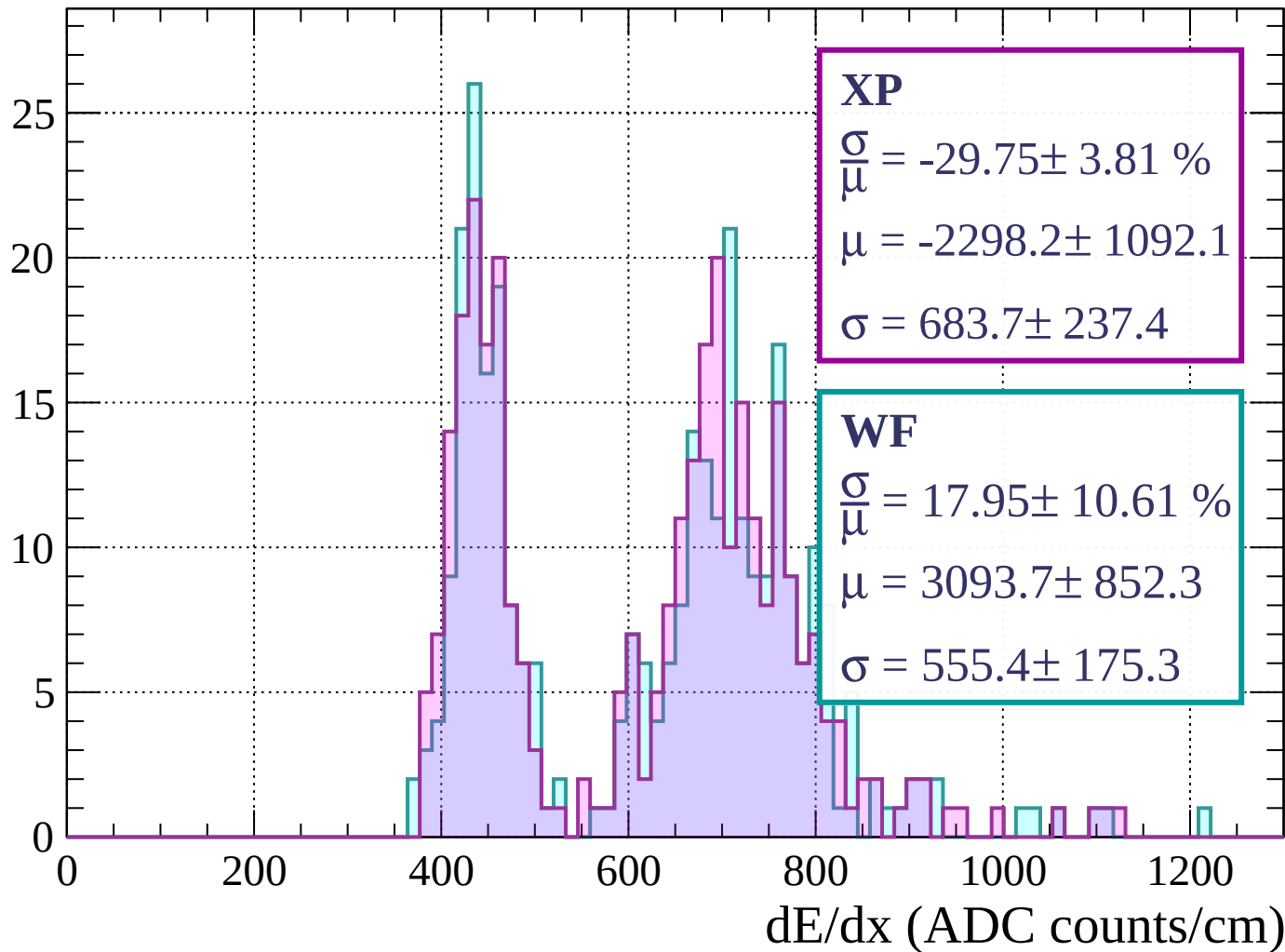
Energy loss | $800 < p < 840$

Count



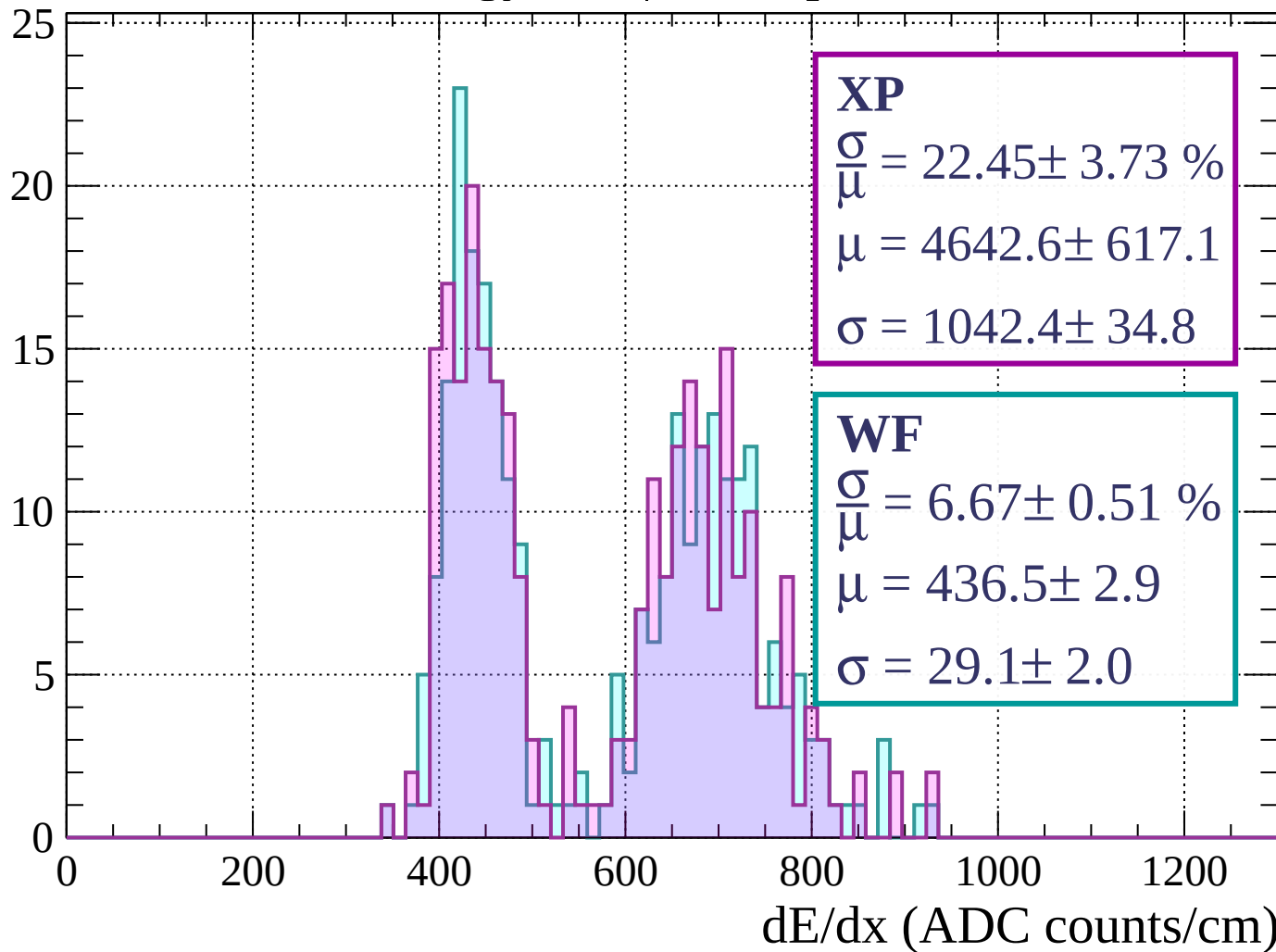
Energy loss | $840 < p < 880$

Count



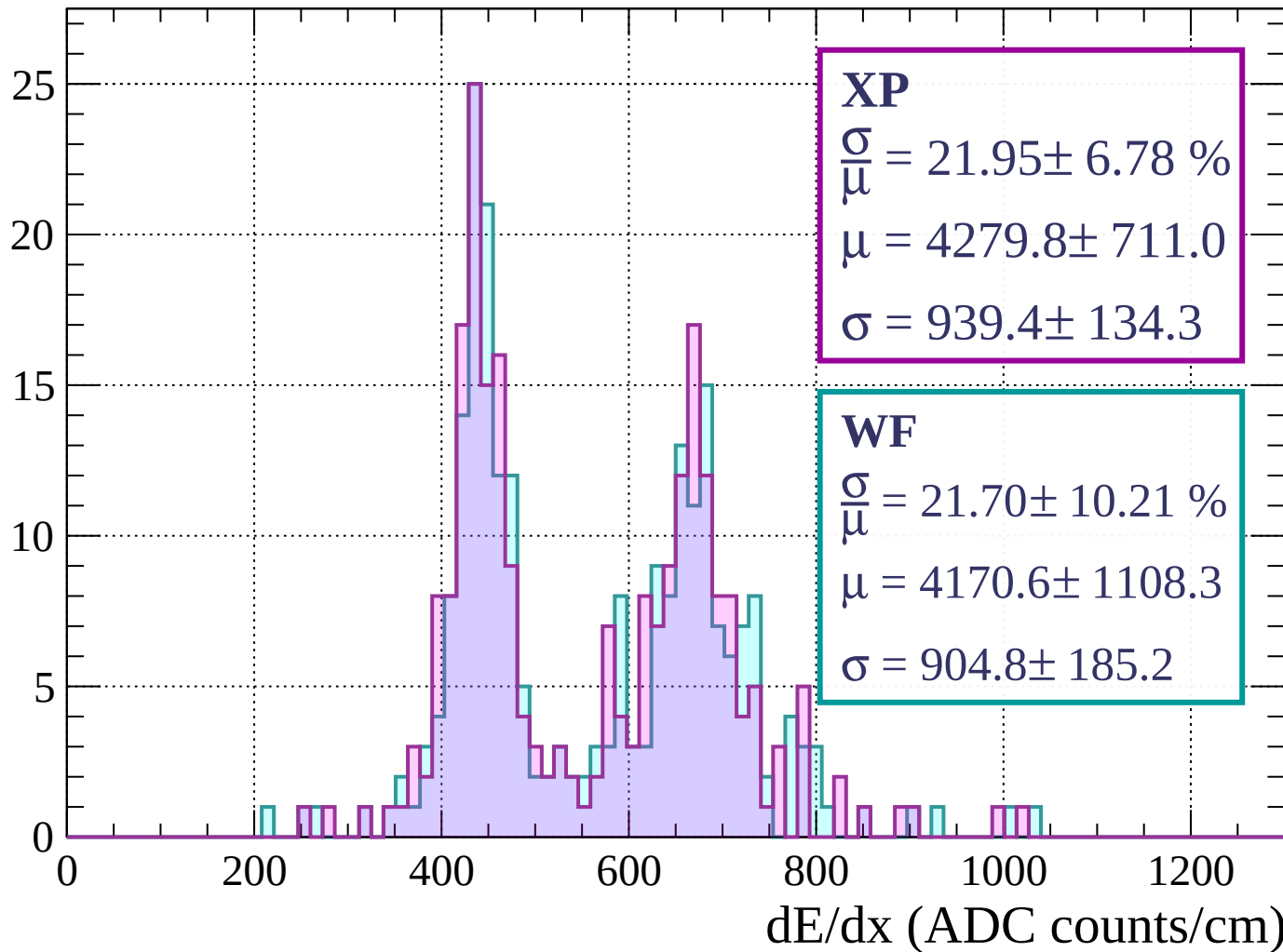
Energy loss | $880 < p < 920$

Count



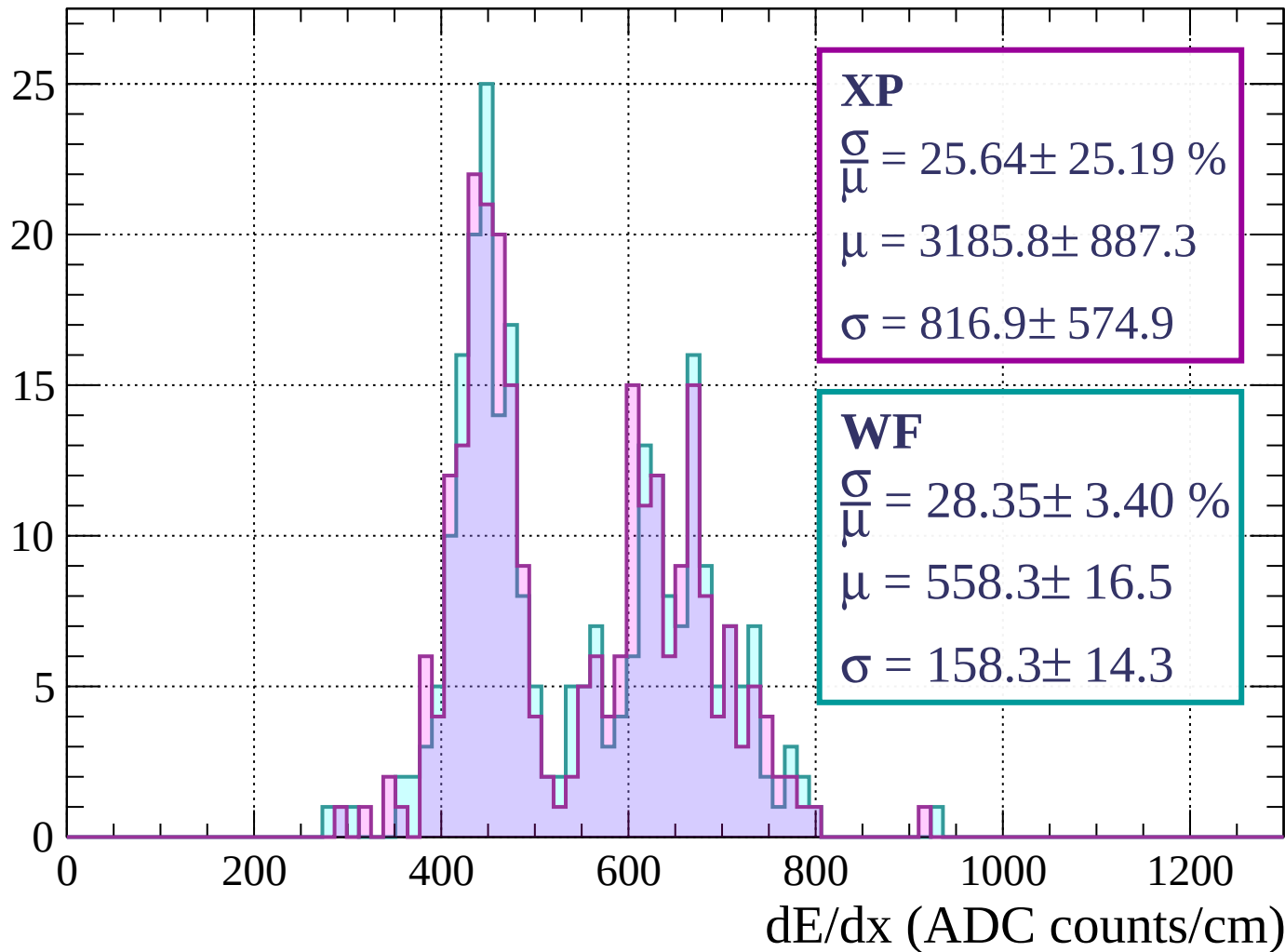
Energy loss | $920 < p < 960$

Count



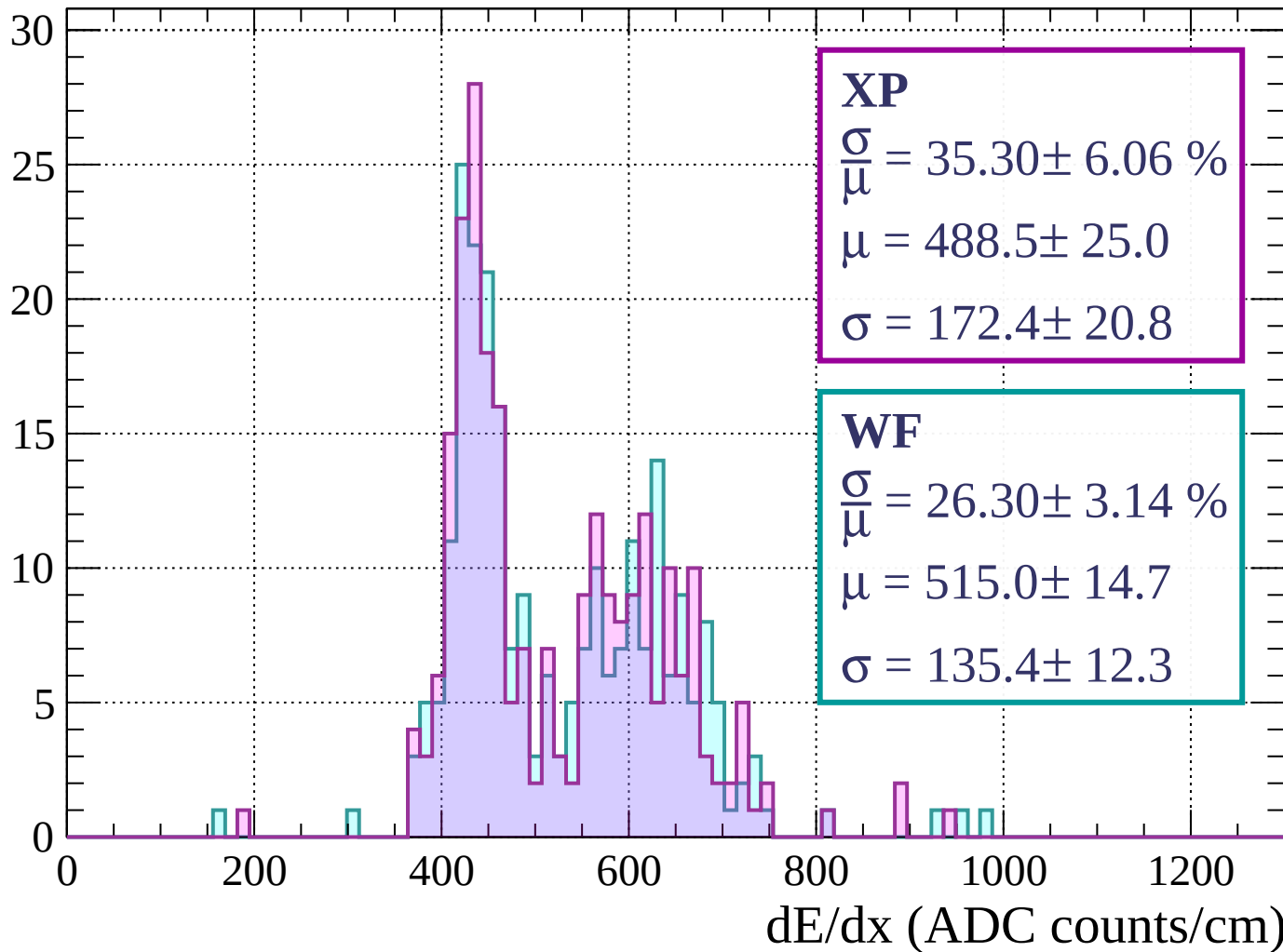
Energy loss | $960 < p < 1000$

Count



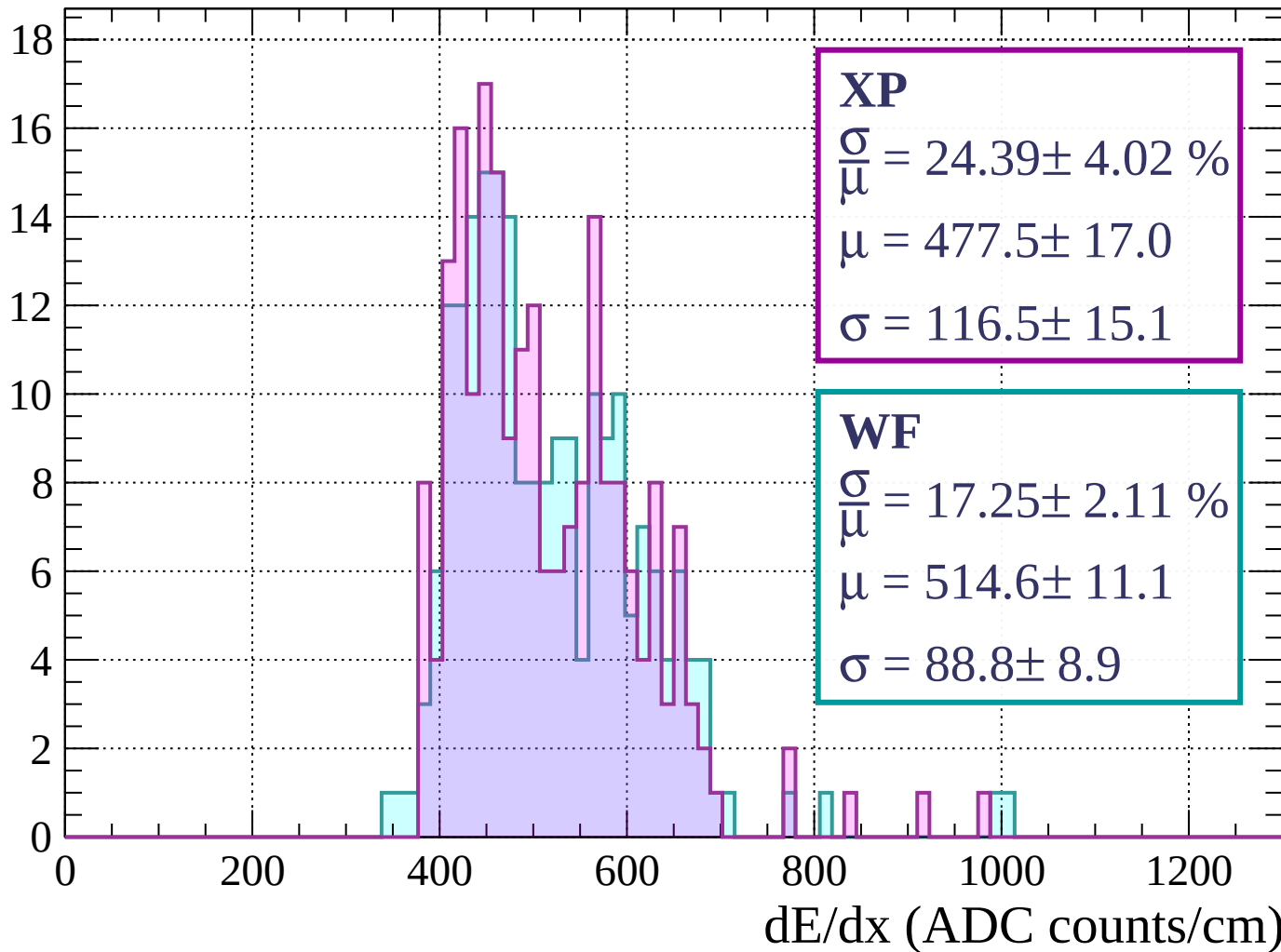
Energy loss | $1000 < p < 1040$

Count



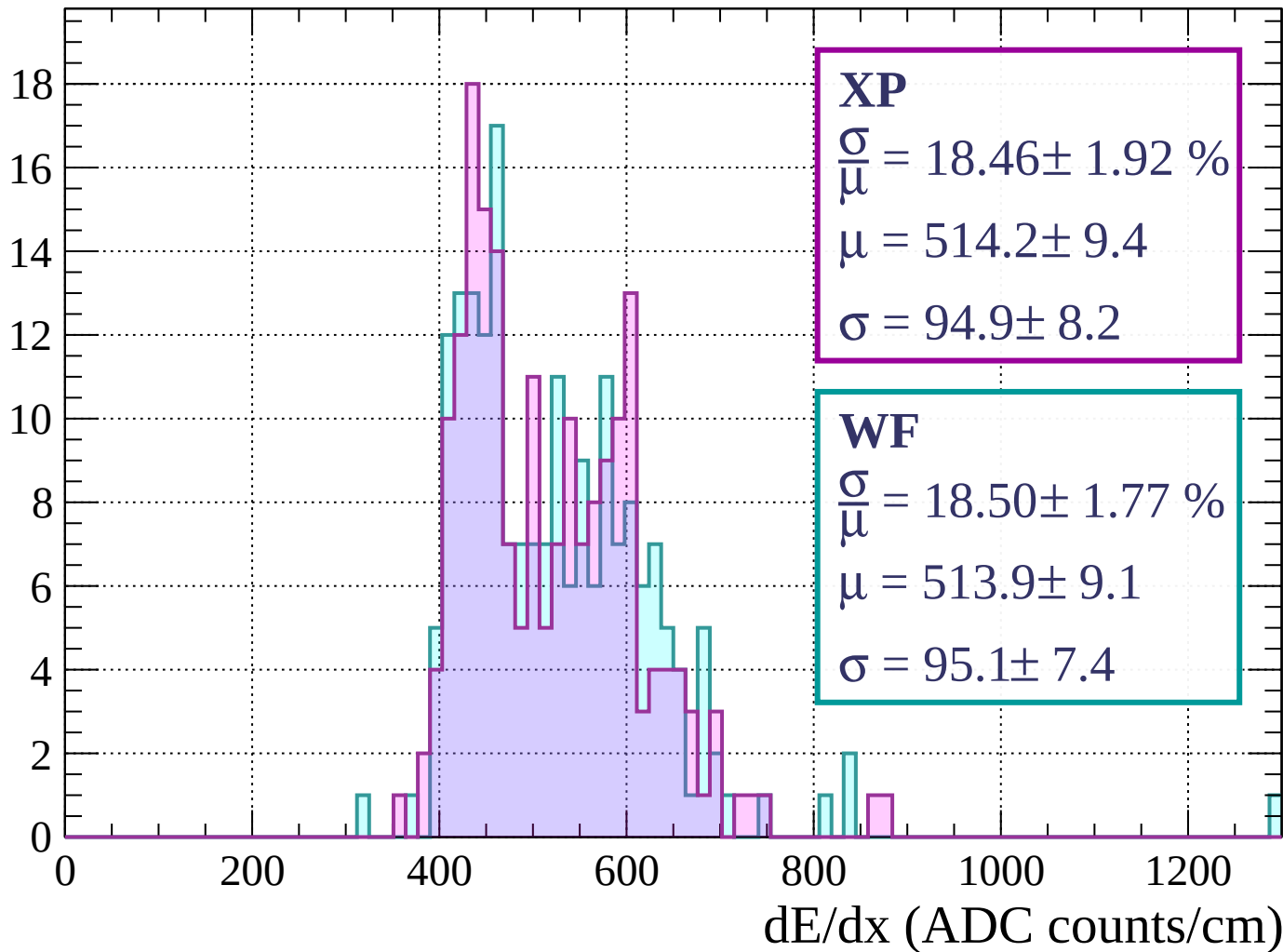
Energy loss | $1040 < p < 1080$

Count



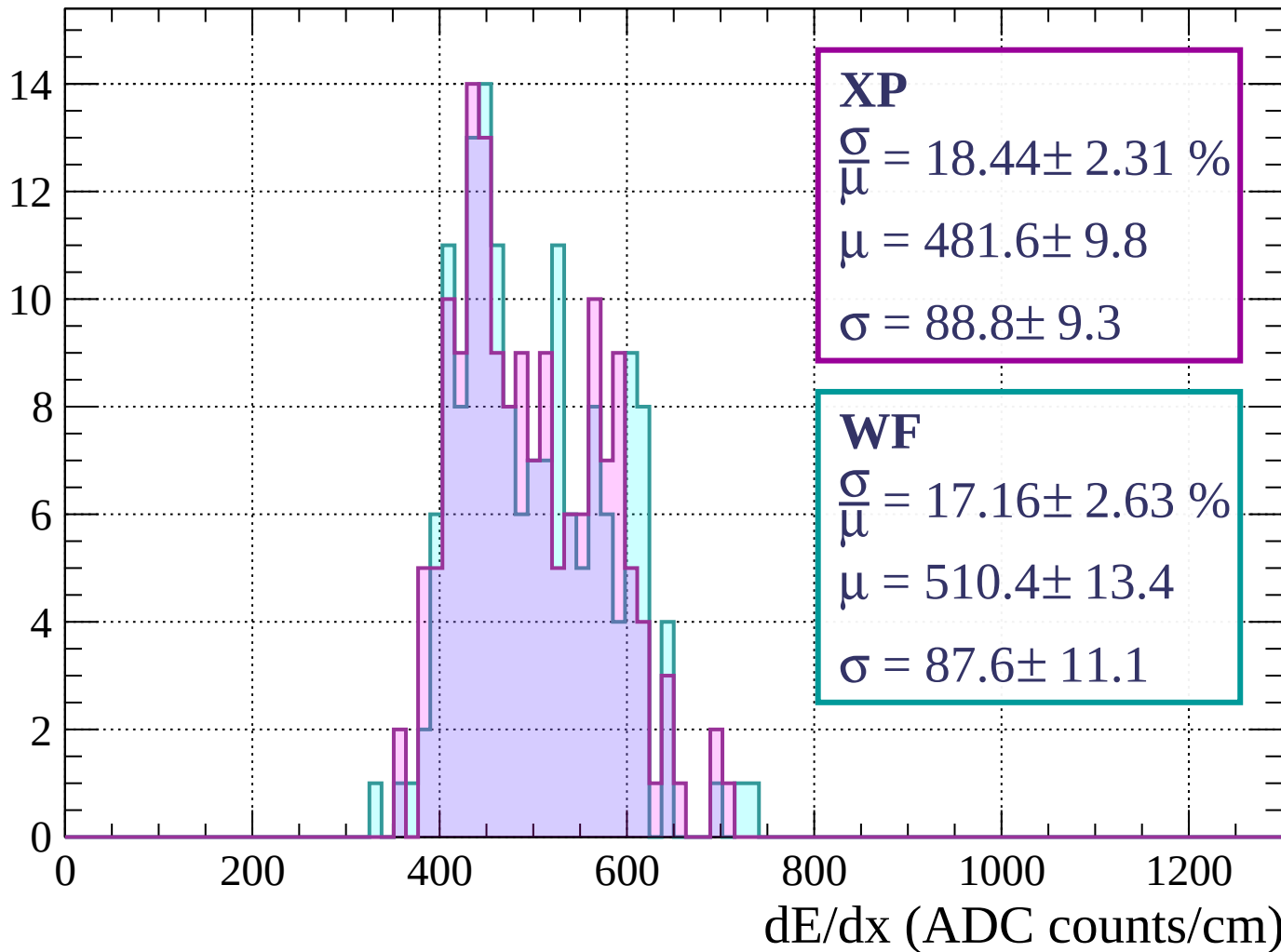
Energy loss | $1080 < p < 1120$

Count



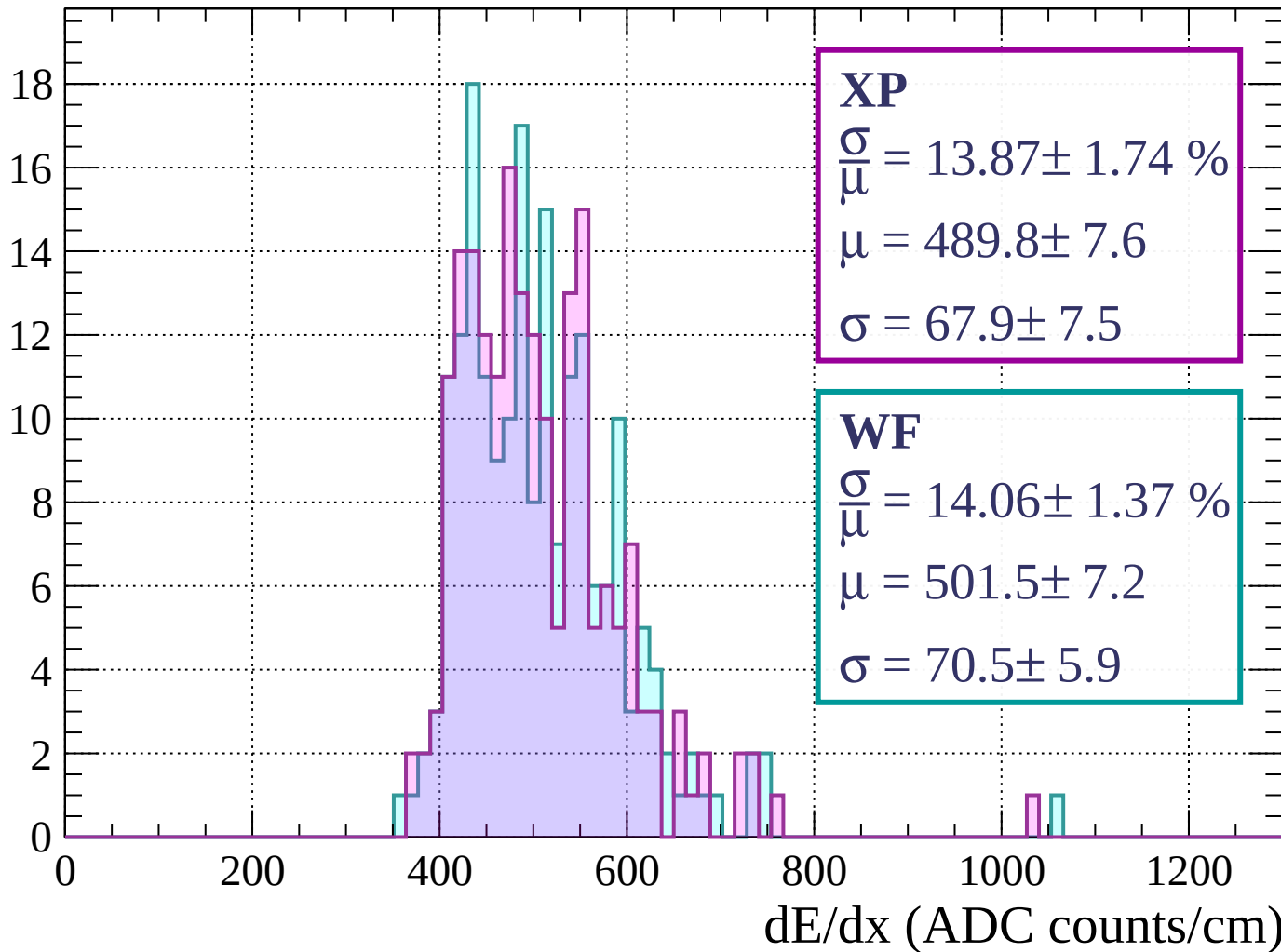
Energy loss | $1120 < p < 1160$

Count



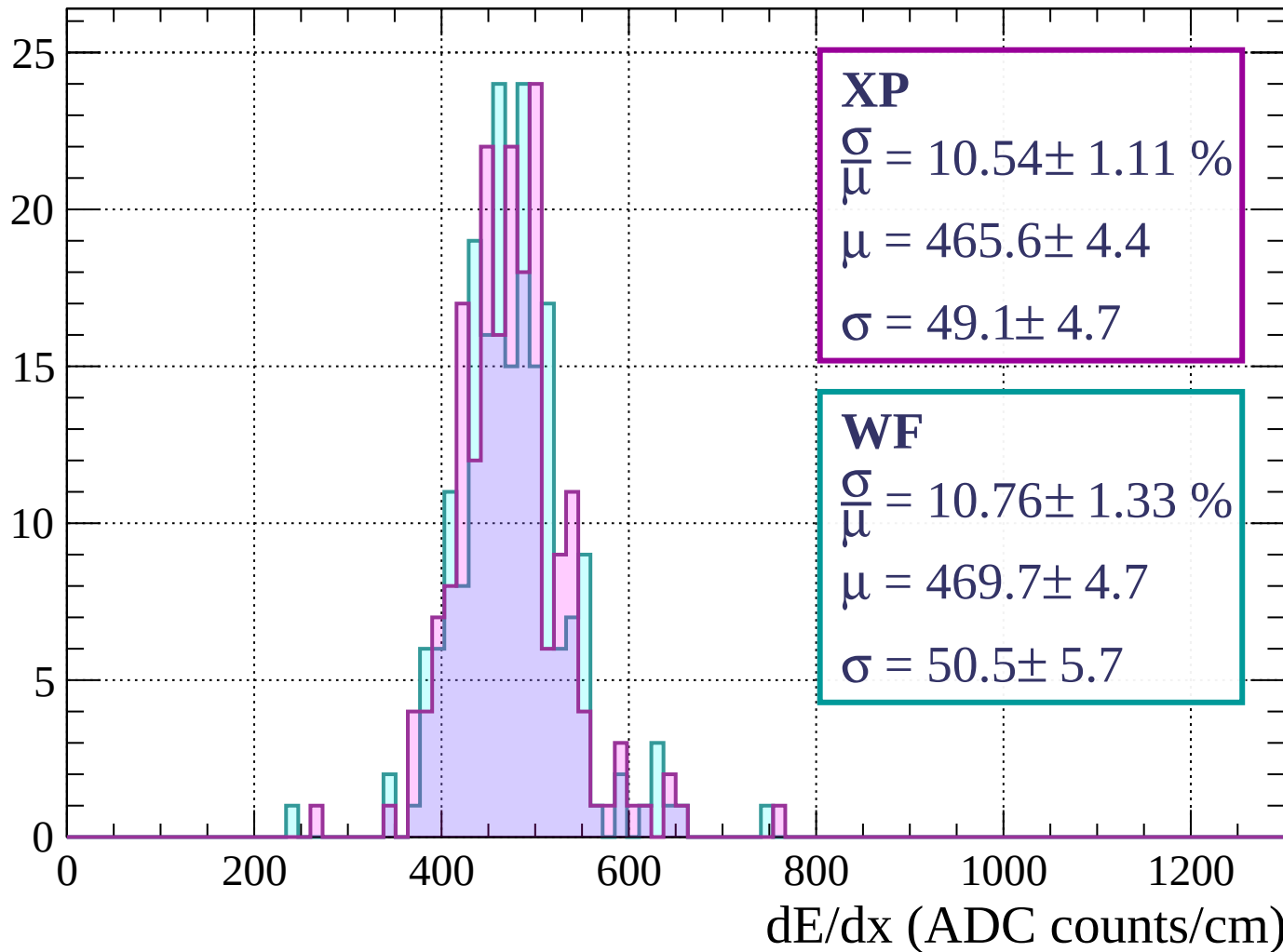
Energy loss | $1160 < p < 1200$

Count



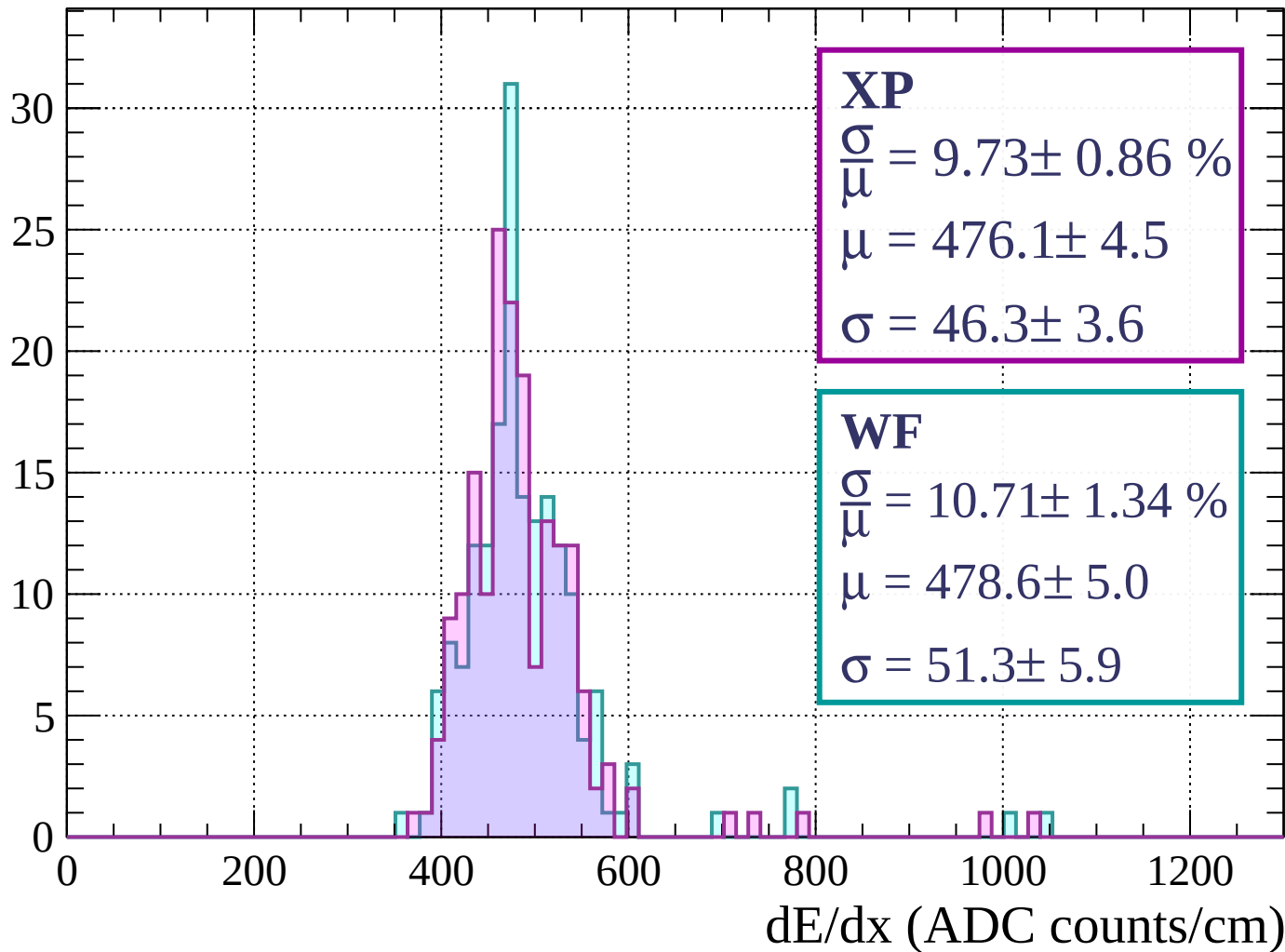
Energy loss | $1200 < p < 1240$

Count



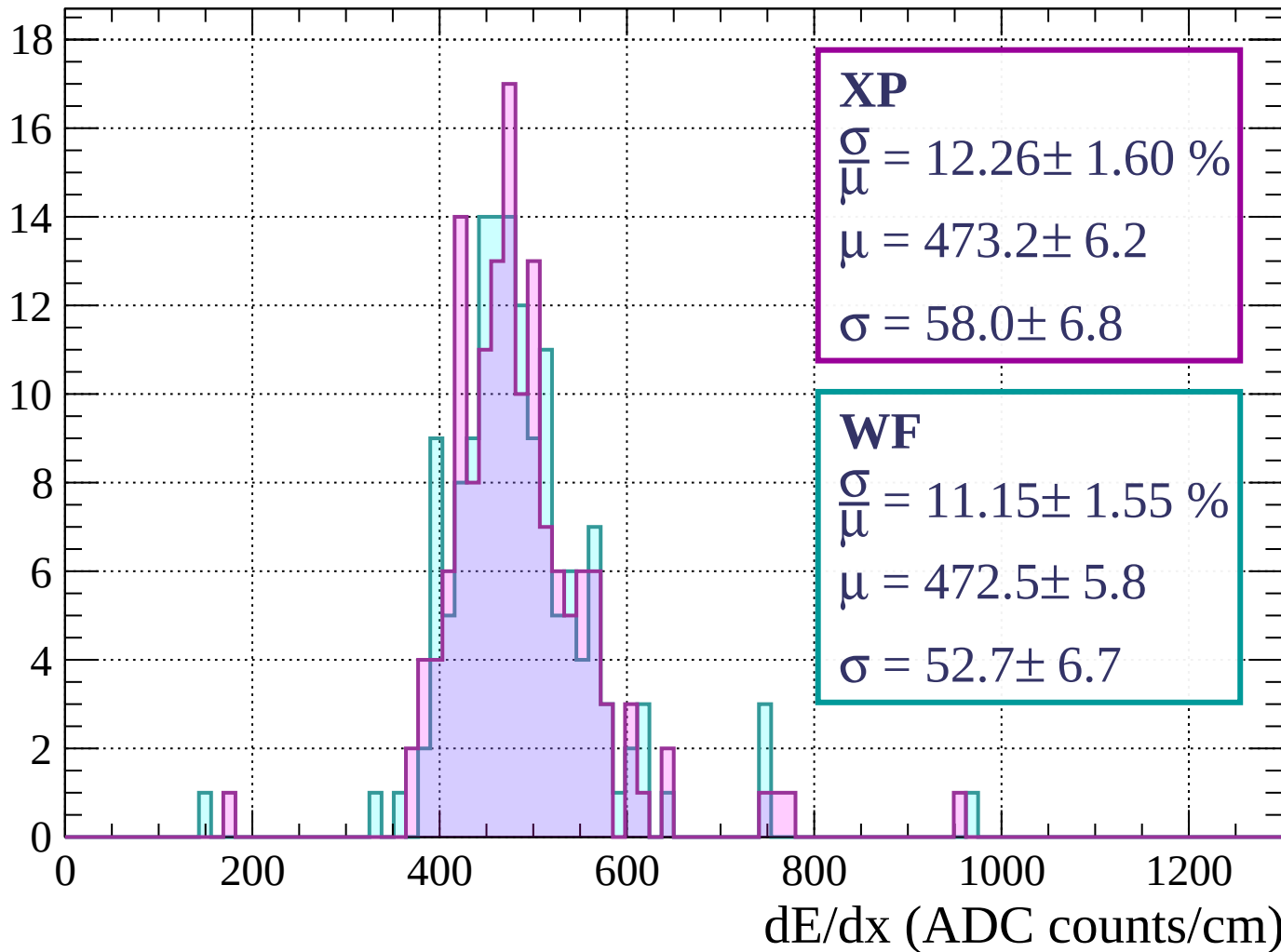
Energy loss | $1240 < p < 1280$

Count



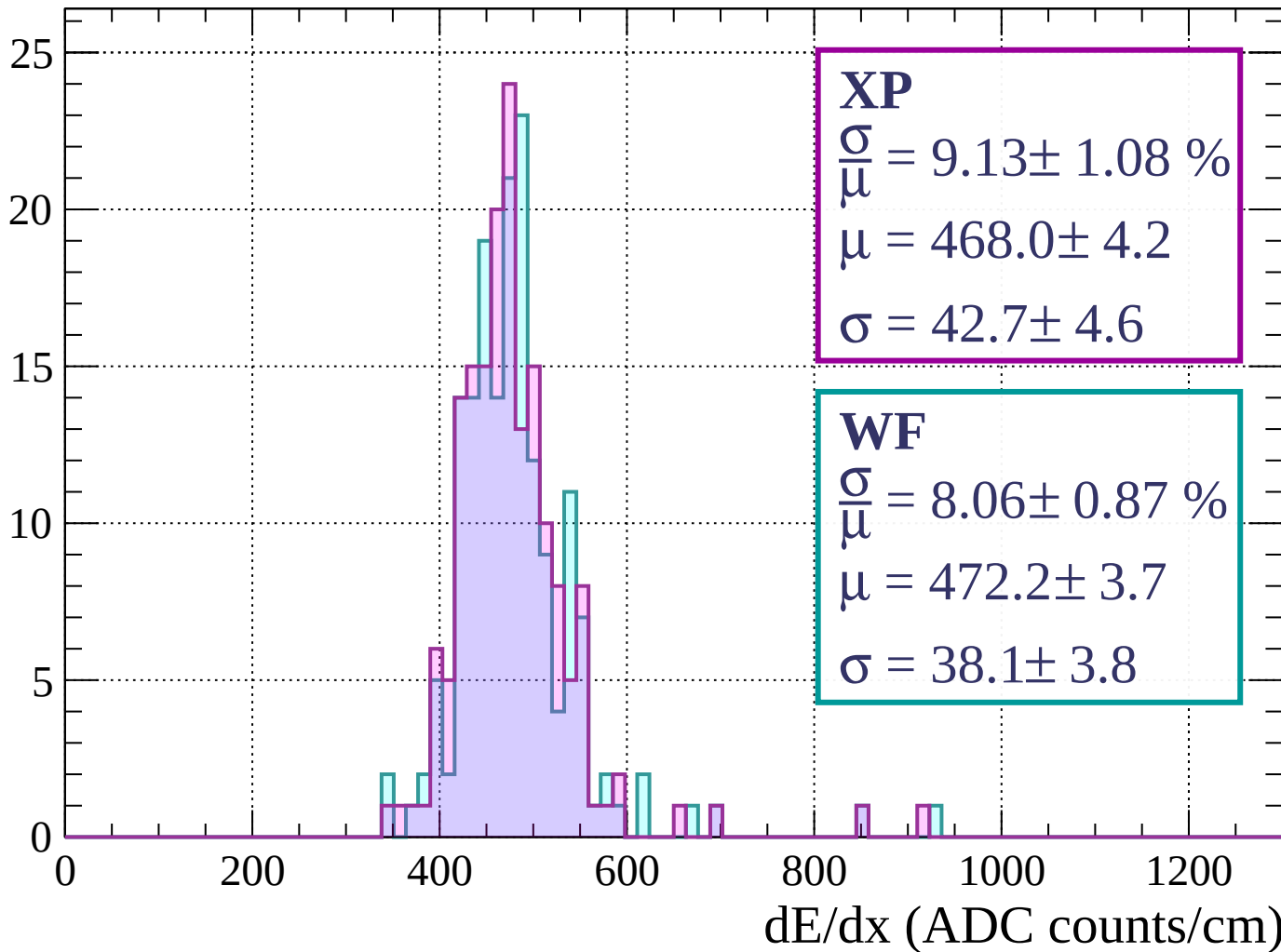
Energy loss | $1280 < p < 1320$

Count



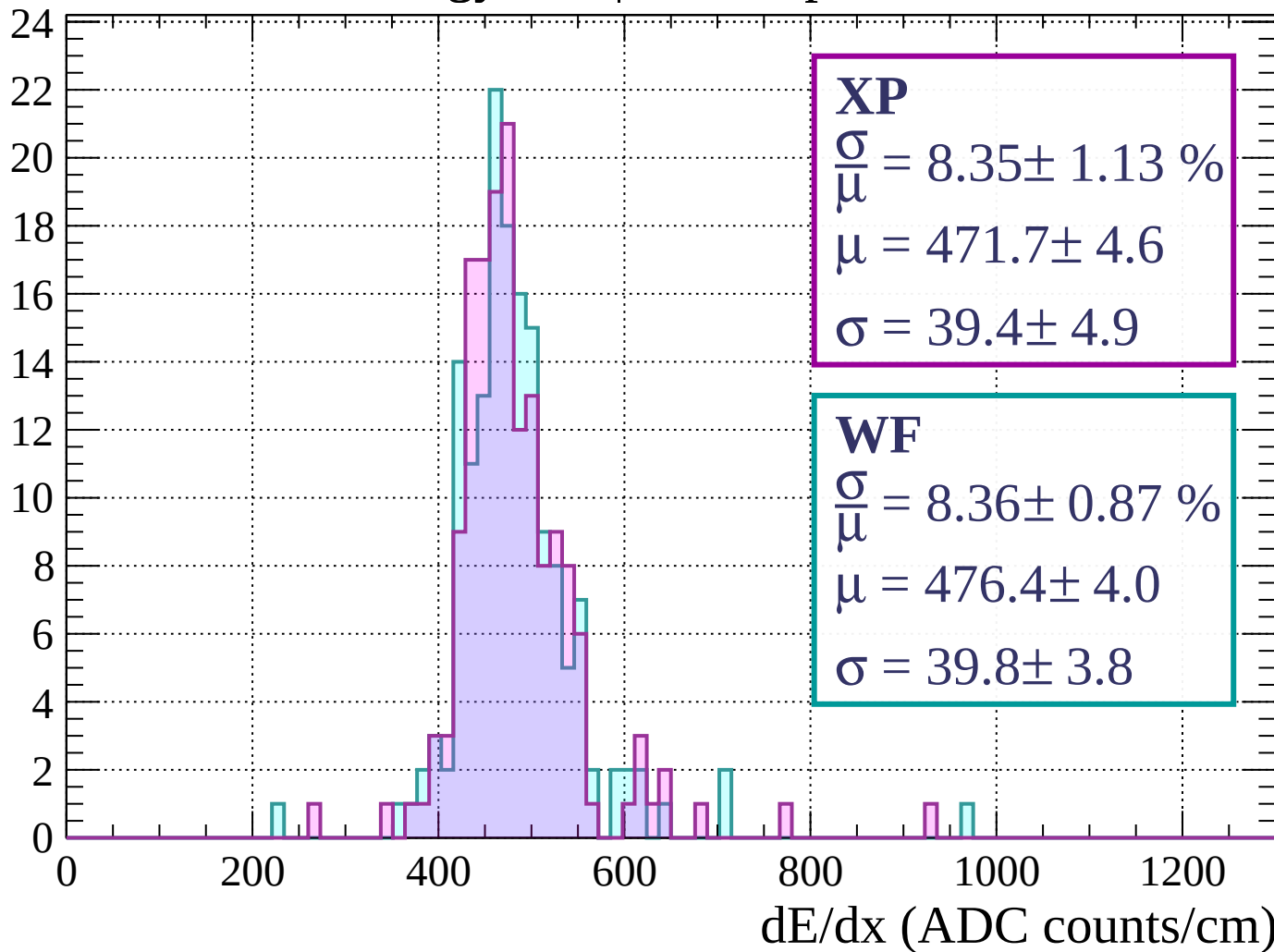
Energy loss | $1320 < p < 1360$

Count



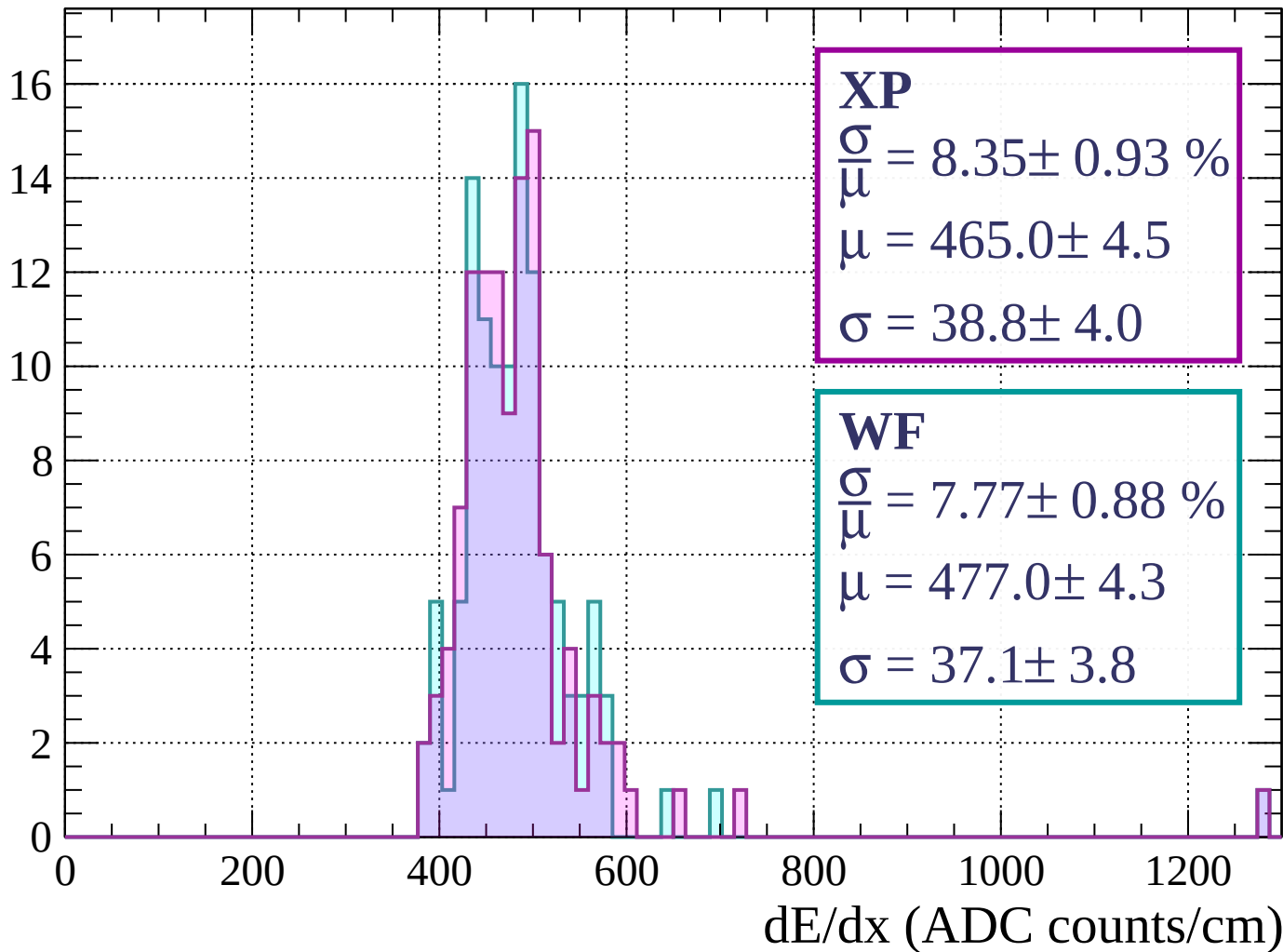
Energy loss | $1360 < p < 1400$

Count



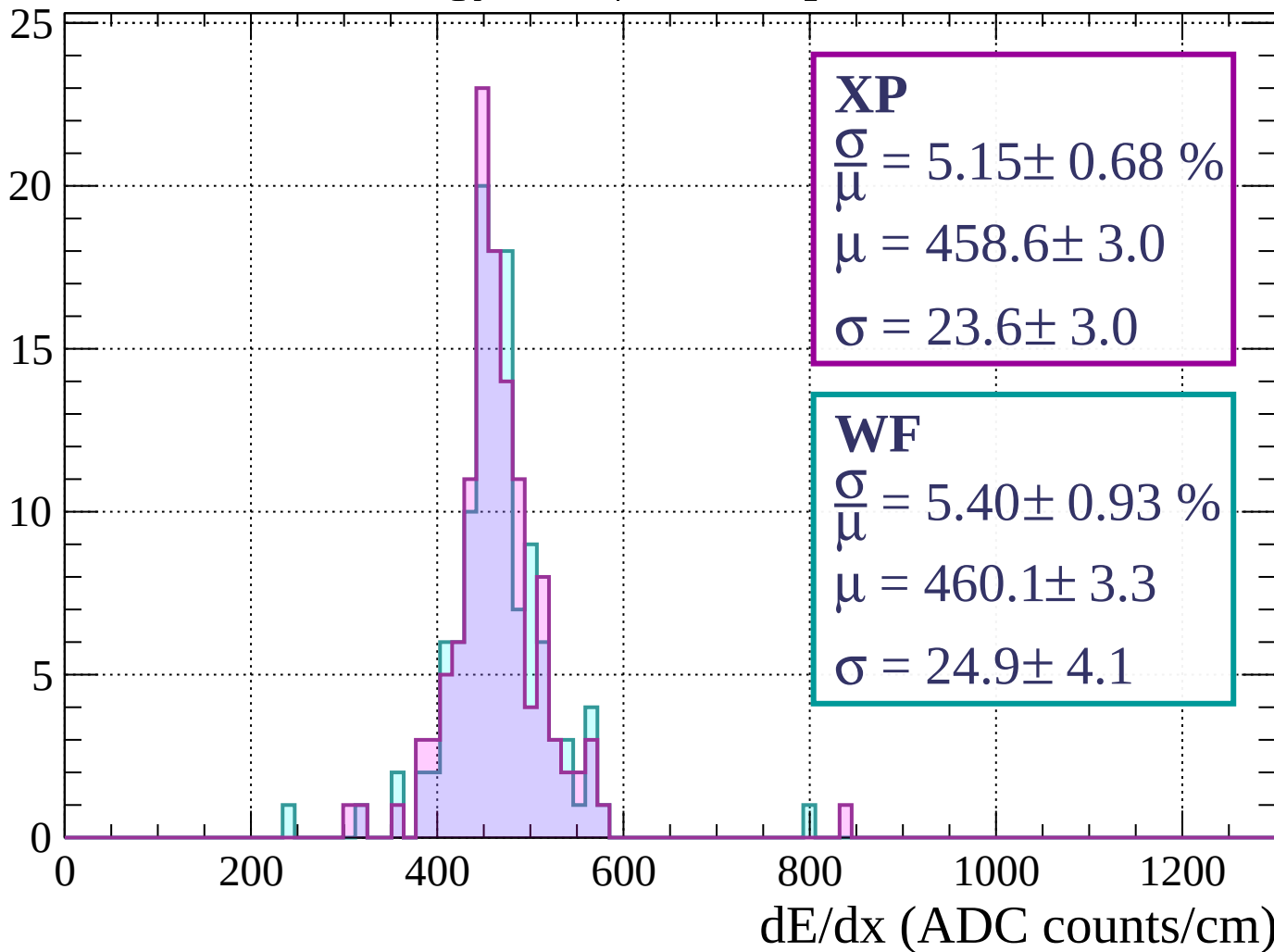
Energy loss | $1400 < p < 1440$

Count



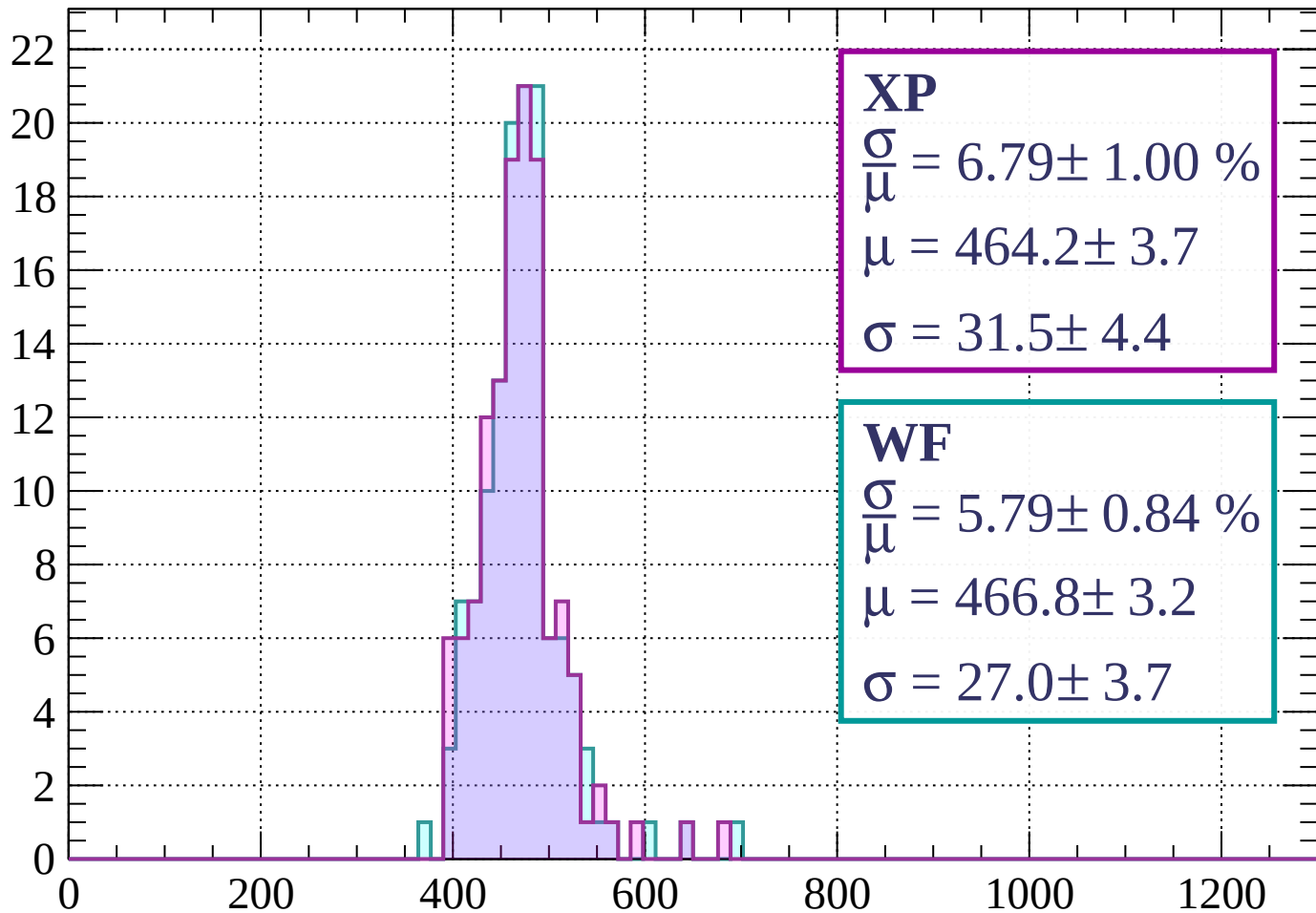
Energy loss | $1440 < p < 1480$

Count



Energy loss | $1480 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 6.79 \pm 1.00 \%$$

$$\mu = 464.2 \pm 3.7$$

$$\sigma = 31.5 \pm 4.4$$

WF

$$\frac{\sigma}{\mu} = 5.79 \pm 0.84 \%$$

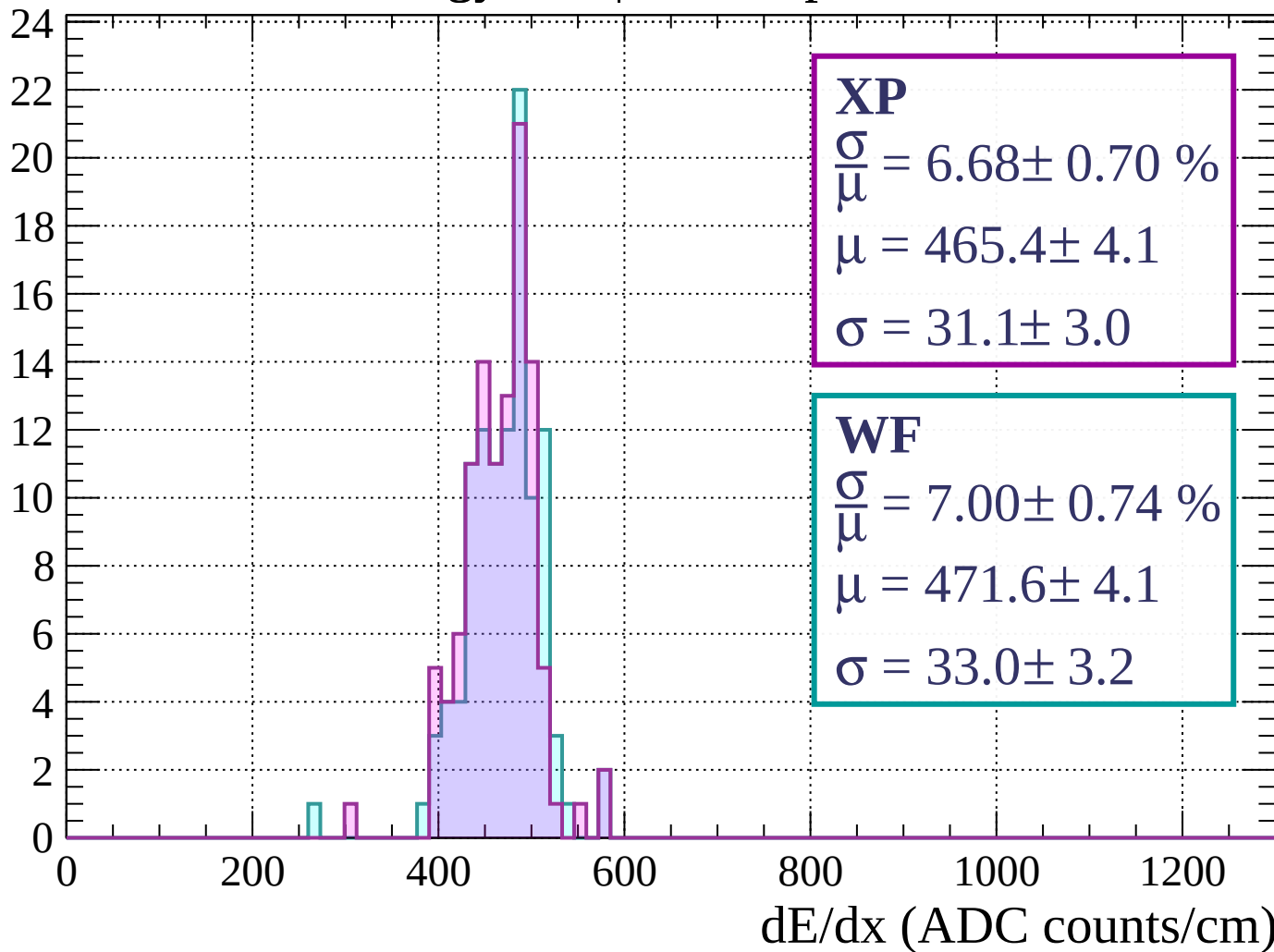
$$\mu = 466.8 \pm 3.2$$

$$\sigma = 27.0 \pm 3.7$$

dE/dx (ADC counts/cm)

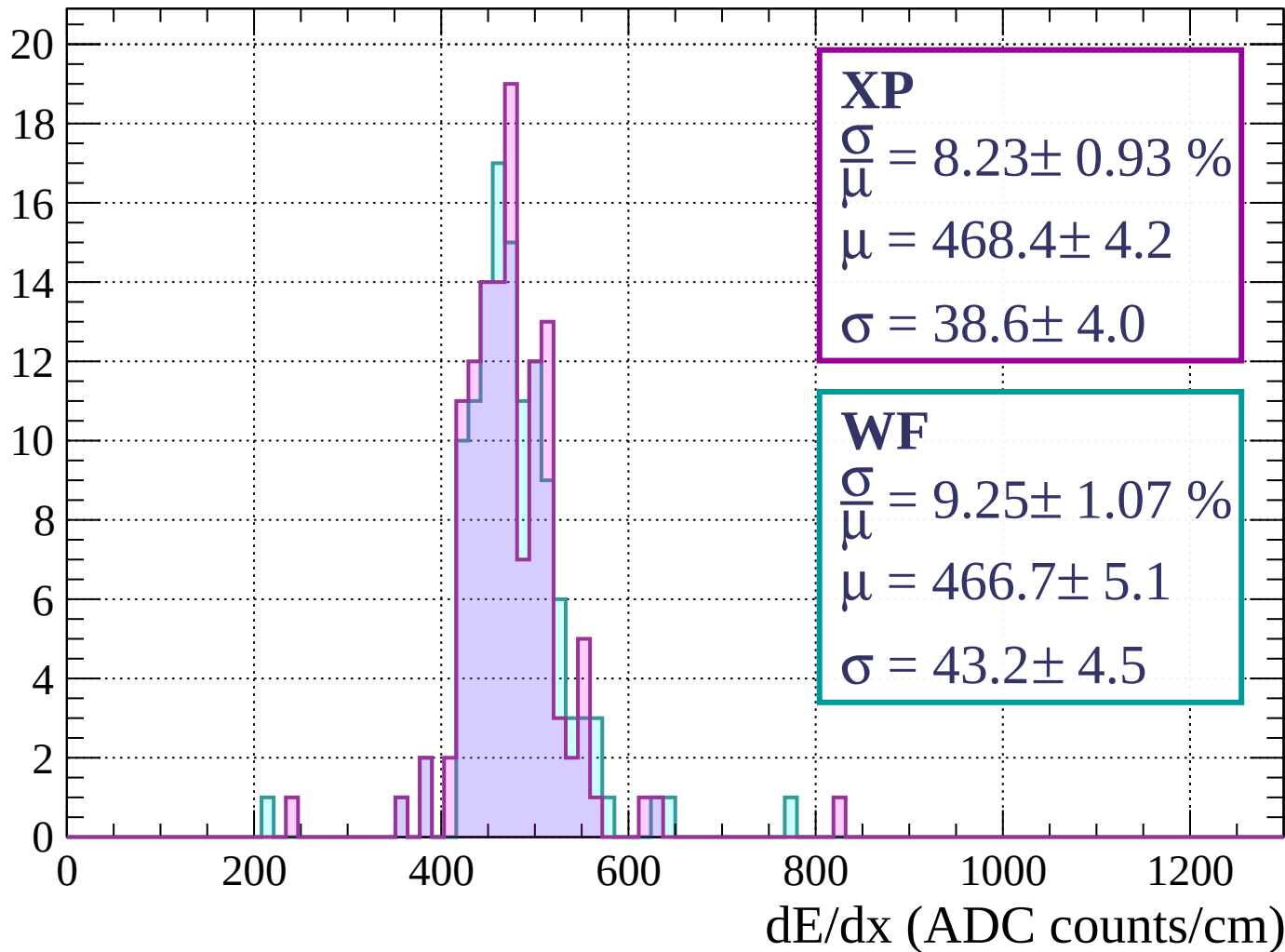
Energy loss | $1520 < p < 1560$

Count



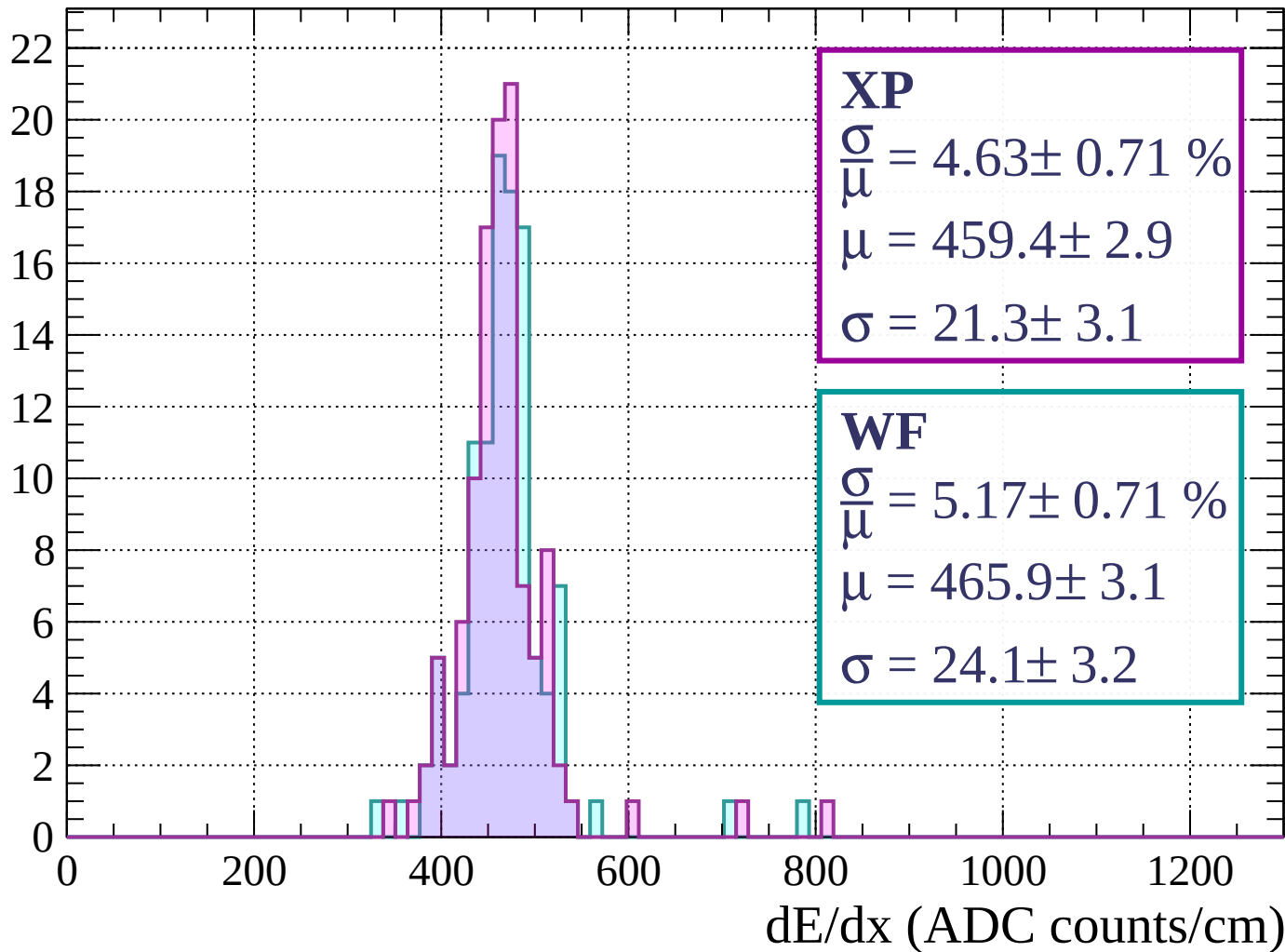
Energy loss | $1560 < p < 1600$

Count



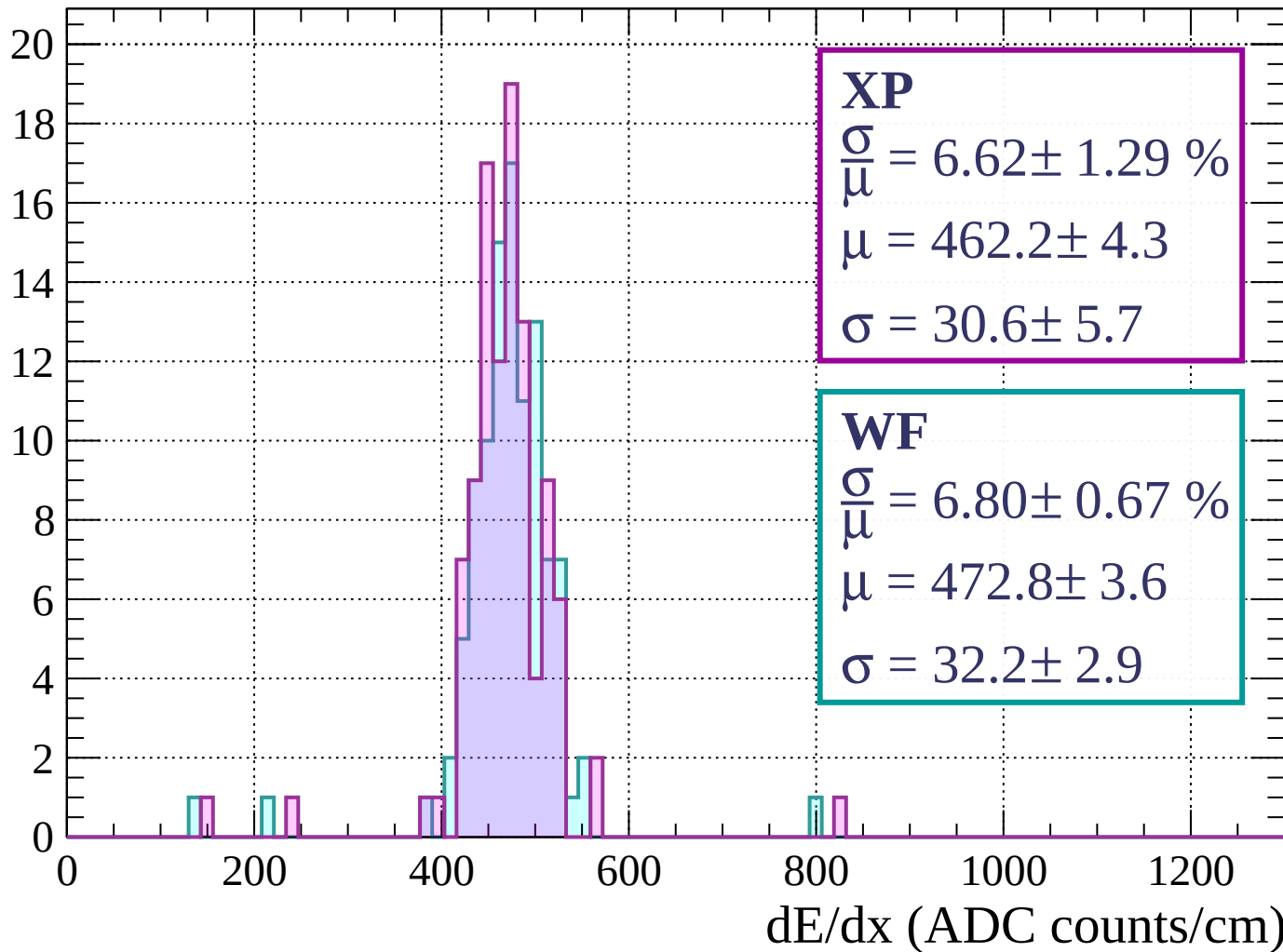
Energy loss | $1600 < p < 1640$

Count



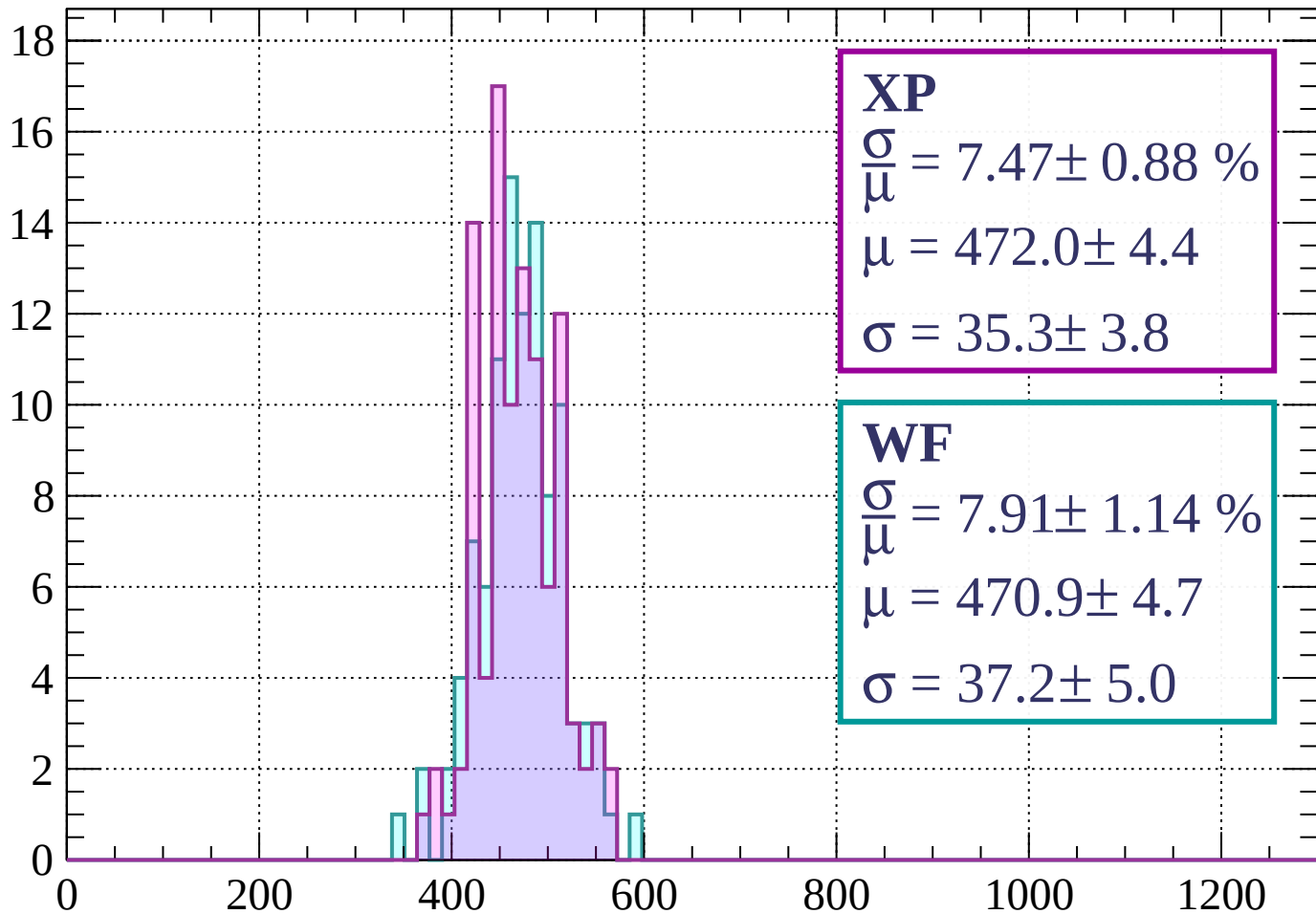
Energy loss | $1640 < p < 1680$

Count



Energy loss | $1680 < p < 1720$

Count



XP

$$\frac{\sigma}{\mu} = 7.47 \pm 0.88 \%$$

$$\mu = 472.0 \pm 4.4$$

$$\sigma = 35.3 \pm 3.8$$

WF

$$\frac{\sigma}{\mu} = 7.91 \pm 1.14 \%$$

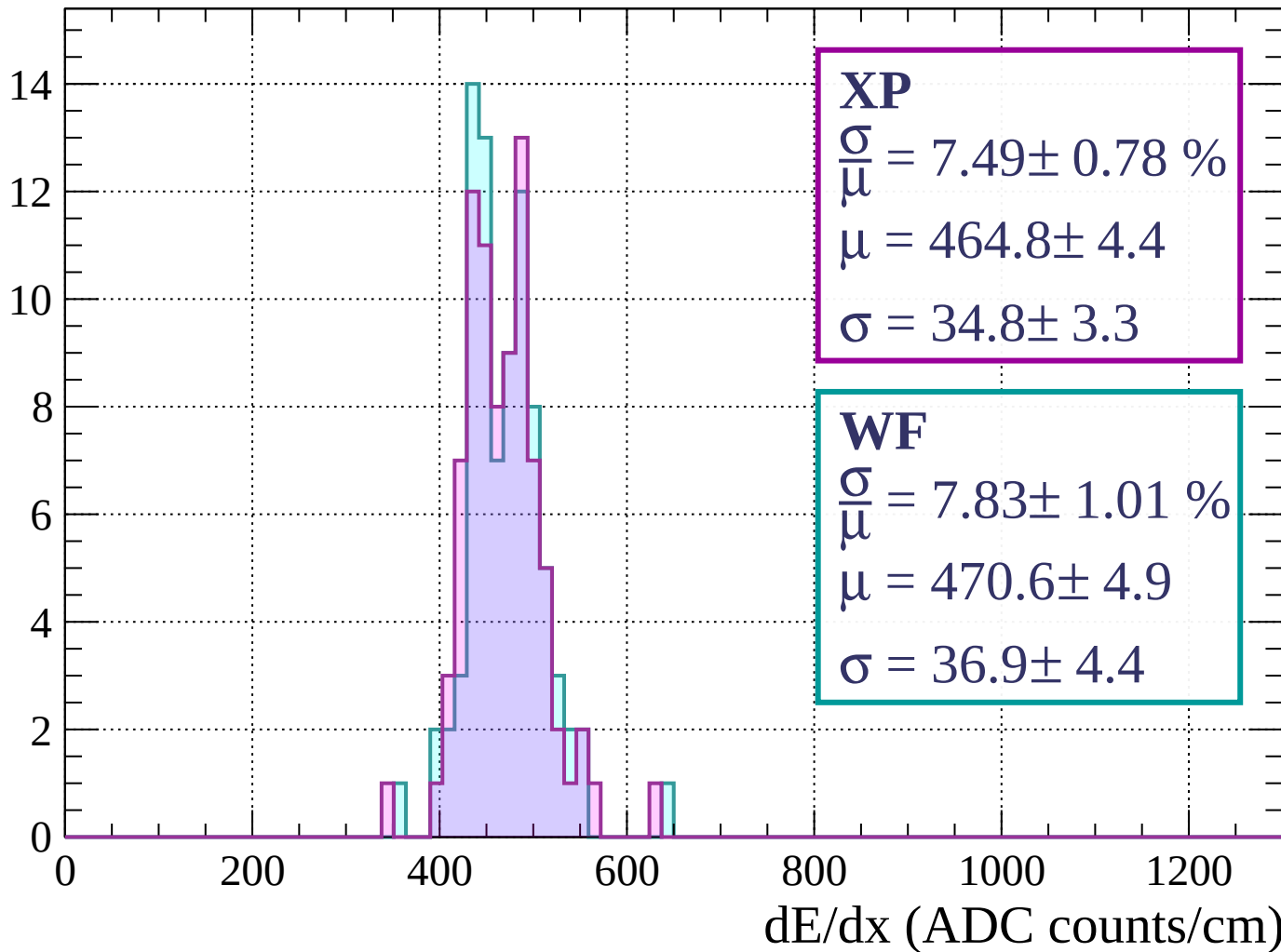
$$\mu = 470.9 \pm 4.7$$

$$\sigma = 37.2 \pm 5.0$$

dE/dx (ADC counts/cm)

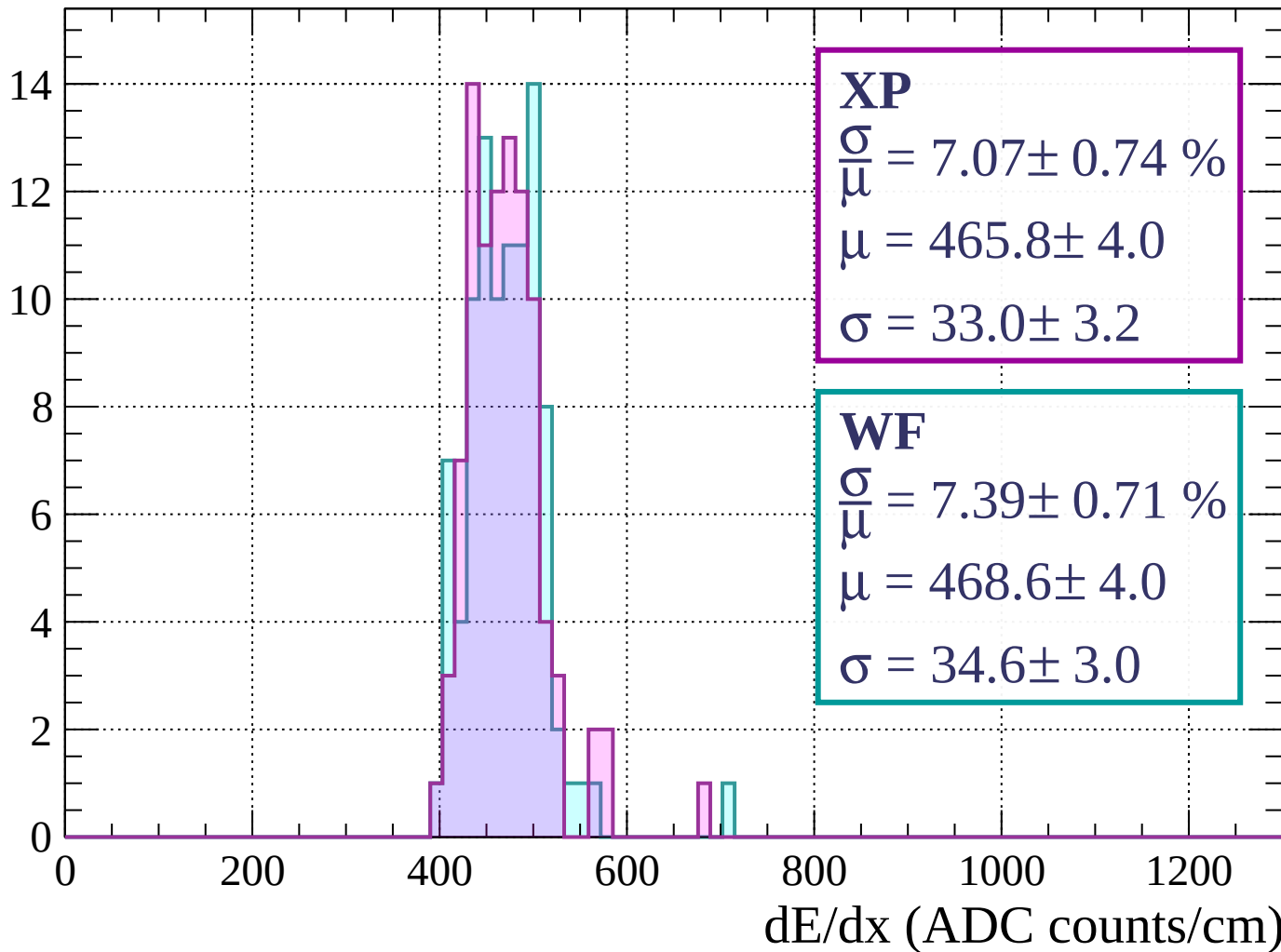
Energy loss | $1720 < p < 1760$

Count



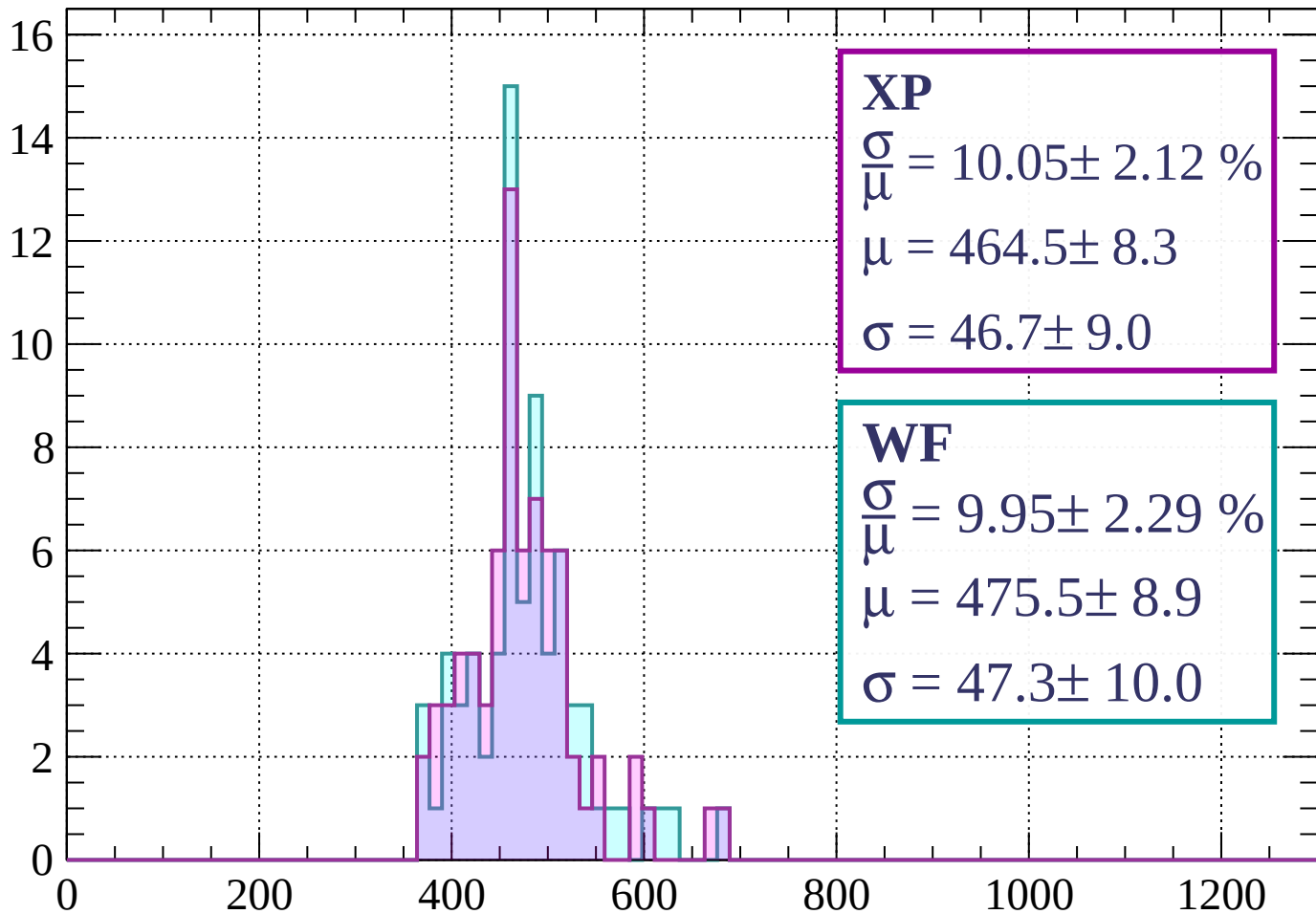
Energy loss | $1760 < p < 1800$

Count



Energy loss | $1800 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 10.05 \pm 2.12 \%$$

$$\mu = 464.5 \pm 8.3$$

$$\sigma = 46.7 \pm 9.0$$

WF

$$\frac{\sigma}{\mu} = 9.95 \pm 2.29 \%$$

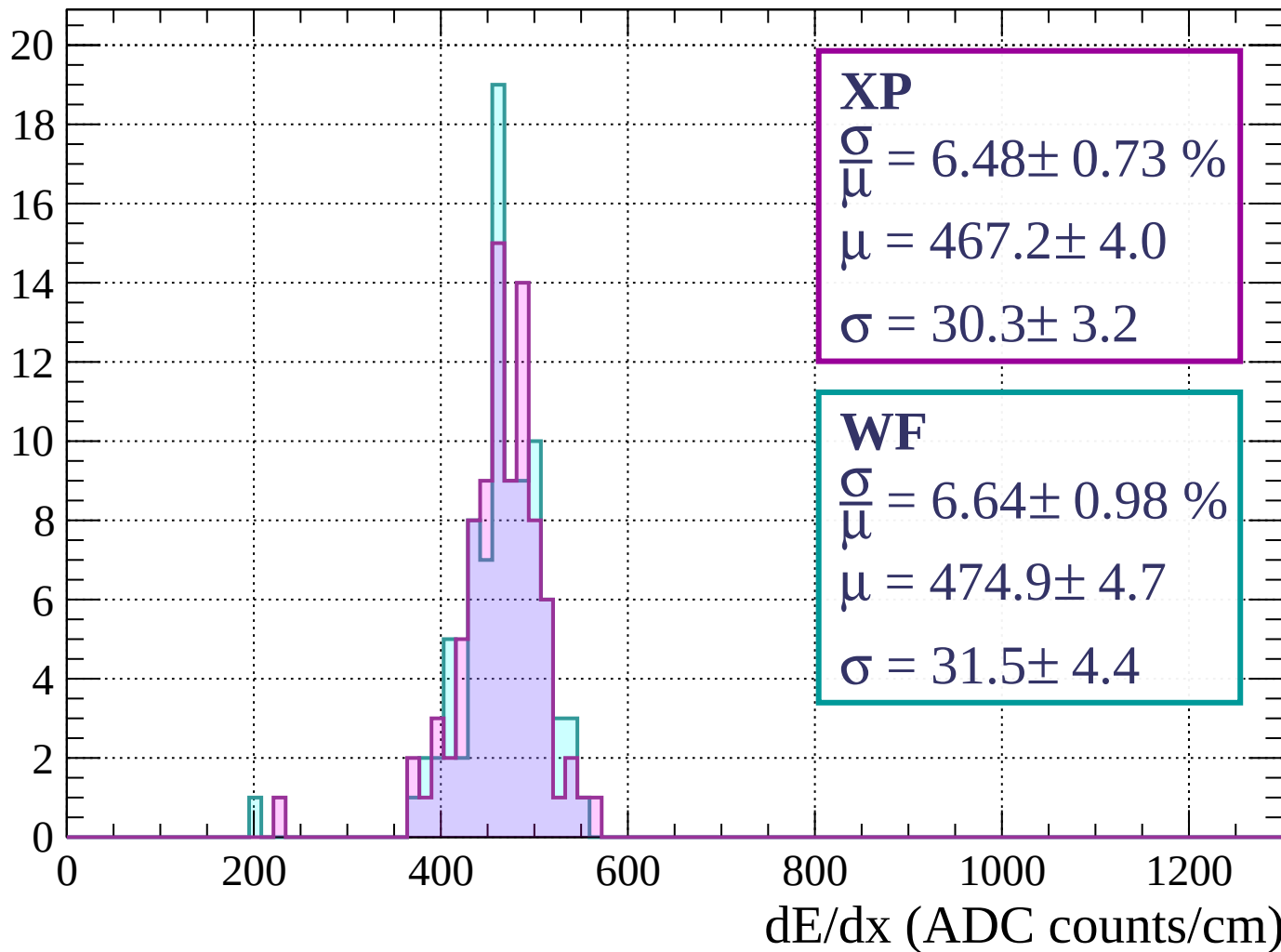
$$\mu = 475.5 \pm 8.9$$

$$\sigma = 47.3 \pm 10.0$$

dE/dx (ADC counts/cm)

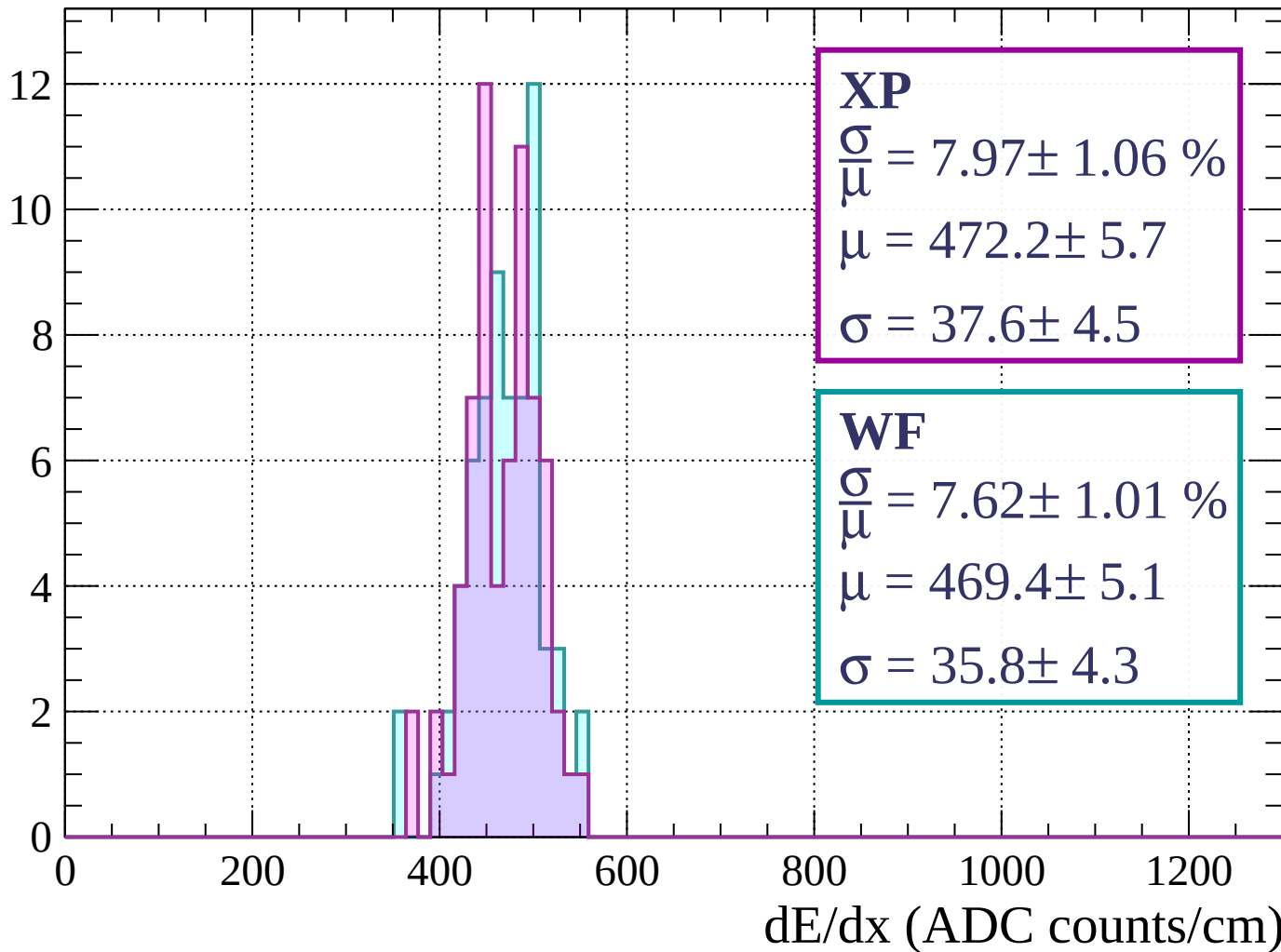
Energy loss | $1840 < p < 1880$

Count



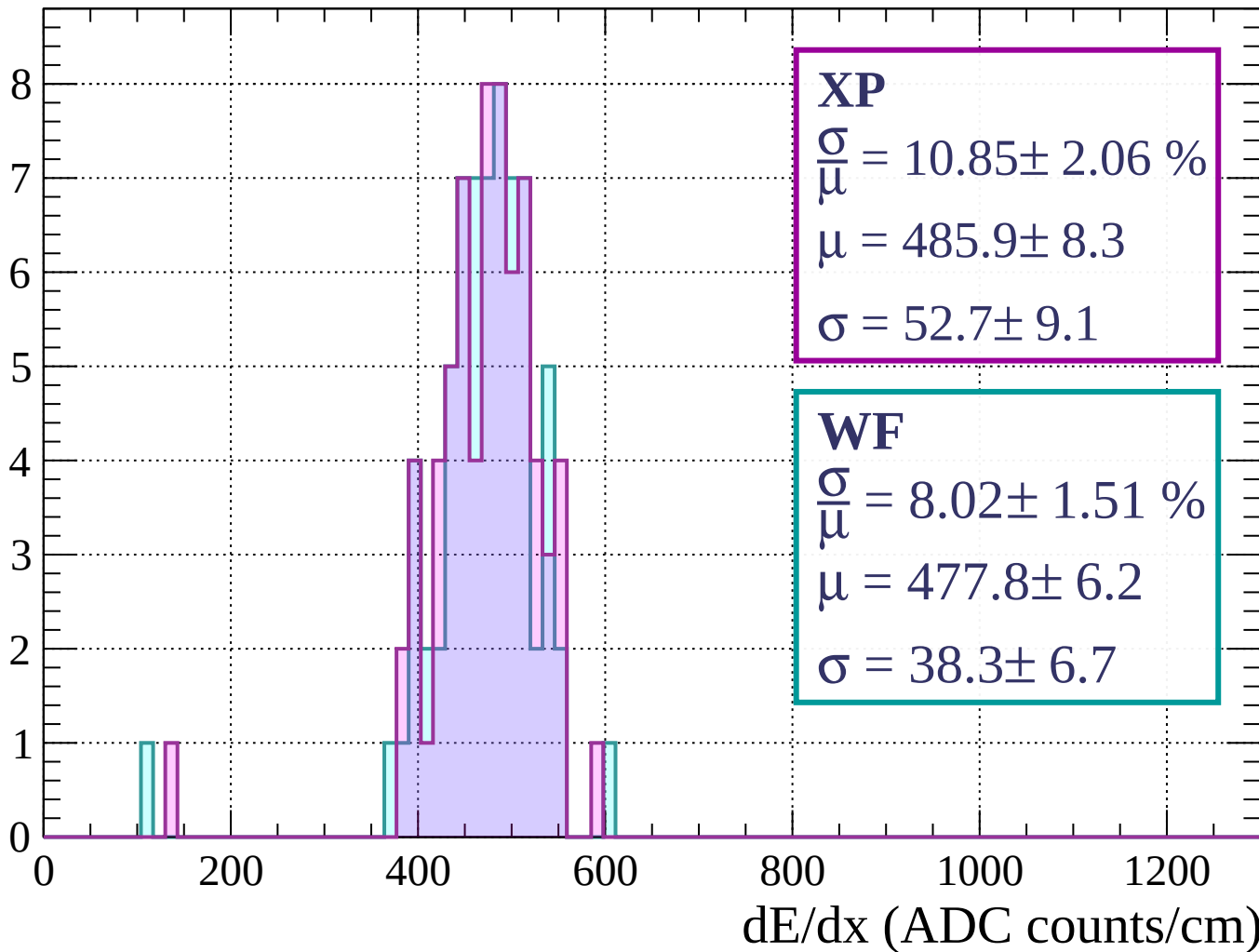
Energy loss | $1880 < p < 1920$

Count



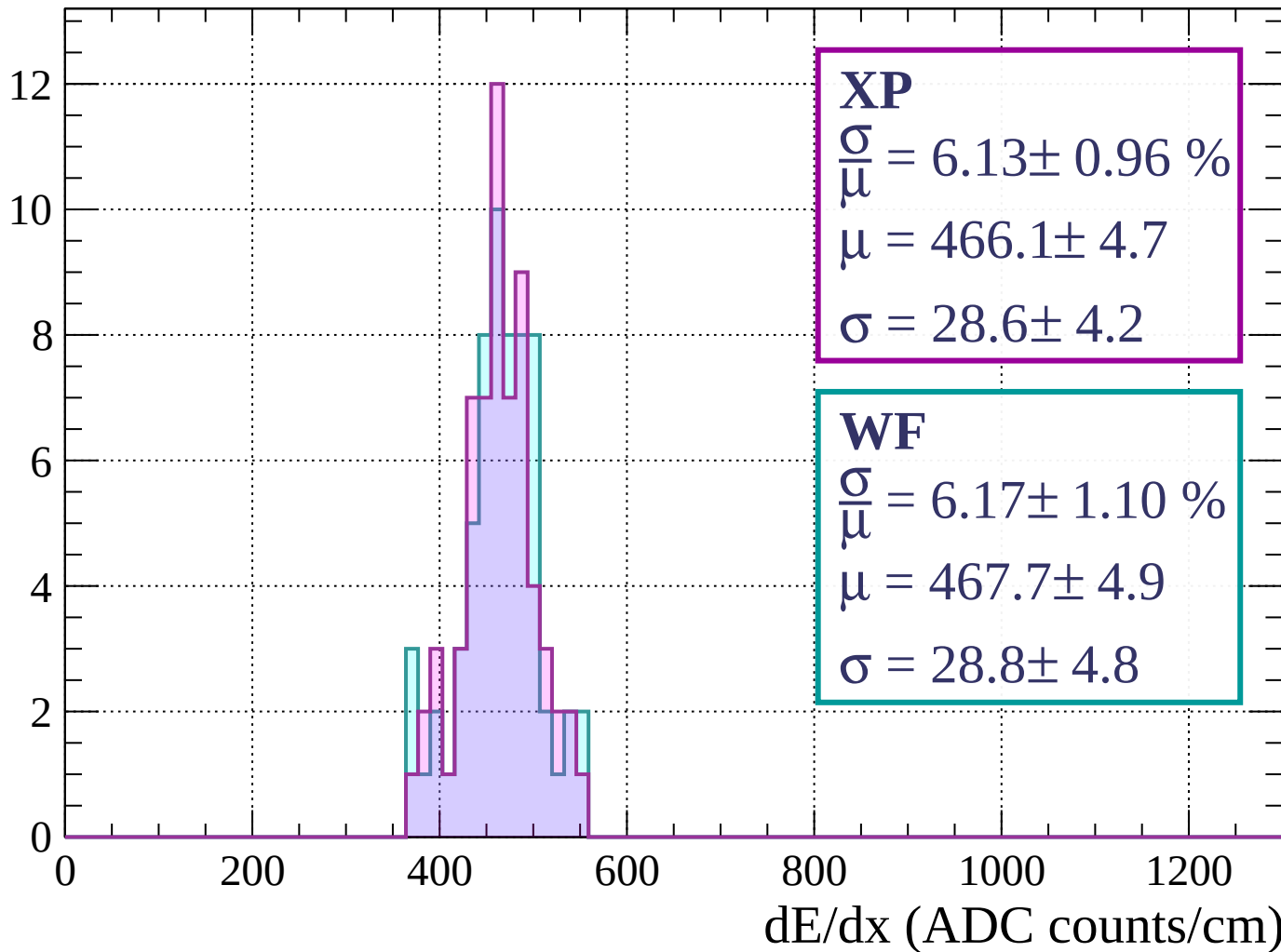
Energy loss | $1920 < p < 1960$

Count



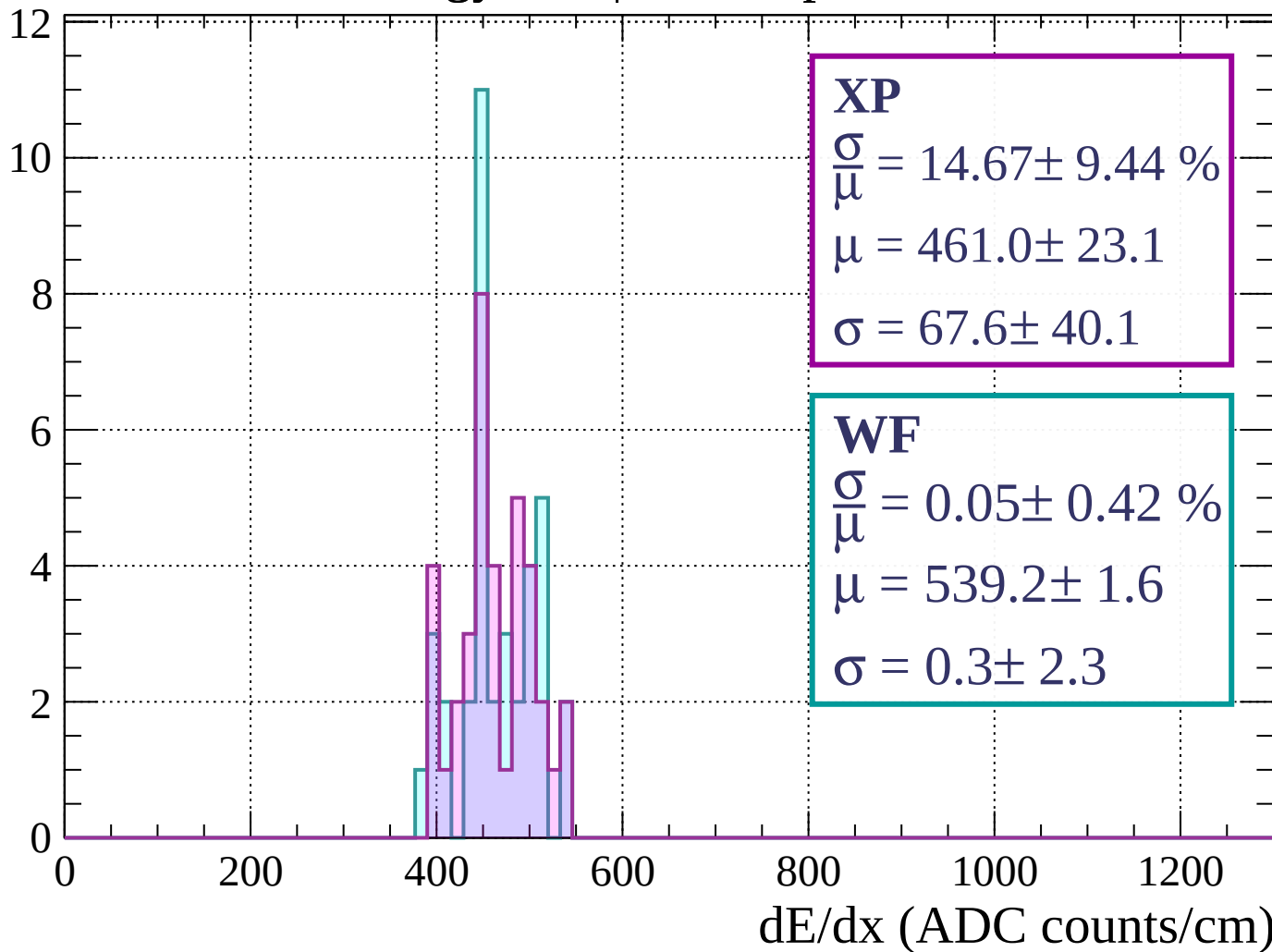
Energy loss | $1960 < p < 2000$

Count



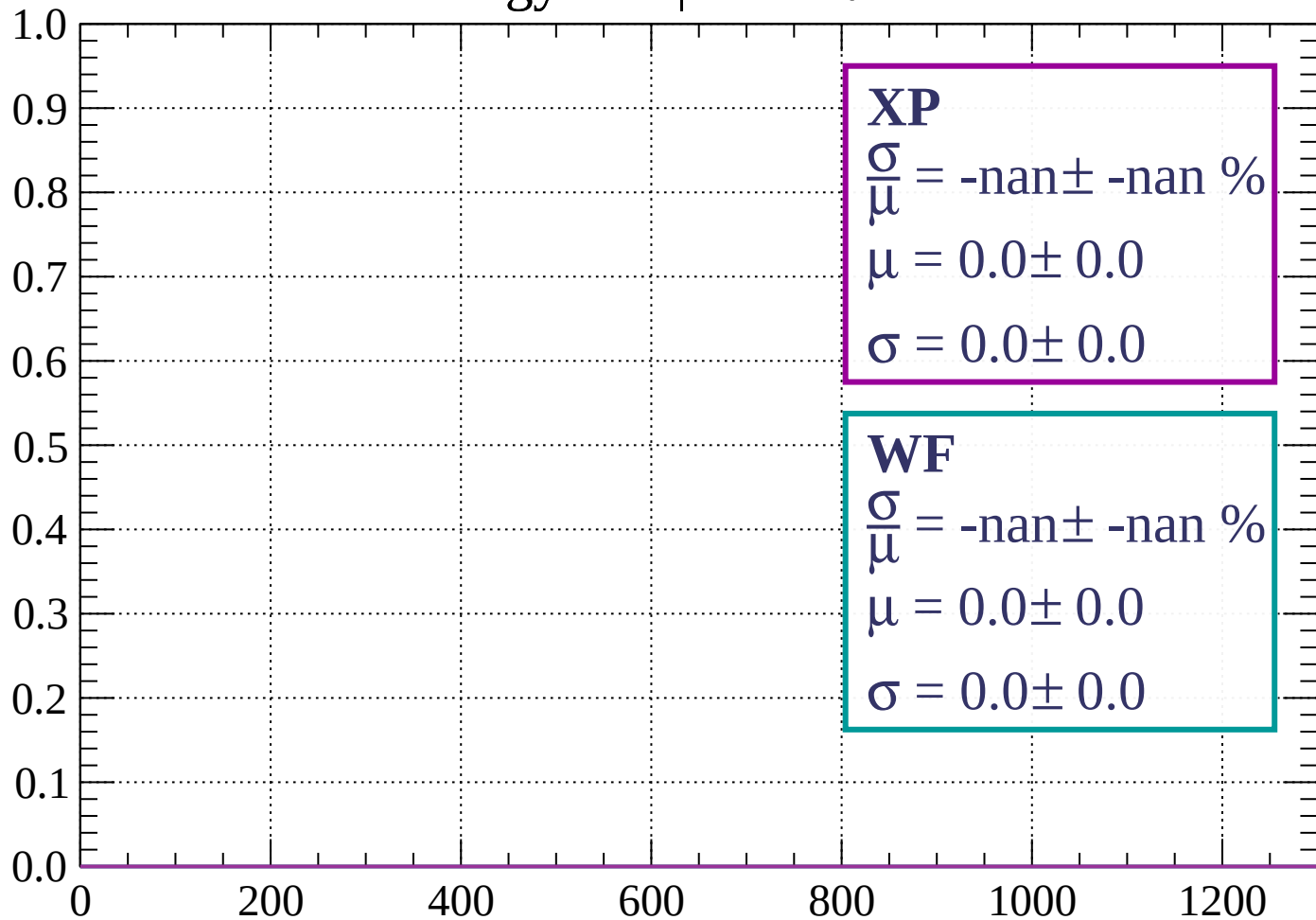
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -84$

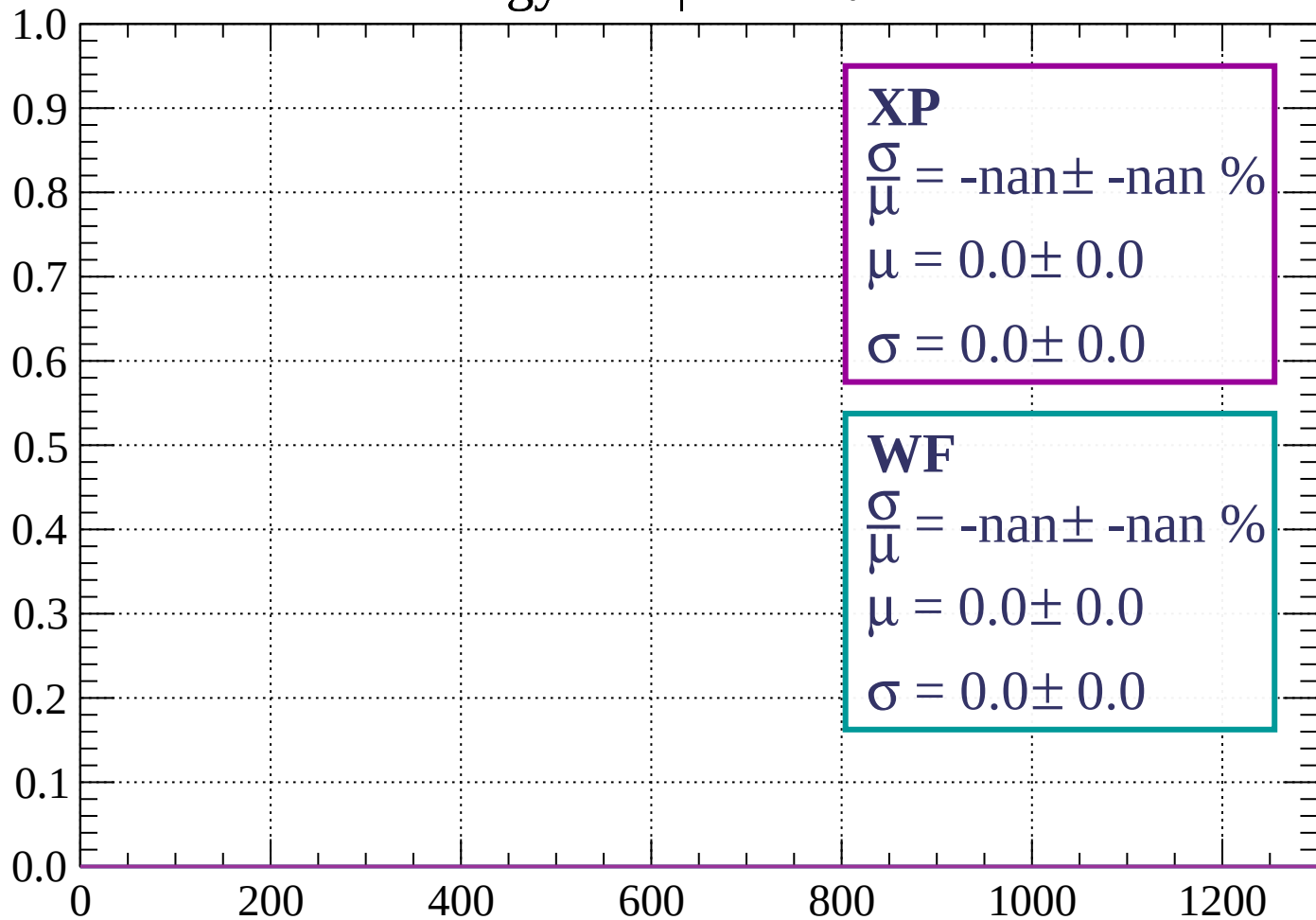
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -78$

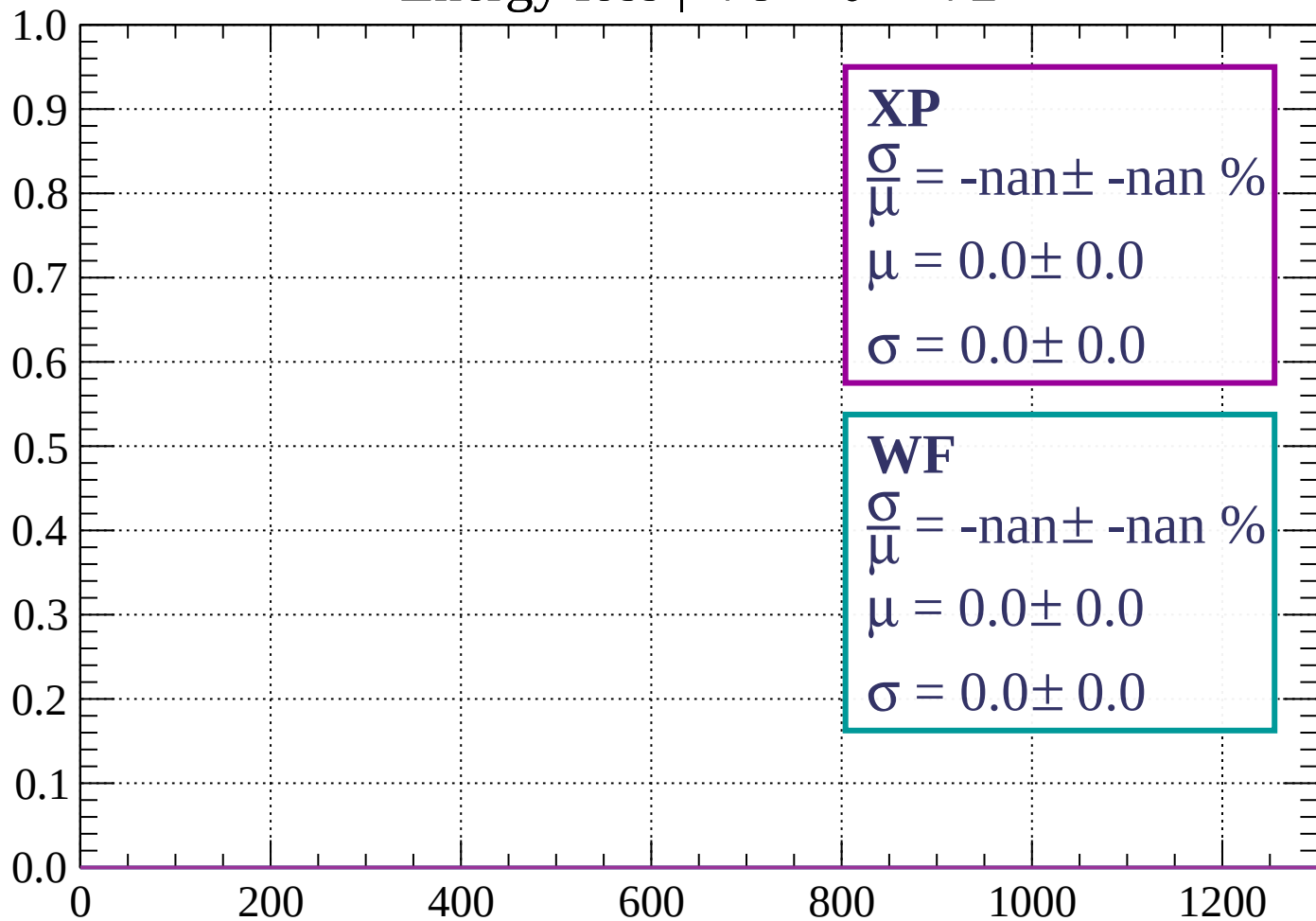
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -72$

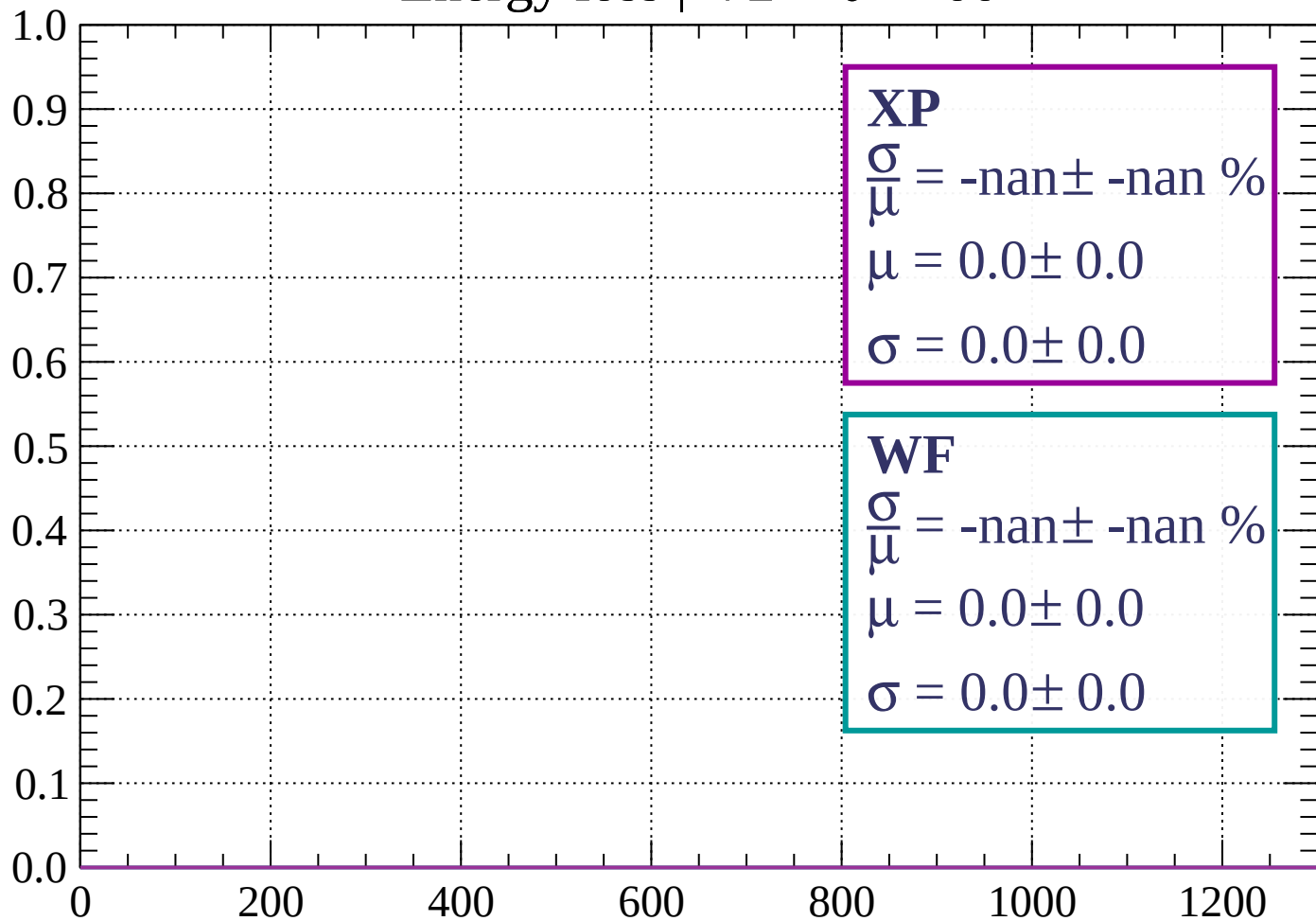
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -66$

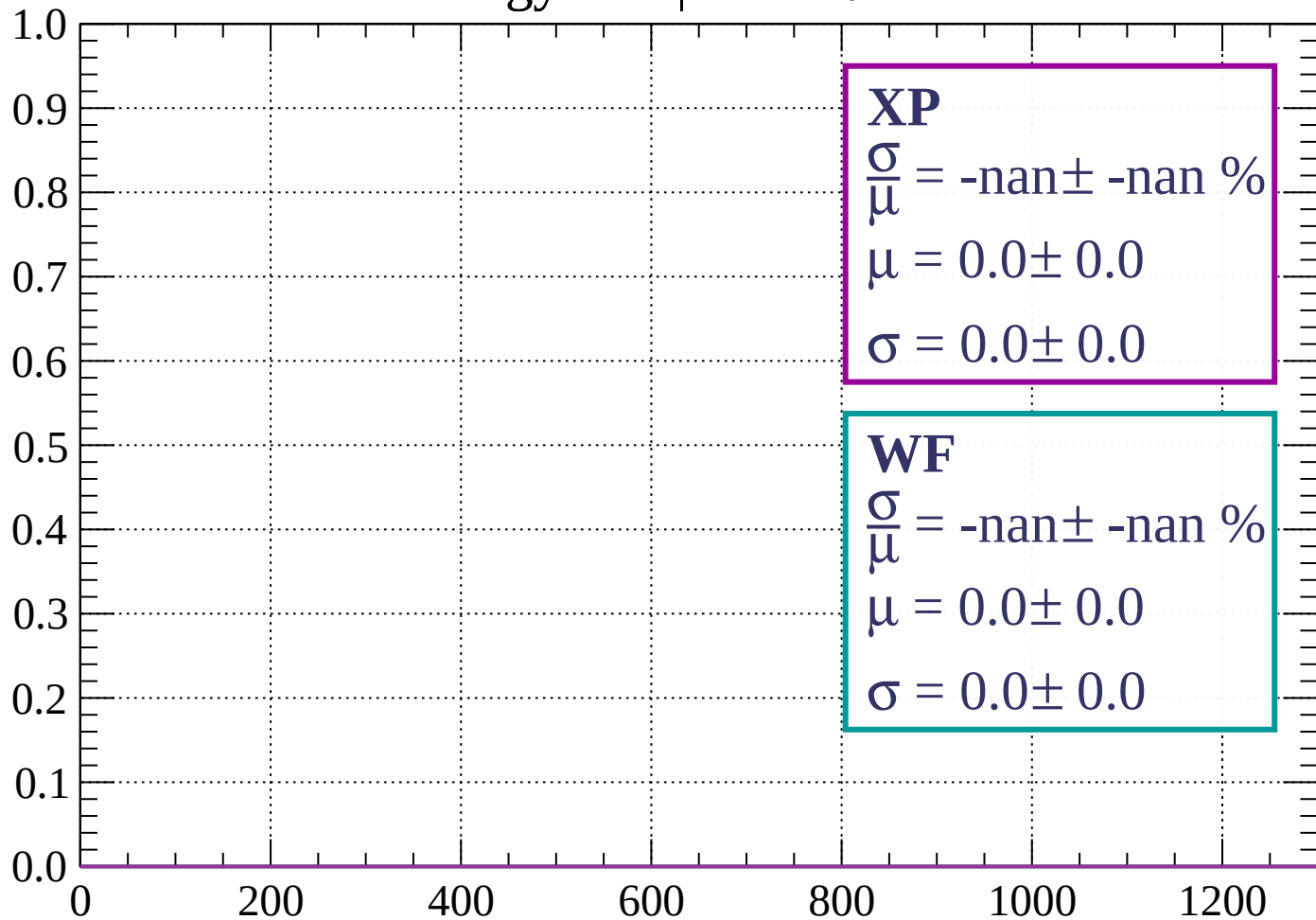
Count



dE/dx (ADC counts/cm)

Energy loss | $-66 < \theta < -60$

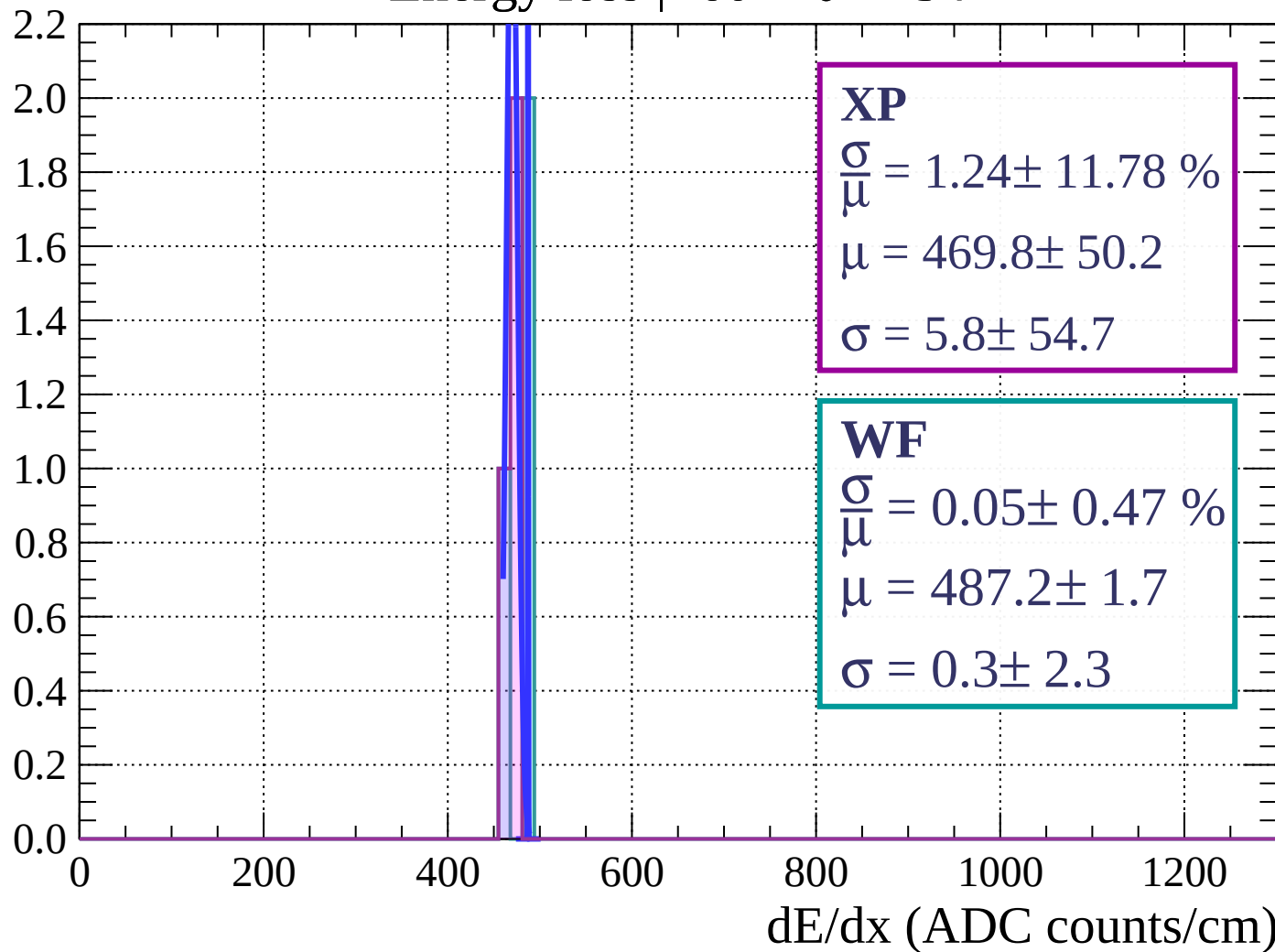
Count



dE/dx (ADC counts/cm)

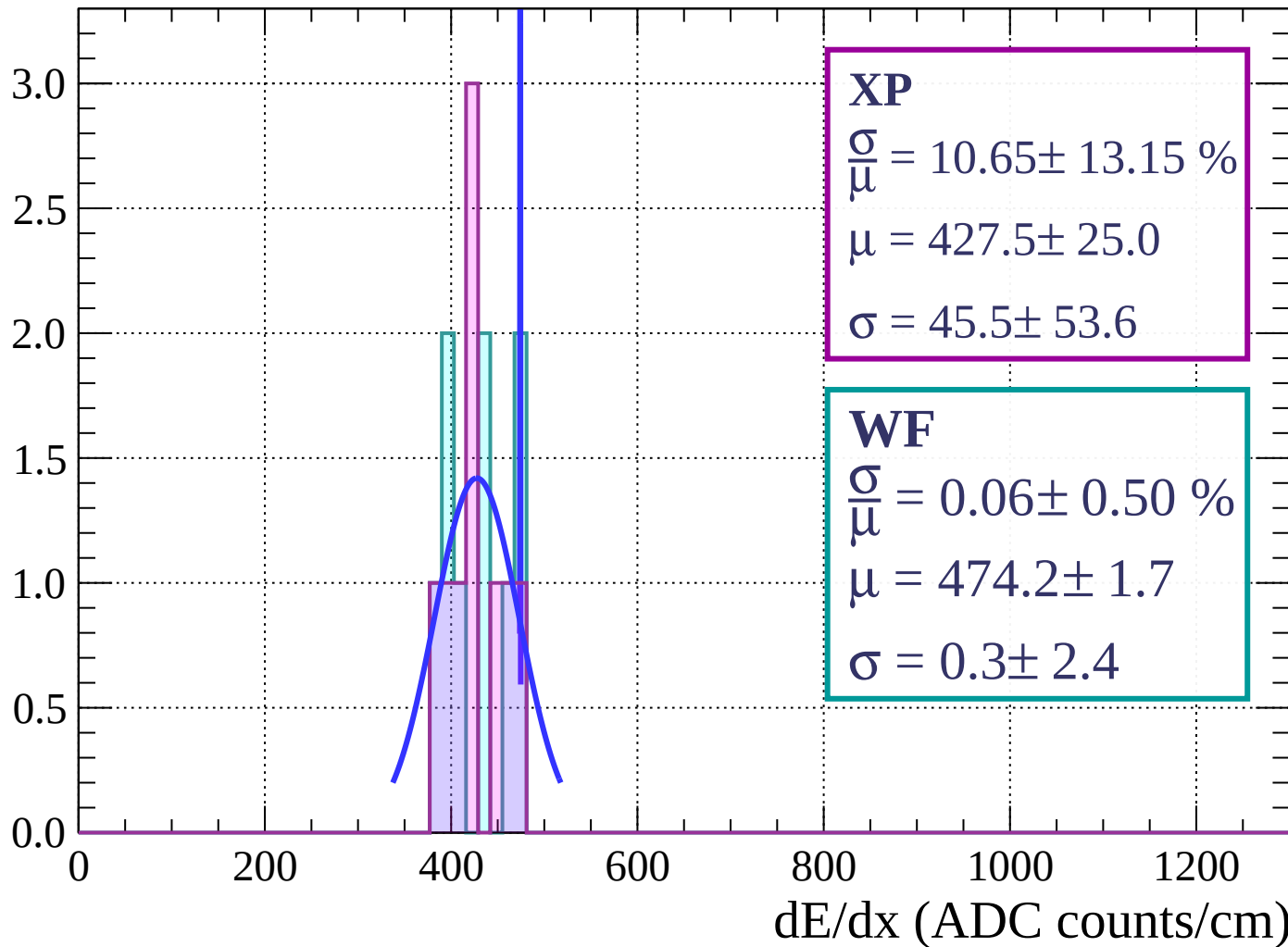
Energy loss | $-60 < \theta < -54$

Count



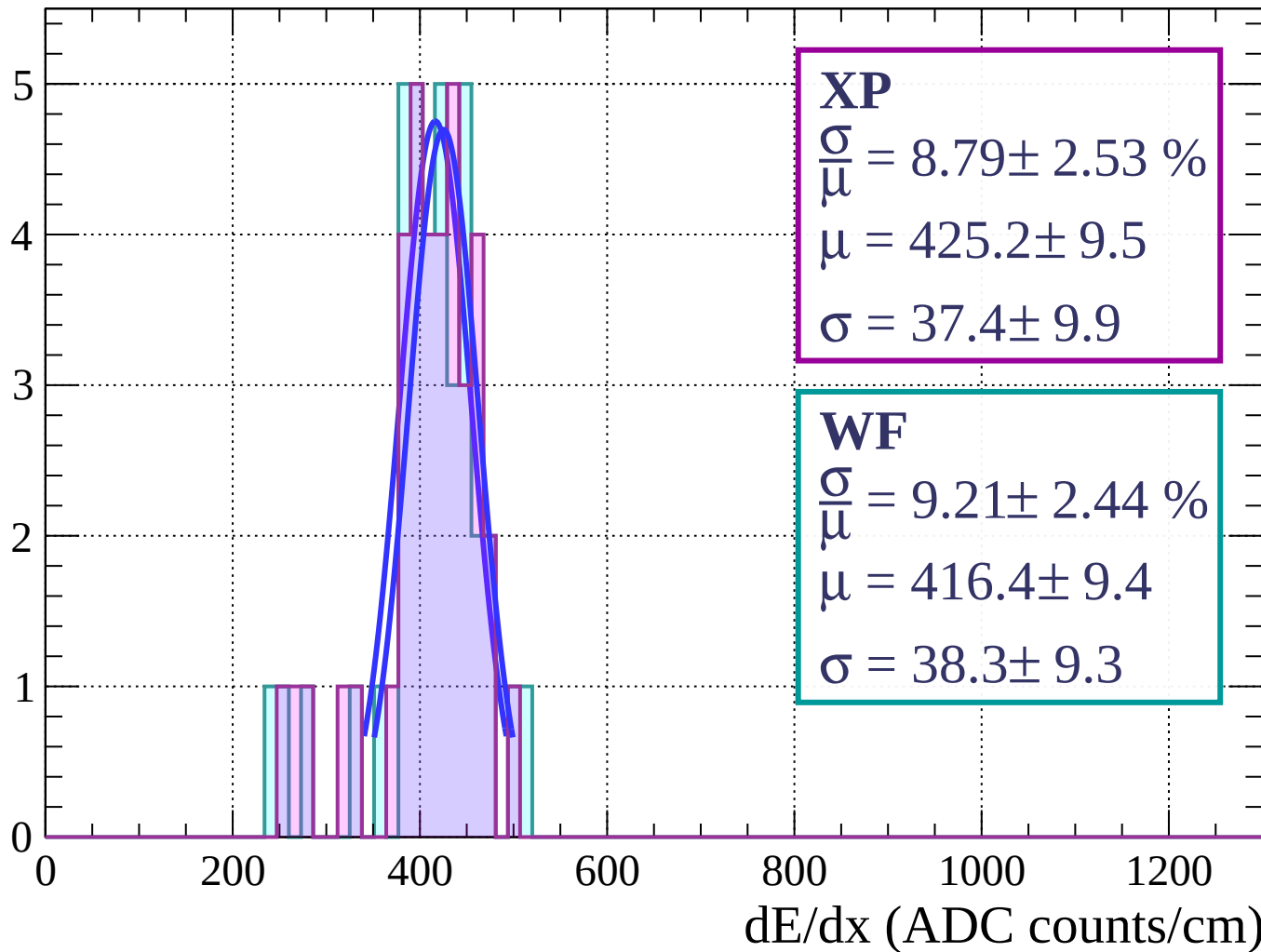
Energy loss | $-54 < \theta < -48$

Count



Energy loss | $-48 < \theta < -42$

Count



Energy loss | $-42 < \theta < -36$

Count

XP

$$\frac{\sigma}{\mu} = 8.48 \pm 1.20 \%$$

$$\mu = 409.4 \pm 4.9$$

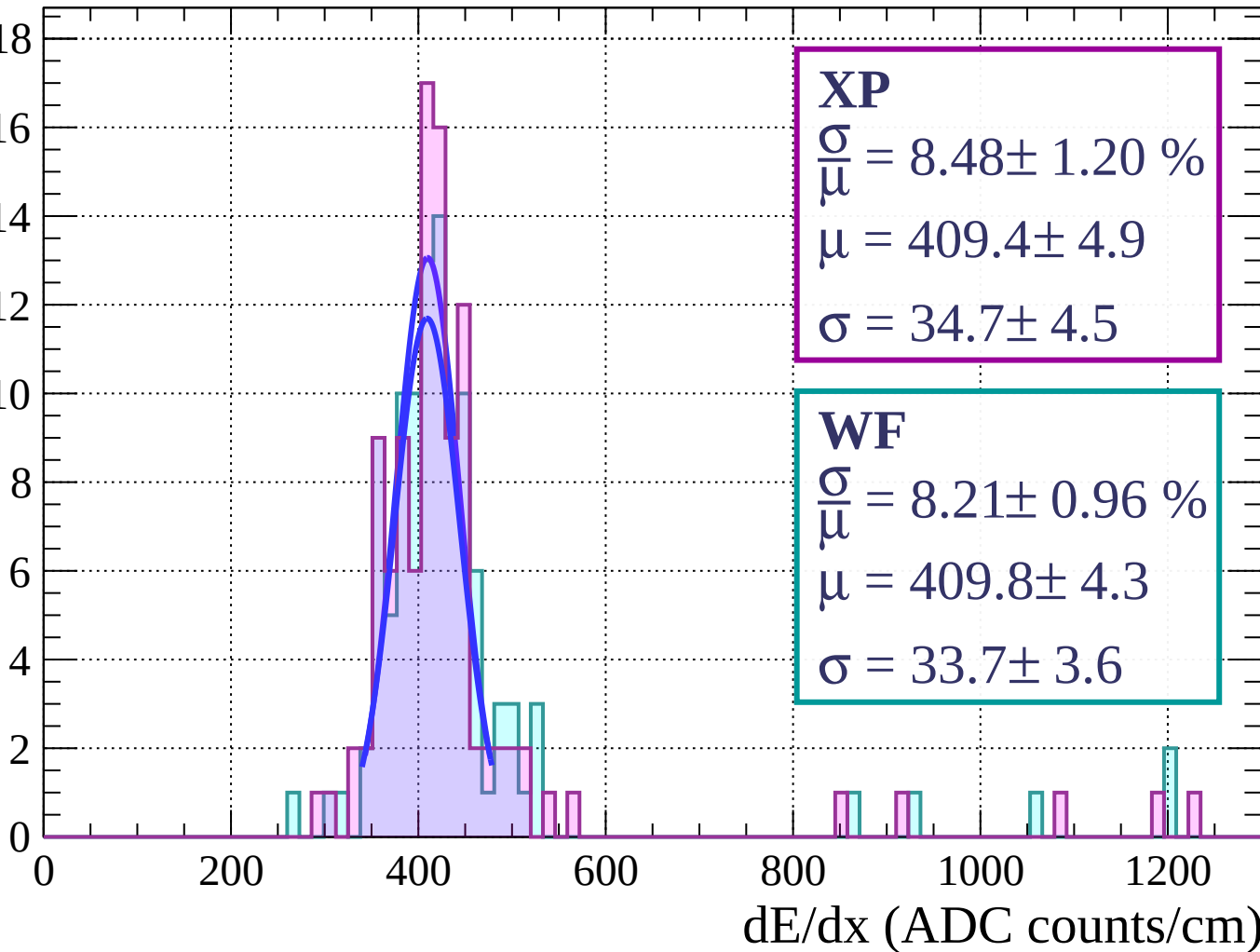
$$\sigma = 34.7 \pm 4.5$$

WF

$$\frac{\sigma}{\mu} = 8.21 \pm 0.96 \%$$

$$\mu = 409.8 \pm 4.3$$

$$\sigma = 33.7 \pm 3.6$$



Energy loss $|-36 < \theta < -30$

Count

XP

$$\frac{\sigma}{\mu} = 8.18 \pm 0.71 \%$$

$$\mu = 414.9 \pm 2.8$$

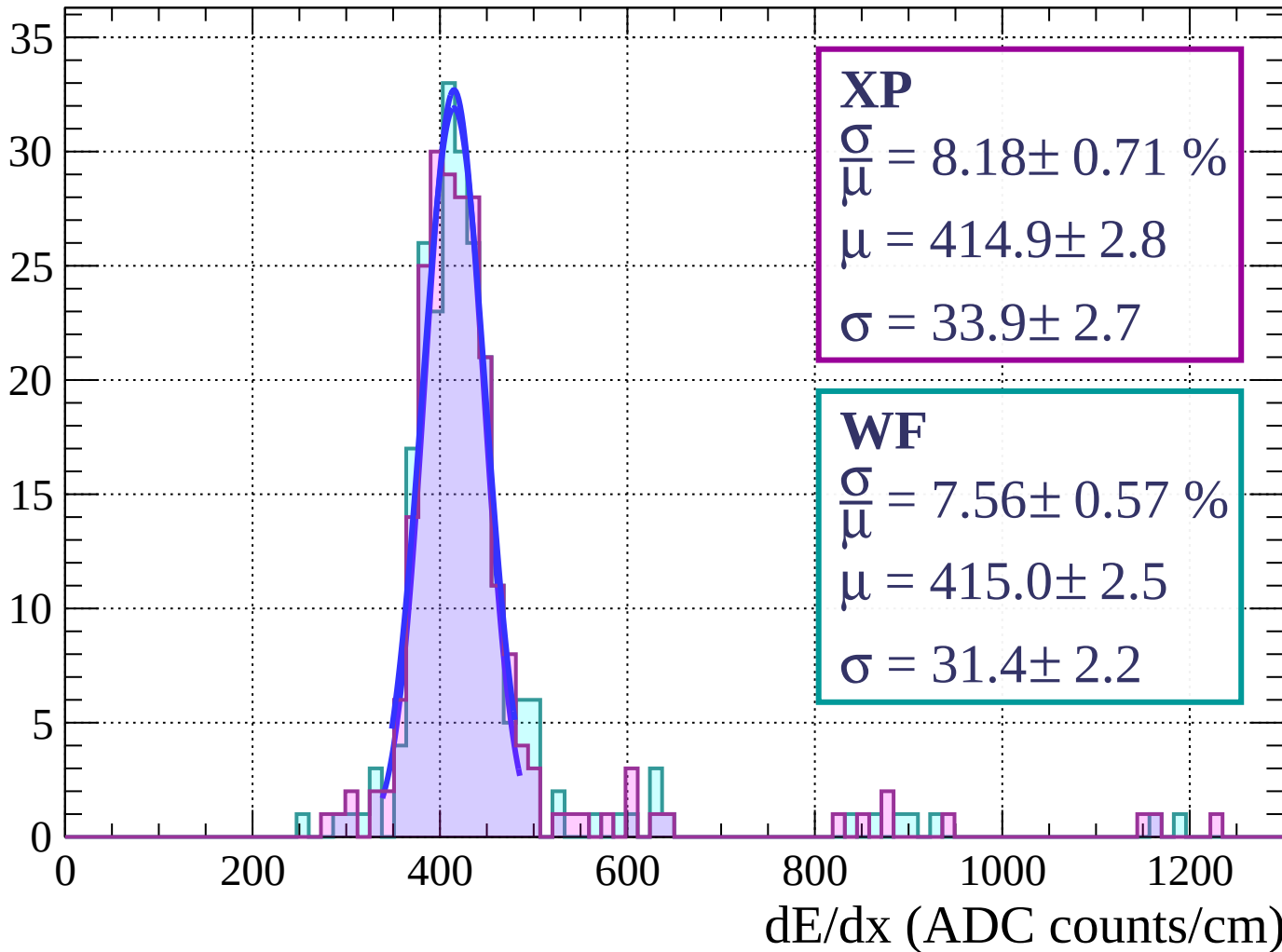
$$\sigma = 33.9 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 7.56 \pm 0.57 \%$$

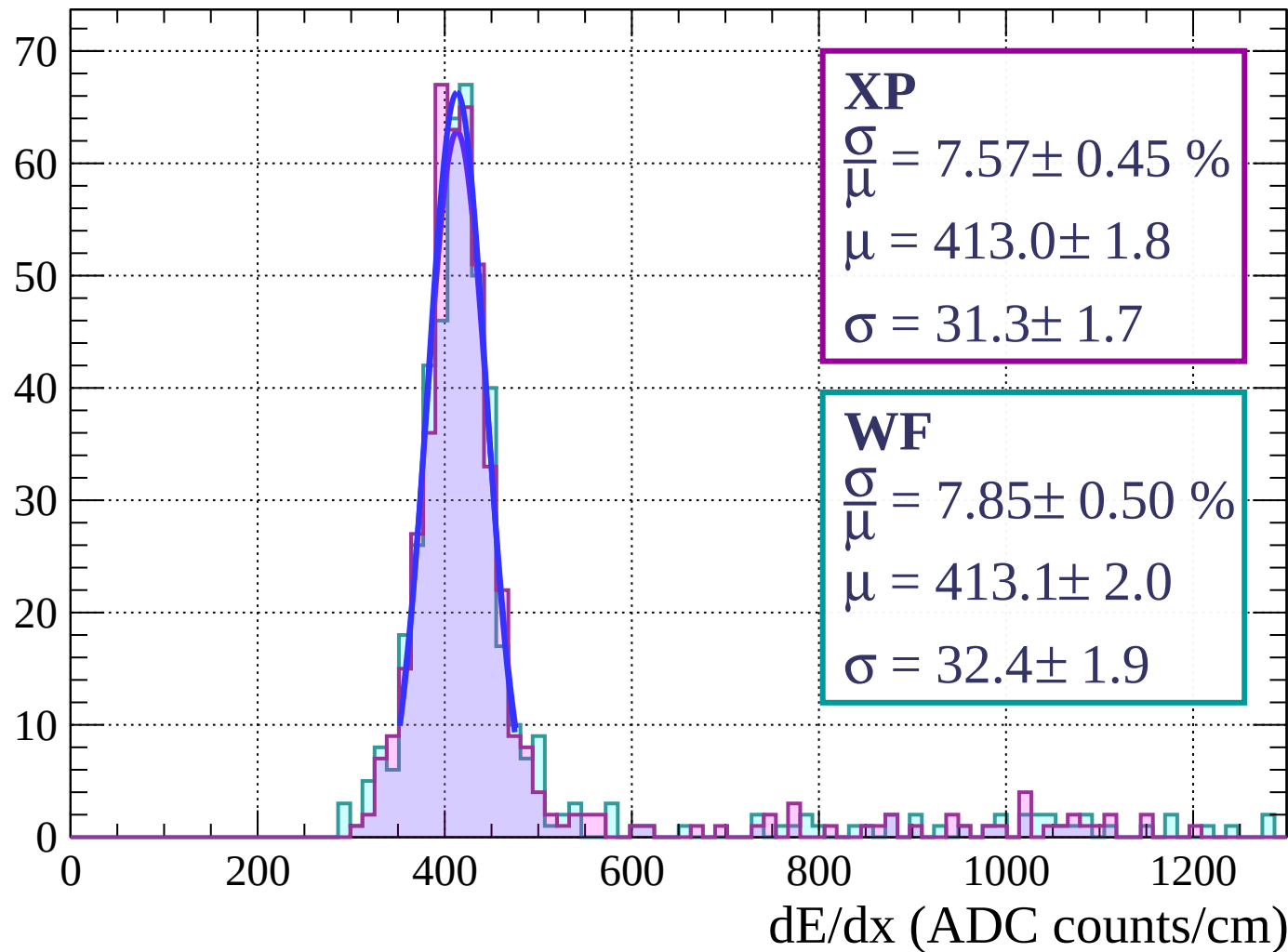
$$\mu = 415.0 \pm 2.5$$

$$\sigma = 31.4 \pm 2.2$$



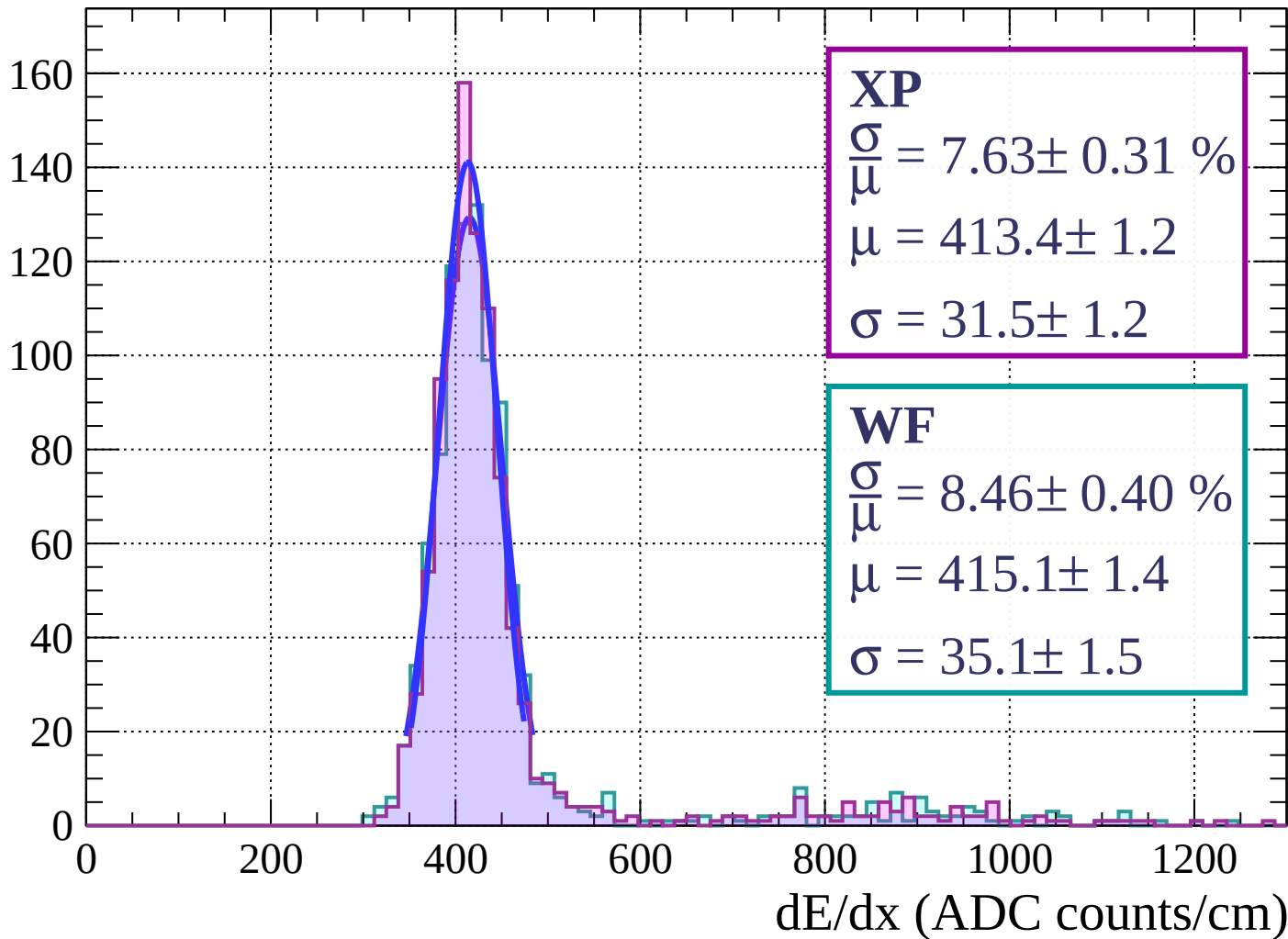
Energy loss | $-30 < \theta < -24$

Count

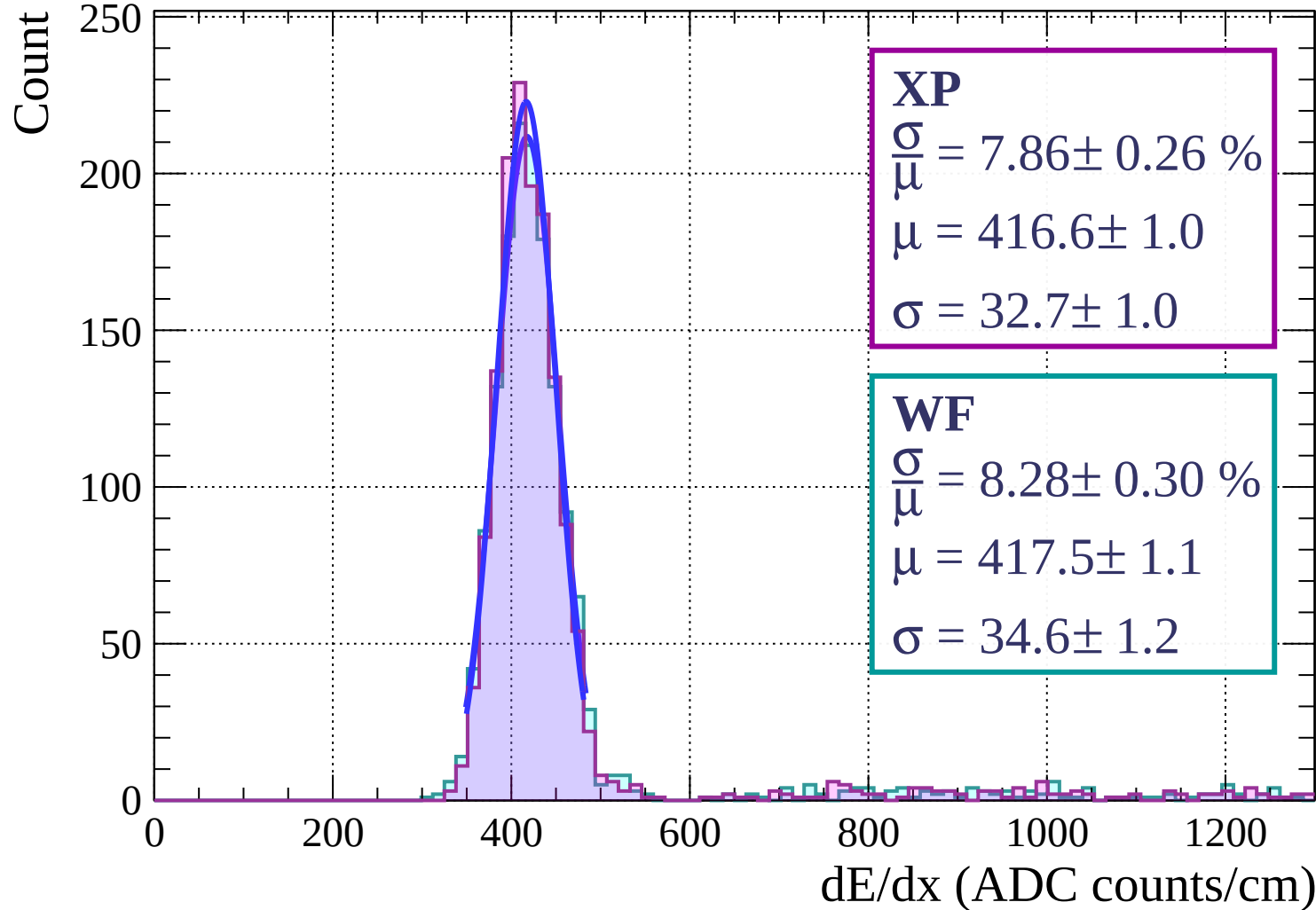


Energy loss | $-24 < \theta < -18$

Count

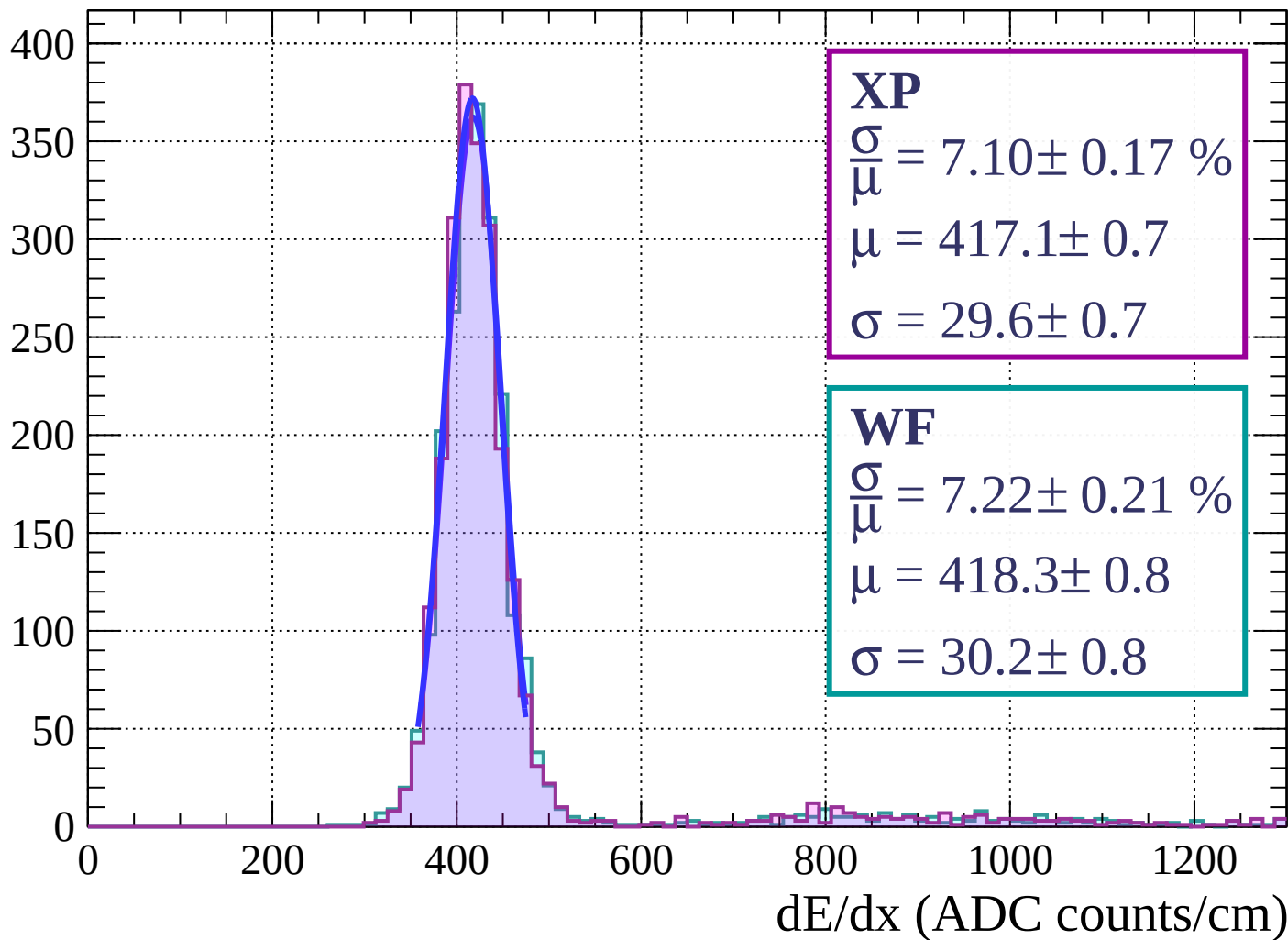


Energy loss | $-18 < \theta < -12$

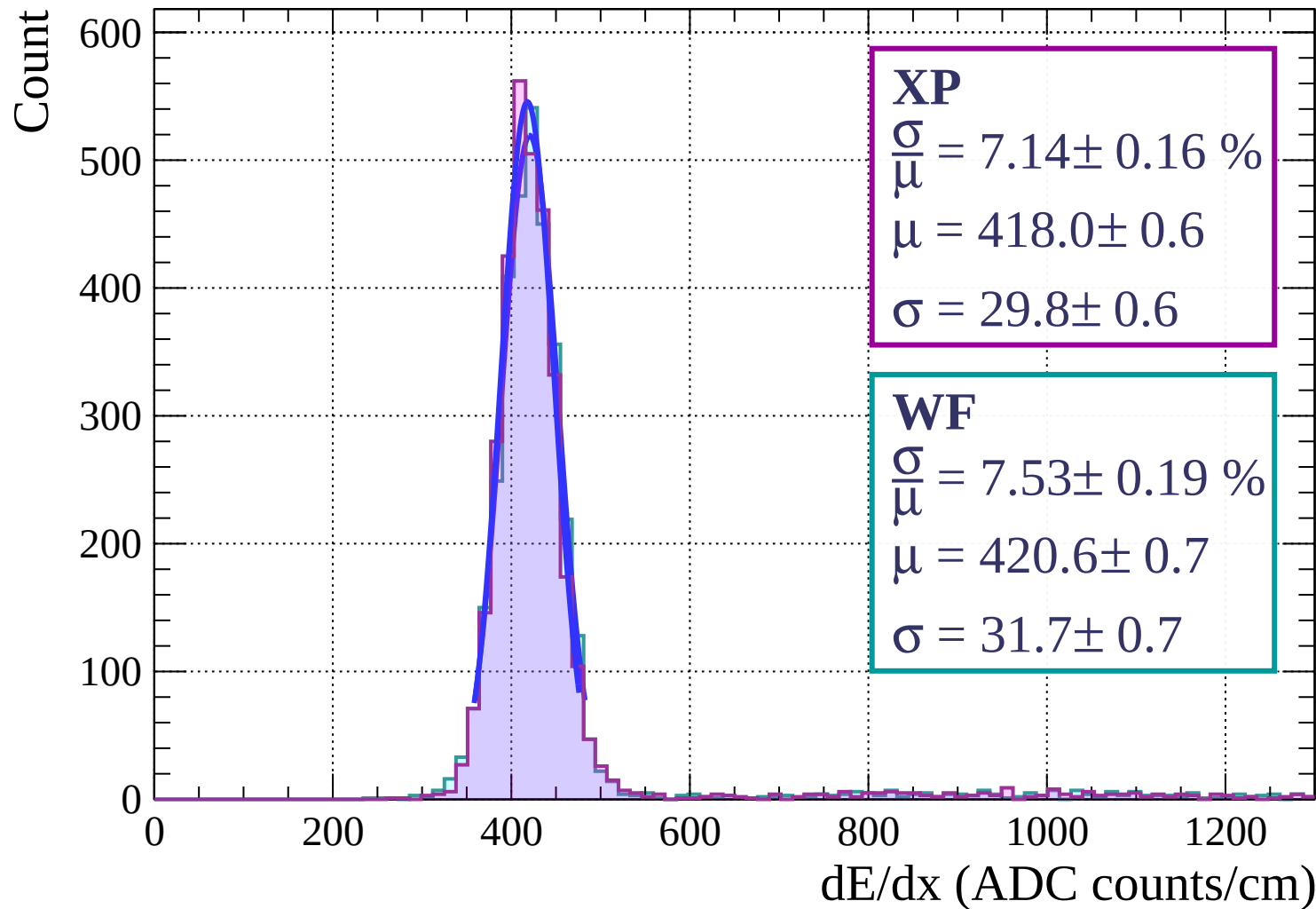


Energy loss | $-12 < \theta < -6$

Count

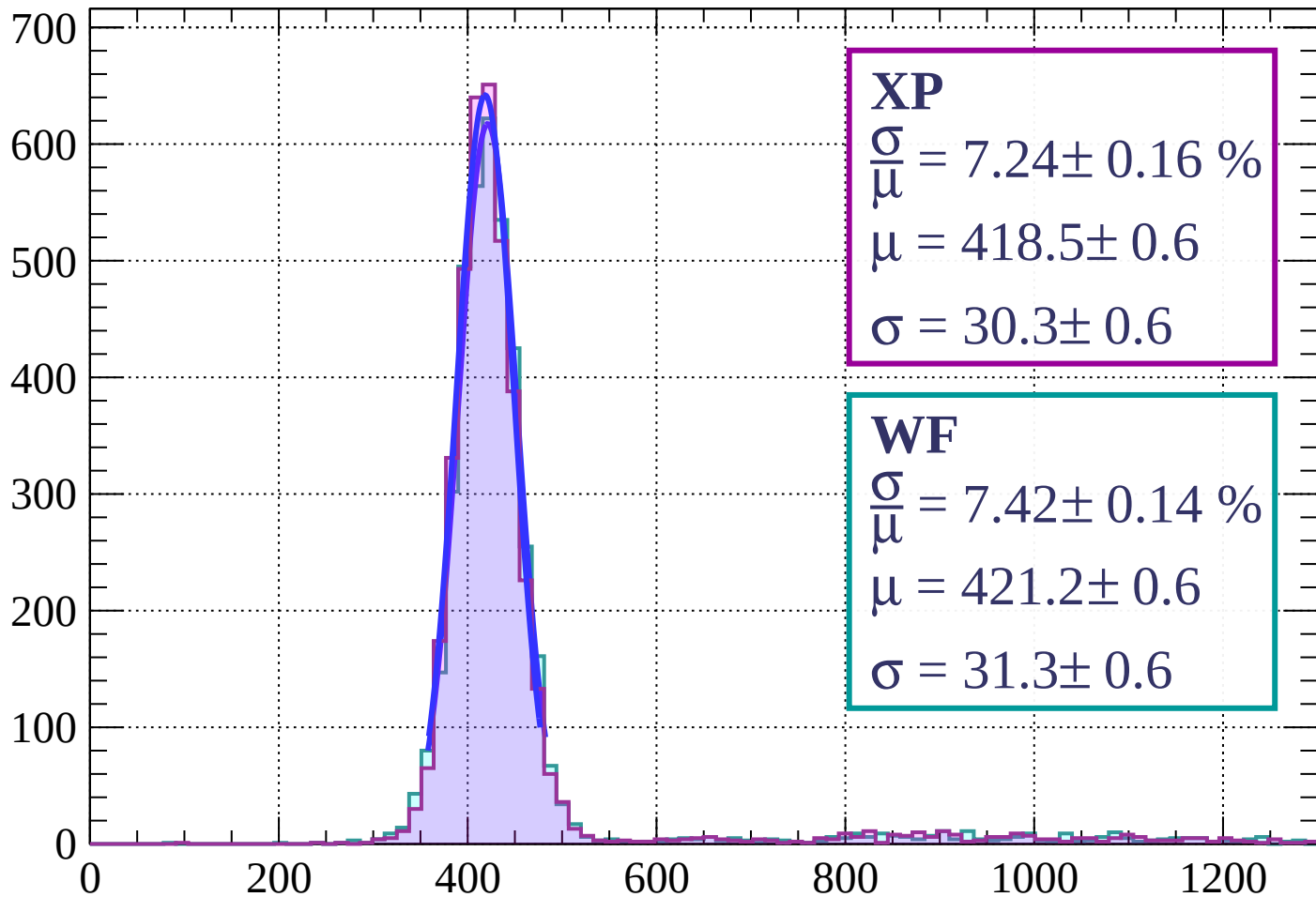


Energy loss | $-6 < \theta < 0$



Energy loss | $0 < \theta < 6$

Count



XP

$$\frac{\sigma}{\mu} = 7.24 \pm 0.16 \%$$

$$\mu = 418.5 \pm 0.6$$

$$\sigma = 30.3 \pm 0.6$$

WF

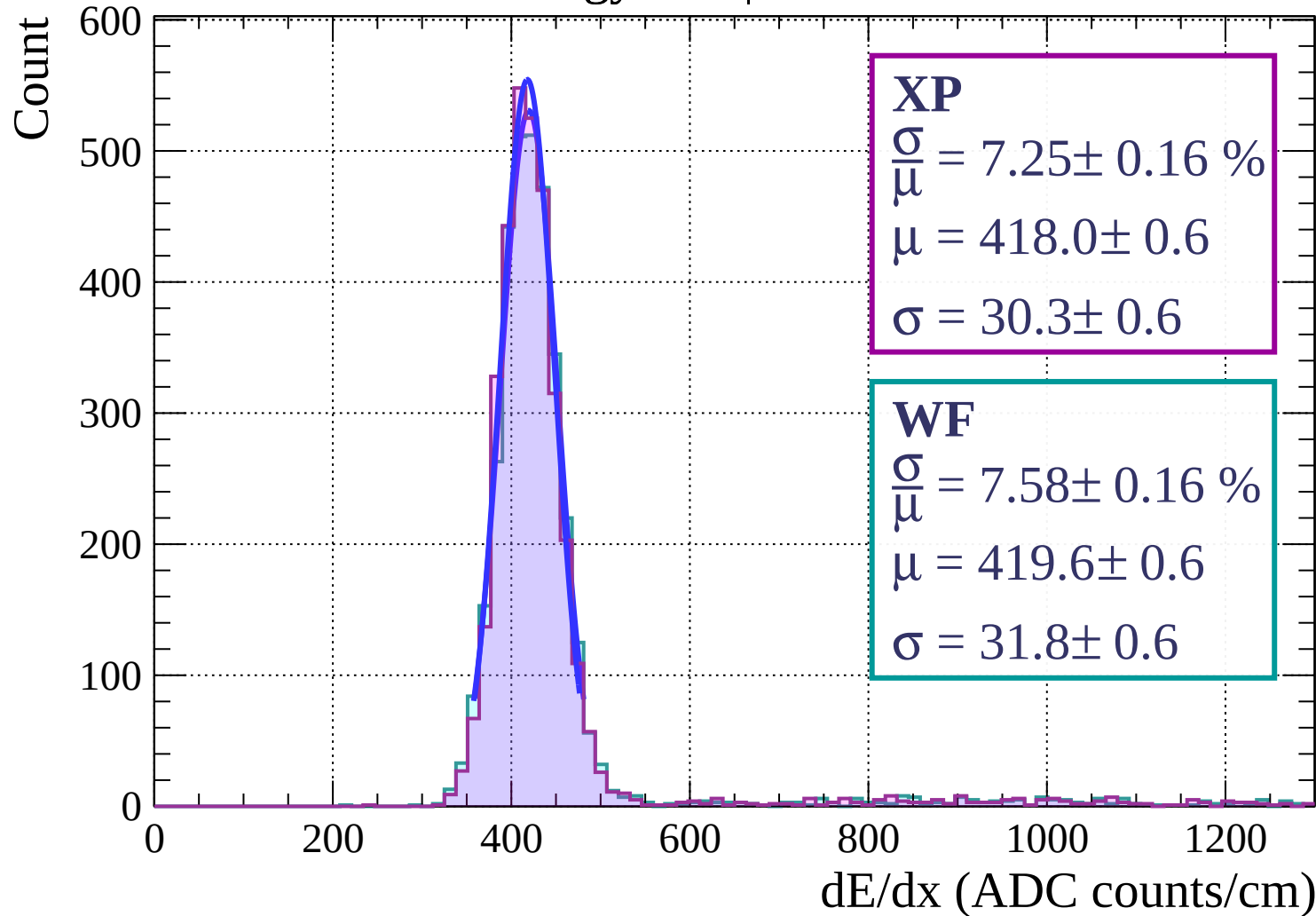
$$\frac{\sigma}{\mu} = 7.42 \pm 0.14 \%$$

$$\mu = 421.2 \pm 0.6$$

$$\sigma = 31.3 \pm 0.6$$

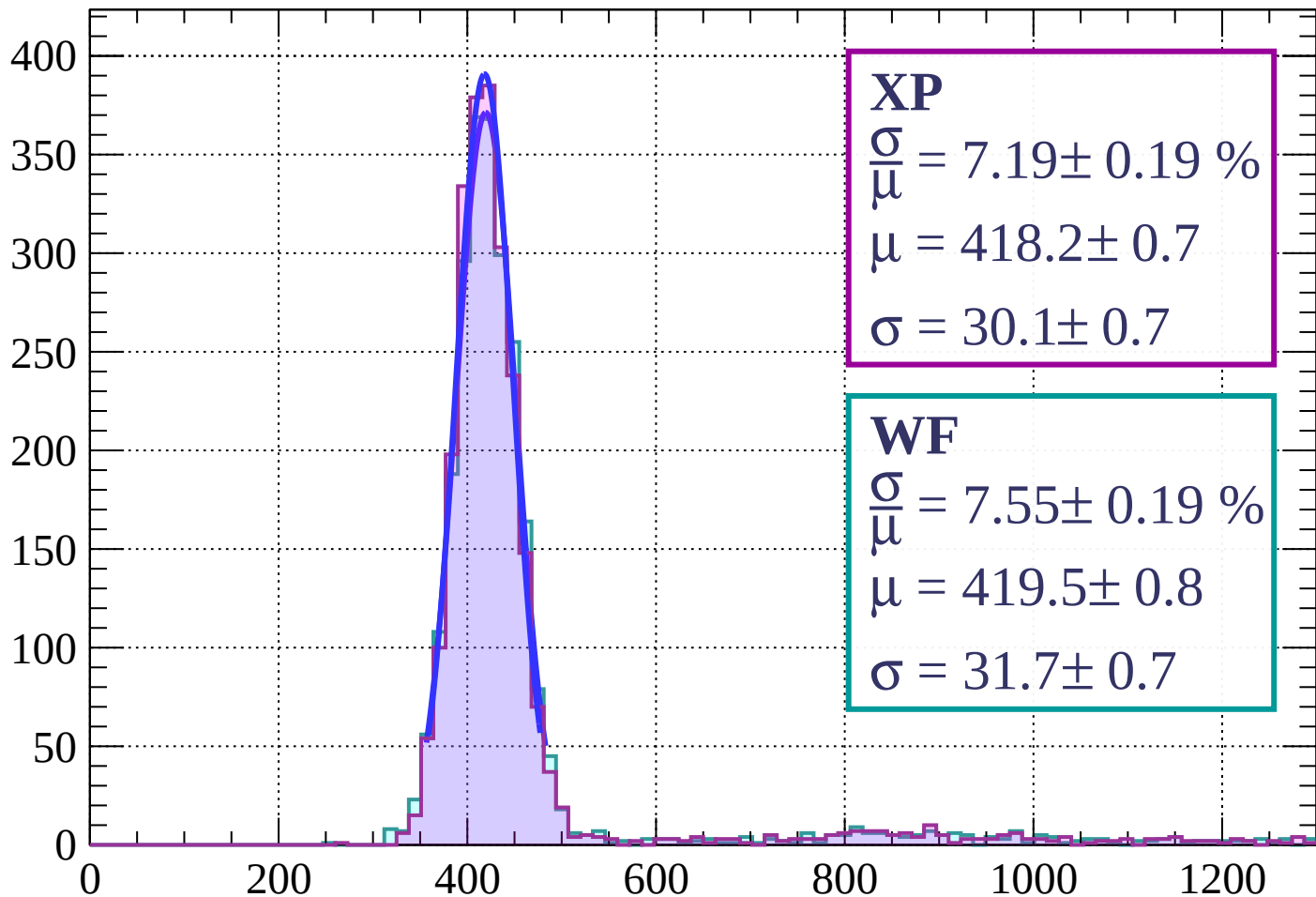
dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 12$



Energy loss | $12 < \theta < 18$

Count



XP

$$\frac{\sigma}{\mu} = 7.19 \pm 0.19 \%$$

$$\mu = 418.2 \pm 0.7$$

$$\sigma = 30.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.55 \pm 0.19 \%$$

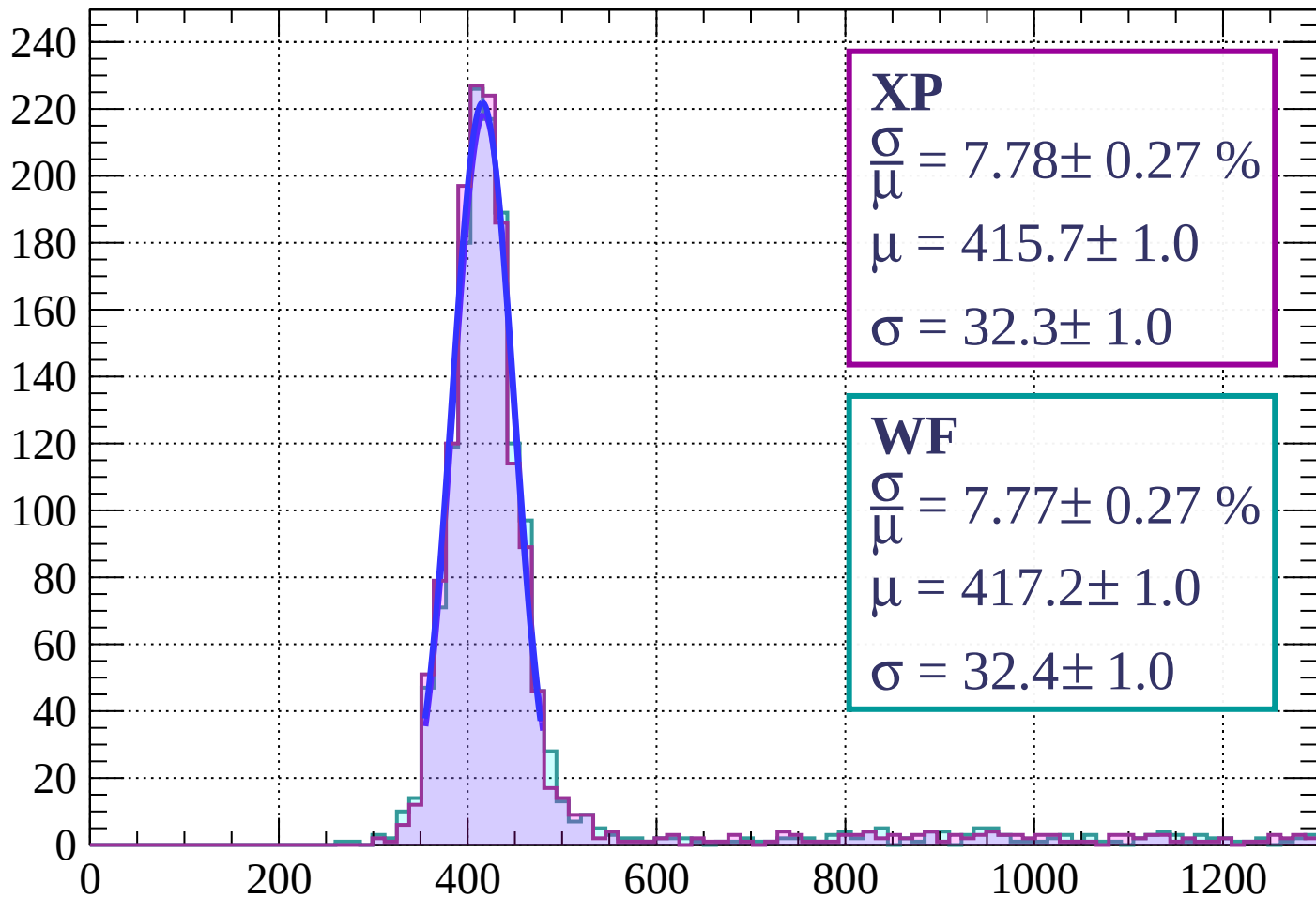
$$\mu = 419.5 \pm 0.8$$

$$\sigma = 31.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 24$

Count



XP

$$\frac{\sigma}{\mu} = 7.78 \pm 0.27 \%$$

$$\mu = 415.7 \pm 1.0$$

$$\sigma = 32.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 7.77 \pm 0.27 \%$$

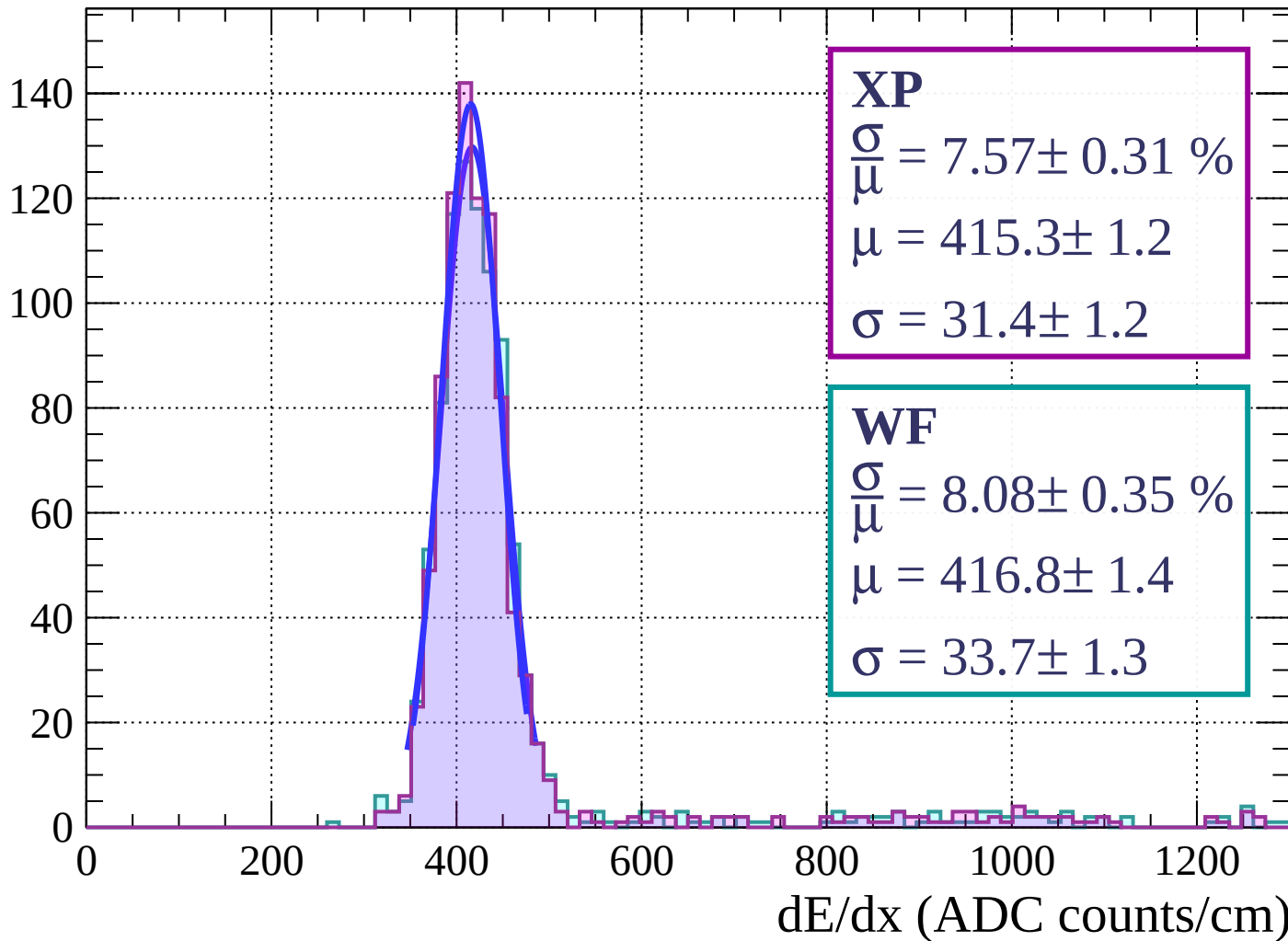
$$\mu = 417.2 \pm 1.0$$

$$\sigma = 32.4 \pm 1.0$$

dE/dx (ADC counts/cm)

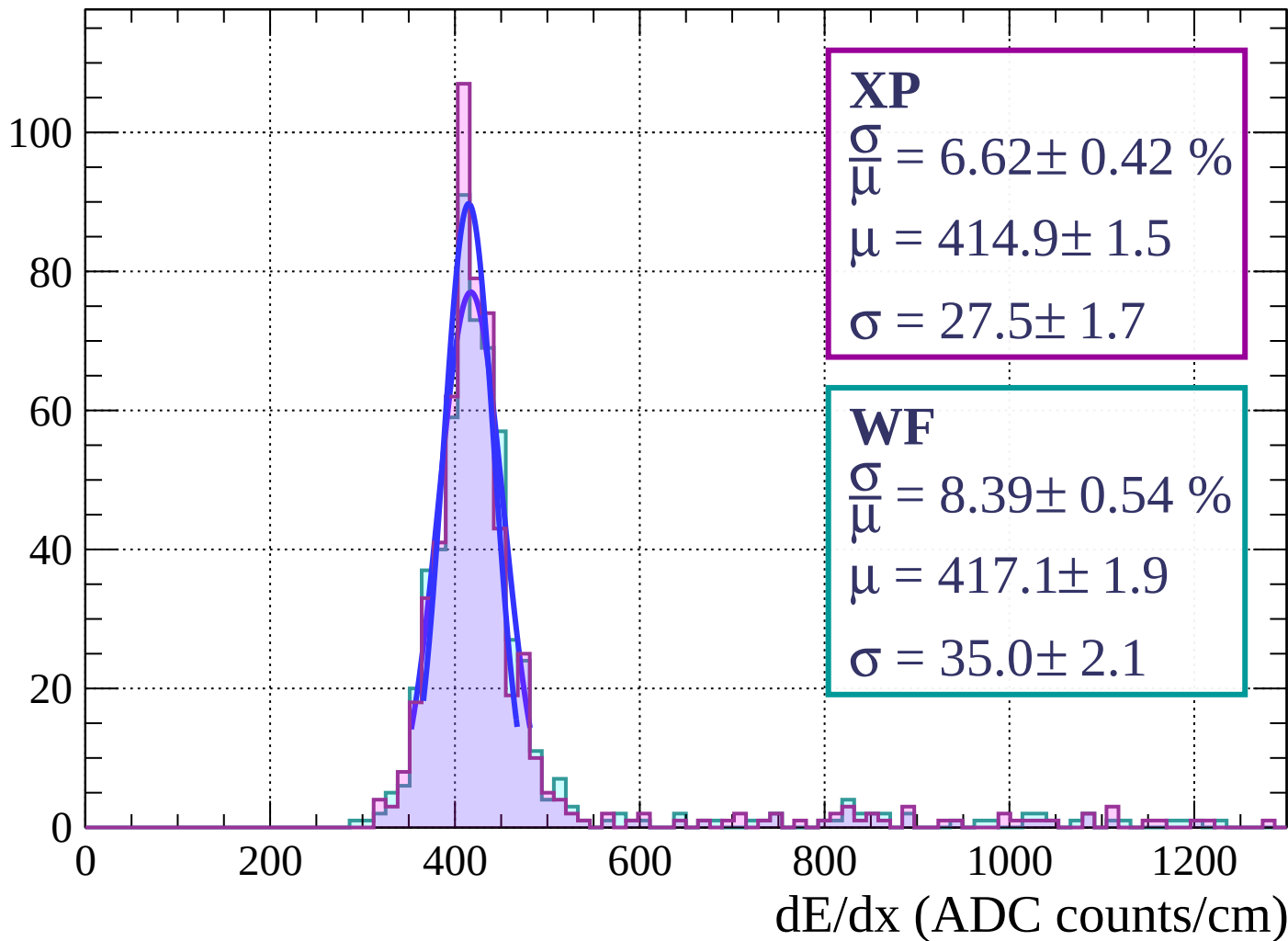
Energy loss | $24 < \theta < 30$

Count



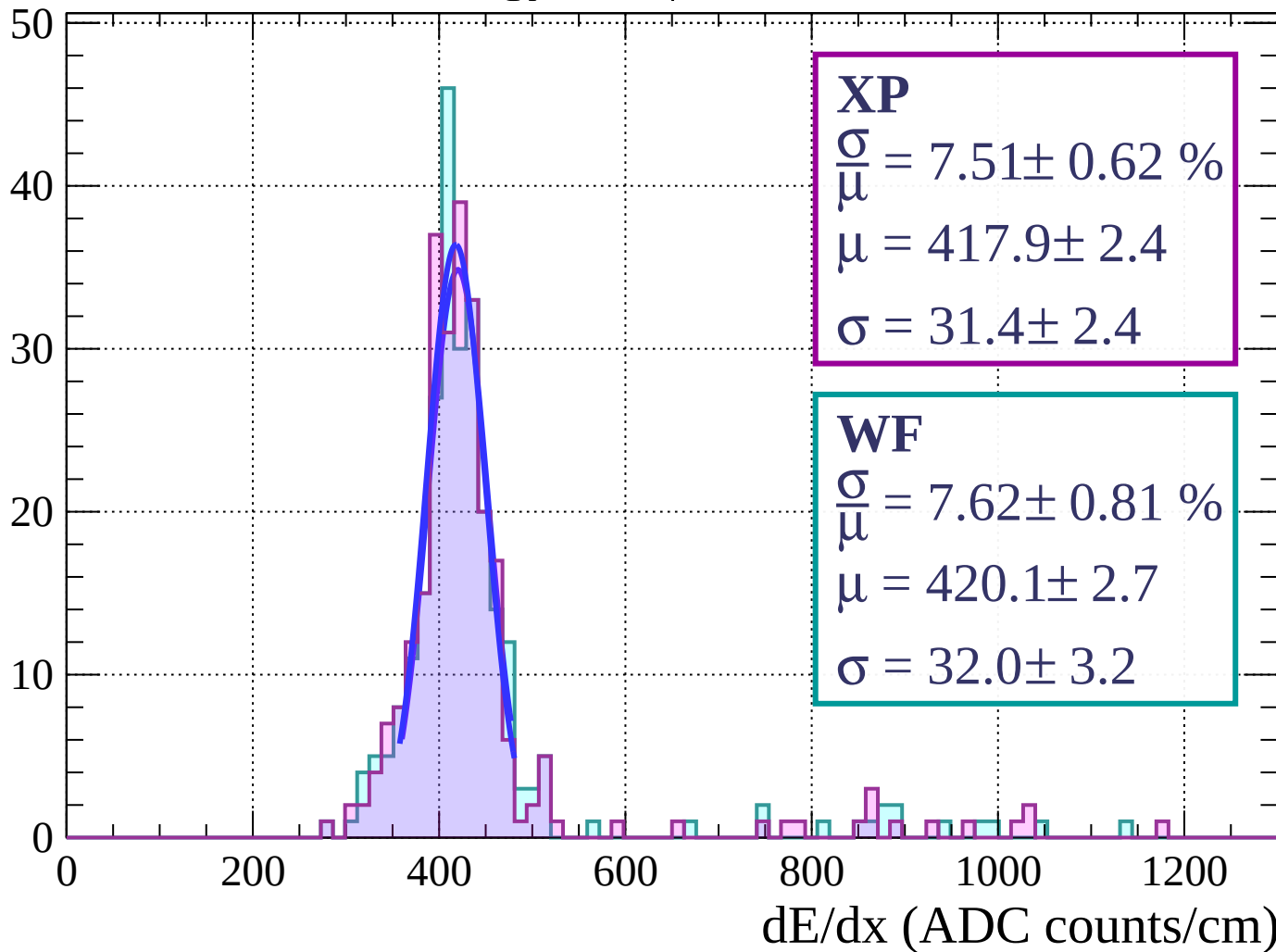
Energy loss | $30 < \theta < 36$

Count



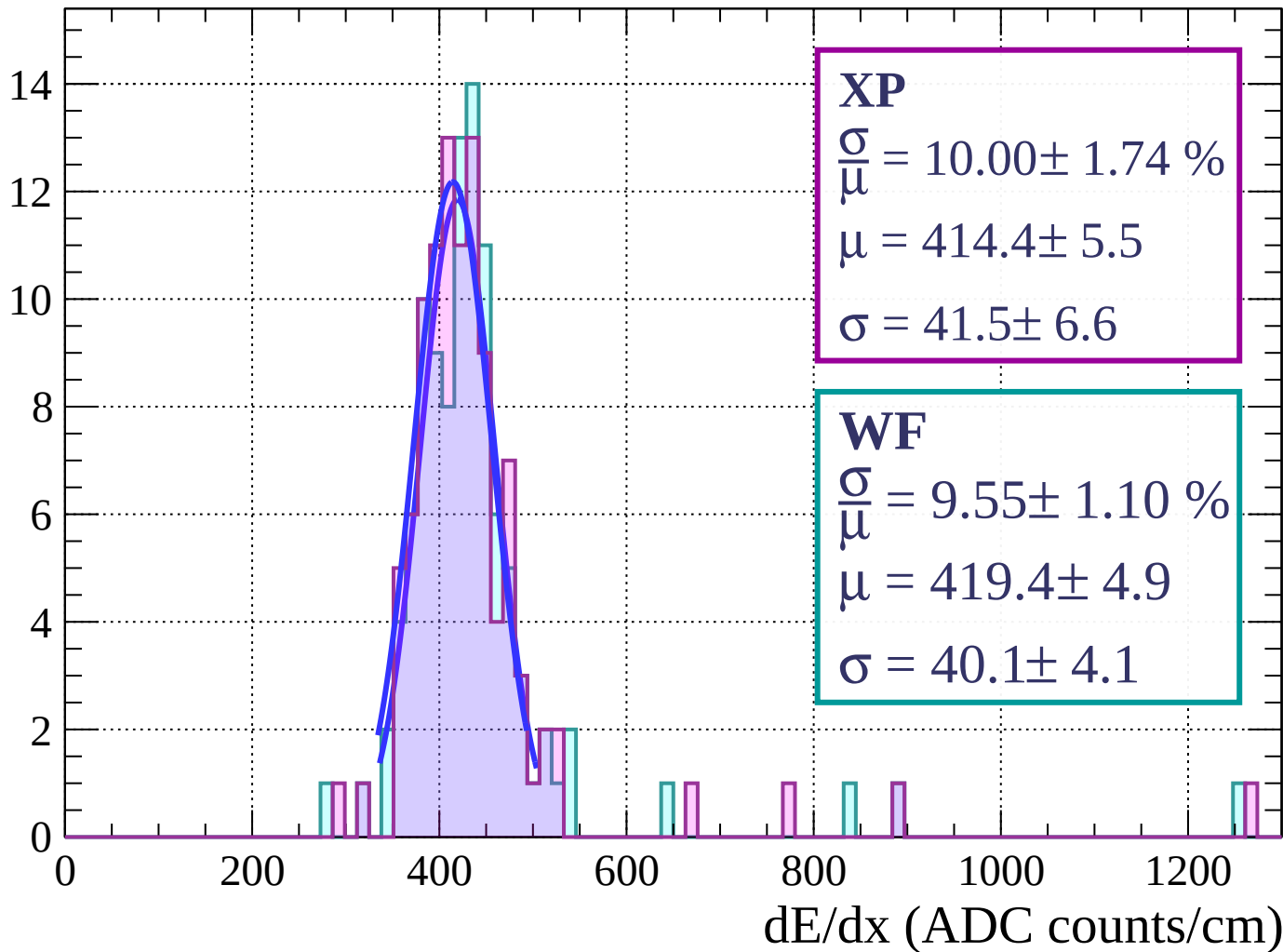
Energy loss | $36 < \theta < 42$

Count



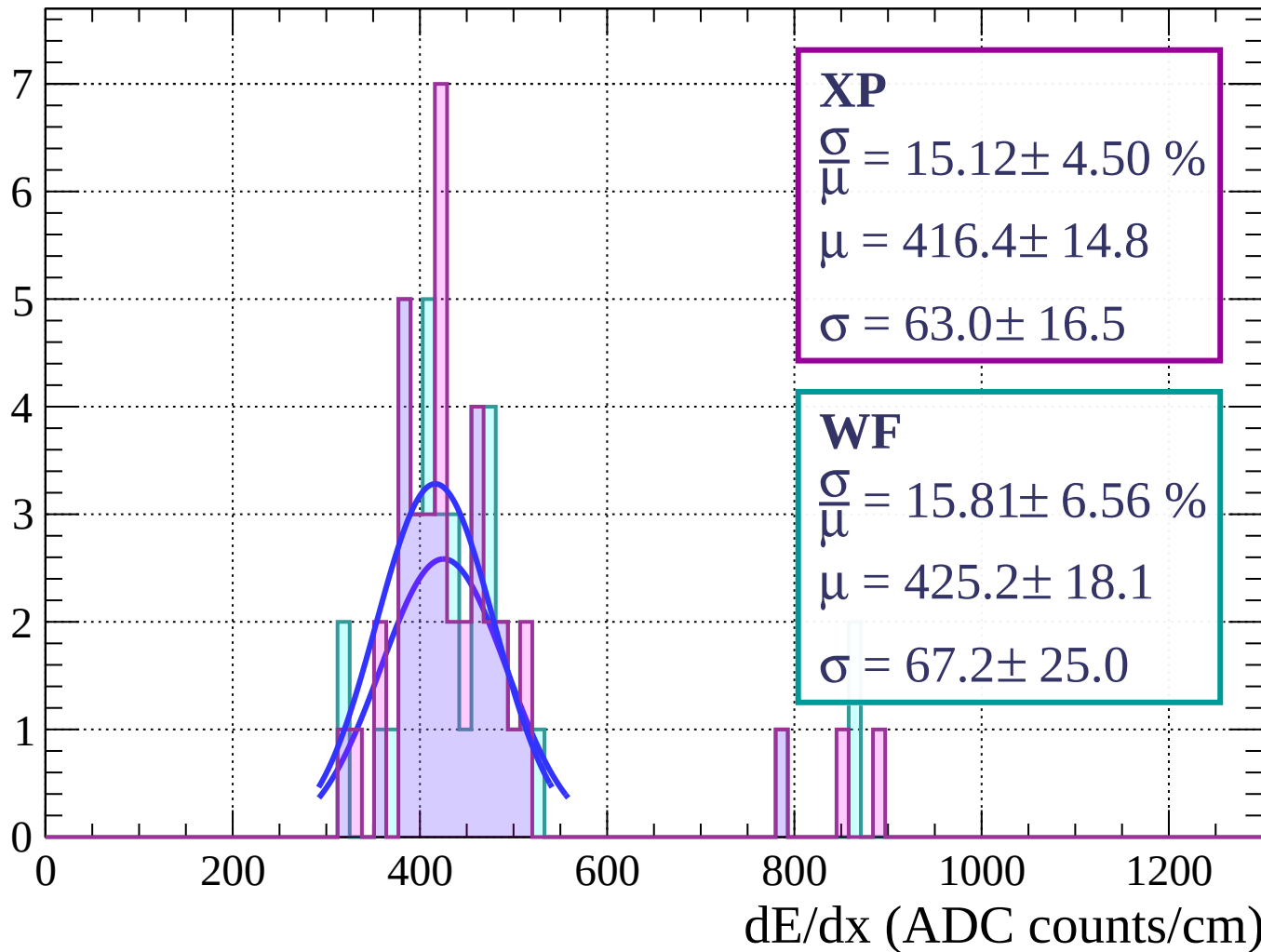
Energy loss | $42 < \theta < 48$

Count



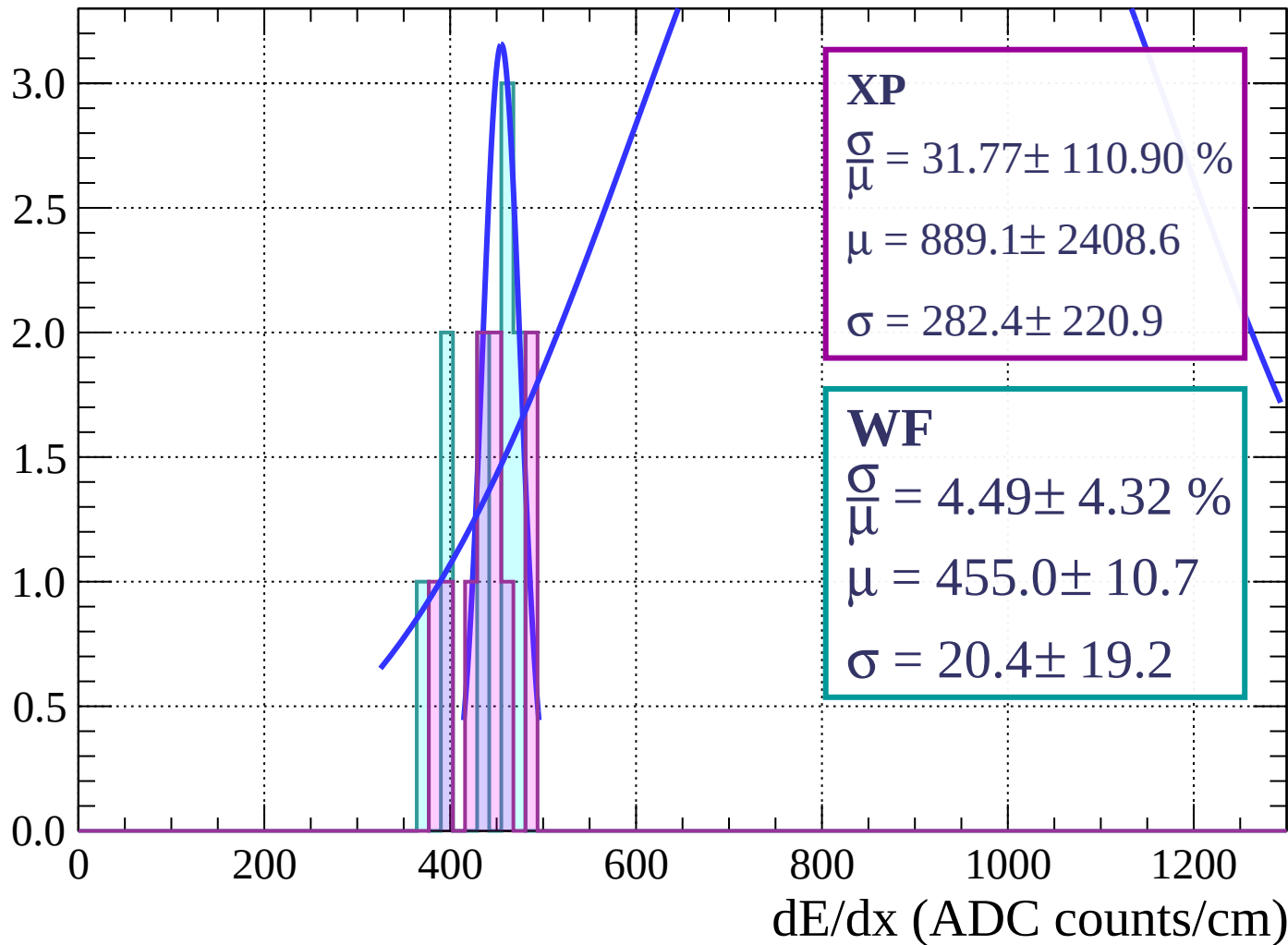
Energy loss | $48 < \theta < 54$

Count



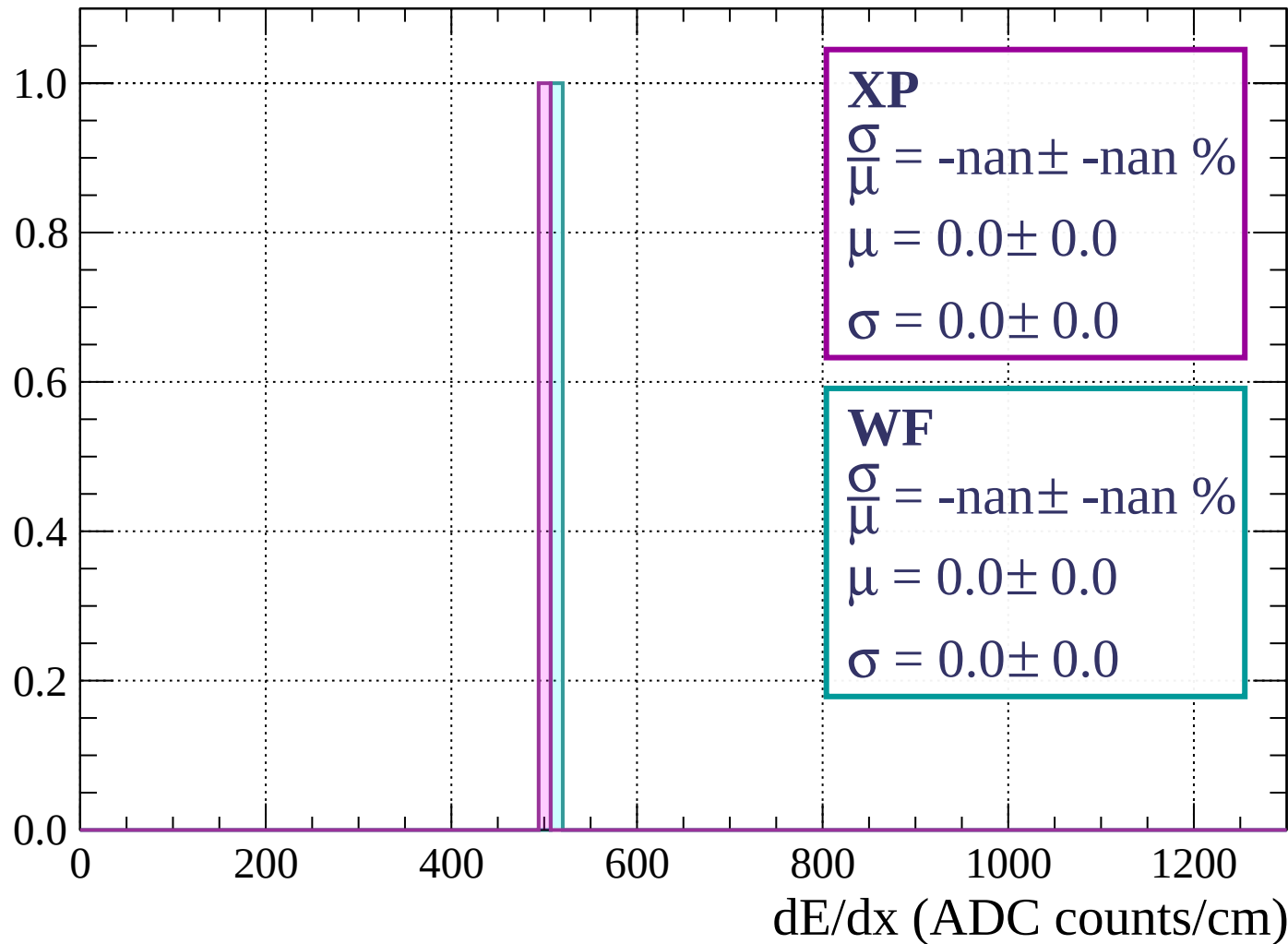
Energy loss | $54 < \theta < 60$

Count



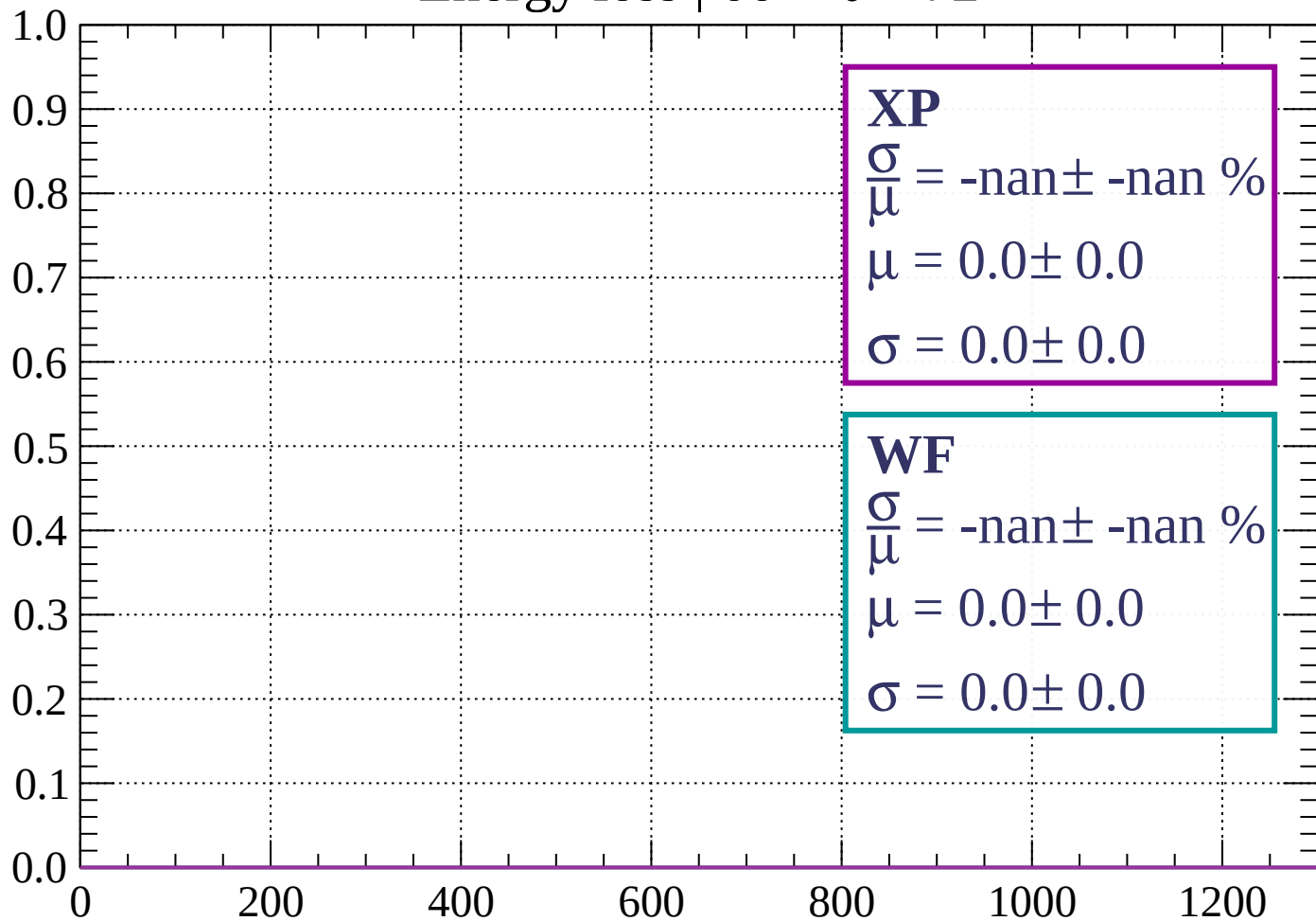
Energy loss | $60 < \theta < 66$

Count



Energy loss | $66 < \theta < 72$

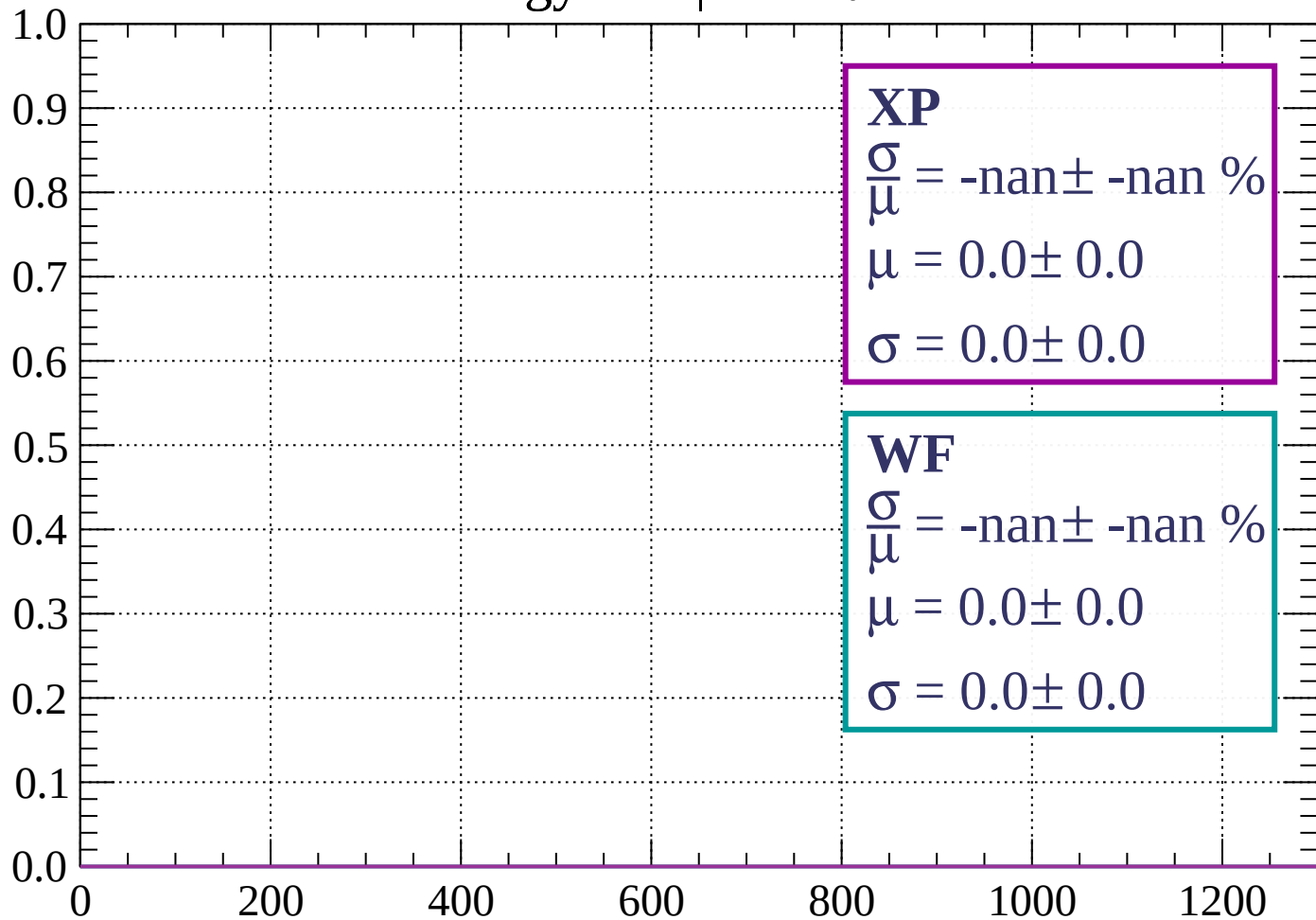
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 78$

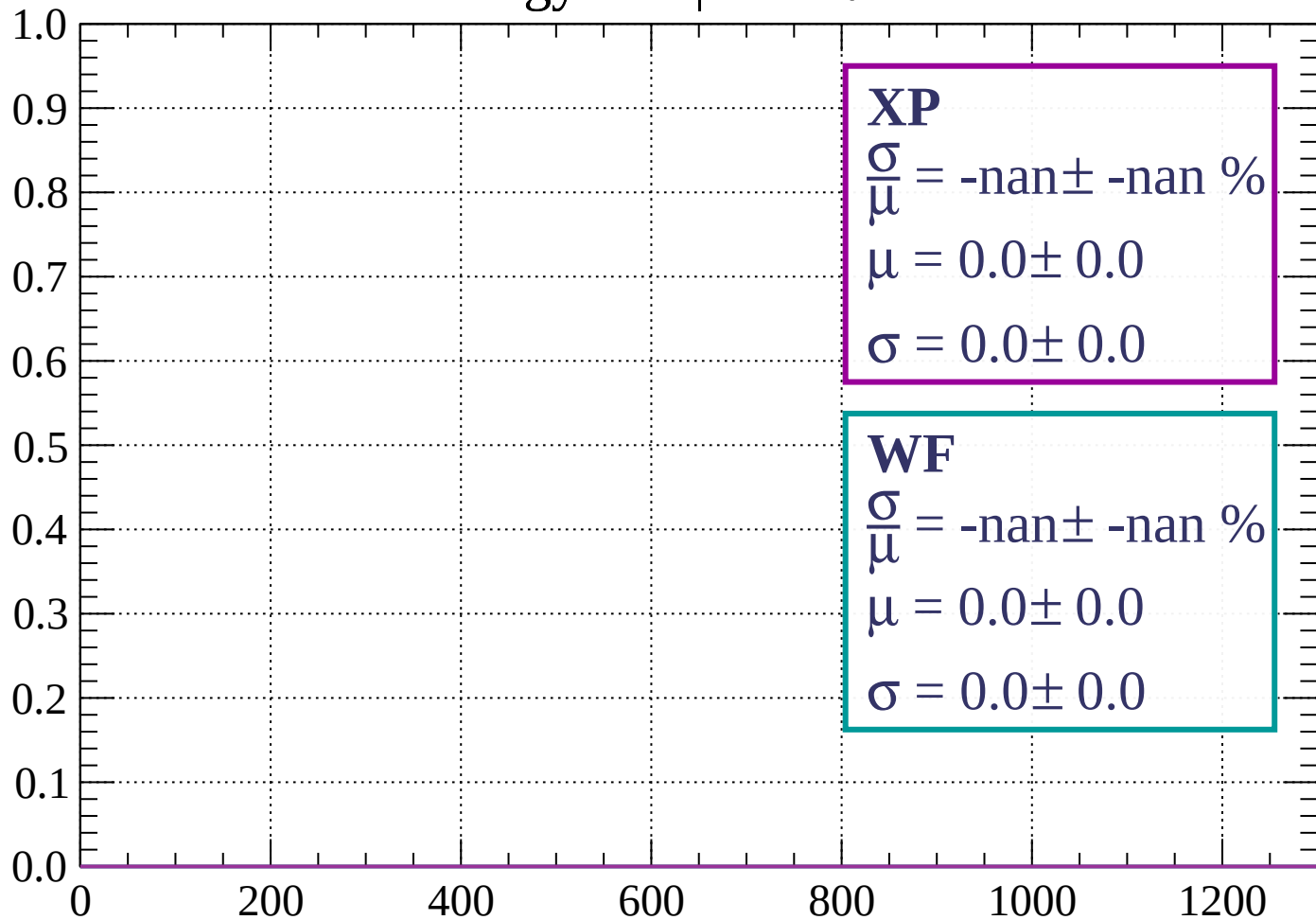
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 84$

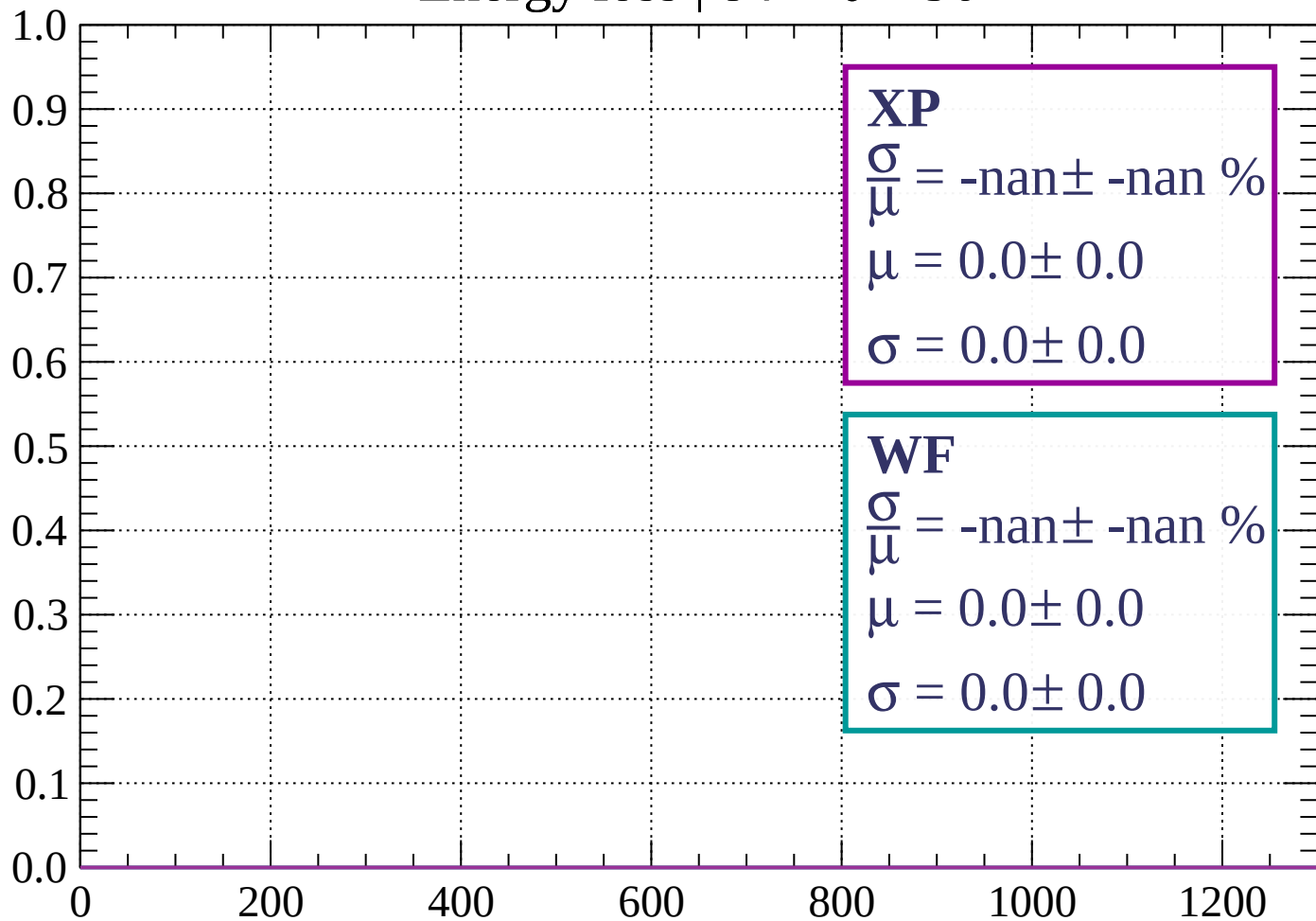
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 90$

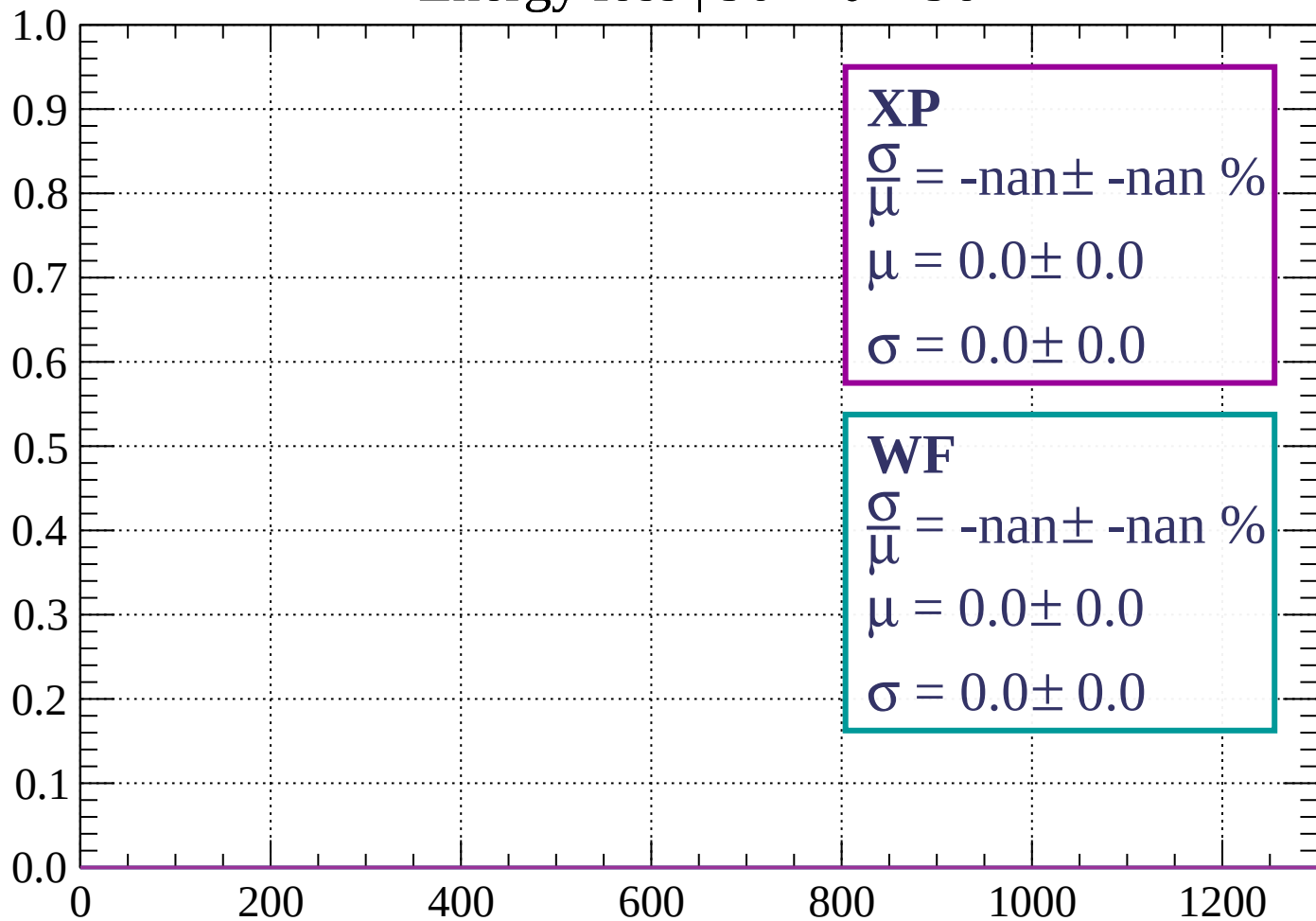
Count



dE/dx (ADC counts/cm)

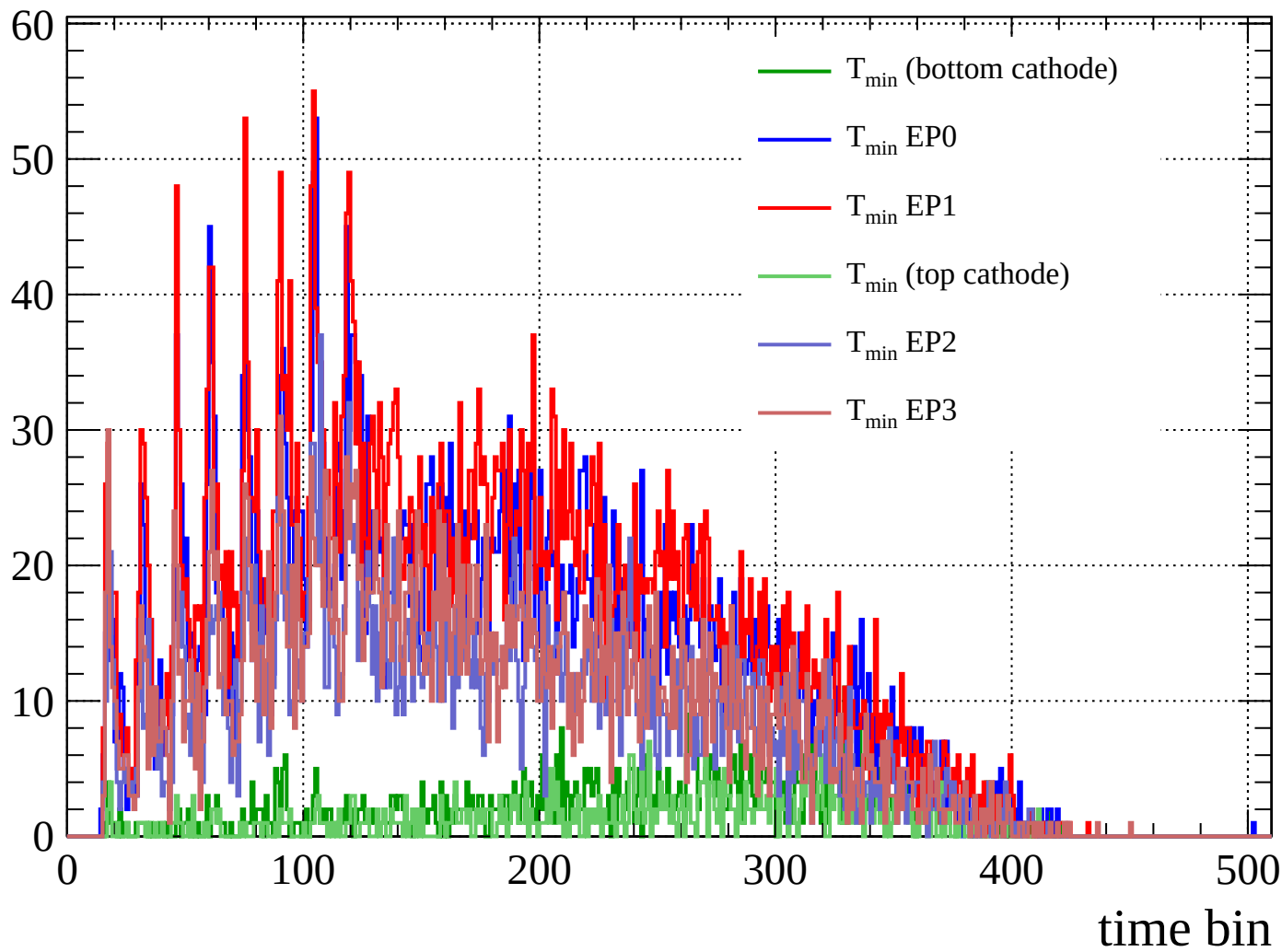
Energy loss | $90 < \theta < 96$

Count



dE/dx (ADC counts/cm)

Count



Count

